Applications of Combinatorial Stochastic Processes to Phylogenetics and Elsewhere.

by

Zachary McNulty

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Committee in charge:

Professor Steven N. Evans, Chair
Professor Shirshendu Ganguly
Professor Fraydoun Rezakhanlou

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University of California, Berkeley

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Abstract

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Abstract; Invasive brag; forbearance.

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Part I

Completed Projects

Chapter 1

An Improved Bipartition Cover Bound for the Multispecies Coalescent Model

Abstract

Bipartition cover probabilities quantify whether a collection of gene trees contains every bipartition of the underlying species tree, a condition used by some species-tree inference methods to restrict the search space without excluding the true tree. We study this problem under the multispecies coalescent (MSC) model and derive topology-free upper bounds on the number of loci needed to obtain a bipartition cover with prescribed confidence, improving significantly upon existing bounds. Practically, our bounds remain below biologically realistic numbers of loci across a substantially broader range of parameter settings, expanding their usefulness for empirical datasets. Theoretically, our analysis sharpens our understanding of coalescence under the MSC model and develops new asymptotics for these bounds and absorption times under Kingman's coalescent in the natural short branch regime. Simulations across several species-tree topologies confirm that the new bounds substantially improve upon existing bounds, although their tightness varies with topology.

1.1 Introduction

Consensus methods have grown in popularity in phylogenetics as large genomic datasets have revealed the limits of single-gene tree inference. Evolutionary processes such as incomplete lineage sorting, gene duplication and loss, and horizontal gene transfer can cause individual gene trees to differ from the true species tree. Consensus and summary-based methods aim to reconcile this discordance by aggregating information across many loci, providing statistically consistent estimates of species relationships under realistic models of gene tree variation.

However, even for a relatively small number of species, the space of possible species tree topologies is intractably large. Because of this, it is often necessary to reduce the search space somehow. One common strategy is to restrict to species trees whose set of bipartitions is contained in the set of bipartitions generated by the gene trees, possibly expanding this set with some collection of heuristics. This is the case for many modern phylogenetic algorithms, including FastRFS [64] and the widely used ASTRAL family: ASTRAL [46], ASTRAL-II [45], ASTRAL-III [69], and wASTRAL [68].

In the case the true species tree lies in this restricted space, we say the gene trees form a *bipartition cover* of the species tree. When this occurs, ASTRAL comes with some strong finite-sample guarantees [59]. When it does not occur, ASTRAL provides no guarantees on how the estimated tree will relate to the true species tree. As a result, it is important to know how likely such a bipartition cover is to occur when assessing whether these finite-sample guarantees apply. Moreover, since an experimenter must decide how many loci to collect before the underlying species tree topology is known, they cannot calibrate the required sample size to that topology. A topology-free upper bound provides a conservative benchmark for the sample size required to obtain a bipartition cover with a prescribed probability. The paper by Uricchio, Warnow, and Rosenberg [63] develops such a bound as a function only of two key parameters: the number of species k and the minimum branch length $T_{\min}$ of the tree.

In this paper, we identify a few areas where the bound of Uricchio, Warnow, and Rosenberg [63] is lossy and develop some topology-free improvements. Our main contributions come from careful analysis of various "worst cases" of the MSC model: species tree topologies that somehow make coalescence difficult and certain bipartitions challenging to recover. Beyond this, since all these bounds are complicated functions of the underlying parameters, we develop some asymptotics to describe the growth rates of these bounds as functions of these parameters in a few natural regimes.

The Multispecies Coalescent Model

We start with a brief discussion of the *multispecies coalescent (MSC) model*, which is the probabilistic model underlying all the theoretical guarantees of ASTRAL and many other phylogenetic inference algorithms. The MSC model describes how gene tree topologies are generated from a given species tree [67, 55]. It is in a sense the simplest model that can explain the phenomenon of *incomplete lineage sorting (ILS)*, the observation that the most common gene tree topology can differ from the species tree topology. This phenomenon makes species tree inference difficult: the intuitive "majority vote" estimator is not consistent.

The MSC model is essentially just a constrained version of Kingman's coalescent: going backwards in time along each branch of the species tree, we assume the surviving gene lineages coalesce according to Kingman's coalescent. Namely, each pair of lineages coalesce at rate one. Figure 1.1 gives an example of the MSC and illustrates how gene tree topologies can differ from the species tree. More concretely, let

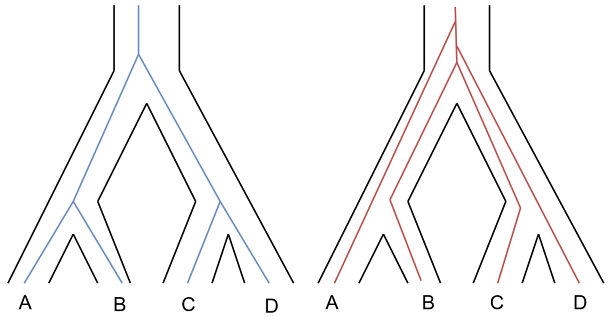

Figure 1.1: Two different gene trees generated from the same species tree under the multi-species coalescent model. The black lines outline the underlying species tree topology, and the colored lines indicate the gene lineages

$$g_{ij}(T) := \sum_{k=j}^{i} \frac{\exp(-\binom{k}{2}T)(2k-1)(-1)^{k-j} j_{(k-1)} i_{[k]}}{j!(k-j)! i_{(k)}} \tag{1.1.1}$$

be the probability that i lineages coalesce into exactly j lineages in time T under Kingman's coalescent. Here $a_{(k)} = a(a+1)\cdot \ldots \cdot (a+k-1)$ and $a_{[k]} = a(a-1)\cdot \ldots \cdot (a-k+1)$ denote the rising and falling factorial respectively, and the time T is measured in coalescent units. The function $g_{ij}(T)$ is the basis of the original bound in Uricchio, Warnow, and Rosenberg [63] and all the new bounds we develop. In the setting of the MSC model, T represents the length of a given branch of the species tree, and g_{ij} describes the distribution of the number of lineages that remain uncoalesced at the end of that branch.

The Original Bound and Main Results

Under the MSC model, Uricchio, Warnow, and Rosenberg [63] show:

Theorem 1.1.1 (Original Bipartition Cover upper bound)**.** *If we have a species tree with k species and minimum branch length $T_{\min}$ (in coalescent units) and a set $\mathcal{G}$ of at least*

$$M_o(k, T_{\min}) := \frac{\log\left(\frac{1-q}{k-3}\right)}{\log\left(1 - g_{k-2,1}(T_{\min})\right)} \tag{1.1.2}$$

gene trees sampled independently under the MSC model, then $\mathcal{G}$ forms a bipartition cover with probability at least q.

In this work, we aim to develop a topology-free improvement of this upper bound. Our work relies on careful analysis of various "worst-case" topologies when it comes to phylogenetic inference. For this purpose, we define two tree types: a *caterpillar tree* and a *balanced tree.*

A *balanced tree* is any rooted tree so that for every vertex the number of leaves in its two subtrees differs by at most one. A *caterpillar tree* is a tree where every non-leaf vertex has one child who is a leaf. Figure 1.2 provides a simple illustration.

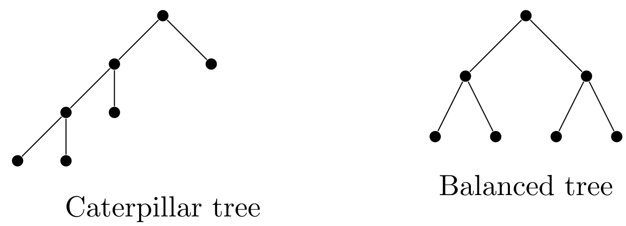

Figure 1.2: The two main extremal tree topologies: caterpillar and balanced trees

These two tree topologies represent opposite extremes for coalescent behavior: one maximally balanced and the other maximally unbalanced. Caterpillar trees present a combinatorial bottleneck for phylogenetic inference, as many of their bipartitions involve very large subsets of taxa. Balanced trees, however, present a more subtle and severe difficulty: they induce a *coalescent bottleneck*, in which lineages are so evenly dispersed that coalescence is systematically delayed.

This distinction is made precise in Lemma 1.2.1, which identifies caterpillar trees as extremal for increasing functions of descendant counts, and in Lemma 1.2.8, which shows that balanced trees are stochastically worst-case for the number of surviving lineages under the multispecies coalescent. Together, these lemmas underpin our main result, which appears as Theorem 1.2.9 below:

Theorem (Main result)**.** For $\ell \geqslant 2$, let $p_\ell(T)$ denote the probability that the lineages descending from a balanced ℓ-taxon tree, with all branch lengths equal to T, coalesce to a single lineage. Then n independently sampled gene trees form a bipartition cover with probability at least q whenever

$$\sum_{\ell=2}^{k-2} \big(1 - p_\ell(T_{\min})\big)^n \leqslant 1 - q.$$

In particular, it suffices that

$$n \geqslant \frac{\log\left(\frac{k-3}{1-q}\right)}{-\log(1 - p_{k-2}(T_{\min}))}.$$

Due to the recursive structure of a balanced tree, these probabilities p_ℓ are easy to compute. Empirically, we observe this improves over Theorem 1.1.1 by several orders of magnitude in some regimes, especially when the minimum branch length $T_{\min}$ is small. Moreover, through careful asymptotic analysis of the coalescent probabilities, we show that, in the fixed-T, large-k regime, the improvement factor is bounded below asymptotically by $\pi^2/(2T)$ as $T \downarrow 0$. Lemma 1.7.4 below makes this precise.

1.2 Results and Discussion

To guide our work, we outline the proof of the original bound, and identify areas where it is lossy.

Proof Outline of Theorem 1.1.1. Every tree contains the trivial bipartitions $\{x\}|(S-\{x\})$, so we may restrict attention to nontrivial bipartitions. There are $k-3$ nontrivial bipartitions in a species tree: one for each internal edge[1].

1. By union bound we can just bound the probability a specific species tree bipartition occurs in the gene trees.

2. By independence of gene trees we can reduce to the case of a single gene tree.

3. Suppose φ_e is the bipartition associated to edge e in a species tree. One way φ_e can occur in the gene tree is if all lineages below e coalesce by the time they exit e. If there are k_e such lineages, we can lower bound this probability by the probability k_e lineages coalesce down to one along just the single edge e.

4. The worst-case for the previous step is if edge e is as short as possible (length $T_{\min}$) and the number of lineages that have to coalesce is as large as possible ($k_e = k-2$ since φ_e is nontrivial).

If we let E_n be the event all bipartitions occur in n gene trees and $E_{i,n}$ be the event bipartition φ_i occurs in at least one of the n gene trees, then combining the above gives

$$\begin{aligned}
\mathbb{P}(E_n) &\geqslant 1 - \sum_{i=1}^{k-3}(1-\mathbb{P}(E_{i,n})) \\
&\geqslant 1-(k-3)\cdot\max_i(1-\mathbb{P}(E_{i,n})) \\
&= 1-(k-3)(1-\min_i \mathbb{P}(E_{i,1}))^n \\
&\geqslant 1-(k-3)(1-g_{k-2,1}(T_{\min}))^n
\end{aligned}$$

from which we can invert to find the bound on n. $\square$

Bookkeeping of Descendant Counts

One step at which the proof of Theorem 1.1.1 is lossy is that we essentially bound

$$\mathbb{P}(E_{i,1}) \geqslant g_{\alpha_i,1}(T_{e_i}) \geqslant g_{k-2,1}(T_{\min})$$

[1]There's a slight subtlety: there are $k-2$ internal edges, but the two edges adjacent to the root induce the same, possibly trivial, bipartition

where α_i is the size of the part of bipartition φ_i that is "below the root". Namely, if φ_i is generated by cutting edge e_i, then α_i is the number of leaves below e_i, in the part not containing the root. Since both edges adjacent to the root generate the same bipartition, there is some ambiguity in the definition of our descendant count set $\{\alpha_i\}$. To resolve this, we make the decision to always choose the edge adjacent to the root with the least descendants. An example is given in Figure 1.3.

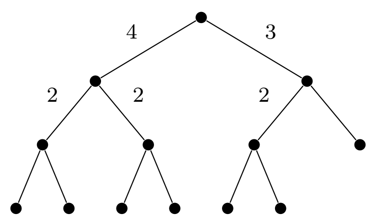

Figure 1.3: Example of the descendant counts $\{\alpha_i\}$ associated to the nontrivial bipartitions/edges of a phylogenetic tree. Note there is some subjectivity in deciding which edge adjacent to the root to take: depending on our choice our descendant counts would either be $\{\alpha_i\} = \{4, 2, 2, 2\}$ or $\{\alpha_i\} = \{3, 2, 2, 2\}$. To remove ambiguity, we always choose the edge with the least descendants.

For most edges e_i, the number of descendants α_i is far smaller than the total number of species k. By Lemma 1.5.4, we know $\ell \mapsto g_{\ell 1}(T_{\min})$ is a decreasing function of ℓ: the more lineages we start with, the harder it is to coalesce down to one. As a result, this second approximation $g_{\alpha_i,1} \geqslant g_{k-2,1}$ can be quite lossy. Nonetheless, to derive a topology-free bound we must assume these α_i are in a sense as large as possible. The following makes this precise.

Lemma 1.2.1 (Caterpillars Maximize Increasing Sums)**.** *Suppose $\{\alpha_i\}_{i=1}^{k-3}$ are the descendant counts of our rooted binary tree T. Moreover, suppose $f : \mathbb{N} \to \mathbb{R}$ is nondecreasing. Then $\sum_i f(\alpha_i)$ is maximized when T is the caterpillar tree, in which case $\{\alpha_i\} = \{2, 3, ..., k-2\}$.*

The proof of Lemma 1.2.1 can be found in Appendix 1.6. This result implies:

Corollary 1.2.2. *Suppose $h : \mathbb{N} \to [0, 1]$ is decreasing and $\mathbb{P}(E_{i,1}) \geqslant h(k_i)$. Then*

$$\mathbb{P}(E_n) \geqslant 1 - \sum_{\ell=2}^{k-2} \Big[1 - h(\ell)\Big]^n.$$

Proof of 1.2.2. By union bound and independence of the gene trees

$$1 - \mathbb{P}(E_n) \leqslant \sum_{i=1}^{k-3} (1 - \mathbb{P}(E_{i,n})) = \sum_{i=1}^{k-3} \Big[1 - \mathbb{P}(E_{i,1})\Big]^n \leqslant \sum_{i=1}^{k-3} \Big[1 - h(k_i)\Big]^n.$$

As h is decreasing and bounded in $[0, 1]$, the function $f(x) = [1 - h(x)]^n$ is increasing. Applying Lemma 1.2.1 to this f gives the desired result. □

From this, the main bound follows shortly.

Theorem 1.2.3. *Suppose $h : \mathbb{N} \to [0,1]$ is a decreasing function and $\mathbb{P}(E_{i,1}) \geqslant h(k_i)$. If we have a species tree with k species and a set $\mathcal{G}$ of at least*

$$n_h := \inf\left\{n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - h(\ell)]^n \leqslant 1 - q\right\} \tag{1.2.1}$$

gene trees sampled independently under the MSC model, then $\mathcal{G}$ forms a bipartition cover of the species tree with probability at least q.

Note that if h is independent of the species tree topology, then so is the bound in Theorem 1.2.3. If we let $h(x) = g_{x,1}(T_{\min}) \in [0,1]$, which is decreasing by Corollary 1.5.5, then Theorem 1.2.3 and the original bound $\mathbb{P}(E_{i,1}) \geqslant g_{k_i,1}(T_{\min})$ of Uricchio, Warnow, and Rosenberg [63] give a new topology-free bound.

Corollary 1.2.4 (Caterpillar Bipartition Cover Bound). *If we have a species tree with k species and a set $\mathcal{G}$ of at least*

$$M_c(k, T_{\min}) := \inf\left\{n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - g_{\ell,1}(T_{\min})]^n \leqslant 1 - q\right\} \tag{1.2.2}$$

gene trees sampled independently under the MSC model, then $\mathcal{G}$ forms a bipartition cover of the species tree with probability at least q. Here $T_{\min}$ is the minimum branch length (in coalescent units) in the species tree.

As with the original bound 1.1.1, Equation (1.2.2) depends on the species tree only through k and $T_{\min}$. Heuristically, this just allows us to replace the worst-case $g_{k-2,1}(T_{\min})$ with more of an average case across all of $\{g_{\ell,1}(T_{\min})\}_{\ell=2}^{k-2}$.

Accounting for Coalescent Events Below an Edge

Local balancing

Theorem 1.2.3 suggests that one avenue for improvement is to obtain a tighter bound $\mathbb{P}(E_{i,1}) \geqslant h(k_i)$ than the one obtained by choosing $h(x) = g_{x,1}(T_{\min})$. For small values of $T_{\min}$ the bound in Theorem 1.2.4 is dominated by the terms $g_{k_i,1}$ for which k_i are quite large. These are the bipartitions corresponding to edges e close to the root of the tree which have many descendants. In Uricchio, Warnow, and Rosenberg [63], the authors essentially assume *none* of the descendants coalesce before they reach edge e. This is conservative, especially when e is far from the leaves: in this setting, descendants have a long distance to travel through the tree and potentially coalesce.

Hence to improve this bound we should try to account for coalescent events that occur below the edge e. Namely, we ought to somehow replace $g_{k_e,1}(T_{\min})$ by $\mathbb{E}g_{X_e,1}(T_{\min})$ where X_e

is the (random) number of remaining lineages entering edge e. Clearly $\mathbb{P}(E_{i,1}) \geqslant \mathbb{E}g_{X_{e_i},1}(T_{\min})$ by the same logic that Theorem 1.1.1 employs. However, X_e will of course depend on the topology of the species tree below the edge e. Hence if we hope to develop a topology-free bound, we must somehow develop a notion of the worst-case topology in this setting.

Recall that random variable X is said to *first-order stochastically dominate* Y, and denote this by $X \geqslant_{st} Y$, if $\mathbb{P}(X > t) \geqslant \mathbb{P}(Y > t)$ for all t, or equivalently if $\mathbb{E}u(X) \geqslant \mathbb{E}u(Y)$ for all nondecreasing functions u. The random variable X is said to *second-order stochastically dominate* Y if $\mathbb{E}u(X) \geqslant \mathbb{E}u(Y)$ for all nondecreasing concave functions, and we denote this by $X \geqslant_{sst} Y$. First-order stochastic dominance naturally implies second-order.

Since $x \mapsto g_{x,1}(T)$ is decreasing, convex (Corollary 1.5.5, Lemma 1.5.7), we immediately get

Lemma 1.2.5. *If $U_{e_i} \geqslant_{sst} X_{e_i}$ then $\mathbb{P}(E_{i,1}) \geqslant \mathbb{E}g_{U_{e_i},1}(T_{\min})$.*

Remark 1.2.6. If U_e is a function of k_e alone and is (second order) stochastically increasing in this parameter, applying Theorem 1.2.3 with $h(\ell) = \mathbb{E}g_{U_\ell,1}(T_{\min})$ yields a new topology-free bound.

To apply this observation, we begin by analyzing the coalescent events occurring in the two edges directly below e.

Let Z_i^T be the random variable with distribution $\mathbb{P}(Z_i^T = j) = g_{ij}(T)$. Namely, if we start i lineages then Z_i^T represents the (random) number of lineages that remain uncoalesced after running Kingman's coalescent for time T. When we drop the T superscript, it is implicitly assumed that $T = T_{\min}$. For a positive integer-valued random variable A, let Z_A^T denote the independent mixture

$$\mathbb{P}(Z_A^T = i) = \sum_{\ell \geqslant 1} \mathbb{P}(A = \ell)\mathbb{P}(Z_\ell^T = i).$$

When we write expressions such as $Z_A^T + Z_B^T$ including multiple such mixtures, we implicitly mean underlying families $\{Z_\ell\}_{\ell \geqslant 1}, \{Z'_\ell\}_{\ell \geqslant 1}$ are independent of each other.

Suppose below the edge e our tree splits into two subtrees of size $m, k-m$. Clearly then

$$X_e \leqslant_{st} S := Z_m^{T_1} + Z_{k-m}^{T_2},$$

where T_1, T_2 are the lengths of the branches connecting the two subtrees to e. Moreover

$$Z_m^{T_1} + Z_{k-m}^{T_2} \leqslant_{st} Z_m^{T_{\min}} + Z_{k-m}^{T_{\min}},$$

by Lemma 1.5.4. Figure 1.4 illustrates the setup.

Lemma 1.2.7 (Deterministic Balancing Lemma)**.** *Let k and T be fixed. Then*

$$Z_i^T + Z_{k-i}^T \leqslant_{st} Z_j^T + Z_{k-j}^T, \quad \forall 1 \leqslant i \leqslant j \leqslant \frac{k}{2}.$$

In particular, the stochastic maximizer is $i = \lfloor k/2 \rfloor$.

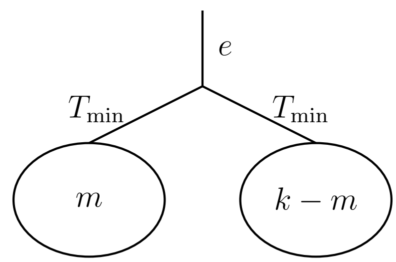

Figure 1.4: Coalescent tree structure showing edge e with two subtrees of sizes m and $k-m$. The subtrees are connected to e via a branch with minimal length $T_{\min}$.

The proof can be found in Appendix 1.6.

Heuristically, $i = \lfloor k/2 \rfloor$, where the two subtrees in Figure 1.4 are as evenly balanced as possible, is the maximizer because it minimizes the total number of pairs $\binom{m}{2} + \binom{k-m}{2}$ available in the two subpopulations, hampering coalescence. Lemma 1.2.7 implies a suitable upper bound

$$X_e \leqslant_{st} Z_{\lfloor k_e/2 \rfloor} + Z_{\lceil k_e/2 \rceil} \tag{1.2.3}$$

This upper bound motivates applying the same balancing principle recursively below e.

From Local Balancing to Balanced Trees

By iterating the ideas from the previous section, it is natural to believe the topology minimizing coalescence is the one where *all* splits are even, namely when the topology below edge e is the balanced tree. This is true in the following sense.

Lemma 1.2.8. *Given a binary tree $\mathcal{T}$ (with branch lengths) let $X_{\mathcal{T}}$ denote the number of lineages that have not coalesced by the time they reach the root. If $\mathcal{B}_k$ is the balanced tree with k leaves and all branch lengths equal to $T_{\min}$ then $X_{\mathcal{B}_k} \geqslant_{st} X_{\mathcal{T}}$ for any other tree $\mathcal{T}$ with k leaves and minimum branch length $T_{\min}$.*

The proof is postponed to Appendix 1.6, reducing the result to our *balancing lemma* 1.6.3. This extension of Lemma 1.2.7 allows for random subpopulation sizes, and essentially by iteratively applying it we can transform any tree $\mathcal{T}$ to the corresponding balanced tree $\mathcal{B}_k$.

Lemma 1.2.8 suggests a natural way of upper-bounding the number of lineages X_e entering a given edge e: $X_e \leqslant_{st} X_{\mathcal{B}_{k_e}}$. From this, we get our final bound.

Theorem 1.2.9 (Balanced Bipartition Cover Bound)**.** *If we have a species tree with k species and a set $\mathcal{G}$ of at least*

$$M_b(k, T_{\min}) := \inf \left\{ n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - w_\ell]^n \leqslant 1 - q \right\} \tag{1.2.4}$$

gene trees sampled independently under the MSC model, where

$$w_\ell := \mathbb{P}(W_\ell = 1) \tag{1.2.5}$$

and where the distributions of $W_\ell := Z_{X_{\mathcal{B}_\ell}}$ *are defined recursively by* $W_1 \equiv 1$ *and*

$$\mathbb{P}(W_\ell = j) = \sum_{a \geqslant 1} \sum_{b \geqslant 1} \mathbb{P}\left(W_{\lfloor \ell/2 \rfloor} = a\right) \mathbb{P}\left(W_{\lceil \ell/2 \rceil} = b\right) g_{a+b,j}(T_{\min}), \qquad \ell \geqslant 2. \tag{1.2.6}$$

Then $\mathcal{G}$ *forms a bipartition cover with probability at least* q*. Here* $T_{\min}$ *is the minimum branch length (in coalescent units) in the species tree.*

Reduction to Lemma 1.2.8. First, we show the recursion above does produce the distribution of $W_\ell = Z_{X_{\mathcal{B}_\ell}}$. We do this by induction on ℓ. The base case of $\ell = 1$ is trivial as $W_1 = Z_{X_1} = Z_1 \equiv 1$. For the inductive step, note that the balanced tree with ℓ leaves has balanced subtrees with $\lceil \ell/2 \rceil$ and $\lfloor \ell/2 \rfloor$ leaves respectively. By induction, $W_{\lceil \ell/2 \rceil}$ and $W_{\lfloor \ell/2 \rfloor}$ give the distributions of the numbers of lineages exiting these subtrees, respectively. These counts are independent under the MSC. Conditional on their values being a and b, the resulting $a + b$ lineages traverse one more branch of length $T_{\min}$, giving Equation (1.2.6).

Lemma 1.2.8 implies $X_e \leqslant_{st} X_{\mathcal{B}_{k_e}}$ so by the remark below Lemma 1.2.5 it suffices to show that $U_\ell := X_{\mathcal{B}_\ell}$ is stochastically increasing in ℓ. This follows from induction as

$$U_\ell \overset{d}{=} Z_{U_{\lfloor \ell/2 \rfloor}} + Z'_{U_{\lceil \ell/2 \rceil}}.$$

When ℓ increases by one, exactly one of $\lfloor \ell/2 \rfloor$ and $\lceil \ell/2 \rceil$ increases by one. Hence the inductive hypothesis and Corollary 1.5.10 imply $U_{\ell+1} \geqslant_{st} U_\ell$. It follows that

$$h(\ell) := \mathbb{E} g_{U_\ell,1}(T_{\min}) = \mathbb{P}(W_\ell = 1) = w_\ell$$

is decreasing. Applying Theorem 1.2.3 with this choice of h gives the result. $\square$

As the subtrees of a balanced tree are balanced, we can calculate w_ℓ in a recursive fashion. In our asymptotic analysis below, we will discuss the following convenient upper bound

$$M_b(k, T) \leqslant \frac{\log\left(\frac{k-3}{1-q}\right)}{-\log\left(1 - g_{\mathbb{E} X_{\mathcal{B}_{k-2}},1}(T)\right)}.$$

Hence our analysis has allowed us to replace $k - 2$ by $\mathbb{E} X_{\mathcal{B}_{k-2}}$, which depending on T may be a large improvement.

1.3 Simulations

In this section we compare the performance of our new bounds both to the original bound of Uricchio, Warnow, and Rosenberg [63] and to empirical results coming from simulations of the MSC model under various species tree topologies. All relevant code can be found on GitHub [42].

Bound Growth Rates

First, we empirically explored how all the bounds grow as functions of the two key parameters: the minimum branch length $T_{\min}$ and the species count k. Figure 1.5 highlights our results. We can observe that all bounds increase with k and decrease with $T_{\min}$ as is natural. However, our newer bounds are dramatically lower in essentially all regimes.

While it varies species to species, the maximal number of independent genes is typically on the order of 10^3 to 10^5 ([38], [53]). One drawback of the original bipartition cover bound 1.1.1 that Figure 1.5 highlights is that it dips above this threshold even for relatively few species k, especially when the minimum branch length is short. This can limit its practicability because scientists might not have access to enough genes to achieve their desired coverage probability.

Thus, one attractive feature of our new bound is that it remains below these biologically reasonable thresholds for a much wider choice of the relevant parameters $k, T_{\min}$, greatly improving its utility.

Improvement Ratios

To quantify how much these new bounds improve on the bound of [63] and over each other, we plot the ratios $I := M_{\text{old}}/M_{\text{new}}$ again as functions of the two nontrivial parameters $T_{\min}, k$. Values of $I > 1$ indicate an improvement over the original bound, and values $I < 1$ a degradation. Figure 1.6 compares the various bounds we have constructed throughout this paper.

As we discussed above, the balanced bound improves on the caterpillar bound, which improves on the original bound of Uricchio, Warnow, and Rosenberg [63]. As such, all improvement ratios $I \geqslant 1$. But as Figure 1.6 highlights, most of the improvement comes from the more sophisticated balanced bound (Theorem 1.2.9), which improves on the original bound by several orders of magnitude in the challenging high k and low $T_{\min}$ regimes.

On the other hand, the caterpillar bound only improves by a small constant factor. As we discuss in Section 1.3 below, this is likely due to the fact that most terms in the sum $\sum_{\ell=2}^{k-2}(1 - g_{\ell,1}(T))^n$ quickly saturate for large ℓ, so we gain little by replacing the maximum $g_{k-2,1}(T)$ by $g_{\ell,1}(T)$.

Quantifying Overestimation

In this section, we aim to quantify the level to which our bounds overestimate the number of gene trees required to obtain a cover. Namely, if n_e is the true q-quantile and n_b is one of our upper bounds, define the *overestimation ratio* to be the quantity n_b/n_e. To estimate n_e, we empirically generate gene trees from the MSC model until we get a bipartition cover, and repeat this for N independent trials to get an estimate of the desired quantile. In this study, we chose $N = 10^4$ trials and $q = 0.9$.

We explore a few different species tree topologies: the caterpillar tree with equal branch lengths, the balanced tree with equal branch lengths, and trees generated randomly from the Yule model. For Yule trees we normalize the branch lengths so they have the desired minimum length $T_{\min}$. We sample 100 different Yule trees in this way in order to study the distribution of overestimation ratios.

Since caterpillar and balanced trees represent the two worst cases we explored in this paper, they in a sense measure how close our bound is to optimal in a topology-free sense. We expect our bound to be tighter in these scenarios as compared to the more average case of Yule trees. Our results for caterpillar and balanced trees are displayed in Figures 1.7 and 1.8 respectively. The overestimation ratio is significantly higher for balanced trees as compared to caterpillars. Since our bound is topology-free, this aligns with the empirical observation that caterpillar trees are difficult to recover under the MSC. Overall, the new bound is still fairly far from tight, suggesting incorporating even partial topological information might be necessary.

Examining its dependence on the model parameters, we observe that the overestimation ratio of our new bound varies less with the species parameter k as compared to the original bound, indicating that the balanced bound should scale more favorably to larger values of k compared to the original bound. On the other hand, performance appears to deteriorate significantly as $T_{\min}$ decreases, suggesting less favorable scaling in the small $T_{\min}$ regime.

Since most trees differ substantially from these worst-case configurations, the performance on Yule trees provides a more realistic measure of the typical behavior of the bound. To assess this, we sample a range of species tree topologies from the Yule model and compute the corresponding overestimation ratios. Figure 1.9 shows the resulting distribution of these ratios across various choices of k and $T_{\min}$.

As expected, performance in this setting is noticeably worse than in the structured worst-case scenarios, but it remains substantially better than that of the original bound. Increasing the species parameter k appears to have a much larger impact than in the previous settings, suggesting worse scaling. This highlights that incorporating even partial topological information may be essential for achieving further improvements.

Estimated Growth Rates

Since the bounds are complicated functions of $T_{\min}$ and k, we develop estimates of their scaling with respect to these two parameters.

Here we confirm our intuition that our new bounds drastically improve performance in the small T regime. For fixed T, we show that all the bounds are $\Theta_T(\log(k))$, and that this is essentially the best one can do while relying on the union bound. We start with a result that specifies the asymptotics of the function $g_{k,1}(T)$ underlying all our bounds.

Lemma 1.3.1 (Asymptotics of $g_{k,1}(T)$)**.** *The function $T \mapsto g_{k,1}(T)$ has asymptotics*

$$\begin{cases} 1 - g_{k,1}(T) \sim \frac{3(k-1)}{k+1} \exp(-T), & T \uparrow \infty, \\ g_{k,1}(T) \sim \frac{k!}{2^{k-1}} T_k^{k-1}, & k^2 T_k = o(1). \end{cases} \tag{1.3.1}$$

Moreover, for fixed T, we have $g_{k,1}(T) \downarrow \mathbb{P}(S_\infty \leqslant T) =: s(T)$ where $S_\infty := \sum_{i=2}^\infty \tau_i$ for $\tau_i \sim \mathrm{Exp}\left(\binom{i}{2}\right)$ independent. Furthermore, the gap $\Delta_k(T) := g_{k,1}(T) - s(T)$ satisfies

$$\Delta_k(T) = \frac{2 f_{S_\infty}(T)}{k} + o\left(\frac{1}{k}\right) \sim \frac{2 f_{S_\infty}(T)}{k}.$$

The proof can be found in Appendix 1.7. Using this, we develop the following asymptotics for our two bounds. Recall that $M_o(k, T)$ denotes the original bound of Uricchio, Warnow, and Rosenberg [63] and $M_b(k, T)$ our balanced bound (Theorem 1.2.9).

Lemma 1.3.2 (Bound Asymptotics)**.** *Let $\kappa_{q,k} := \log((k-3)/(1-q))$ and $u(T) := \frac{2-\exp(-T/2)}{1-\exp(-T/2)}$. Then*

$$M_o(k, T) \sim \begin{cases} \frac{\kappa_{q,k}}{T}, & T \uparrow \infty, \\ \frac{2^{k-3}\kappa_{q,k}}{(k-2)! T^{k-3}}, & T = o(k^{-2}), \\ \frac{\kappa_{q,k}}{-\log(1-s(T))}, & T \text{ fixed}, k \to \infty \end{cases}$$

$$M_b(k, T) \lesssim \frac{\kappa_{q,k}}{-\log(1 - g_{u(T),1}(T))}, \qquad T \text{ fixed}, k \to \infty$$

Moreover, in the fixed T regime, the improvement factor satisfies

$$\liminf_{k \to \infty} \frac{M_o(k, T)}{M_b(k, T)} \geqslant \beta_T := \frac{\log(1 - g_{u(T),1}(T))}{\log(1 - s(T))}, \qquad \beta_T \sim \frac{\pi^2}{2T} \quad \text{as } T \downarrow 0.$$

The proof can be found in Appendix 1.7.

To see why the $\Theta(\log(k))$ growth rate is natural for the k asymptotics, note that $g_{k,1}(T)$ is roughly constant and equal to $s(T) > 0$ for all large k. Hence in large caterpillar trees, where most of the edges have large descendant counts, most of their bipartitions have $g_{k,1}(T)$ saturating in this way.

In this setting, we are essentially saying each bipartition corresponds to a Geometric$(s(T))$ random variable which specifies how many gene trees we need for that specific bipartition to show up. Each such variable has tail $\mathbb{P}(\mathrm{Geo}(s) > m) = (1 - s)^m$. Solving for $p(m) = k^{-1}$ we see $m = \log(k)/[-\log(1 - s)]$, so the maximum of k independent such geometric random variables ought to be roughly of this order.

1.4 Conclusion

We improved the bipartition cover bound of Uricchio, Warnow, and Rosenberg [63] by identifying two topology-free extremal features of species trees. Caterpillar trees maximize the descendant-count profiles that enter the union-bound argument, whereas balanced trees stochastically maximize the number of lineages that remain uncoalesced for a fixed number of descendants. Combining these observations yields a recursively computable balanced bound that can improve on the original bound by several orders of magnitude across the parameter regimes considered in our simulations.

In analyzing coalescence under the MSC, our key technical tools are the balancing lemmas, Lemmas 1.2.7 and 1.6.3. These formalize the intuition that, when a fixed collection of lineages is distributed more evenly across descendant populations, more lineages tend to remain uncoalesced. While short internal branches and anomaly-zone behavior are familiar obstacles to species-tree inference, our analysis identifies a distinct topology-dependent mechanism: balanced splits systematically delay coalescence, allowing ancestral lineages to persist longer and thereby increasing the potential for incomplete lineage sorting.

For fixed branch length, the new bound grows as $\Theta_T(\log k)$, which is the best possible order within the union-bound framework used here. The bound need not be globally tight, however, because its two extremal ingredients are attained by different tree topologies. Our simulations likewise show that some conservativeness remains, particularly for typical rather than extremal trees. Incorporating partial information about the species tree topology therefore provides a natural direction for obtaining sharper finite-sample guarantees.

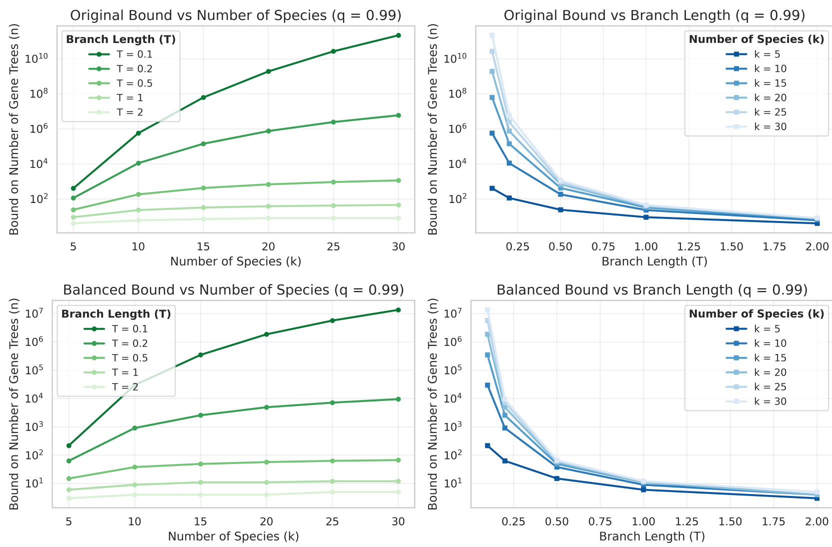

Figure 1.5: Bipartition cover bounds as functions of the two key parameters: k, $T_{\min}$. (top) The original bound [63]. (bottom) Our balanced bound (Theorem 1.2.9)

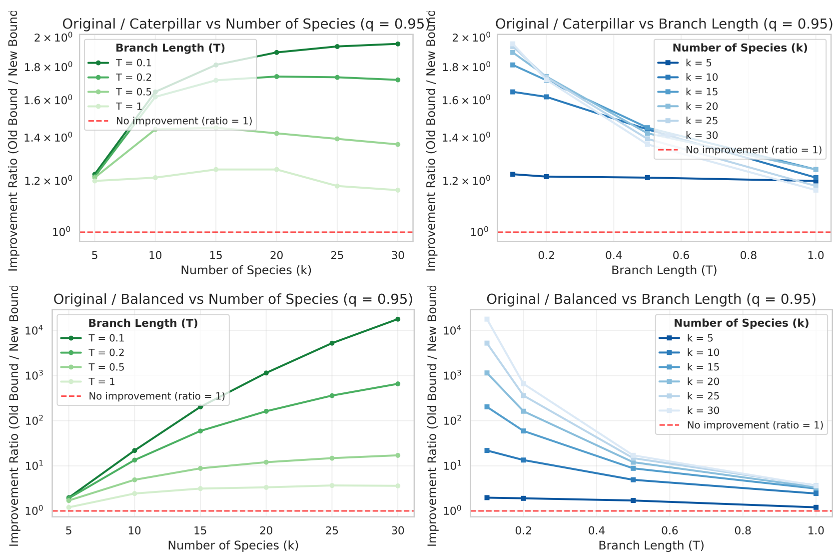

Figure 1.6: Improvement ratios. (top) Improvement of caterpillar bound (Theorem 1.2.3) over the original bound (Theorem 1.1.1). (bottom) Improvement of the balanced bound over original

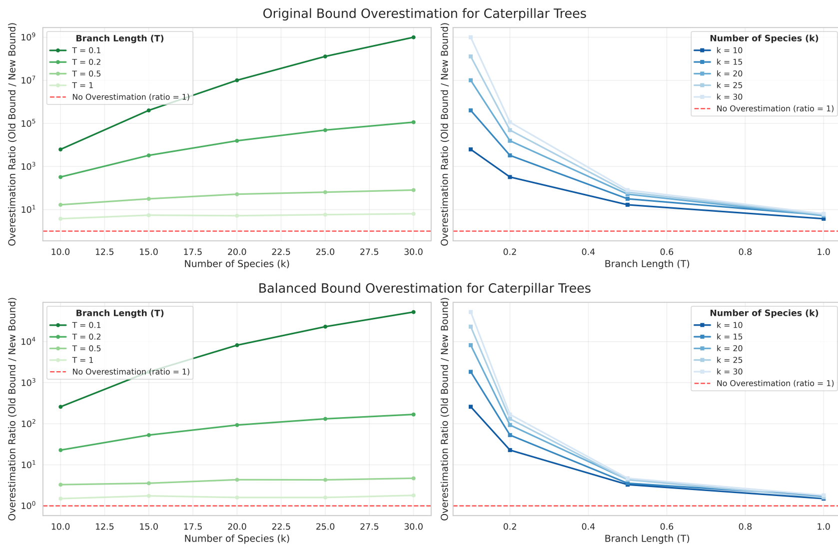

Figure 1.7: Empirical overestimation ratios for caterpillar trees. (top) The original bound (Theorem 1.1.1) and (bottom) our balanced bound (Theorem 1.2.9)

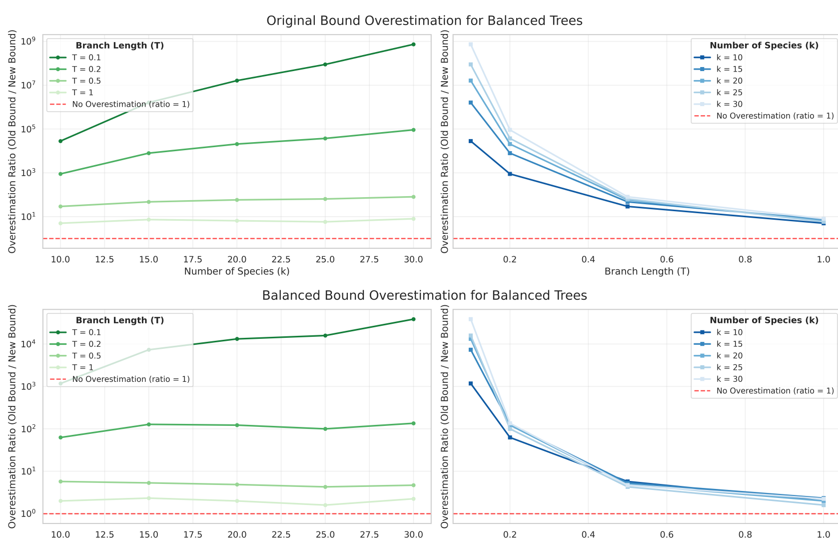

Figure 1.8: Empirical overestimation ratios for balanced trees. (top) The original bound (Theorem 1.1.1) and (bottom) our balanced bound (Theorem 1.2.9)

1.5 Appendix: Basic Results

Basic Properties of Log-Concavity

A sequence $\{x_k\}$ is *log-concave* if $x_k^2 \geqslant x_{k-1}x_{k+1}$. We say a random variable X is log-concave if the sequence $x_k = \mathbb{P}(X = k)$ is log-concave.

In order to apply Theorem 1.5.12, we will need to determine whether various variables of interest are log-concave. For this purpose, we introduce a few basic closure properties of log-concave random variables. It will turn out to be useful to work with a strengthening of log-concavity. We say a sequence $\{x_k\}$ is *ultra log-concave (ULC)* if the sequence $\{k! \cdot x_k\}$ is log-concave, or equivalently $kx_k^2 \geqslant (k+1)x_{k-1}x_{k+1}$. Clearly ultra log-concavity implies log-concavity. The following result, and much more, can be found in Chapter 4 of [56].

Theorem 1.5.1 (Prékopa)**.** *If X, Y are independent, (ultra) log-concave, integer-valued random variables, then $X+Y$ is (ultra) log-concave. Equivalently, the class of (ultra) log-concave distributions is closed under convolution.*

The following shows us that Kingman's coalescent preserves log-concavity in our setting of interest.

Theorem 1.5.2 (Preservation of Ultra Log-Concavity)**.** *Suppose P_t is the transition semigroup of a pure-death process with death rates λ_i so that (a) λ_i is convex (b) $\eta_i := \lambda_i/i$ is concave and (c) at least one of these is strictly convex/concave. Moreover, suppose that π is ultra log-concave with either $\operatorname{supp}(\pi) = \{2, ..., M\}$ or $\operatorname{supp}(\pi) = \{1, ..., M\}$ for some $M \in \mathbb{N} \cup \{\infty\}$. Then $\pi(t) := \pi P_t$ is ultra log-concave for all $t \geqslant 0$. In particular, this is true for Kingman's coalescent with $\lambda_i = \binom{i}{2}$.*

Proof of Theorem 1.5.2. Recall Kolmogorov's forward equations imply

$$\frac{d}{dt}\pi_i(t) = \lambda_{i+1}\pi_{i+1}(t) - \lambda_i\pi_i(t).$$

First, we handle the interior. For $i \in \operatorname{Int}(\operatorname{supp}(\pi))$ we can define the ratios

$$v_i := \lambda_i \cdot \frac{\pi_i}{\pi_{i-1}}, \qquad r_i := \frac{i\pi_i^2}{(i+1)\pi_{i-1}\pi_{i+1}} = \frac{v_i}{v_{i+1}} \cdot \frac{\eta_{i+1}}{\eta_i}.$$

From here, using the forward equations we can see

$$\begin{aligned}
\frac{\dot{r}_i}{r_i} &= \frac{d}{dt}\log(r_i) \\
&= 2\frac{\dot{\pi}_i}{\pi_i} - \frac{\dot{\pi}_{i-1}}{\pi_{i-1}} - \frac{\dot{\pi}_{i+1}}{\pi_{i+1}} \\
&= (-2\lambda_i + \lambda_{i-1} + \lambda_{i+1}) + 2\lambda_{i+1}\frac{\pi_{i+1}}{\pi_i} - \lambda_i\frac{\pi_i}{\pi_{i-1}} - \lambda_{i+2}\frac{\pi_{i+2}}{\pi_{i+1}}
\end{aligned}$$

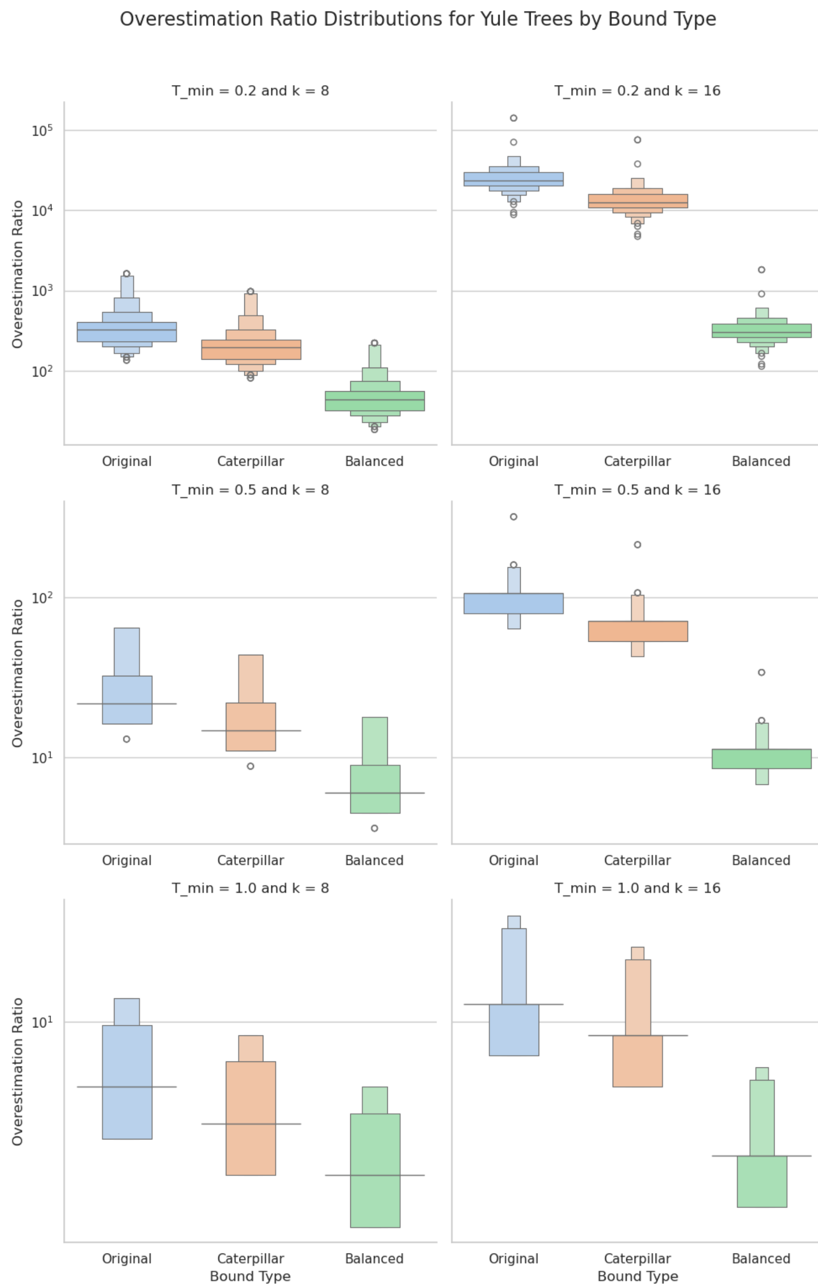

Figure 1.9: Boxenplots for the distribution of (log) overestimation ratios for Yule trees across different choices of our parameters T_{min}, k. Each row corresponds to a specific choice of $T_{\text{min}} \in \{0.2, 0.5, 1\}$ and each column a choice of $k \in \{8, 16\}$. A boxenplot is a boxplot variant that displays additional nested quantile boxes in the tails, making the shape and tail behavior of the distribution more visible.

$$= (\lambda_{i-1} - 2\lambda_i + \lambda_{i+1}) + 2v_{i+1} - v_i - v_{i+2}.$$

By assumption, $r_i(0) \geqslant 1$ for all i. Since $r_i(t)$ is continuous in t, the only way for $r_i(t) < 1$ to occur is if $r_i(s) = 1$ for some $0 \leqslant s < t$. We will show at this time that $\dot{r}_i(s) > 0$, and hence we can never drop below one. Suppose $r_i(s) = 1$ and $r_k(s) \geqslant 1$ for all other k. From these, we see respectively

$$v_i = \frac{\eta_i}{\eta_{i+1}} \cdot v_{i+1}, \qquad v_{i+2} \leqslant \frac{\eta_{i+2}}{\eta_{i+1}} \cdot v_{i+1}.$$

Using these two bounds, we can see

$$\frac{\dot{r}_i}{r_i} \geqslant (\lambda_{i-1} - 2\lambda_i + \lambda_{i+1}) - [\eta_i - 2\eta_{i+1} + \eta_{i+2}] \cdot \frac{v_{i+1}}{\eta_{i+1}}.$$

Since λ_i is convex, the first term is nonnegative. Since η_i is concave, the second term is nonnegative. Moreover, by assumption one of these must be strictly positive, and thus $\dot{r}_i/r_i > 0$ as desired.

Now, we handle the boundaries. In the case $M < \infty$, we trivially have

$$M\pi_M^2(t) \geqslant (M+1)\pi_{M-1}(t)\pi_{M+1}(t) = 0$$

as $\pi_{M+1}(t) = 0$ for all t – in a pure-death process on $\mathbb{N}$ the support can never increase. For the lower boundary, first consider the case $\mathrm{supp}(\pi) = \{1, ..., M\}$. Then again we trivially have $\pi_1(t) \geqslant 2\pi_0(t)\pi_2(t) = 0$ for all t as we can never drop below one in a pure-death process on $\mathbb{N}$. In the case $\mathrm{supp}(\pi) = \{2, ..., M\}$ note that $2\pi_2(0) > 3\pi_1(0)\pi_3(0) = 0$. By continuity, we must have $2\pi_2(t) > 3\pi_1(t)\pi_3(t)$ for small enough $t > 0$. For such a t, we see that $\pi(t)$ is ultra log-concave. Moreover, since $\pi_1(t) > 0$ for any positive t we see $\mathrm{supp}(\pi(t)) = \{1, ..., M\}$ so we can now apply the previous case to show ultra log-concavity is preserved for all further times. □

Basic Properties of $g_{i,j}(T)$ and Z_i^T

We will start by proving some basic properties about the function $g_{i,j}(T)$ and the associated random variables Z_i^T with distribution $\mathbb{P}(Z_i^T = j) = g_{i,j}(T)$. Recall that Z_i^T gives the distribution of the number of lineages remaining after starting with i lineages and running Kingman's coalescent for T seconds.

Moreover, recall $X_{\mathcal{B}_k}$ denotes the number of lineages reaching the root of a balanced tree with k leaves and branch lengths all equal to T. For convenience, we denote $X_k := X_{\mathcal{B}_k}$. Using the above, we will show that all the variables of interest in this paper are log-concave.

Lemma 1.5.3. *The random variables X_m, Z_{X_m} are ultra log-concave, and hence log-concave.*

Proof. We will prove this by induction on m. Trivially $X_1 = Z_1 \equiv 1$ are ultra log-concave. Now suppose for induction that X_i, Z_{X_i} are log-concave for $i < m$. Note

$$X_m = Z_{X_{\lfloor m/2 \rfloor}} + Z_{X_{\lceil m/2 \rceil}}.$$

By Prékopa's Theorem 1.5.1, the sum of independent ultra log-concave distributions is ultra log-concave, so X_m is ultra log-concave. Since $\text{supp}(X_m) = \{2, ..., m\}$, by Theorem 1.5.2, Kingman's coalescent preserves ultra log-concavity, so Z_{X_m} is ultra log-concave. This completes the induction. $\square$

Now we show our variables Z_i^T have some simple monotonicity properties.

Lemma 1.5.4. *Let $i \leqslant j$ and $t \leqslant T$. Then $Z_i^T \leqslant_{st} Z_j^T$ and $Z_i^T \leqslant_{st} Z_i^t$.*

Heuristically this just states the obvious: the more lineages we start with at the beginning of a branch and the less time we have to coalesce, the more lineages we tend to end with.

Proof of Lemma 1.5.4. Again the fact the function is decreasing in T follows immediately via a coupling argument. To see that it is increasing in i, we just condition on the first coalescence event. Assume $z \leqslant i$, otherwise both $\mathbb{P}(Z_i^T \geqslant z) = \mathbb{P}(Z_{i-1} \geqslant z) = 0$. Note if $\tau \sim \exp\left(\binom{i}{2}\right)$ is the time of the first coalescent event

$$\mathbb{P}(Z_i^T \geqslant z) = \int_0^\infty \mathbb{P}(Z_{i-1}^{T-t} \geqslant z)\tau(dt) \geqslant \int_0^\infty \mathbb{P}(Z_{i-1}^T \geqslant z)\, \tau(dt) = \mathbb{P}(Z_{i-1}^T \geqslant z)$$

where we let $\mathbb{P}(Z_{i-1}^T \geqslant z) = 1$ if $T < 0$. This gives $Z_i^T \geqslant Z_{i-1}^T$ stochastically as desired. $\square$

Note that $g_{i1}(T) = 1 - \mathbb{P}(Z_i^T \geqslant 2)$, from which we get the immediate corollary:

Corollary 1.5.5. *The function $g_{i1}(T)$ is decreasing in i and increasing in T.*

Thus $g_{i,j}(T)$ has some natural monotonicity properties. The following Lemmas show that most of the variability in $g_{i,j}(T)$ occurs when both T and i are small.

Lemma 1.5.6. *For fixed n, k, the function $\mathbb{P}(Z_n^T = k) = g_{n,k}(T)$ is log-concave in T.*

Proof of Lemma 1.5.6. We prove this by induction on n. For $n = 1$ we see $g_{nk}(T) = \mathbf{1}\{k = 1\}$ which is trivially log-concave. For the inductive step, recall that by conditioning on the first coalescent event and using the memoryless property of Kingman's coalescent, we observed

$$g_{n,k}(T) = \int_0^T g_{n-1,k}(T-t)\binom{n}{2}\exp\left(-\binom{n}{2}t\right)\, dt.$$

Namely, we are just convolving the log-concave function $g_{n-1,k}$ with the log concave function $\exp(-\binom{n}{2}t)$. By Prékopa's theorem 1.5.1, convolutions preserve log-concavity. Thus $g_{n,k}$ is log-concave. $\square$

Lemma 1.5.7. *For fixed T, the function $i \mapsto g_{i,1}(T)$ is convex.*

Proof of Lemma 1.5.7. It is a classical result of Karlin that the transition matrix $P_T := [g_{i,j}(T)]_{i,j\in\mathbb{N}}$ is totally positive (of any order) for any fixed T. See for example [33]. Theorem 2 of the paper [18], again a result of Karlin, then shows such matrices preserve convexity. Since $e_1 = (1, 0, 0, ...)$ is convex and $(g_{i,1}(T)) = P_T e_1$ this implies the sequence $\{g_{i,1}(T)\}_{i\in\mathbb{N}}$ is convex. □

Next, we study the expected number of lineages that survive under Kingman's coalescent when the number of starting lineages is large. Since the initial coalescent rate $\binom{i}{2}$ grows so fast in the number of lineages, eventually adding more lineages has barely any influence on the number remaining at time T. This is essentially the well-known observation that Kingman's coalescent can come down from infinity in any positive time.

Lemma 1.5.8 (Upper Bound for Expected Number of Lineages)**.** *For any $i \in N$ and $T > 0$*

$$\mathbb{E}Z_i^T \leqslant \frac{1}{1-(1-1/i)\exp(-\frac{T}{2})} \leqslant \frac{1}{1-\exp\left(-\frac{T}{2}\right)}.$$

In particular, for any fixed $T > 0$ we see $\mathbb{E}Z_i^T = \Theta_T(1)$.

Note that in the small T regime, where $\exp(-T/2) \approx 1 - T/2$, this bound is roughly $2/T$, recovering the well-known asymptotics of Kingman's coalescent as it comes down from infinity. Hence in a sense this bound is roughly tight in that regime.

Proof of Lemma 1.5.8. Kolmogorov's backwards equations followed by Jensen's inequality implies

$$\frac{d}{dt}\mathbb{E}Z_i^t\bigg|_{t=T} = \mathbb{E}\left(-\binom{Z_i^T}{2}\right) \leqslant -\binom{\mathbb{E}Z_i^T}{2}.$$

Hence letting $z(t) := \mathbb{E}Z_i^t$ we see $z(t)$ satisfies

$$\dot{z}(t) \leqslant -\frac{z(z-1)}{2}, \qquad z(0) = i.$$

Let $\dot{y} = -y(y-1)/2$ with $y(0) = i$. Applying separation of variables

$$y(t) = \frac{1}{1-(1-1/i)\exp(-t/2)}.$$

Since $\dot{z} \leqslant \dot{y}$ and $z(0) = y(0)$, applying the comparison principle yields

$$z(t) \leqslant y(t) \leqslant \frac{1}{1-\exp(-t/2)},$$

as we desired. □

Corollary 1.5.9. *Suppose that $k \in (2^{m-1}, 2^m]$. Then*

$$\mathbb{E}X_{\mathcal{B}_k}^T \leqslant \frac{2-\exp(-T/2)}{1-\exp(-T/2)+\exp(-mT/2)/2^m} \leqslant \frac{2-\exp(-T/2)}{1-\exp(-T/2)}$$

Proof of Corollary 1.5.9. Fix T and let

$$u(n) := \frac{1}{1-(1-1/n)\alpha}, \qquad \alpha := \exp(-T/2)$$

be from our upper bound in Lemma 1.5.8. Then u is increasing and concave in n. Let $Y_m := X_{\mathcal{B}_{2^m}}$ and $a_m := \mathbb{E}Y_m$. Clearly by a coupling argument $\mathbb{E}X_{\mathcal{B}_k} \leqslant a_m$. Moreover, by concavity

$$a_{m+1} = \mathbb{E}(Z_{Y_m} + Z_{Y_m}) \leqslant 2\mathbb{E}u(Y_m) \leqslant 2u(a_m)$$

In particular, this implies

$$\frac{1}{a_{m+1}} \geqslant \frac{1-\alpha}{2} + \frac{\alpha}{2} \cdot \frac{1}{a_m}.$$

Iterating this, we see

$$a_m \leqslant \frac{2-\alpha}{1-\alpha+(\alpha/2)^m} \leqslant \frac{2-\alpha}{1-\alpha}$$

□

Basic Properties of Stochastic Orderings

The following results are just simple general properties of stochastic orderings which we have sometimes reinterpreted in our current setting. The book [58] contains more information on properties of such stochastic orderings, far beyond what we cover here.

First-Order Stochastic Dominance

In this section, we prove some lemmas relating to first-order stochastic dominance and the random variables Z_i^T. Below we typically suppress the T notation unless it is explicitly necessary. The following generalizes Lemma 1.5.4 by allowing for the number of initial lineages to be random.

Corollary 1.5.10 (Stochastic Dominance of Compositions)**.** *If $X \geqslant_{st} Y$ then*

$$Z_X \geqslant_{st} Z_Y.$$

Moreover, if $X' \geqslant_{st} Y'$ and $X \perp\!\!\!\perp Y, X' \perp\!\!\!\perp Y'$, or more generally if $(X, Y) \geqslant_{st} (X', Y')$ coordinate-wise then

$$Z_X + Z_Y \geqslant_{st} Z_{X'} + Z_{Y'}.$$

In our context, the first statement above just says that if we tend to have more lineages entering a given edge of our tree, then we tend to also have more lineages exiting it. The second statement says that if an edge e tends to have more lineages uncoalesced in both its subtrees, then the edge e tends to have more lineages entering it.

Proof of Corollary 1.5.10. For the first statement, note that

$$\mathbb{P}(Z_X \geqslant t) = \mathbb{E}_{i \sim X}\mathbb{P}(Z_i \geqslant t) \geqslant \mathbb{E}_{i \sim Y}\mathbb{P}(Z_i \geqslant t) = \mathbb{P}(Z_Y \geqslant t),$$

as by Lemma 1.5.4 the function $i \mapsto \mathbb{P}(Z_i \geqslant t)$ is increasing. Hence $Z_X \geqslant Z_Y$ stochastically. The second claim follows immediately by conditioning on Z_X

$$\begin{aligned}\mathbb{P}(Z_X + Z_Y \geqslant t) &= \mathbb{E}_{z \sim Z_X}\mathbb{P}(Z_Y \geqslant t - z)\\ &\geqslant \mathbb{E}_{z \sim Z_X}\mathbb{P}(Z_{Y'} \geqslant t - z)\\ &\geqslant \mathbb{E}_{z \sim Z_{X'}}\mathbb{P}(Z_{Y'} \geqslant t - z)\\ &= \mathbb{P}(Z_{X'} + Z_{Y'} \geqslant t)\end{aligned}$$

so that $Z_X + Z_Y \geqslant_{st} Z_{X'} + Z_{Y'}$. The joint statement follows similarly just from the fact $\varphi(a, b) := \mathbb{P}(Z_a + Z_b \geqslant t)$ is trivially increasing coordinate-wise. □

The following tells us that increasing mixtures of increasing variables are again increasing.

Lemma 1.5.11. *Suppose that we have collections of random variables $\{V_i\}_{i \in \mathbb{N}}, \{\Theta_i\}_{i \in \mathbb{N}}$ so that for each $i \leqslant j$ we have:*

a) $\Theta_i \leqslant_{st} \Theta_j$,

b) $V_i \mid \{\Theta_i = \theta\} \leqslant_{st} V_j \mid \{\Theta_j = \theta\}$ *for every fixed* θ,

c) $V_i \mid \{\Theta_i = \theta\} \leqslant_{st} V_i \mid \{\Theta_i = \theta'\}$ *for every* $\theta \leqslant \theta'$ *and* i.

Then $V_i \leqslant_{st} V_j$ *for* $i \leqslant j$.

Proof of Lemma 1.5.11. For $i < j$ note by conditioning on Θ_i we get

$$\begin{aligned}\mathbb{P}(V_i \geqslant x) &= \int_{\mathbb{R}} \mathbb{P}(V_i \geqslant x \mid \Theta_i = \theta)\ \Theta_i(d\theta)\\ &\leqslant \int_{\mathbb{R}} \mathbb{P}(V_j \geqslant x \mid \Theta_j = \theta)\ \Theta_i(d\theta)\\ &\leqslant \int_{\mathbb{R}} \mathbb{P}(V_j \geqslant x \mid \Theta_j = \theta)\ \Theta_j(d\theta)\\ &= \mathbb{P}(V_j \geqslant x)\end{aligned}$$

by first using property (b) and then properties (a),(c) together. For the latter, we are using the fact that (c) implies $\mathbb{P}(V_j \geqslant x|\Theta_j = \theta)$ is an increasing function in θ, and since $\Theta_i \leqslant_{st} \Theta_j$ this ordering is preserved when you take the expectation with respect to any increasing function. □

Likelihood Ratio Ordering

We say a random variable X is less than Y with respect to the *likelihood ratio ordering (LRO)*, and denote this by $X \leqslant_{lr} Y$, if the likelihood ratios $f_Y(t)/f_X(t)$ are non-decreasing in t. Here, f_X, f_Y are the mass functions of X, Y respectively. Equivalently, we require

$$f_X(s) f_Y(t) \geqslant f_X(t) f_Y(s), \quad \forall s \leqslant t$$

It is not hard to see that $X \leqslant_{lr} Y$ implies $X \leqslant_{st} Y$, so this is a strictly stronger ordering than that of first-order stochastic dominance. The LRO has some convenient closure properties. The following appear as Theorems 1.C.9 and 1.C.17 of [58] respectively:

Theorem 1.5.12 (LRO preserved by Convolutions)**.** *Suppose we have two collections of independent, log-concave random variables,* $\{X_i\}_{i=1}^n$ *and* $\{Y_i\}_{i=1}^n$*. If* $X_i \leqslant_{lr} Y_i$ *for all* i*, then* $\sum_{i=1}^n X_i \leqslant_{lr} \sum_{i=1}^n Y_i$*.*

Theorem 1.5.13 (LRO preserved by Mixtures)**.** *Suppose we have a collection* $\{V_i\}_{i \in \mathbb{N}}$ *of random variables so that* $V_i \leqslant_{lr} V_j$ *for* $i \leqslant j$*. Moreover, suppose that we have random variables* M, N *independent of* $\{V_i\}$ *with* $M \leqslant_{lr} N$*. Then* $V_M \leqslant_{lr} V_N$*.*

Using these, we can see our random variables of interest are increasing with respect to the LRO:

Lemma 1.5.14. *For* $i \leqslant j$ *we have* $Z_i \leqslant_{lr} Z_j$*,* $X_i \leqslant_{lr} X_j$ *and* $Z_{X_i} \leqslant_{lr} Z_{X_j}$*.*

Proof. First, we show $\{Z_i\}_{i \in \mathbb{N}}$ is increasing in the LRO, namely

$$\mathbb{P}(Z_i = m) \cdot \mathbb{P}(Z_j = n) \geqslant \mathbb{P}(Z_i = n) \cdot \mathbb{P}(Z_j = m), \quad \forall i \leqslant j, \, \forall m \leqslant n.$$

But this is a consequence of the total positivity of the transition kernels for birth-death processes (see for example [33]).

We will prove the other statements by induction. Trivially $X_1 \equiv 1 \leqslant_{lr} 2 \equiv X_2$. Thus suppose for induction that $X_1 \leqslant_{lr} ... \leqslant_{lr} X_i$. Then Theorem 1.5.13 implies $Z_{X_1} \leqslant_{lr} ... \leqslant_{lr} Z_{X_i}$. Using this, Theorem 1.5.12 implies

$$X_i = Z_{X_{\lfloor i/2 \rfloor}} + Z_{X_{\lceil i/2 \rceil}} \leqslant_{lr} Z_{X_{\lfloor (i+1)/2 \rfloor}} + Z_{X_{\lceil (i+1)/2 \rceil}} = X_{i+1}$$

since Z_{X_m} are log-concave by Lemma 1.5.3. Hence the result follows from induction.

□

1.6 Appendix: Proofs of the Main Upper-Bound Results

Proof of Lemma 1.2.1: Caterpillars Maximize Increasing Sums

The following is true regardless of our convention for choosing the edge adjacent to the root. Moreover, it is clearly also true if we use *all* descendant counts $\{\alpha_i\}_{i=1}^{2k-2}$ rather than just those corresponding to nontrivial bipartitions: all but one of the additional α_i are equal to one, and it is clear from the proof below we can handle the exceptional value as well.

Lemma: Suppose $\{\alpha_i\}_{i=1}^{k-3}$ are the descendant counts of our rooted binary tree T corresponding to our nontrivial bipartitions. Moreover, suppose $f : \mathbb{N} \to \mathbb{R}$ is nondecreasing. Then $\sum_i f(\alpha_i)$ is maximized when T is the caterpillar tree, for which $\{\alpha_i\} = \{2, 3, ..., k-2\}$.

Proof Lemma 1.2.1. Let's start by proving the caterpillar is the maximizer. Since f is nondecreasing, it suffices to show we can reindex $\{\alpha_i\}_{i=1}^{k-3}$ so that $\alpha_i \leqslant i+1$. Heuristically, we are just pairing up each edge e in our tree T with an edge e' in our caterpillar tree so that e has at most as many descendants as e'. We do so by induction on the number of leaves/species.

Base Case: $k = 4$
In this case, the two unique rooted trees, shown in Figure 1.2, both have $\alpha_1 = 2$ for their only nontrivial bipartition, so the statement is trivially true. Again we are ignoring trivial bipartitions and possibly one of the edges adjacent to the root.

Inductive Step (Caterpillar)
Suppose we have a tree T with $k > 4$ leaves/species. There are two cases:

(a) *One child of root is a leaf.* If T' is the non-leaf subtree of the root, apply the inductive hypothesis to T'. Let $\{\alpha_i\}_{i=1}^{k-4}$ be the descendant counts of T'. By induction, we can reindex $\{\alpha_i\}_{i=1}^{k-4}$ so that $\alpha_i \leqslant i+1$. Moreover, as all nontrivial bipartitions have at most $k-2$ leaves in one part, trivially $\alpha_{k-3} \leqslant k-2$, so we are done. Namely, we can just pair the remaining nontrivial edge of T, with the top nontrivial edge of our caterpillar tree.

(b) *Neither child of root is a leaf.* Let T', T'' be the two subtrees of the root. If these have r, m leaves respectively, where $r, m \in \{2, 3, ..., k-2\}$ and $r + m = k$, let $\{\kappa_i\}_{i=1}^{r-3}$ and $\{\beta_j\}_{j=1}^{m-3}$ be their descendant counts. By induction, these can be ordered so that $\kappa_i \leqslant i+1$ and $\beta_j \leqslant j+1$. Clearly then defining

$$\alpha_i = \begin{cases} \kappa_i, & 1 \leqslant i \leqslant r-3, \\ \beta_{i-r+3}, & r-3 < i \leqslant k-6. \end{cases}$$

Namely, we just list the edges of T' first, and then the edges of T''. This accounts for all but three of the edges of T: one of the edges e adjacent to the root of T, one edge

e' in T', and one edge e'' in T''. Note that the descendant counts $\alpha_{e'}, \alpha_{e''}$ of e', e'' are at most $r-1, m-1$ respectively. Moreover

$$r - 1 = k - m - 1 \leqslant k - 3, \qquad m - 1 = k - r - 1 \leqslant k - 3$$

as both T', T'' have at least two leaves. Equality holds if $m = 2$ or $r = 2$ respectively, so for $k > 4$ equality cannot hold for both simultaneously. Hence without loss of generality assume $\alpha_{e'} \leqslant k - 4$. So we can let $\alpha_{k-5}, \alpha_{k-4}$ correspond to edges e', e'' respectively. Trivially e has descendant count at most $k-2$, as all nontrivial bipartitions do. Thus we can let α_{k-3} correspond to edge e. This completes the proof of Lemma 1.2.1.

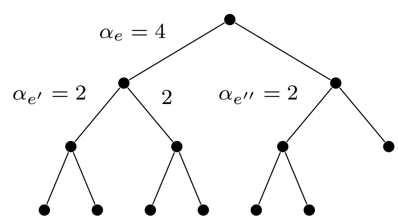

Figure 1.10: Example of the inductive step in Lemma 1.2.1

□

We can provide a partial complement to the above Lemma 1.2.1 analyzing the other extreme. More concretely, we will show that the more balanced the tree is, the smaller these sums tend to be. We will not use this result in this paper, but it adds to the idea that the balanced tree and the caterpillar tree are in a sense the two extreme points in our space of trees.

Lemma 1.6.1. *Suppose $\{\alpha_i\}_{i=1}^{2k-2}$ are the descendant counts of our rooted binary tree T corresponding to all the edges of our tree (not just ones corresponding to nontrivial bipartitions). Moreover, suppose $f : \mathbb{N} \to \mathbb{R}$ is nondecreasing and convex. Then $\sum_i f(\alpha_i)$ is minimized when T is the balanced tree.*

Proof of Lemma 1.6.1. We will describe a natural greedy balancing operation on the tree and prove it decreases $\sum f(\alpha_i)$. Recall that a *cherry* in a binary tree refers to a subtree consisting of two leaves.

Say that a given vertex $v \in V(T)$ is *unbalanced* if the number of leaves in its two subtrees differs by more than one. In order to make the tree more balanced, we will construct a new tree T' by pruning one of the cherries of the larger subtree and attaching it to one of the leaves of the smaller subtree.

To locate which subtree to prune, start at v and traverse the edge leading to the larger subtree. Iteratively descend the tree in this fashion until you reach a leaf ℓ. At this point, necessarily ℓ and its sibling form a cherry. If this were not the case, then the sibling of ℓ forms a larger subtree, and we would have traversed into that subtree rather than towards ℓ_p. This will be the cherry we prune.

To locate the leaf at which to reattach, start at v and traverse the edge leading to the smaller subtree. Iterate this procedure until you reach a leaf ℓ_a. Attach the cherry at this leaf ℓ to form a new tree T' with the same number of leaves overall as T. An illustration of this algorithm is provided in Figure 1.11.

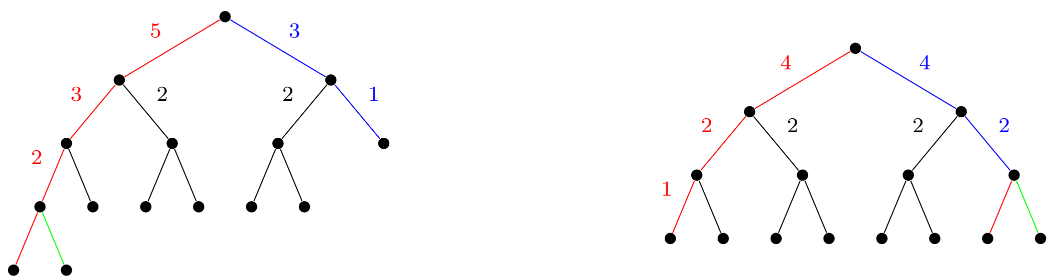

Figure 1.11: Side-by-side illustration of the initial tree T (left) and the tree T' after balancing (right). The pruning path $\{e_i^p\}$ in red and the attachment path $\{e_i^a\}$ in blue are highlighted

Suppose that $\{\alpha_i\}_{i=1}^{2k-2}, \{\alpha_i'\}_{i=1}^{2k-2}$ are the descendant counts of T, T' respectively. We claim that $\sum f(\alpha_i') \leqslant \sum f(\alpha_i)$, namely that this balancing operation can only decrease the overall sum. To see why, let

$$v = u_0 \xrightarrow{e_1^p} \dots \xrightarrow{e_n^p} u_n = \ell_p, \quad v = w_0 \xrightarrow{e_1^a} \dots \xrightarrow{e_m^a} w_m = \ell_a$$

be the paths we traverse the tree in our pruning (from $v \to \ell_p$) and grafting (from $v \to \ell_a$) operations respectively. Note that $\alpha_e = \alpha_e'$ for all edges $e \notin \{e_i^p\} \cup \{e_i^a\}$ not along either of these paths. Moreover, every edge along the pruning path loses exactly one descendant leaf and every edge along the attachment path gains exactly one leaf. The edges in T, T' are exactly the same, except for the relocation of the cherry, and hence we pair the descendant counts α_e, α_e'

$$\sum_i (f(\alpha_i) - f(\alpha_i')) = \sum_{i=1}^{n-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)] + \sum_{i=1}^{m-1} [f(\alpha_{e_i^a}) - f(\alpha_{e_i^a} + 1)].$$

Moreover, by construction $m \leqslant n$ and $\alpha_{e_i^p} > \alpha_{e_i^a}$. This is because we always choose the larger subtree for pruning and the smaller for attaching. Thus $\alpha_{e_1^p} > \alpha_{e_1^a}$ and hence by induction

$$\alpha_{e_i}^p \geqslant \left\lceil \frac{\alpha_{e_{i-1}^p}}{2} \right\rceil > \left\lfloor \frac{\alpha_{e_{i-1}^a}}{2} \right\rfloor \geqslant \alpha_{e_i^a}$$

so $\alpha_{e_i^p} > \alpha_{e_i^a}$ is true for all i. Using this, we can see

$$= \sum_{i=1}^{m-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)] + [f(\alpha_{e_i^a}) - f(\alpha_{e_i^a} + 1)] + \sum_{i=m}^{n-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)].$$

Each term in the first summand is nonnegative as f is convex, and each term in the second summand is nonnegative as f is increasing. Hence $\sum(f(\alpha_i) - f(\alpha_i')) \geqslant 0$ and the result follows.

Lastly, we argue that any tree can be converted into a balanced tree by iterating this procedure. If the subtrees of a given unbalanced vertex have size L, R respectively where $L - R \geqslant 2$ then clearly after this procedure they have sizes $L-1, R+1$ respectively with imbalance $(L-1)-(R+1) = (L-R)-2$. Hence the imbalance decreases by two, and if we iterate this procedure enough times we can get the imbalance to be $(L-R)\ mod(2)$, namely the vertex is balanced. Moreover, the descendant counts of any vertices not contained in one of the subtrees of v are left unchanged. Hence we can start at the root and iteratively apply the procedure until the root is balanced. Then we can work our way down the tree. $\square$

This type of iterative subtree balancing is reminiscent of measures of tree balance in phylogenetics, where metrics such as the Colless or Sackin indices quantify how balanced a tree is relative to the maximally balanced or maximally imbalanced (caterpillar) configurations. Tree balance statistics have long been used in phylogenetics to characterize the shapes of evolutionary trees, compare them to null models, and infer evolutionary processes such as speciation or extinction biases [35].

Classical tree-balancing procedures in computer science—such as AVL trees [1] and weight-balanced trees [48]—use local rotations to enforce constraints on subtree depths or sizes. Our algorithm shares the same conceptual goal of redistributing leaves to reduce imbalance, but it is purely combinatorial and does not maintain any search property.

Proof Lemma 1.2.7: Deterministic Balancing Lemma

Recall we aim to prove:

Lemma: Let $k \geqslant 4$ be fixed. Then for any fixed T

$$Z_i^T + Z_{k-i}^T \leqslant_{st} Z_j^T + Z_{k-j}^T, \quad 1 \leqslant i \leqslant j \leqslant \frac{k}{2}.$$

Namely, the more balanced the subtrees are below edge e, the more lineages typically enter e.

Proof Lemma 1.2.7. The statement is vacuously true for $k = 2, 3$, so for induction assume it is true for all $k' < k$. Let $S_{k,m}^T := Z_m^T + Z_{k-m}^T$ denote the total number of lineages entering an edge e with subtrees of size $m, k-m$, the setup in Figure 1.4. Let $\tau_{k,m}$ be the time of the first coalescent event in either of our two subpopulations of size $m, k-m$, allowing the possibility $\tau_{k,m} > T$.

The basic idea is that if we start with k lineages, we can reduce to $k-1$ lineages by just waiting for the first coalescence event to occur in either subpopulation. There is of course the chance no coalescent event occurs, so we must handle this separately. The proof then relies on a few main ideas. Firstly, for more balanced splits we tend to wait longer until the first coalescent event.

Claim 1. For fixed k, $\tau_{k,m}$ is stochastically increasing in m for $1 \leqslant m \leqslant k/2$.

Next, conditional on a specific first coalescent time, the more balanced our split is, the more lineages tend to remain uncoalesced by the end of the branch.

Claim 2. For fixed k, x, t, $\mathbb{P}(S_{k,m}^T \geqslant x \mid \tau_{k,m} = t)$ is an increasing function of m for $1 \leqslant m \leqslant k/2$.

Lastly, the more time we wait until the first coalescent event, the less time the remaining lineages have to coalesce and the more lineages we tend to have by the end of the branch.

Claim 3. For fixed k, x, m, $\mathbb{P}(S_{k,m}^T \geqslant x \mid \tau_{k,m} = t)$ is an increasing function of t.

Proof of Claim 1. Let $h_{k,m} := \binom{m}{2} + \binom{k-m}{2}$. Then $\tau_{k,m} \sim \operatorname{Exp}(h_{k,m})$. This comes from the fact that coalescents occur at rate $\binom{m}{2}$ in the population of size m and at rate $\binom{k-m}{2}$ in the population of size $k-m$. Together, these produce coalescent events at rate $h_{k,m}$. Since $\tau_{k,m} \sim \exp(h_{k,m})$

$$\mathbb{P}(\tau_{k,m} > t) = \exp(-t h_{k,m})$$

As $h_{k,m}$ is decreasing in m for $1 \leqslant m \leqslant k/2$, the above is increasing in m. Hence $\tau_{k,m}$ is stochastically increasing in m. This proves Claim 1. $\square$

Proof of Claim 2. For $t > T$ then we $\mathbb{P}(S_{k,m}^T \geqslant x | \tau_{k,m} = t) = 1$ so the claim is trivially true. For $t < T$, by the strong memoryless property of the exponential distribution underlying Kingman's coalescent, we know

$$\mathbb{P}(S_{k,m}^T \geqslant x \mid \tau_{k,m} = t) = p_{k,m} \cdot \mathbb{P}(S_{k-1,m-1}^{T-t} \geqslant x) + (1 - p_{k,m}) \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geqslant x) \tag{1.6.1}$$

where

$$p_{k,m} = \mathbb{P}\left[\exp\left(\binom{m}{2}\right) \leqslant \exp\left(\binom{k-m}{2}\right)\right] = \frac{\binom{m}{2}}{\binom{m}{2} + \binom{k-m}{2}}$$

is the probability the first coalescence occurs in the population of size m. This essentially just says that after the first coalescent event occurs in either subpopulation, what remains is still a Kingman's coalescent just with one less lineage. By our inductive hypothesis, for $1 \leqslant m \leqslant k/2 - 1$

$$\begin{aligned}
&\mathbb{P}(S_{k,m}^T \geqslant x \mid \tau_{k,m} = t) \\
&= p_{k,m} \cdot \mathbb{P}(S_{k-1,m-1}^{T-t} \geqslant x) + (1 - p_{k,m}) \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geqslant x) \\
&\leqslant \mathbb{P}(S_{k-1,m}^{T-t} \geqslant x) \\
&\leqslant p_{k,m+1} \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geqslant x) + (1 - p_{k,m+1}) \cdot \mathbb{P}(S_{k-1,m+1}^{T-t} \geqslant x) \\
&= \mathbb{P}(S_{k,m+1}^T \geqslant x \mid \tau_{k,m+1} = t)
\end{aligned}$$

so that $\mathbb{P}(S_{k,m}^T \geqslant x \mid \tau_{k,m} = t)$ is increasing in m for each fixed t. This proves Claim 2. □

Proof of Claim 3. By equation (1.6.1) it suffices to show for fixed k, m that $S_{k,m}^T$ is stochastically decreasing in T. But this follows from a simple coupling argument. For $s \leqslant T$, couple $S_{k,m}^T$ to $S_{k,m}^s$ by using the same exponential clocks for the coalescent events. At time s the number of lineages remaining agree, and $S_{k,m}^T$ can only decrease from here if more coalescents occur between times s and T. This proves Claim 3. □

With the claims proved, we are ready to finally finish our proof of Lemma 1.2.7. Note that our above claims are exactly the conditions of Lemma 1.5.11 where $\{S_{k,m}^T\}_{m=1}^{\lfloor k/2 \rfloor}$ are playing the role of our V_i and $\{\tau_{k,m}\}_{m=1}^{\lfloor k/2 \rfloor}$ are playing the role of our Θ_i. Hence the result just follows from applying this lemma. This completes the proof of Lemma 1.2.7.

□

Proof of Lemma 1.2.8: Balanced is Worst-Case

Recall we aim to prove:

Lemma: If $\mathcal{B}_k$ is the balanced tree with k leaves and all branches length $T_{\min}$, then $X_{\mathcal{B}_k} \geqslant_{st} X_{\mathcal{T}}$ for any other tree $\mathcal{T}$ with k leaves and minimum branch length $T_{\min}$.

To simplify notation a bit, denote $X_m := X_{\mathcal{B}_m}$. To start, we will prove two extensions of our deterministic balancing lemma 1.2.7. The first allows the difference between the subpopulation sizes to be random. Heuristically, it says that conditional on the total number of species in the two subtrees below a given vertex, the more even the split tends to be, the more lineages tend to reach the vertex.

Lemma 1.6.2. *Suppose for $s \in \mathbb{N}$ that $0 \leqslant_{st} D \leqslant_{st} D' < s$ and $D, D' \equiv s \bmod(2)$. Then*

$$Z_{\frac{s+D}{2}} + Z_{\frac{s-D}{2}} \geqslant_{st} Z_{\frac{s+D'}{2}} + Z_{\frac{s-D'}{2}}.$$

Proof of Lemma 1.6.2. Since the sum $s = (s+d)/2 + (s-d)/2$ is fixed, our deterministic balancing lemma 1.2.7 implies that

$$Y_d := Z_{\frac{s+d}{2}} + Z_{\frac{s-d}{2}}$$

is decreasing in d with respect to $\leqslant_{st}$ for $0 \leqslant d < s$. Lemma 1.5.11 then implies the result. □

The next extension generalizes this even further by allowing the sum of the subpopulation sizes to be random as well. Heuristically, it captures the same idea as before: more balanced subpopulations tend to produce less coalescence.

Lemma 1.6.3 (Balancing Lemma)**.**
If A, A', B, B' are mutually independent, positive integer-valued random variables so that $A \leqslant_{lr} A'$ and $B \leqslant_{lr} B'$, then

$$Z_{A+B} + Z_{A'+B'} \leqslant_{st} Z_{A+B'} + Z_{A'+B}.$$

Note when A, B, A', B' are all deterministic, this reduces to Lemma 1.2.7.

Proof of Balancing Lemma 1.6.3. For convenience, define

$$\Delta_A := A' - A, \quad \Delta_B := B' - B, \quad S_A := A' + A, \quad S_B = B' + B.$$

Now, note in both $Z_{A+B}+Z_{A'+B'}$ and $Z_{A+B'}+Z_{A'+B}$ the total number of lineages $S = S_A+S_B$ is the same. Hence by conditioning on $S_A, S_B, |\Delta_A|$, and $|\Delta_B|$, Lemma 1.6.2 implies it suffices to show $D' \geqslant_{st} D$ where D, D' are the conditional differences

$$D' = |\Delta_A + \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}, \qquad D = |\Delta_A - \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}.$$

After this conditioning, $\{D, D'\} = \{|\Delta_A| + |\Delta_B|, ||\Delta_A| - |\Delta_B||\}$. Let

$$q = \mathbb{P}(D' = |\Delta_A| + |\Delta_B|) = \mathbb{P}(\Delta_A\Delta_B \geqslant 0 \mid |\Delta_A|, |\Delta_B|, S_A, S_B)$$

be the probability D' takes on the larger of these two values. Then it suffices to show $q \geqslant 1/2$. Since Δ_A, Δ_B are conditionally independent given S_A, S_B

$$q = p_A p_B + (1 - p_A)(1 - p_B) \geqslant \frac{1}{2} \iff \left(p_A - \frac{1}{2}\right)\left(p_B - \frac{1}{2}\right) \geqslant 0,$$

where $p_A = \mathbb{P}(\Delta_A \geqslant 0 \mid |\Delta_A|, S_A)$ and likewise for p_B. We will show that $p_A, p_B \geqslant 1/2$, completing the proof. Note conditional on $\{|\Delta_A| = m, S_A = s\}$ then $p_A \geqslant 1/2$ occurs precisely when

$$\mathbb{P}\left(A' = \frac{s+m}{2}, A = \frac{s-m}{2}\right) \geqslant \mathbb{P}\left(A' = \frac{s-m}{2}, A = \frac{s+m}{2}\right).$$

Since A, A' are independent, this follows from $A \leqslant_{lr} A'$. Similarly, $p_B \geqslant 1/2$. Thus $D' \geqslant_{st} D$. □

Now we are ready to return to the main result. First, we show that the result follows from the following stochastic extension of Lemma 1.2.7:

Lemma 1.6.4. *For fixed $k \in \mathbb{N}$, then $Z_{X_i} + Z_{X_{k-i}} \leqslant_{st} Z_{X_{\lfloor k/2 \rfloor}} + Z_{X_{\lceil k/2 \rceil}}$ for $1 \leqslant i \leqslant \lfloor k/2 \rfloor$.*

We conjecture that $Z_{X_i} + Z_{X_{k-i}}$ is increasing in i rather than just maximized at the even split. But for our current purposes, the above suffices. Assuming Lemma 1.6.4 for now, it is easy to prove our desired Lemma 1.2.8.

Reduction of Lemma 1.2.8 to Lemma 1.6.4. We induct on the number of species k.

Base Case ($k = 4$)
Again there are only two different tree topologies, the ones in Figure 1.2. Lemma 1.2.7 shows the even split $Z_2 + Z_2' \geqslant_{st} Z_1 + Z_3'$, and clearly $Z_1 + Z_3' \geqslant_{st} Z_1 + Z_{X_3}'$ by Corollary 1.5.10.

Inductive Step
Suppose $\mathcal{T}$ is a tree with $k > 4$ leaves. As usual, the main idea is to reduce to smaller trees by looking at the subtrees of the root. We perform the induction in two steps: we first reduce to the case the two subtrees of the root are themselves balanced trees, and then we show in such a setting the worst-case is when the two subtrees have the same number of leaves.

Let L and R be the two subtrees of the root of $\mathcal{T}$ which are of sizes m and $k - m$ respectively. By our inductive hypothesis, $X_L \leqslant_{st} X_{\mathcal{B}_m}$ and $X_R \leqslant_{st} X_{\mathcal{B}_{k-m}}$. Let $\mathcal{T}'$ be the tree whose two subtrees are $\mathcal{B}_m, \mathcal{B}_{k-m}$, and whose branch lengths are all $T_{\min}$. Then as $X_{\mathcal{T}} = Z_{X_L} + Z_{X_R}$ we see Corollary 1.5.10 implies $X_{\mathcal{T}'} \geqslant_{st} X_{\mathcal{T}}$ stochastically. Hence all that is left to show is that $X_{\mathcal{B}_k} \geqslant_{st} X_{\mathcal{T}'}$. But this is exactly just the content of Lemma 1.6.4. Hence Lemma 1.2.8 follows from Lemma 1.6.4. □

Now, all that is left to do is prove Lemma 1.6.4.

Proof of Lemma 1.6.4. We prove this statement by induction on k. Again the base case is trivial

$$Z_1 + Z_{X_3} \leqslant_{st} Z_1 + Z_3 \leqslant_{st} Z_2 + Z_2 = Z_{X_2} + Z_{X_2}.$$

Now suppose the statement is true for $k' < k$. Note

$$X_m = Z_{X_{\lfloor m/2 \rfloor}} + Z_{X_{\lceil m/2 \rceil}}, \qquad X_{k-m} = Z_{X_{\lfloor (k-m)/2 \rfloor}} + Z_{X_{\lceil (k-m)/2 \rceil}}.$$

By Lemma 1.5.14 we have for $1 \leqslant m \leqslant k/2$

$$Z_{X_{\lfloor m/2 \rfloor}} \leqslant_{lr} Z_{X_{\lceil m/2 \rceil}} \leqslant_{lr} Z_{X_{\lfloor (k-m)/2 \rfloor}} \leqslant_{lr} Z_{X_{\lceil (k-m)/2 \rceil}}.$$

Note that if k is even then

$$\lfloor m/2 \rfloor + \lceil (k-m)/2 \rceil = \lceil m/2 \rceil + \lfloor (k-m)/2 \rfloor = \frac{k}{2}.$$

The balancing lemma 1.6.3 then implies

$$Z_{X_m} + Z_{X_{k-m}} \leqslant_{st} \left(Z_{Z_{X_{\lfloor m/2 \rfloor}} + Z_{X_{\lceil (k-m)/2 \rceil}}} \right) + \left(Z_{Z_{X_{\lceil m/2 \rceil}} + Z_{X_{\lfloor (k-m)/2 \rfloor}}} \right).$$

The inductive hypothesis applied to $k' = k/2$ shows the subscripts of each term are maximized at $m = k/2$. Hence Corollary 1.5.10 implies

$$\leqslant_{st} \left(Z_{Z_{X_{\lfloor k/4 \rfloor}} + Z_{X_{\lceil k/4 \rceil}}} \right) + \left(Z_{Z_{X_{\lceil k/4 \rceil}} + Z_{X_{\lfloor k/4 \rfloor}}} \right) = Z_{X_{k/2}} + Z_{X_{k/2}},$$

as desired. Likewise, if k is odd then

$$\lfloor m/2 \rfloor + \lfloor (k-m)/2 \rfloor = \frac{k-1}{2}, \quad \textit{and} \quad \lceil m/2 \rceil + \lceil (k-m)/2 \rceil = \frac{k+1}{2}$$

so we can rearrange using the balancing lemma 1.6.3 as

$$Z_{X_m} + Z_{X_{k-m}} \leqslant_{st} \left(Z_{Z_{X_{\lfloor m/2 \rfloor}} + Z_{X_{\lfloor (k-m)/2 \rfloor}}} \right) + \left(Z_{Z_{X_{\lceil m/2 \rceil}} + Z_{X_{\lceil (k-m)/2 \rceil}}} \right)$$

Applying induction with $k' = (k-1)/2$ on the left term and $k' = (k+1)/2$ on the right term we see both terms are maximized at $m = \lfloor k/2 \rfloor = (k-1)/2$. Hence

$$\leqslant_{st} \left(Z_{Z_{X_{\lfloor (k-1)/4 \rfloor}} + Z_{X_{\lfloor (k+1)/4 \rfloor}}} \right) + \left(Z_{Z_{X_{\lceil (k-1)/4 \rceil}} + Z_{X_{\lceil (k+1)/4 \rceil}}} \right) = Z_{X_{(k-1)/2}} + Z_{X_{(k+1)/2}}$$

as desired. $\square$

1.7 Appendix: Asymptotic Results

Proof of Lemma 1.3.1: Asymptotics $g_{k,1}(T)$

In this section, we develop some asymptotics for the time $g_{k,1}(T)$ to coalescence of Kingman's coalescent, specialized to the topic of interest. More properties of this coalescent time are discussed in [47] and [11]. The paper [34] further generalizes some of these ideas to Λ-coalescents.

We will work in slightly more generality than this paper requires. Namely, suppose that we have positive **distinct** rates $\{\lambda_i\}_{i=1}^{\infty}$. Define the random variables $\{S_k\}_{k=1}^{\infty}$ by $S_k = \sum_{i=1}^{k} X_i$ for $X_i \sim \operatorname{Exp}(\lambda_i)$ independent (namely X_i has density $\tau_i(x) := \lambda_i \exp(-\lambda_i x)$). Moreover, define $F_k(T) := \mathbb{P}(S_k \leqslant T)$ to be their CDFs and $f_k(T)$ their densities. If we define

$$C_{i,k} := \prod_{\substack{m=1 \\ m \neq i}}^{k} \frac{\lambda_m}{\lambda_m - \lambda_i}$$

then we have the following well-known explicit formulas for the CDFs and densities

$$F_k(T) = 1 - \sum_{i=1}^{k} C_{i,k} \exp(-\lambda_i T), \qquad f_k(T) = \sum_{i=1}^{k} C_{i,k} \lambda_i \exp(-\lambda_i T). \tag{1.7.1}$$

See [66], for example. Moreover, we have the natural recursive relation

$$F_k(T) = (F_{k-1} * \tau_k)(T) := \int_0^T F_{k-1}(x) \lambda_k \exp(-\lambda_k (T - x))\, dx. \tag{1.7.2}$$

Since S_k are monotonically increasing, we see $S_k \uparrow S_\infty := \sum_{i=1}^{\infty} X_i$ almost surely, where the limit may be infinite. To quantify the rate of convergence, define the partial sums $s_k = \sum_{i=1}^{k} \lambda_i$, the remainder $R_k := \sum_{i=k+1}^{\infty} X_i$, and the remainder's expectation $r_k := \mathbb{E} R_k = \sum_{i=k+1}^{\infty} \lambda_i^{-1}$. We start by getting some explicit formulas for the derivatives of the CDF in terms of these rates $\{\lambda_i\}$:

Lemma 1.7.1. *If we have positive, distinct rates $\{\lambda_i\}_{i=1}^{\infty}$:*

(i) At zero, the one-sided derivatives $F_k^{(m)}(0) = 0$ for $0 \leqslant m \leqslant k-1$. The nonzero derivatives take the form

$$F_k^{(k+r)}(0) = (-1)^r \left(\prod_{i=1}^{k} \lambda_i \right) \left(h_r(\lambda_1, \ldots, \lambda_k) \right),$$

where h_r are the complete homogeneous symmetric polynomials

$$h_r(\lambda_1, \ldots, \lambda_k) := \sum_{1 \leqslant i_1 \leqslant i_2 \leqslant \ldots \leqslant i_r \leqslant k} \prod_{j=1}^{r} \lambda_{i_j}.$$

In particular, for $r = 0, 1$ we have

$$F_k^{(k)}(0) = \prod_{i=1}^{k} \lambda_i, \qquad F_k^{(k+1)}(0) = -\left(\sum_{i=1}^{k} \lambda_i\right)\left(\prod_{i=1}^{k} \lambda_i\right).$$

(ii) The derivatives are uniformly bounded. Namely, for fixed m

$$||F_k^{(m)}||_\infty \leqslant M_m = \max_{i=1,\ldots,m} |F_i^{(m)}(0)|.$$

Proof. (i) It is easy to see that for $m \geqslant 1$

$$F_k^{(m)}(0) = (-1)^{m-1} \sum_{i=1}^{k} C_{i,k} \lambda_i^m$$

which vanishes by Lemma 2 of [66] when $m \leqslant k-1$. To calculate the first nonzero derivatives, we will rely on the recurrence in Equation 1.7.2. Note that given two smooth functions $g, h : [0, \infty) \to \mathbb{R}$ and their convolution

$$g * h(T) := \int_0^T g(T-t)h(t)\, dt = \int_0^T g(t)h(T-t)\, dt.$$

Hence, by Leibniz's integral rule we have

$$(g * h)'(T) = (g' * h)(T) + g(0)h(T) = (g * h')(T) + g(T)h(0). \tag{1.7.3}$$

By iteratively applying this to $F_k = F_{k-1} * \tau_k$, we see

$$F_k^{(m)}(T) = (F_{k-1}^{(m)} * \tau_k)(T) + \sum_{j=0}^{m-k} F_{k-1}^{(k-1+j)}(0)\tau_k^{(m-k-j)}(T), \tag{1.7.4}$$

since by (ii) the first $k-2$ derivatives of F_{k-1} vanish. In particular, if $k > m$ we see $F_k^{(m)} = F_{k-1}^{(m)} * \tau_k$. Using this, we can prove $F_k^{(k+m)}(0)$ has the desired form by induction. Note $F_1(T) = 1 - \exp(-\lambda_1 T)$ so $F_1^{(m+1)}(0) = (-1)^m \lambda_1 \cdot \lambda_1^m$ as desired. Moreover, $\tau_k^{(r)}(0) = (-1)^r \lambda_k^{r+1}$. Thus by induction and equation (1.7.4)

$$F_k^{(k+m)}(0) = \sum_{j=0}^{m} F_{k-1}^{(k-1+j)}(0)\, \tau_k^{(m-j)}(0) = (-1)^m \left(\prod_{i=1}^{k} \lambda_i\right) \sum_{j=0}^{m} \lambda_k^{m-j} a_{k-1,j}$$

where $a_{k,m} := h_m(\lambda_1, ..., \lambda_k)$ are our symmetric polynomials. Hence it suffices to show these polynomials satisfy the recurrence

$$a_{k,m} = \sum_{j=0}^{m} \lambda_k^{m-j} a_{k-1,j}.$$

But this is almost immediate, just by grouping terms according to the number of i_ℓ that equal k

$$a_{k,m} = \sum_{1 \leqslant i_1 \leqslant i_2 \leqslant \cdots \leqslant i_m \leqslant k} \prod_{\ell=1}^{m} \lambda_{i_\ell} = \sum_{j=0}^{m} \lambda_k^{m-j} \sum_{1 \leqslant i_1 \leqslant i_2 \leqslant \cdots \leqslant i_j \leqslant k-1} \prod_{\ell=1}^{j} \lambda_{i_\ell} = \sum_{j=0}^{m} \lambda_k^{m-j} a_{k-1,j},$$

where the inner sum and product for $j = 0$ are understood to equal one. Hence

$$F_k^{(k+m)}(0) = (-1)^m a_{k,m} \prod_{i=1}^{k} \lambda_i$$

takes the desired form.
(ii) Since we saw in the previous section that for $k > m$ we have $F_k^{(m)} = F_{k-1}^{(m)} * \tau_k$, by iteratively applying Young's convolution inequality we get

$$||F_k^{(m)}||_\infty \leqslant ||F_{k-1}^{(m)}||_\infty \cdot ||\tau_k||_1 = ||F_{k-1}^{(m)}||_\infty \Longrightarrow ||F_k^{(m)}||_\infty \leqslant ||F_m^{(m)}||_\infty, \quad \forall k \geqslant m.$$

Hence we can see $||F_k^{(m)}||_\infty \leqslant \max_{i=1,\ldots,m} ||F_i^{(m)}||_\infty =: M_m < \infty$.

To quantify M_m, note that using the expression for the derivatives of τ_k, we can see equation (1.7.4) can be re-expressed as

$$F_k^{(m)}(T) = F_{k-1}^{(m)} * \tau_k(T) + F_k^{(m)}(0) \exp(-\lambda_k T).$$

Then Young's convolution inequality implies

$$|F_k^{(m)}(T)| \leqslant ||F_{k-1}^{(m)}||_\infty \cdot (1 - \exp(-\lambda_k T)) + |F_k^{(m)}(0)| \exp(-\lambda_k T) \leqslant \max\{||F_{k-1}^{(m)}||_\infty, |F_k^{(m)}(0)|\}.$$

for any $T \geqslant 0$. Taking the supremum over T yields

$$||F_k^{(m)}||_\infty \leqslant \max\{||F_{k-1}^{(m)}||_\infty, |F_k^{(m)}(0)|\}.$$

By induction we see $M_m = \max_{i=1,\ldots,m} |F_i^{(m)}(0)|$ as desired. □

CDF Asymptotics

Now that we have these bounds, the main results follow. We in fact show something slightly stronger: that the relative error in the Taylor series approximation is negligible in the $T = o((s_k/k)^{-1})$ regime no matter where we truncate it.

Theorem 1.7.2. *If $m = \arg\min_{i=1,\ldots,k} \lambda_i$, then the CDFs have asymptotics*

$$\begin{cases} 1 - F_k(T) \sim C_{m,k} \exp(-\lambda_m T), & T \uparrow \infty \text{ and } k \text{ fixed,} \\ F_k(T) \sim \frac{\prod_{i=1}^{k} \lambda_i}{k!} T^k, & T = o((s_k/k)^{-1}), \end{cases}$$

where $T = o((s_k/k)^{-1})$ denotes any sequence of pairs (T_n, k_n) so that $\lim_{n\to\infty} s_{k_n} T_n / k_n = 0$. More generally, in the same $T = o((s_k/k)^{-1})$ regime and for any finite $m \geqslant 0$

$$\frac{F_k(T) - \sum_{r=0}^{m} a_{r,k}}{a_{m,k}} \to 0, \qquad a_{r,k} := (-1)^r \frac{h_r(\lambda_1, \ldots, \lambda_k) \prod_{i=1}^{k} \lambda_i}{(k+r)!} T^{k+r}.$$

Proof. For k fixed, the exponential in equation 1.7.1 with the smallest rate decays the slowest, so $1 - F_k(T) \sim C_{m,k} \exp(-\lambda_m T)$ as $T \to \infty$. For the other asymptotics, first fix k and let $h_{r,k} := h_r(\lambda_1, ..., \lambda_k)$ be our symmetric polynomials and $p_k = \prod_{i=1}^{k} \lambda_i$. Since $F_k(T)$ is the sum of a finite number of exponentials, its Taylor series at zero converges to $F_k(T)$ for all T. Moreover, (i) of Lemma 1.7.1 above tells us $F_k^{(k+r)}(0) = (-1)^r p_k h_{r,k}$. Thus, let

$$a_{r,k} = (-1)^r \frac{p_k h_{r,k}}{(k+r)!} T^{k+r}$$

denote the $(r+1)^{th}$ nonzero term in the Taylor series expansion about zero. Suppose that we approximate $F_k(T)$ by the first $(m+1)$ nonzero terms of its Taylor series about zero. We can measure the approximation error relative to the last term $a_{m,k}$ to see

$$\frac{F_k(T) - \sum_{r=0}^{m} a_{r,k}}{a_{m,k}} = \sum_{r=1}^{\infty} (-1)^r \frac{h_{m+r}/h_m T^r}{(k+m+r)_r}$$

where $(x)_r = x(x-1) \cdot ... \cdot (x-r+1)$ denotes the falling factorial. We will now bound the right hand side by a geometric series.

Note that trivially $h_{r+1} \leqslant h_r s_k$ as the former includes all monomials of degree $(r+1)$ in $\{\lambda_i\}_{i=1}^{k}$ with coefficient one, and the latter has all the monomials with coefficient at least one. Hence $h_{m+r}/h_m = \prod_{i=0}^{r-1} h_{m+i+1}/h_{m+i} \leqslant s_k^r$. Using this and the simple bound $(k+m+r)_r \geqslant k^r$ we get

$$\left| \sum_{r=1}^{\infty} (-1)^r \frac{h_{m+r}/h_m T^r}{(k+m+r)_r} \right| \leqslant \sum_{r=1}^{\infty} \left(\frac{s_k T}{k} \right)^r = \frac{s_k T/k}{1 - s_k T/k}$$

for $|s_k T/k| < 1$. When $T = o((s_k/k)^{-1})$, this tends to zero. Thus in this regime

$$\frac{F_k(T) - \sum_{r=0}^{m} a_{r,k}}{a_{m,k}} \to 0,$$

as desired. In particular

$$F_k(T) = (1 + o(1)) \frac{\prod_{i=1}^{k} \lambda_i}{k!} T^k.$$

□

Gap Asymptotics

The following statement gives us a sense of how fast our coalescent probabilities $g_{k,1}(T)$ will saturate as $k \to \infty$.

Theorem 1.7.3 (Gap Asymptotics)**.** *If $\sum_{i=1}^{\infty} \lambda_i^{-1} < \infty$, then $F_k \to F_\infty$ uniformly. Quantitatively, the gap $\Delta_k(T) = F_k(T) - F_\infty(T)$ has asymptotics*

$$\Delta_k = (f_\infty + o(1)) r_k.$$

Proof. Dini's theorem already guarantees that $F_k \to F_\infty$ uniformly. Here, we will get quantitative control on the rate of convergence. Note that (ii) of Lemma 1.7.1 implies the functions f_k are uniformly bounded by M_1 and uniformly M_2-Lipschitz. Thus they form a uniformly equicontinuous family which is uniformly bounded. Since $S_k \Rightarrow S_\infty$, as a consequence of [6] converse to Scheffe's lemma, we see $f_k \to f_\infty$ uniformly on $[0, \infty)$. Now, note that by conditioning on the tail R_k, as $S_k \perp\!\!\!\perp R_k$

$$\mathbb{P}(S_\infty \leqslant T) = \int_0^\infty \mathbb{P}(S_k \leqslant T - r)\mathbb{P}(R_k \in dr) = \mathbb{E}F_k(T - R_k).$$

Hence by Taylor expanding the CDF F_k we get

$$\Delta = \mathbb{E}_{R_k}[F_k(T) - F_k(T - R_k)] = f_k(T) \cdot \mathbb{E}R_k + \frac{1}{2}\mathbb{E}\left(f_k'(\xi(R_k))R_k^2\right),$$

where $\xi(r) \in (T - r, T)$ for every r. Note that $\mathrm{Var}(R_k) = \sum_{i=k+1}^\infty \lambda_i^{-2} \leqslant r_k^2$, implying $\mathbb{E}R_k^2 \leqslant 2r_k^2$. Since $||f_k'||_\infty \leqslant M_2$ is uniformly bounded by (iii) and as $f_k(T) \to f_\infty(T)$ uniformly as $k \to \infty$

$$\Delta_k(T) = f_\infty(T)r_k + (f_k(T) - f_\infty(T))r_k + O(r_k^2) = f_\infty(T)r_k + o(r_k).$$

More specifically, since $M_2 = \max\{|F_1''(0)|, |F_2''(0)|\} = \lambda_1 \max(\lambda_1, \lambda_2)$

$$||\Delta_k(T) - f_\infty r_k||_\infty \leqslant ||f_k - f_\infty||_\infty r_k + \lambda_1 \max(\lambda_1, \lambda_2) r_k^2 = o(r_k)$$

so the rate of convergence is uniform in T. □

Lemma 1.3.1 then just follows from the special case of Kingman's coalescent, where we have rates $\tilde{\lambda}_i = \binom{i+1}{2}$. Note the indexing is slightly different here than in the main paper: it is shifted down by one, but otherwise the same. Here we have

$$r_k = \sum_{i=k+1}^\infty \frac{1}{\binom{i+1}{2}} = \frac{2}{k+1}, \qquad s_k = \sum_{i=1}^k \binom{i+1}{2} = \frac{k(k+1)(k+2)}{6} = \Theta(k^3).$$

Moreover, a simple induction shows that $C_{1,k} = 3k/(k+2)$. Hence we get asymptotics

$$F_k(T) \sim \begin{cases} 1 - \frac{3k}{k+2}\exp(-T), & T \uparrow \infty \text{ and k fixed,} \\ \frac{(k+1)!}{2^k}T^k, & T = o(k^{-2}). \end{cases} \qquad \Delta_k(T) = (f_\infty(T) + o(1)) \cdot \frac{2}{k+1}$$

Improvement Ratio Asymptotics

In Section 1.3 above, we obtained the following asymptotic lower bound on the improvement factor of our balanced bound (Theorem 1.2.9) over the original bound:

$$\beta_T := \frac{-\log(1 - g_{u(T),1}(T))}{-\log(1 - s(T))}$$

In this section, we develop small T asymptotics for β_T. In particular:

Lemma 1.7.4 (Small T Improvements)**.** *As $T \downarrow 0$, we have*

$$\beta_T \sim \frac{\pi^2}{2T}$$

Proof of Lemma 1.7.4. Since $-\log(1-x) \sim x$ as $x \downarrow 0$

$$\beta_T \sim 1 + \frac{\Delta(T)}{s(T)}, \qquad \Delta(T) := g_{u(T),1}(T) - s(T).$$

Lemma 1.3.1 and Corollary 1.5.9 imply as $T \downarrow 0$

$$\frac{\Delta(T)}{s(T)} \sim \frac{2f_{S_\infty}(T)}{s(T)} \cdot \frac{1}{u(T)} \sim T \cdot r(T),$$

where $r := d/dt \log(s(t))$ is the reversed hazard rate of S_∞. To control this, we develop asymptotics for $K(x) := \log \mathbb{E} \exp(-xS_\infty)$ and use a classical Tauberian theorem to relate this to the log CDF. By independence, for $x > 0$

$$K(x) = \log\left(\prod_{n=2}^{\infty} \frac{\lambda_n}{\lambda_n + x}\right) = -\log\left(\prod_{n=2}^{\infty}\left[1 + \frac{x}{\lambda_n}\right]\right).$$

Since $(n-1)^2 \leqslant 2\lambda_n \leqslant n^2$, using the infinite product expansion of sinh, we see

$$0 \leqslant K(x) + \log\left(\frac{\sinh(\pi\sqrt{2x})}{\pi\sqrt{2x}}\right) \leqslant \log(1+2x).$$

Hence as $\log(y^{-1}\sinh(y)) \sim y$ as $y \to \infty$, we see $K(x) \sim -\pi\sqrt{2x}$ as $x \to \infty$. Applying Corollary 2 of [9] with $A = 1, P(x) = s(x), B = -\pi^2/2$, and $\beta = -1$ we see this implies that $\log(s(T)) \sim -\pi^2/2T^{-1}$ as $T \downarrow 0$.

Let $h(t) := \log s(t)$. Prékopa's theorem implies that S_k is log-concave for all k and, as log-concavity is preserved under weak limits, so is S_∞. Hence its distribution function $s(T)$ is log-concave, and thus $h(t)$ is concave. Thus for fixed $\epsilon \in (0,1)$, its derivative is bounded by the two secant slopes

$$\frac{h((1+\epsilon)t) - h(t)}{\epsilon t} \leqslant h'(t) \leqslant \frac{h(t) - h((1-\epsilon)t)}{\epsilon t}.$$

Our asymptotics for h then imply

$$\frac{1 + o_\epsilon(1)}{1+\epsilon} \leqslant \frac{h'(t)}{\pi^2/2t^2} \leqslant \frac{1+o_\epsilon(1)}{1-\epsilon},$$

and thus

$$\frac{1}{1+\epsilon} \leqslant \liminf_{t\downarrow 0} \frac{h'(t)}{\pi^2/2t^2} \leqslant \limsup_{t\downarrow 0} \frac{h'(t)}{\pi^2/2t^2} \leqslant \frac{1}{1-\epsilon}$$

Sending $\epsilon \downarrow 0$ shows $r(t) \sim \pi^2/(2t^2)$ as desired. $\square$

Note that above we showed

$$s(T) = \exp\left(-\frac{\pi^2}{2}T^{-1}(1+o(1))\right).$$

Since for typical trees (e.g. Yule trees) we have $T_{\min} = O(k^{-1})$, this heuristically tells us the original bound is $O(\log(k)\exp(k))$.

Bound Asymptotics (Lemma 1.3.2)

Now that we have analyzed the core function $g_{k,1}(T)$ we are ready to analyze our bounds. Let $\kappa_{q,k} = -\log((1-q)/(k-3))$ be the coefficient that occurs in the numerator of all our bounds. Moreover, let $s(T) := \mathbb{P}(S_\infty \leqslant T)$ be our limiting CDF, $\Delta_k(T) := g_{k,1}(T) - s(T)$ our gap, and $c(T) := 2f_{S_\infty}(T)$ our gap's asymptotic constant. Recall that $M_o(k,T)$ denotes the original bound from Theorem 1.1.1 and $M_b(k,T)$ our balanced bound Theorem 1.2.9.

We start with M_o. Note Lemma 1.3.1 implies

$$\log(1-g_{k,1}(T)) \sim \begin{cases} -T, & T \uparrow \infty, \\ -\frac{k!}{2^{k-1}}T^{k-1}, & T = o(k^{-2}). \end{cases} \tag{1.7.5}$$

From this

$$M_o(k,T) \sim \begin{cases} \frac{\kappa_{q,k}}{T}, & T \uparrow \infty, \\ \frac{2^{k-3}\kappa_{q,k}}{(k-2)!T^{k-3}}, & T = o(k^{-2}). \end{cases}$$

In particular, we recover the small T bound found in Appendix B of [59], although here we exactly quantify the asymptotic constant.

For the fixed T and asymptotic k bound, we Taylor expand $\log(x)$ about $1-s$ to observe:

$$\log(1-g_{k,1}(T)) = \log(1-s) - \sum_{n=1}^{\infty} \frac{\Delta^n}{n(1-s)^n} = \log(1-s) - \frac{\Delta}{1-s} - O(\Delta^2).$$

Hence this and Lemma 1.3.1 imply as $k \to \infty$

$$M_o(k,T) = \frac{\kappa_{q,k}}{\left(\frac{c(T)}{1-s} + o(1)\right)\cdot\frac{1}{k} - \log(1-s)} \sim \frac{\kappa_{q,k}}{-\log(1-s)}.$$

For $M_b(k,T)$, note as $g_{k,1}(T)$ is decreasing in k we have

$$\sum_{\ell=2}^{k-2}(1-w_\ell(T))^n \geqslant (k-3)(1-g_{2,1}(T))^n = (k-3)\exp(-nT).$$

Hence $M_b(k,T) \geqslant \kappa_{q,k}T^{-1}$. As $k \mapsto g_{k,1}(T)$ is decreasing and convex by Lemma 1.5.7

$$\sum_{\ell=2}^{k-2}(1-\mathbb{E}g_{U_\ell,1}(T))^n \leqslant (k-3)(1-g_{\mathbb{E}U_{k-2},1}(T))^n.$$

Applying Corollary 1.5.9

$$M_b(k,T) \leqslant \frac{\kappa_{q,k}}{-\log(1-g_{u(T),1}(T))}, \quad u(T) := \frac{2-\exp(-T/2)}{1-\exp(-T/2)}.$$

Combining these two previous results

$$\frac{\kappa_{q,k}}{T} \leqslant M_b(k,T) \leqslant \frac{\kappa_{q,k}}{-\log(1-g_{u(T),1}(T))}.$$

Since $M_b(k,T) \leqslant M_o(k,T)$, the lower bound shows us $M_b(k,T) \sim \kappa_{q,k}T^{-1}$ as $T \to \infty$. Since $u(T) \sim 2/T$ as $T \downarrow 0$, the upper bound does not provide any provable improvement in the regime $k^2T = o(1)$ that we analyzed previously. More careful analysis of the regime $kT = o(1)$ is likely necessary to observe a noticeable improvement, but this is outside the scope of our current analysis.

For fixed T, the upper bound implies that the asymptotic improvement factor of $M_b(k,T)$ over $M_o(k,T)$ is at least

$$\beta_T := \frac{\log(1-g_{u(T),1}(T))}{\log(1-s(T))}.$$

In Lemma 1.7.4, we show that $\beta_T \sim \pi^2/(2T)$ as $T \downarrow 0$. Thus, in this regime, the improvement factor is bounded below by a quantity asymptotic to $\pi^2/(2T)$.

Repeating our above analysis shows that *any* bound built using the union bound and the coalescent function $g_{k,1}(T)$ cannot be asymptotically better than $\log(k)/T$. This is true, *even if the descendant counts α_i are known*, as we can still upper bound $g_{\alpha_i,1}(T) \leqslant g_{2,1}(T)$. Hence the asymptotics in k are fundamentally limited to $\Theta(\log(k))$ by our union bound argument, and we can only hope to improve upon the constant down to the limit T^{-1}. Still, we see some finite-sample improvements empirically, as we observed in Figure 1.6.

Statements and Declarations

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Generative AI use and AI-assisted technologies: During the preparation of this work the author used OpenAI Codex to assist with debugging and reviewing simulation code. Additionally, this tool was used in the above document to aid in typesetting, copy-editing, and improving the exposition. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Chapter 2

Self-Balancing Sampling

This project was joint work with Daniel Raban.

Abstract

We study a family of sequential sampling rules that adaptively biases samples in order to achieve faster convergence of the empirical distribution to a desired target law, both in an average and worst-case sense. The resulting self-balancing sampler is simple to implement and arises naturally among a class of Markovian samplers sharing a certain invariance property. Moreover, it is unique solution to a natural entropy-regularized control problem, which in a sense balances the convergence rate of the empirical law and the unpredictability of the samples. In the weak-biasing regime, we show the properly centered counts process converges to an Ornstein–Uhlenbeck process in the diffusive limit, and use this limit to study how the behavior of the sampler depends on the choice of biasing parameter.

2.1 Introduction

Consider the problem of sequentially sampling X_i from d outcomes in a way that allows the empirical distributions

$$\mu(n) \coloneqq \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}$$

to converge to the uniform distribution. Two naive ways of doing this are:

- *Uniformly at random*: Sample $X_i \overset{\text{IID}}{\sim} \text{Uniform}([d])$.
- *Deterministically/Greedily*: Sample $X_i = i \pmod d + 1$.

For these two sampling schemes, there's a trade-off between the *speed* of convergence and the *predictability* of the samples. Namely, the former converges at rate $\Theta(n^{-1/2})$ in the total variation distance, while the latter converges at the much faster rate $\Theta(n^{-1})$. On the other hand, the former process is as unpredictable as possible, maximizing the entropy at each sampling step, while the latter is perfectly predictable.

This tradeoff becomes readily apparent in the problem of scheduling audits or inspections, such as testing for doping in professional sports. A sports regulatory body is tasked with routinely testing teams for the use of performance enhancing drugs, but any deterministic schedule has the potential to be taken advantage of, since the players can temporarily cease doping in anticipation of the tests. On the other hand, sampling teams IID to receive doping tests can lead to wasteful behavior, such as spending resources to test a team multiple times in relatively short succession, while letting other teams remain temporarily untested; in effect, temporary imbalances in the empirical distribution of audited teams reflect inefficiencies in the order of teams chosen.

A closely related problem is how to sample from a population while keeping the realized sample balanced across several overlapping strata. For example, suppose we are constructing a citizen's assembly of 100 members out of a volunteer pool of 1000, and we would like our assembly to be demographically representative across multiple overlapping characteristics, such as gender, age, and education level. We may attempt to balance these characteristics using a deterministic or quota-constrained selection algorithm for choosing a committee, but, as noted in [21], some algorithms can lead to somewhat undemocratic outcomes in which certain members of the volunteer pool have a miniscule chance of being chosen for the commmittee. On the other hand, a simple random sample can result in a committee which is not properly representative for each characteristic. In a similar setting, [28] explores an analogous trade-off between covariate balance and robustness.

In this paper, we aim to explore the trade-off between these two aims for a natural family of sampling rules. Heuristically, we show that by adaptively biasing our sampling process, we can limit predictability while maintaining fast convergence rates.

For instance, consider the following sequential sampling rule, informally described as "downweight the outcome's probability and renormalize": Fix some $\alpha \in (0, 1)$, and start with some initial distribution $p(0) \in \Delta^d$. At step n, recursively do the following:

Step 1: Sample an outcome $X_n \sim p(n-1)$.

Step 2: Define $p(n)$ by downweighting $p_{X_n}(n-1)$ by a factor of α and renormalizing the probability vector, i.e.

$$p_i(n) := \begin{cases} \frac{\alpha \cdot p_i(n-1)}{1-(1-\alpha)\cdot p_{X_n}(n-1)} & \text{if } i = X_n, \\ \frac{p_i(n-1)}{1-(1-\alpha)\cdot p_{X_n}(n-1)} & \text{if } i \neq X_n. \end{cases}$$

See Figure 2.1 for a visual schematic of the update rule.

This sequential sampling rule is still random, and intuitively, it generates empirical distributions which are closer to uniform because it de-emphasizes outcomes which are currently

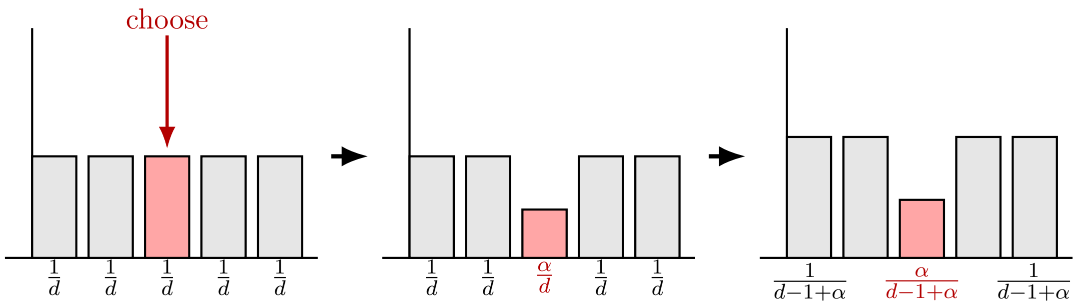


Figure 2.1: Schematic of the self-balancing update starting from a uniform distribution: choose an outcome, downweight it by α, then renormalize.

over-represented in the empirical distribution. More precisely, if we denote the counts process $N_i = N_i(n) := \#\{1 \leqslant j \leqslant n : X_j = i\}$, then this rule generates samples according to

$$p_i = p_i(n) := \frac{p_i(0) \cdot \alpha^{N_i(n)}}{\sum_{j=1}^d p_j(0) \cdot \alpha^{N_j(n)}}.$$

Hence we can see that this sampler interpolates the two naive sampling methods: when $\alpha = 1$, we recover the uniform sampler, and when $\alpha \downarrow 0$, we essentially recover the deterministic sampler.

We outline the paper as follows. Section 2.2.2 shows that such samplers arise naturally by enforcing an invariance property on the sampling probabilities $p(n)$, namely that they depend only on *relative* differences in the counts N_i. We show that after properly centering the counts process, it descends to an irreducible and positive recurrent Markov chain on the centered counts.

Then, in Section 2.2.3 we show that for any choice of α, the self-balancing sampler converges at rate $\Theta(1/n)$ in expectation, matching the fastest possible rate up to constants, while in the worst case it remains within $O(\log n/n)$ almost surely, which is optimal up to a logarithmic factor.

In Section 2.2.4, we show that the self-balancing sampler admits a natural interpretation through entropy-regularized optimization: it arises as the unique solution to a one-step problem that balances minimizing the mean-squared imbalance in the counts with maximizing entropy. Appendix 2.5.2 shows this fits within the broader framework of entropy-regularized optimal control.

In an effort to quantify the trade-off between convergence and unpredictability, we develop diffusion limits in Section 2.3 and an adversarial notion of predictability in Section 2.3.2. In particular, we study a *weak-biasing regime*, in which $\alpha \uparrow 1$. Using these tools, we study how the behavior of the sampler changes as we vary the bias parameter α. In this weak-biasing regime, the centered counts are well-approximated by an Ornstein–Uhlenbeck process with stationary fluctuations of order $(\log(1/\alpha))^{-1/2}$ and mixing time of order $(\log(1/\alpha))^{-1}$, quantifying how stronger biasing enforces mean-reversion while reducing randomness. This perspective also refines the convergence behavior, leading to an average-case error of order $1/(n\sqrt{\log(1/\alpha)})$.

Complementing this, the adversarial analysis translates these fluctuations into prediction accuracy: when p^* does not have a unique maximizer, the leading correction vanishes and predictability improves at rate $O(\log(1/\alpha))$, whereas in the presence of ties, Gaussian fluctuations dominate and the improvement is only $O(\sqrt{\log(1/\alpha)})$.

2.2 Basic results

2.2.1 Definitions

A *sequential sampling rule* on $[d]$ consists of a sequence $(\rho_n)_{n\geqslant 0}$ of random distributions on $[d]$ and random variables $(X_n)_{n\geqslant 1}$ on $[d]$ so that $X_{n+1} \mid \rho_n \sim \rho_n$, $X_{n+1} \mid \rho_n$ are conditionally independent, and $\rho_n \in \sigma(X_0, X_1, ..., X_{n-1})$. Namely, it selects its samples based solely on the previously observed samples, plus some independent randomness.

We say such a rule is *Markovian* if it selects its next sample $X_{n+1} \in [d]$ based solely on the current selection counts $\{N_i(n)\}_{i=1}^d$ where:

$$N_i(n) := |\{1 \leqslant j \leqslant n : X_j = i\}|.$$

Formally, such a sampling rule is given by a measurable function $\pi : \mathbb{N}^d \to \Delta^d$ where Δ^d is the unit simplex in $\mathbb{R}^d$, and samples are generated via:

$$\mathbb{P}\left(X_{n+1} = i \mid \{X_j\}_{j\leqslant n}\right) = \pi_i(N(n)).$$

The self-balancing sampler is then a Markovian sampling rule with sampling function $\pi_i = p_i$. In this paper, we will show that this sampler (and more generally samplers with $\alpha_i \in (0,1)$ possibly different for each outcome) arises naturally within this class of sampling rules after enforcing a specific symmetry on the sampling function π, and that it exhibits nice optimality properties. Note we can express this sampling rule as a softmax

$$p_i = p_i(n) = \frac{p_i(0) \cdot e^{-\beta_i N_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\beta_j N_j(n)}},$$

where $\beta_i = -\log \alpha_i$. As we will see below, these β_i are in some sense a more natural parameterization. Heuristically, they can be interpreted as follows: for $i, j \in [d]$, if $\frac{\beta_i^{-1}}{\beta_j^{-1}} = \frac{p}{q}$ is rational, then p samples of i are weighted the same as q samples of j under this sampling scheme. Thus β_i^{-1} measures the relative weight assigned to sampling $i \in [d]$. Higher weighted indices tend to be selected more often.

In this paper, we are interested in how the empirical sampling distributions

$$\mu(n) := \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$$

evolve over time under the self-balancing sampler. We show the limiting distribution is

$$p_i^* := \frac{\beta_i^{-1}}{\sum_{j=1}^d \beta_j^{-1}}.$$

Namely, asymptotically we just select each i proportional to its relative weight β_i^{-1}.

Since the counts $N_i(n)$ drift off to infinity over time, we will require some form of centering to properly study this process. There are two natural centering schemes. The first, which we call the *mean-centered counts*, just centers each count by its expected limiting proportion:

$$Y_i(n) := N_i(n) - np_i^*.$$

The second, which we call the *projection-centered counts*, reweights the counts according to their relative weights β_i^{-1}, fixes the coordinate d as a baseline, and compares the counts against this baseline:

$$\widetilde{Y}_i(n) := \beta_i N_i(n) - \beta_d N_d(n) = \beta_i \beta_d \cdot [\beta_d^{-1} N_i(n) - \beta_i^{-1} N_d(n)].$$

We can express the sample probabilities p_i directly in terms of these centered counts:

$$p_i = \frac{p_i(0) \cdot e^{-\beta_i Y_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\beta_j Y_j(n)}} = \frac{p_i(0) \cdot e^{-\widetilde{Y}_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\widetilde{Y}_j(n)}}.$$

2.2.2 Potential-based sampling

In this section, we show that the self-balancing sampler arises naturally by enforcing a certain invariance property on our class of Markovian sampling rules, and that under this relation the counts process $(N(n))_{n\in\mathbb{N}}$ descends to a stationary Markov chain $(\tilde{Y}(n))_{n\in\mathbb{N}}$ on our projection-centered counts.

For a Markovian sampling rule, one natural choice for the sampling function π is to define a potential $\varphi_i : \mathbb{N} \to (0, \infty)$ for each coordinate and to select a sample proportional to its potential:

$$\pi_i^\varphi(x) := \frac{\varphi_i(x_i)}{\sum_{j=1}^d \varphi_j(x_j)}, \quad \forall x \in \mathbb{N}^d.$$

We say a Markovian sampling rule π is *translation invariant* if:

$$\pi(x + \mathbf{1}) = \pi(x), \quad \forall x \in \mathbb{N}^d.$$

Translation-invariance expresses the opinion that our next sample X_{n+1} should depend only on the relative differences in the sampling counts $\{N_i(n)\}_{i=1}^d$ rather than their absolute sizes. More generally, we say π is *v-invariant* for some $v \in \mathbb{N}^d$ if:

$$\pi(x + v) = \pi(x), \quad \forall x \in \mathbb{N}^d.$$

Here v-invariance expresses the heuristic that sampling i a total of v_i times ought to be equivalent to sampling j a total of v_j times. Thus (v_i) plays an analogous role to the relative weights (β_i^{-1}).

Note that we can view v-invariance as inducing an equivalence relation $\sim_v$ on $\mathbb{N}^d$:

$$x \sim_v y \iff v_d x_i - v_i x_d = v_d y_i - v_i y_d, \quad \forall i, j \in [d].$$

Under this relation, our counts process $(N(n))_{n\in\mathbb{N}}$ descends to a Markov chain on $\mathbb{Z}^{d-1}$:

$$N(n) \mapsto v_d N_i(n) - v_i N_d(n) = \beta_d^{-1} N_i(n) - \beta_i^{-1} N_d(n),$$

which, up to constants, is just our projection-centered counts $\tilde{Y}(n)$. Alternatively, we can express the above equivalence as:

$$\frac{x_i}{v_i} - \frac{1}{d}\sum_{j=1}^{d} \frac{x_j}{v_j} = \frac{y_i}{v_i} - \frac{1}{d}\sum_{j=1}^{d} \frac{y_j}{v_j}, \quad \forall i \in [d],$$

which shows the projection-centered counts $\tilde{Y}(n)$ center the reweighted counts $\beta_i N_i(n)$ by their mean, rather than the long-term average np^* as $Y(n)$ does. Now, we show the self-balancing sampler is essentially the only v-invariant sampling rule generated by a potential.

Proposition 2.2.1. *If π^φ is v-invariant, then:*

$$\varphi_i(mv_i + k) = c_{i,k} b^m, \quad \forall m \in \mathbb{N}, k \in \{0, 1, \ldots, v_i - 1\}$$

Moreover, if φ_i is either log-convex or log-concave, then $\varphi_i(n) = c_i \alpha_i^n$ where:

$$(\log(\alpha_1), \ldots, \log(\alpha_d)) \propto (v_1^{-1}, \ldots, v_d^{-1}).$$

In the case of translation-invariance, the log-convexity/concavity condition is unnecessary and we recover the self-balancing sampler in the case of uniform $\beta_i \equiv \beta$. Note that the sampling function for the self-balancing sampler is v-invariant only in the case the ratios $\beta_i^{-1}/\beta_j^{-1}$ are all rational, enforcing a relatively minor constraint on β.

In a sense, the projection-centered counts are more natural because they live on a countable space $\mathbb{Z}^{d-1}$. As the self-balancing sampler has strong mean-reversion properties, its natural to expect the projection-centered counts are positive recurrent, and hence have some stationary distribution. This is indeed the case.

Lemma 2.2.2. *If π is v-invariant, the projection-centered chain $\tilde{Y}^v(n) = v_d N_i(n) - v_i N_d(n)$ is irreducible, positive recurrent, and has a unique stationary distribution η on some subset of $\mathbb{Z}^{d-1}$.*

In general, determining the stationary distribution η is challenging, as the chain is non-reversible for $d > 2$. However, in the special case $d = 2$, we can calculate η explicitly, which we do in Lemma 2.3.3 below. Positive recurrence is just the first manifestation of the samplers mean-reversion properties, which we will explore more in the following sections.

Unsurprisingly, at stationarity the sampler generates samples according to the target p^*.

Lemma 2.2.3.

Suppose π is v-invariant and the projection-centered chain $\tilde{Y}^v(n) = v_d N_i(n) - v_i N_d(n)$ has stationary distribution η on $\mathbb{Z}^{d-1}$. If we sample $y \sim \eta$ and $X \mid y \sim \pi(y)$, then:

$$\mathbb{P}(X_i = i) = \frac{v_i}{\sum_j v_j} = p_i^*$$

.

2.2.3 Fast convergence of the empirical distribution for self-balancing sampling

In this section, we show that for any choice of $\alpha_i \in (0,1)$, the self-balancing sampler converges to the desired limiting distribution at a rate faster than IID sampling, and in fact converges as fast as possible.

As a preliminary result, we note that the empirical distribution associated to *any* sequential sampling rule generally converges, which is a consequence of its close relationship with the Césaro means of the sampling distributions $p(n)$. However, this relationship is at the scale of $O(1/\sqrt{n})$, which is the best one can hope for in general.

Proposition 2.2.4.

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant d}\left|\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m)\right|\right] \lesssim \frac{d}{\sqrt{n}}.$$

Moreover, if the projection-centered count process for a sequential sampling rule is irreducible and positive recurrent, then both of these quantities converge almost surely.

Any empirical distribution sampled on $d \geqslant 2$ values (each with nonzero probability) converges at a rate of $\Omega(1/n)$, because for any distribution $q \in \Delta^d$,

$$\|\mu_n - q\|_{\mathrm{TV}} = \frac{1}{2}\sum_{i=1}^{d}\left|\frac{N_i}{n} - q_i\right| \geqslant \frac{1}{2n}\sum_{i=1}^{d}\min_{z\in\mathbb{Z}}|z - nq_i|.$$

If q_i is irrational for some i, then $\{nq_i \pmod 1\}_{n=1}^{\infty}$ is dense in $[0,1]$, so there are infinitely many n such that $\min_{z\in\mathbb{Z}}|z - nq_i| \geqslant 1/3$, giving an overall lower bound of $\|\mu_n - q\|_{\mathrm{TV}} \geqslant \frac{1}{6n}$ for infinitely many n. On the other hand, if all q_i are rational with least common denominator

m, then for any n not divisible by m, $\min_{z\in\mathbb{Z}} |z - nq_i| \geqslant 1/m$ for some i. Thus, for infinitely many n, we get $\| \ mu_n - q\|_{\mathrm{TV}} \geqslant \frac{1}{2mn}$.

IID sampling converges at rate $\Theta(1/\sqrt{n})$ in expectation, as for samples from a distribution q,

$$\mathbb{E}[\|\mu_n - q\|_{\mathrm{TV}}] = \frac{1}{2\sqrt{n}} \sum_{i=1}^{d} \mathbb{E}\left[\left|\frac{N_i - nq_i}{\sqrt{n}}\right|\right],$$

where each expectation term is $O(1)$ by the central limit theorem. However, for the self-balancing sampler, the empirical distribution converges much faster than IID sampling! Theorem 2.2.5 shows that in expectation, the convergence actually matches the fastest possible rate of $\Theta(1/n)$, regardless of the choice of $\alpha_i \in (0, 1)$! Additionally, Theorem 2.2.7 shows even in the worst case the empirical distribution stays within distance $O(\log n/n)$ of its target p^* almost surely.

Theorem 2.2.5 (Average-case convergence rate)**.** *For the self-balancing sampling rule, the empirical distribution μ_n converges at a rate of $1/n$:*

$$\mathbb{E}\|\mu(n) - p^*\|_{\mathrm{TV}} = \Theta(1/n).$$

The proof reduces to a statement about how far the count vector strays from being balanced.

Lemma 2.2.6. *Let $Y = (Y_1, \ldots, Y_d)$ and $Y_i = Y_i(n) \coloneqq N_i(n) - np_i^*$ be the centered counts. Then $\mathbb{E}[\|Y\|_1]$ is uniformly bounded in n.*

Proof of Theorem 2.2.5. Write

$$\begin{aligned}
\|\mu_n - p^*\|_{\mathrm{TV}} &= \frac{1}{2}\sum_{i=1}^{d}\left|\frac{N_i}{n} - p_i^*\right| \\
&= \frac{1}{2n}\sum_{i=1}^{d}|N_i - np_i^*| \\
&= \frac{\|Y(n)\|_1}{2n}.
\end{aligned}$$

Applying Lemma 2.2.6 finishes the proof. □

Similarly, we can relate the total variation to the imbalance in the projection-centered counts:

$$||\mu_n - p^*||_{\mathrm{TV}} = \frac{1}{2n}\sum_{i=1}^{d}\frac{1}{\beta_i}\left|\tilde{Y}_i(n) - \langle \tilde{Y}(n), p^*\rangle\right| = O_\beta\left(\frac{||\tilde{Y}(n)||_1}{n}\right).$$

In Section 2.3, we will study how $\sup_n \mathbb{E}[\|Y(n)\|_1]$ depends on the choice of β, quantifying its effect on the convergence rate.

The following shows even in the worst-case, the self-balancing sampler still improves upon the $\Theta(n^{-1/2})$ convergence rate of IID sampling.

Theorem 2.2.7 (Worst-case convergence rate)**.** *For the self-balancing sampling rule, the empirical distribution*

$$\|\mu(n) - p^*\|_{\mathrm{TV}} \xrightarrow{a.s.} 0$$

as $n \to \infty$. Moreover, almost surely,

$$\|\mu(n) - p^*\|_{\mathrm{TV}} \lesssim \frac{d^2(\log n + O(1))}{n} \qquad \forall n \geqslant 1.$$

Hence, even in the worst-case realization, the empirical distribution converges at a rate of $O(\log n/n)$, which is only a logarithmic factor worse than the optimal $O(1/n)$ rate, and drastically improves over the rate of $O(1/\sqrt{n})$ for IID sampling.

2.2.4 Interpretation as an optimization problem

Here, we formalize the interpretation of the self-balancing sampler as optimizing the trade-off between the convergence rate of the empirical distribution and its unpredictability at each step. To this end, we show that this sampling rule solves a greedy optimization problem, attempting to maximize both these quantities at every step.

Let $H(q) \coloneqq -\sum_{i=1}^d q_i \log q_i$ denote the Shannon entropy, let $H(q,p) \coloneqq -\sum_{i=1}^d q_i \log p_i$ denote the cross-entropy, and let $\mathrm{KL}(q\|p) \coloneqq H(q,p) - H(q) = \sum_{i=1}^d q_i \log \frac{q_i}{p_i}$.

Theorem 2.2.8. *Let $L(n) \coloneqq \frac{1}{2}\sum_{i=1}^d (N_i(n) - \frac{n}{d})^2$ denote the quadratic loss for the empirical counts and $D(q) \coloneqq \mathbb{E}[L(n+1) - L(n) \mid N(n), q]$ denote the one-step increase in L when sampling $X_{n+1} \sim q$. Then the unique solution to the optimization problem*

$$\arg\max_{q \in \Delta^d} H(q) - \beta D(q)$$

is the self-balancing sampling method $p_i(n) \propto e^{-\beta N_i(n)}$, initialized at the uniform distribution.

Theorem 2.2.8 is a special case of the following more general version, which allows the α_i (and hence the β_i) to be non-constant in i.

Theorem 2.2.9. *Let $L_\beta(n) \coloneqq \frac{1}{2}\sum_{i=1}^d \beta_i(N_i(n) - np_i^*)^2$ denote the weighted quadratic loss for the empirical counts and $D_\beta(q) \coloneqq \mathbb{E}[L_\beta(n+1) - L_\beta(n) \mid N(n), q]$ denote the one-step increase in L_β when sampling $X_{n+1} \sim q$. Then the unique solution to the optimization problem*

$$\arg\max_{q \in \Delta^d} H(q) - H(q, p(0)) - D_\beta(q) = \arg\min_{q \in \Delta^d} \mathrm{KL}(q\|p(0)) + D_\beta(q)$$

is the $1/2$-offset self-balancing sampling method $p_i(n) \propto p_i(0)e^{-\beta_i(N_i(n)+1/2)}$, initialized at the distribution $p_i(0)$.

The entropy-regularized optimization problem found in Theorem 2.2.9 appears naturally in the context of optimal control and reinforcement learning. In this setting, such regularization is used to quantify the classical exploration versus exploitation trade-off [60], [65], [27]. Moreover, as [62] observes, such exponential-reweighting naturally appear as a result of this entropy penalty. We briefly summarize some of the main results in discrete-time in Appendix 2.5.2 below, and show that the self-balancing sampler and the above entropy-regularized optimization problem arise naturally in this framework when you penalize imbalance using mean-squared error.

The curious offset of 1/2 in the counts appears to be an effect of greedy optimization over *discrete* time steps with a quadratic cost. Indeed, $\frac{\beta_i}{2}((x+1)^2 - x^2) = \beta_i x + \frac{\beta_i}{2}$. When all β_i are equal, this factor is absorbed into the normalization of the density. Theorem 2.2.10 gives an "instantaneous" optimization perspective, which does not include the 1/2 offset.

Theorem 2.2.10. *Let $L_\beta(x) \coloneqq \frac{1}{2}\sum_{i=1}^d \beta_i(x_i - np_i^*)^2$ denote the weighted quadratic loss as a function of counts x. Then the unique solution to the optimization problem*

$$\arg\min_{q\in\Delta^d} \langle \nabla L_\beta(N(n)), q\rangle + \mathrm{KL}(q\|p(0))$$

is the self-balancing sampling method $p_i \propto p_i(0)e^{-\beta_i N_i(n+1)}$, initialized at the distribution $p_i(0)$.

Finally, we observe that the self-balancing sampler implements stochastic mirror descent, minimizing the objective $\ell_\beta(x) \coloneqq \frac{1}{2}\sum_{i=1}^d \beta_i(p_i - p_i^*)^2$. The sampler uses the unbiased stochastic gradient $\beta_{X_n} e_{X_n} - \frac{1}{\sum_{j=1}^d \beta_j^{-1}}\mathbf{1}$, where $e_1, \ldots, e_d$ are the standard basis vectors:

$$\mathbb{E}\left[\beta_{X_n} e_{X_n} - \tfrac{1}{\sum_{j=1}^d \beta_j^{-1}}\mathbf{1} \mid p(n)\right] = \sum_{i=1}^d \beta_i(p_i(n) - p_i^*)e_i = \nabla \ell_\beta(p(n)).$$

Theorem 2.2.11 (Stochastic mirror descent formulation)**.** *Let $\ell_\beta(x) \coloneqq \frac{1}{2}\sum_{i=1}^d \beta_i(p_i - p_i^*)^2$ denote the weighted quadratic imbalance in the sampling probabilities. The unique solution to the optimization problem*

$$\arg\min_{q\in\Delta^d} h\left\langle \beta_{X_n} e_{X_n} - \tfrac{1}{\sum_{j=1}^d \beta_j^{-1}}\mathbf{1}, q\right\rangle + \mathrm{KL}(q\|p(n))$$

is the self-balancing sampler $p_i(n) \propto p_i(0)e^{-h\beta_i N_i(n)}$.

From an optimization perspective, the constant step-size h may seem strange, since it generally means that the stochastic mirror descent algorithm may not converge. In our case, this is expected, since the sequence $p(n)$ does not converge. Considering a step size decreasing in n could lead to a version of the sampler in which $p(n)$ does converge.

2.2.5 Expected cover time

Beyond asymptotic convergence of the empirical distribution, it is also important in applications that the sampler does not neglect any index for too long. A natural way to quantify this is via the cover time, namely the number of steps required before every outcome has been sampled at least once. In audit and inspection settings, this measures how long a team can go entirely untested; in sampling and representation problems, it measures how long a category can remain completely unrepresented in the realized sample.

For deterministic sampling, the number of steps to see all d outcomes is d, and for IID sampling, the expected number of steps to see all the outcomes, known as the "coupon collector problem," is $\Theta(d \log d)$. The self-balancing sampler has behavior which interpolates between these two extremes, as we now show.

Theorem 2.2.12. *Let $C := \max_{1 \leqslant i \leqslant d} \min\{n \geqslant 1 : X_n = i\}$ be the number of steps until the self-balancing sampler sees every outcome at least once. Then, starting from the initial condition $p_i(0) = 1/d$,*

$$\mathbb{E}[C] = \sum_{i=1}^{d} \frac{\log(1 + \beta_i \log d)}{\beta_i} + O(d).$$

For fixed β_i, the cover time is $\Theta(d \log \log d)$. Moreover, in the extremal cases $\beta \uparrow \infty$ and $\beta \downarrow 0$, the self-balancing sampler's behavior converges to the deterministic and IID behaviors, respectively:

$$\lim_{\beta \uparrow \infty} \mathbb{E}[C] = \Theta(d), \qquad \lim_{\beta \downarrow 0} \mathbb{E}[C] = \Theta(d \log d).$$

As a result, the self-balancing sampler only marginally increases the expected cover time compared to deterministic sampling.

2.3 Diffusion Approximation

In this section, we explore a few different diffusion approximations of our self-balancing sampler, highlighting its mean-reversion and other asymptotic properties. Later, we use these approximations to attempt to quantify how the sampler depends on the bias parameter β.

First, we show that, after properly centering and rescaling, each coordinate is in a sense just a time-changed Ornstein–Uhlenbeck (OU) process.

Proposition 2.3.1. *Fix $i \in [d]$ and let $\tilde{Y}_i^\beta(n) := \beta_i N_i(n) - \beta_d N_d(n)$ be the projection-centered version of the self-balancing sampler with the given β. Let:*

$$\tau_k^\beta := \inf\left\{ n > \tau_{k-1}^\beta \ : \ \tilde{Y}_i^\beta(n) \neq \tilde{Y}_i^\beta(\tau_{k-1}^\beta) \right\}, \quad \tau_0^\beta = 0.$$

For $h > 0$, define the continuous-time chain:

$$Z_t^h := h^{-1/2}\left(\tilde{Y}_i^{h\beta}(\tau_{\lfloor t/h \rfloor}^{\beta h}) - c_\beta\right), \quad c_\beta := \log\left(\frac{\beta_i}{\beta_d}\right)$$

If $Z_0^h \to z_0$, then we have $Z_t^h \Rightarrow Z_t$ as $h \downarrow 0$ where Z_t is the diffusion:

$$dZ_t = -\frac{\beta_i\beta_d}{\beta_i + \beta_d} Z_t dt + \sqrt{\beta_i\beta_d} \cdot dW_t, \quad Z_0 = z_0.$$

Here $\Rightarrow$ denotes convergence in the Skorokhod space $D([0,T], \mathbb{R})$ for any $T < \infty$.

In the uniform case $\beta_i \equiv \beta$, the correction c_β vanishes and our diffusion simplifies to:

$$dZ_t = -\frac{\beta}{2} Z_t dt + \beta dW_t.$$

Namely, the above says if we look at a single coordinate $\tilde{Y}_i(n)$ of our projection-centered counts and we follow only the time points where it changes, then properly rescaled it is just an Ornstein–Uhlenbeck process, so it has strong mean-reversion properties.

Since we showed for fixed β the original mean and projection-centered processes are basically $O(1)$, we must also scale-down the bias β to allow the process to fluctuate at scale $h^{-1/2}$ if we hope to get a nontrivial diffusive limit. We call this βh scaling the *weak-biasing limit*. To see why the correction c_β is necessary when β is not constant, note the projection-centered counts have drift $(\beta_i - \beta_d)/2 \neq 0$ when $\tilde{Y}_i(n) = 0$. The shift c_β just shifts the starting point of $Y(n)$ to the location where it has zero drift, and expands about this point.

Note that only in the uniform case $\beta_i \equiv \beta$ can we avoid centering by this c_β term, and thus can we choose $\tilde{Y}_0^{h\beta} = 0$ to get a diffusion starting at $Z_0 = 0$. Otherwise, we must choose $\tilde{Y}_0^{h\beta} = c_\beta + o(h^{-1/2})$, or equivalently $N(n) = O(h^{-1/2})$, to get Z_t to start at zero.
.

We develop an analogous diffusion-approximation for the mean-centered counts, this time without needing a time-change. Here, the interaction between the different coordinates becomes visible, and again the process exhibits a mean-reversion property

Proposition 2.3.2. *Let $M^\beta(n) := (\beta_i[N_i(n) - np_i^*])_{i=1}^d$ be the (weighted) mean-centered version of the self-balancing sampler with the given β. For $h > 0$, define the continuous-time chain:*

$$Z_t^h := h^{-1/2}\left(M^{h\beta}(\lfloor t/h \rfloor) - c_\beta\right), \quad (c_\beta)_i := \log(\beta_i) - \frac{\sum_j \frac{1}{\beta_j}\log(\beta_j)}{\sum_j \frac{1}{\beta_j}}$$

If $Z_0^h \to z_0$, then $Z_t^h \Rightarrow Z_t$ as $h \downarrow 0$ where Z_t is the diffusion:

$$dZ_t = -\frac{1}{S_\beta} Z_t dt + \Sigma^{1/2} dW_t, \quad Z_0 = z_0,$$

for $S_\beta = \sum_j \beta_j^{-1}$ and $\Sigma = S_\beta^{-1}\mathrm{diag}(\beta) - S_\beta^{-2}\mathbf{1}\mathbf{1}^T$.

Hence we again see our limiting process is some kind of OU process. Note as a consequence of the mean-centering, the above diffusion is singular in the sense it is constrained to the hyperplane $\{z : \langle z, p^* \rangle = 0\}$.

Again, the situation simplifies considerably in the uniform $\beta_i \equiv \beta$ case. Here, the centering $c_\beta = 0$ and our diffusive limit simplifies to:

$$dZ_t = -\frac{\beta}{d} Z_t dt + \frac{\beta}{d}(dI_d - \mathbf{1}\mathbf{1}^T)^{1/2} dW_t.$$

2.3.1 Interpreting Diffusive Limits

Now that we have developed our diffusive limits, let's try to see how they suggest our two centered processes behave in this weak-biasing regime. This section is non-rigorous and mainly serves to provide some intuition on the role β plays in the self-balancing sampler. We start with the projection centered case. Recall the one-dimensional OU process:

$$dZ_t = -\theta Z_t dt + \sigma dW_t, \quad Z_0 = z_0,$$

has stationary distribution $N(0, \sigma^2/2\theta)$ and mixing time $t_{mix}(\epsilon) \asymp \frac{1}{\theta} \log\left(\frac{1}{\epsilon}\right)$. Moreover at stationarity it has autocorrelation:

$$\text{Corr}(Z_t, Z_{t+s}) = e^{-\theta s}.$$

For small β, where sampling is close to uniform, the waiting times $\tau_k^\beta - \tau_{k-1}^\beta \approx \text{Geo}(d^{-1})$, so we wait roughly time d between samples. Hence using Proposition 2.3.1, we heuristically see that in the uniform β case the (time-changed) projection-centered counts are well-approximated by:

$$N_i(n) - N_d(n) \approx \frac{Z_{nh/d}}{\beta\sqrt{h}}.$$

which is an OU process with stationary distribution $N(0, (\beta h)^{-1})$. At stationarity:

$$\text{Corr}(N_i(n+k) - N_d(n+k), N_i(n) - N_d(n)) \approx \exp\left(-\frac{\beta h k}{2d}\right),$$

and $n_{mix}(\epsilon) \approx 2d(\beta h)^{-1} \log(\epsilon^{-1})$. Since βh was just our rescaled bias, this essentially says in the small β regime the process mixes at rate $O(\beta^{-1})$ and its auto-correlations decay exponentially at rate $O(\beta)$. Since $p^*(\beta) = p^*(c\beta)$ for any $c > 0$, scaling β has no effect on the limiting distribution. Instead, it merely controls this trade-off between unpredictability and convergence.

For the mean-centered counts, the story is similar. Specifically, for arbitrary $d \times d$ matrices Θ, Σ the $d-$dimensional OU process:

$$dZ_t = -\Theta Z_t dt + \Sigma dW_t,$$

has stationary distribution $N(0, \omega)$ for ω solving the Lyapunov equation:

$$\Theta\omega + \omega\Theta^T = \Sigma\Sigma^T.$$

Moreover, its mixing time and stationary covariances are again given by:

$$t_{mix}(\epsilon) \asymp \frac{1}{\lambda_{min}(\Theta)} \log(1/\epsilon), \quad \mathrm{Cov}(Z_{t+s}, Z_t) = e^{-\Theta s}\omega$$

In the context of Proposition 2.3.2, it is easy to see $\Theta = S_\beta^{-1} I_d$ and:

$$\omega = \frac{1}{2}[D_\beta - S_\beta^{-1} 1 1^T] = \frac{1}{2} D_\beta [I_d - 1(p^*)^T],$$

for $D_\beta = \mathrm{diag}(\beta_i)$. Thus, Proposition 2.3.2 implies:

$$N(n) - np^* \approx D_\beta^{-1} \left[\frac{c_\beta}{h} + \frac{Z_{nh}}{\sqrt{h}}\right] \sim N\left(\frac{D_\beta^{-1} c_\beta}{h}, \ \frac{D_\beta^{-1} \omega D_\beta^{-1}}{h}\right).$$

At stationarity, for any nonzero $a \in \mathbb{R}^d$:

$$\mathrm{Corr}(a^T(N(n+k) - (n+k)p^*), \ a^T(N(n) - np^*)) = e^{-kh/S_\beta}$$

and mixing time:

$$n_{mix}(\epsilon) \approx \frac{S_\beta}{h} \log(1/\epsilon).$$

Hence as before we see the convergence rate scales inversely with β, and the auto-correlation decay rate is proportional to β in this weak-biasing regime.

In the proof of Theorem 2.2.5, we showed the self-balancing sampler has average-case convergence rate given by:

$$\mathbb{E}||\mu_n - p^*||_{TV} = \frac{\mathbb{E}||Y(n)||}{2n}.$$

We showed that $\sup_n \mathbb{E}||Y(n)|| < \infty$ in Lemma 2.2.6 already, but we can use our diffusive limit to get a better sense of how this behaves as a function of β. Using the above stationary distribution and the mean of a folded normal:

$$\mathbb{E}|N(\mu, \sigma^2)| = \sigma\sqrt{\frac{2}{\pi}} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) + \mu\left[1 - 2\Phi\left(-\frac{\mu}{\sigma}\right)\right],$$

we can see that our diffusion approximation suggests:

$$\mathbb{E}||\mu_n - p^*||_{TV} \approx \frac{1}{2n}\left(\sum_{i=1}^{d} \mathbb{E}\left|N\left(\frac{(c_\beta)_i}{\beta_i h}, \ \frac{1 - p_i^*}{2\beta_i h}\right)\right|\right).$$

There are two regimes:

$$\mathbb{E}||\mu_n - p^*||_{TV} \approx \begin{cases} O\left(\frac{d}{n\sqrt{\beta h}}\right) & c_\beta = 0 \\ O\left(\frac{1}{nh}\sum \frac{|(c_\beta)_i|}{\beta_i}\right) & c_\beta \neq 0 \end{cases}$$

The former corresponds to the uniform β case. Since $(c_\beta)_i = O(\log(\beta_i))$, this suggests a shift from $O(\beta^{-1/2})$ to $O(\log(\beta)/\beta)$. In Appendix 2.5.1, we provide some additional empirical that there is indeed a fundamental shift in behavior as β become non-uniform.

2.3.2 Adversarial Interpretation of Predictability

In order to develop a notion of how "predictable" such a sampler is, consider the case where an adversary attempts to guess the next sample given the current sample history. Then one way of quantifying the predictability of the sampler is to study how accurate such an adversary can be if they behave optimally. In this section, we formalize this idea. Using our diffusive limits, we show if β is small that the predictability is of the order $\max_i p_i^* + O(\sqrt{\beta})$ when p^* does not have a unique maximizer. When it does, the predictability is roughly of order $\max_i p_i^* + O\left(\exp(-\beta^{-1})\right)$.

Clearly, since the sampler exponentially favors under-sampled indices, the best strategy for the adversary is to just guess the most under-sampled index. Under this strategy, we define the adversary's asymptotic error-rate as:

$$\limsup_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} \mathbb{P}\left(X_k \neq \arg\max_{i\in[d]} p_i(N(k-1))\right) = 1 - \liminf_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} \mathbb{E}_{y\sim\tilde{Y}(k)}\left[\max_{i\in[d]} p_i(y)\right]$$

where $\tilde{Y}(k)$ denote our projection-centered counts. Lemma 2.2.2 shows these centered counts form an ergodic Markov chain, and hence by the ergodic theorem this error-rate ought to converge to the error-rate at stationarity.

In general, it is hard to determine what the stationary distribution of the projection-centered counts $\tilde{Y}$ is. However, in the special case of $d = 2$, then $\tilde{Y}$ becomes a simple birth-death process on $\mathbb{Z}$ which makes this a lot more tractable.

Lemma 2.3.3. *When $d = 2$ and uniform $\beta_1 = \beta_2 = \beta > 0$, the projection-centered counts $\tilde{Y}(n)$ form a reversible Markov chain on $\beta\mathbb{Z}$ with stationary distribution:*

$$\eta(\beta z) \propto \frac{p_2(0) + p_1(0)e^{-\beta z}}{p_1(0) + p_2(0)} \left(\frac{p_1(0)}{p_2(0)}\right)^z e^{-\beta\binom{z}{2}}$$

In the case $p_1(0) = p_2(0) = \frac{1}{2}$, we get the symmetric and especially simple form:

$$\eta(\beta z) \propto \cosh\left(\frac{\beta z}{2}\right) \cdot e^{-\frac{\beta z^2}{2}}.$$

In the $d = 2$ case, we can exactly quantify how this error-rate depends on β.

Lemma 2.3.4. *When $d = 2$ and uniform β, as $\beta \downarrow 0$ we have:*

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}\Big(X_k \neq \arg\max_{i=1,2} p_i(N(k-1)) \Big) = \frac{1}{2} - \Theta(\sqrt{\beta}).$$

More precise constants can be found in Appendix 2.5.3. Even when $d = 2$, when β is non-uniform we lose this simple birth-death structure. Instead, to analyze the general β case we rely on our diffusive limits. The following section is non-rigorous, but aims to add some heuristic evidence for why this $\Theta(\sqrt{\beta})$ rate appears.

We saw in Section 2.3 that in the weak-biasing regime the weighted mean-centered counts are approximated at stationarity by

$$M^\beta(n) = D_\beta[N(n) - np^*] \approx c_\beta + \sqrt{h}Z, \qquad Z \sim N(0, \omega),$$

where ω is the stationary covariance from Section 2.3. Since

$$p_i(M) = \frac{p_i(0)e^{-M_i}}{\sum_{j=1}^d p_j(0)e^{-M_j}}, \qquad p(c_\beta) = p^*,$$

this suggests

$$\mathbb{E}\left[\max_i p_i(n)\right] \approx \mathbb{E}\left[\max_i p_i(c_\beta + \sqrt{h}Z)\right].$$

Let J_β denote the Jacobian of the softmax map at c_β. A direct calculation gives

$$J_\beta = -\left[\mathrm{diag}(p^*) - p^*(p^*)^T\right],$$

and hence the first-order Taylor expansion yields

$$p(c_\beta + \sqrt{h}Z) = p^* + \sqrt{h}J_\beta Z + O(h).$$

so heuristically

$$\mathbb{E}\left[\max_i p_i(n)\right] \approx \mathbb{E}\left[\max_i \left(p_i^* + \sqrt{h}G_i\right)\right], \quad G = J_\beta Z \sim N(0, J_\beta \omega J_\beta^T).$$

If the maximizer of p^* (equivalently minimizer of β) is unique, then the max function is differentiable at p^*, and since $\mathbb{E}[G] = 0$ the first-order correction vanishes:

$$\mathbb{E}\left[\max_i p_i(n)\right] = \max_i p_i^* + O(h).$$

In contrast, if several coordinates tie for the maximum of p^*, then the max is no longer differentiable and the first-order Gaussian fluctuation contributes. Writing $I^* = \{i : p_i^* = \max_j p_j^*\}$, standard analysis of maxima of correlated normals gives

$$\mathbb{E}\left[\max_i p_i(n)\right] = \max_i p_i^* + \sqrt{h}\,\mathbb{E}\left[\max_{i\in I^*} G_i\right] + O(h).$$

Using the fact $\langle Z, p^* \rangle = 0$, we see:

$$G = -\text{diag}(p^*)Z \stackrel{d}{=} -\text{diag}(p^*)(W - \langle W, p^* \rangle 1)$$

for $W \sim N(0, D_\beta/2)$. Hence as the second term is constant on I^* and mean-zero:

$$\mathbb{E}\left[\max_i p_i(n)\right] - \max_i p_i^* = \sqrt{\frac{h}{2\beta_{min} S_\beta^2}} \cdot \mathbb{E} \max_{i \leqslant |I^*|} B_i + O(h) \asymp \sqrt{\frac{h \log |I^*|}{\beta_{min} S_\beta^2}} + O(h).$$

for $B_i \stackrel{iid}{\sim} N(0,1)$. The second approximation comes from classical asymptotics for the maximum of iid Gaussians, at least when $|I^*|$ is large. Hence the error rate at stationarity is approximately:

$$1 - \max_i p_i^* - \sqrt{\frac{h \log |I^*|}{\beta_{min} S_\beta^2}} + O(h).$$

In the uniform β case, this becomes:

$$1 - \frac{1}{d} - \frac{\sqrt{\beta h \log d}}{d} + O(h)$$

Again, this is reasonable when d is large. Thus, the diffusion limit suggests in the weak-biasing regime, the predictability of the sampler is governed primarily by the maximum of p^*, and by the number of coordinates sharing this maximum value. Increasing either increases the predictability of the sampler. Moreover, for fixed p^*, it shows this predictability grows at rate $\Theta(\beta^{1/2})$ as $\beta \downarrow 0$, which matches the $d = 2$ case from Lemma 2.3.4.

Somewhat interestingly, predictability can increase when several coordinates share the minimal value of β_i, since the adversary can exploit the largest realized fluctuation among these tied coordinates.

Lastly, we briefly return to the case where p^* has a unique maximizer. Suppose $m := \arg\max_i p_i^*$ and let:

$$\Delta^* := p_m^* - \max_{j \neq m} p_i^* > 0,$$

be the gap between the two most likely outcomes. Then we see $m \neq \arg\max p(c_\beta + \sqrt{h}Z)$ essentially only when some coordinate $Z_i = O(\Delta^*/\sqrt{h})$ experiences a large deviation. Roughly, using $\mathbb{P}(N(0,1) > t) \asymp t^{-1} e^{-t^2/2}$, this has probability $O((ch)^{-1/2} \cdot e^{-1/2ch})$ exponentially small in c^{-1}. Hence in the case the of a unique maximizer, when c is small we are exponentially close to the naive baseline p_{max}^*.

2.4 Concluding Remarks

We introduced a family of self-balancing sequential samplers that adaptively downweight previously sampled outcomes. The resulting process interpolates between IID sampling, which is maximally random but poorly balanced, and deterministic sampling, which is optimally balanced but perfectly predictable. Despite retaining randomness at every step, the self-balancing sampler achieves empirical convergence at the deterministic scale $O(n^{-1})$ in expectation and $O(\log n/n)$ almost surely.

A central theme of the paper is that this sampler is not merely a heuristic. It arises naturally from an invariance principle for potential-based Markovian samplers, is the unique solution to a one-step entropy-regularized imbalance minimization problem, and shows up informally in actual policy decisions. Together, these perspectives give complementary evidence for such exponential sampling rules naturally appear.

The weak-biasing regime makes the convergence–predictability tradeoff explicit. As the bias parameter decreases, the centered counts fluctuate on a larger scale and mix more slowly, preserving more unpredictability. As the bias parameter increases, the process becomes more tightly mean-reverting, improving balance but making the next sample easier to predict. The diffusion approximation quantifies this tradeoff and shows that the geometry of the target law matters: tied maximizers of p^* produce first-order Gaussian corrections to predictability, while unique maximizers with fixed gaps only contribute through rare crossing events.

Several extensions remain open. One natural direction is to replace categorical balance with covariate balance, sampling units according to their contribution to a cumulative covariate imbalance. Namely, suppose each unit $i \in [d]$ has some covariate vector v_i assigned to it. If we have samples $X_1, ..., X_n \in [d]$ define the cummulative (centered) covariate vector as:

$$S_n := \sum_{i=1}^{n} v_{X_i} - n\bar{v}, \qquad \bar{v} := \frac{1}{d}\sum_{i=1}^{d} v_i.$$

We can then generalize our self-balancing sampler so that index i is sampled with probability:

$$p_i(n) \propto \exp\left(-\sum_j \beta_j v_{ij} (S_n)_j\right) = \exp\left(-\langle v_i, \beta \odot S_n\rangle\right).$$

When $v_i = e_i$ are the standard basis vectors, this recovers our original self-balancing sampler. This provides a parallel to the Gram-Schmidt walk of [4], which in the context of [28] provides the key tool for ensuring covariates are balanced across treatment groups in a randomized control trial.

Another is to study risk-weighted or nonstationary versions of the sampler, where the target distribution changes over time or depends on observed outcomes. Such extensions would bring the model closer to applications in inspections, audits, randomized experiments, and other settings where one seeks to balance coverage, responsiveness, and resistance to prediction.

2.5 Appendix

2.5.1 Empirical Results

In this section, we provide some additional empirical evidence for some of the β scalings we developed heuristically in previous sections. Simulation code can be found in the GitHub repository self_balancing_sampler. We will study the behavior under two different unit-norm choices of β when $d = 4$: the uniform β and the non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$, the latter of which also has a unique maximizer of $p^*(\beta)$.

Convergence Rate

Theorem 2.2.5 implies:

$$d_n := \mathbb{E}||\mu_n - p^*||_{TV} = \frac{C(\beta)}{n} + o(n^{-1}).$$

Since $C(\beta) = \lim_{n\to\infty} nd_n$, we can generate an estimate $\hat{C}(\beta)$ of this constant by simulating many trajectories of the self-balancing sampler and taking the average of Nd_N for some large N. To get a sense of how this constant varies with β, we fix some unit-norm β and study how $C(c\beta)$ varies as we vary the scaling $c > 0$. The results are shown in Figure 2.2 below. Since $p^*(\beta) = p^*(c\beta)$ for any $c > 0$, all these samplers have the same limiting distribution. Our heuristic analysis in Section 2.3 suggests:

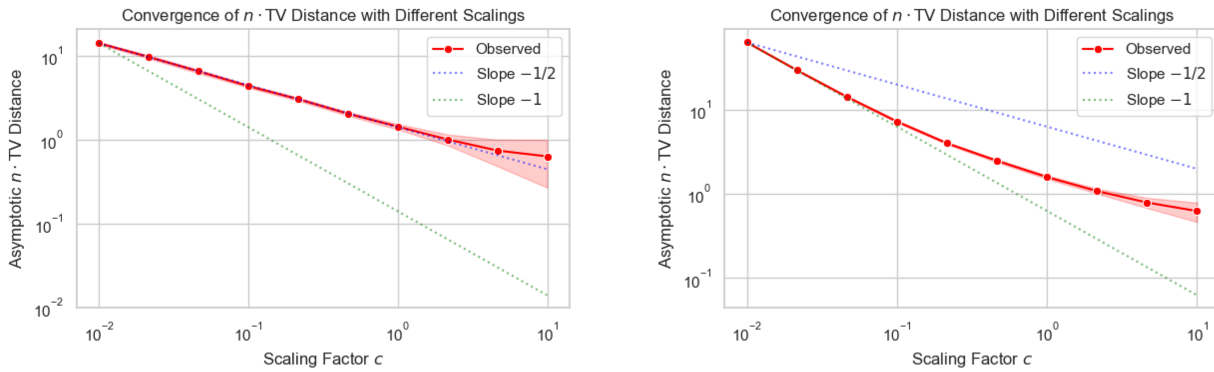
$$C(c\beta) = \begin{cases} \Theta(c^{-1/2}) & \beta \text{ uniform}, \\ \Theta(c^{-1}) & \text{otherwise}. \end{cases}$$

Figure 2.2 shows this is indeed the case, at least in the small c regime where the diffusion approximations are reasonable. To generate these plots, we take $M = 100$ trajectories of our sampler with $N = 10000$ samples each. We estimate the limiting value $C(\beta) = \lim nd_n$ by taking the average of Nd_N over our M trials.

Predictability

In this section, we explore how the parameter β affects the predictability of the sampler, as described in Section 2.3.2. There, we showed that the key factor was whether or not p^* has a unique maximizer. If it does not, then the predictability scales like $\Theta(\sqrt{\beta})$ above the naive baseline p^*_{max} of IID sampling. If the maximizer is unique, we saw this first order correction vanishes, and the gap is exponentially small. We compare it to our two naive baselines: IID sampling and deterministic sampling.

In IID sampling, the adversary will clearly always predict the index i which maximizes p^*_i, and hence its long-term accuracy will be p^*_i. In greedy sampling, we can make the sampler a bit more unpredictable by sampling the next index uniformly from $\arg\min_i N_i(n)$, rather than just the purely deterministically. In the uniform β case, this gives an accuracy



(a) Uniform β.

(b) Non-uniform β.

Figure 2.2: Log-log plots of Nd_N as a function of the scaling factor c for (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$. Dotted lines show the expected $\Theta(c^{-1})$ and $\Theta(c^{-1/2})$ scalings in the uniform and non-uniform case respectively.

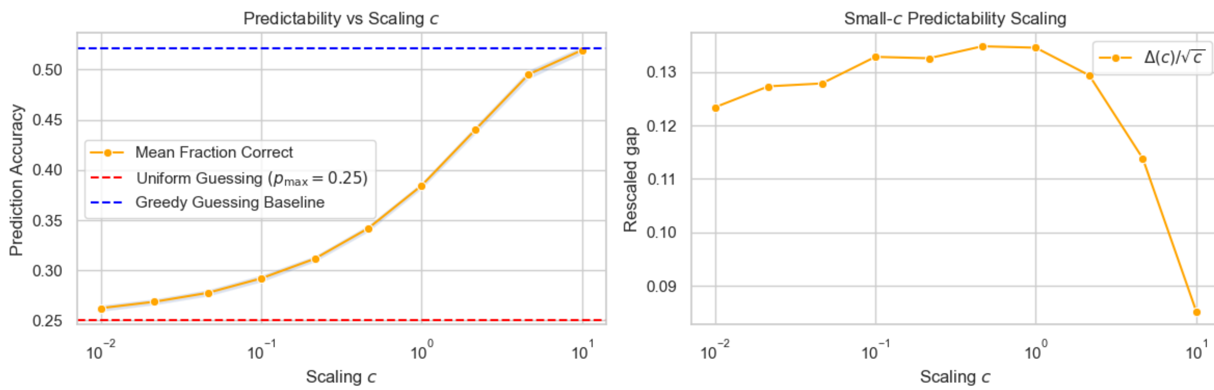
of approximate $\frac{\log(d)}{d}$, compared to the d^{-1} of IID sampling. In the non-uniform β case, we estimate the error rate under the greedy model through a simple Monte-carlo simulation.

Figures 2.3 shows in the uniform β case, there is a gradual increase in the prediction accuracy as we increase the bias c, and in the small c regime we achieve the expected $\Theta(\sqrt{c})$ scaling. In the non-uniform β case, we see prediction accuracy increases much more starkly. Moreover, the gap decays much more quickly as the bias $c \downarrow 0$.

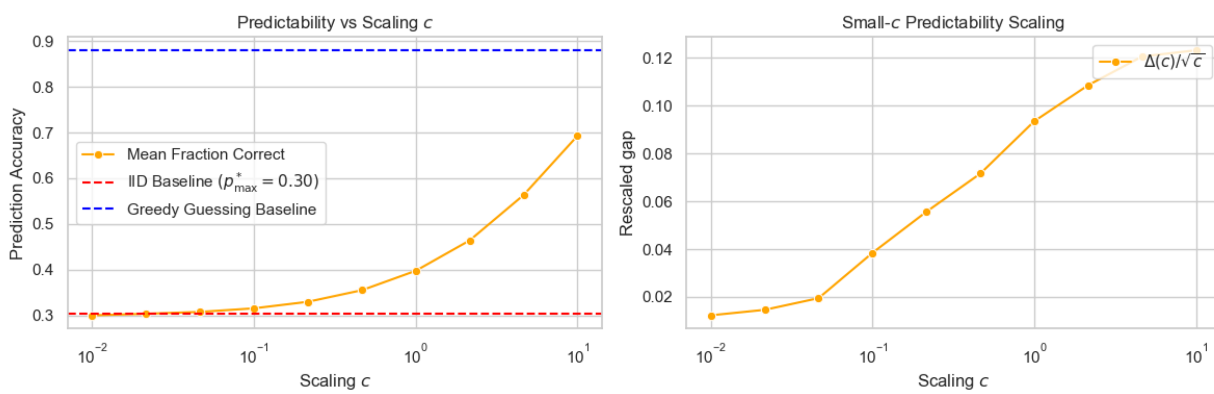
Trade-Off

Lastly, we study the trade-off between predictability and the convergence rate. As discussed in the previous two subsections, we quantify the former using the adversaries prediction accuracy, and the latter using the limit $\lim_{n\to\infty} nd_n$. Figure 2.4 highlights the results.

Note the predictability starts essentially at the baseline p^*_{max}, and increases as we increase the scaling c. It starts smaller in the uniform β case merely because p^*_{max} is smaller than in the non-uniform case. However, we see that in the latter the predictability is much slower to increase initially, but then quickly becomes more predictable as we scale β.



(a) Uniform $\beta = [0.5, 0.5, 0.5, 0.5]$.



(b) Non-uniform, distinct maximizer $\beta \propto [1.1, 0.8, 0.9, 1.2]$.

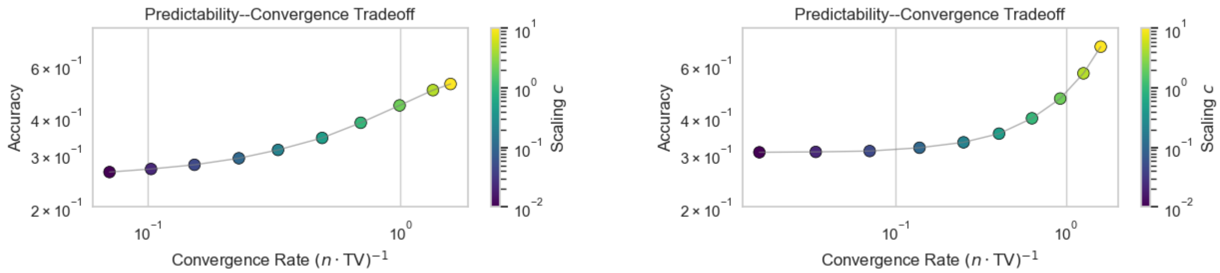
Figure 2.3: Predictability plots for different choices of β. (left) Relationship between prediction accuracy and the scaling c of β. (right) The ratio of the gap $\Delta(c) :=$ accuracy $- p^*_{max}$ from the uniform baseline and the first order $\sqrt{c}$ expected scaling.

2.5.2 Optimal Control

In this section, we briefly review some of basic optimal control theory, and discuss how Theorem 2.2.9 fits into the entropy-regularized optimal control framework. Suppose we have some discrete time dynamical system on some state space S:

$$x_{t+1} = b(x_t, u_t), \quad x_t \in S, u_t \in U$$

where u_t is the input of some controller and U is the set of all allowable controls. In classical control theory, we have some loss function $\ell(x, u)$ and we wish to minimize the cummulative



(a) Uniform β. (b) Non-uniform β.

Figure 2.4: Log-log plots of the trade-off between adversarial accuracy and the convergence rate $(Nd_N)^{-1}$. (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$.

(discounted) loss over the entire trajectory $(x_t)_{t\in\mathbb{N}}$. Namely we aim to solve:

$$V(x) := \inf_{(u_t)} \left[\sum_{s=1}^{\infty} \gamma^s \cdot \ell(x_s^u, u_s) \right],$$

for some discount $\gamma \in (0,1)$ and where $x_0 = x$ is the initial state of the system. Here (x_t^u) is the trajectory of the process when control sequence (u_t) is applied. The *value function* V and optimal control (u_t^*) can be determined using the standard Bellman equation.

$$V(x) = \inf_{u\in U} \left\{ \ell(x,u) + \gamma V(b(x,u)) \right\}.$$

If we let $Q_\gamma(x,u) := \ell(x,u) + \gamma V(b(x,u))$ then we see the optimal policy u^* and its corresponding value function V satisfy:

$$u^*(x) = \arg\min_{u\in U} Q_\gamma(x,u), \quad V(x) = Q_\gamma(x, u^*(x)) \tag{2.5.1}$$

We can generalize the above by allowing the control policy to select its control u_t at random, possibly depending on the current state. This paradigm is useful in the following settings:

(i) *Imperfect Controls*: In many real-world settings, the controller does not have perfect control over the system, and the implemented actions may be subject to noise. By allowing for randomized control policies, we can model this inherent uncertainty in the control process.

(ii) *Adversarial or Strategic Environments*: In some settings, predictability can be exploited by an adversary or the environment. Entropy regularization encourages policies that remain sufficiently randomized, thereby mitigating worst-case or strategic responses to deterministic behavior.

(iii) *Exploration Versus Exploitation*: In reinforcement learning, we often want to encourage the agent to explore the state space in order to learn about the environment and the corresponding reward process. Randomized policies explicitly encode this exploration into the objective function.

(iv) *Robustness and Regularization*: Entropy regularization acts as a convex regularizer on the control problem, smoothing the optimization landscape and preventing degenerate (overly deterministic) solutions. This can significantly improve stability and sample efficiency in practice [27].

(v) *Information Constraints / Bounded Rationality*: Entropy regularization can also be interpreted as penalizing deviations from a reference policy via a Kullback–Leibler divergence. In this view, the controller faces a cost for using highly concentrated policies, reflecting limited information-processing capacity or a preference for staying close to a prior behavior ([50]).

Suppose there is some deterministic function $\pi(\cdot \mid x) : S \to \mathcal{P}(U)$ which maps the state-space into the set of distributions over controls. Then define dynamics:

$$X_{t+1} \mid X_t \sim b(X_t, U_t), \quad U_t \overset{ind}{\sim} \pi(\cdot \mid X_t), \quad X_0 = x_0.$$

This generalizes the Markovian sampler we discuss above. We then force a trade-off between the entropy of the control and the accrued loss:

$$V(x_0) := \inf_{(\pi)} \mathbb{E}\left[\sum_{s=0}^{\infty} \gamma^s \cdot (\ell(X_s^u, U_s) - \tau H(\pi(\cdot \mid X_s)))\right],$$

In this case, Bellman's equation gives:

$$V(x) = \inf_{\mu \in \mathcal{P}(U)} \left\{ \int_U \left[Q_\gamma(x,u) + \tau \log(\mu(u))\right] \ \mu(du) \right\}.$$

This can be optimized to find the optimal policy π pointwise, as well as the value function. In the case U is finite, these solutions can be expressed implicitly as:

$$\pi^*(u \mid x) \propto \exp\left(-\frac{Q_\gamma(x,u)}{\tau}\right), \quad V(x) = -\tau \log\left(\sum_{u \in U} \exp\left[-\frac{Q_\gamma(x,u)}{\tau}\right]\right). \tag{2.5.2}$$

As we can see, the optimal policy is naturally a soft-min / Gibbs distribution with Hamiltonian $-Q_\gamma(x, \cdot)/\tau$, and the value function is essentially the log-partition function of this distribution. Note how entropy-regularization essentially smooths out the strict minimization problem that occurs in equation (2.5.2). While equations (2.5.2) only implicitly define the optimal policy, for certain choices of dynamics and loss functions we can be more explicit. Suppose we have some potential $\Phi : S \to \mathbb{R}$ and we define loss:

$$\ell(x, u) := \Phi(x) + \tau \log \sum_{v \in U} \exp\left(-\frac{\gamma}{\tau} \cdot \Phi(b(x, v))\right),$$

which is independent of u. Note that the second term is a simple soft-minimum. That is, we penalize spending time in states x that have high-potential $\Phi(x)$ and that do not have neighboring low-potential states. In this setting:

$$\pi^*(u \mid x) \propto \exp\left(-\frac{\gamma(\Phi(b(x, u)) - \Phi(x))}{\tau}\right), \qquad V(x) = \Phi(x).$$

The self-balancing sampler fits nicely into this framework. If we choose state-space $S = \mathbb{N}^d$, controls $u \in [d]$, dynamics $b(x, u) = x + \delta_u$, and potential $\Phi(x) = \frac{1}{2}||x||^2$ then we have $\pi^*(u|x) \propto e^{-\gamma x_u/\tau}$. Hence we recover optimization problem Theorem 2.2.8, the uniform $\beta_i \equiv \beta$ case. Analogously, if we instead choose $\Phi(x) = \frac{1}{2}\sum_i \beta_i(x_i - np_i^*)^2$ we recover optimization problem 2.2.9.

The former gives us another way of interpreting the role of β in the self-balancing sampler: $\beta = \gamma/\tau$ measures the trade-off between minimizing future imbalance through the value term Q_γ and maintaining randomness in the sampling policy via entropy-regularization.

Alternatively, if we choose a linear loss $\ell(x, u) = \beta_u x_u$, and myopic discounting $\gamma = 0$, we see the policy $\pi^*(u \mid x) \propto \exp(-\beta_u x_u)$ is the minimizer of:

$$V(x) := \inf_\pi \left\{ \sum_{i=1}^d [\pi(i|x) \cdot \beta_i x_i] - H(\pi(\cdot \mid x)) \right\},$$

Hence the self-balancing sampler can be seen as a greedy mechanism for minimizing the weighted-L^1 norm of our counts $N(n)$, analogous to Theorem 2.2.10.

We showed above that the self-balancing sampler converges at rate $O(n^{-1})$ on average. The theory of entropy-regularized optimal control provides us a convenient framework for extending these ideas to a broader class of samplers. The following shows as long as under-sampled indices are at least exponentially favored, we get the same convergence rate.

Theorem 2.5.1. *Suppose we have sampler $\pi_i(x) \propto a_i \exp(-H_i(x))$ so that for $P = I - p^* 1^T$ then $\pi(x) = \pi(x')$ whenever $Px = Px'$. Moreover, suppose there are constants $c, C > 0$ so for all y with $\sum y_i = 0$ and $y_j \leqslant y_i$ we have:*

$$H_i(y) - H_j(y) \geqslant c(y_i - y_j) - C.$$

Then if μ_n is the empirical distribution associated to sampler π, we have:

$$\mathbb{E}||\mu_n - p^*||_{TV} \leqslant O(n^{-1}).$$

As in the previous sections, p^* plays the role of the limiting distribution. Our assumptions on π simply state that it is a function of the mean-centered counts alone, and the proper centering p^* determines the limiting distribution.

If we have a sampler generated by potential Φ as above, then the assumptions of Theorem 2.5.1 reduce to the constraints:

$$\Phi(x + \delta_i) - \Phi(x) = \Phi(Px + \delta_i) - \Phi(Px) + \kappa,$$

$$\Phi(x + \delta_i) - \Phi(x + \delta_j) \geqslant c((Px)_i - (Px)_j) - C.$$

for some κ independent of x, i. The self-balancing sampler is the special case $H_i(y) = \beta_i y_i$ which is globally linear and arises from potential $\Phi(x) = \frac{1}{2}||x||^2$. Alternatively, any separable potential $\Phi(x) = \sum_i \varphi_i((Px)_i)$ for φ_i strongly-convex satisfies these constraints.

2.5.3 Proof Appendix

In this subsection, we collect the formal proofs of all the main results of the paper.

Proof of Proposition 2.2.1

Proposition. If π^φ is v-invariant, then for each i there exist constants $b > 0$ and $c_{i,k}$ such that $\varphi_i(mv_i + k) = c_{i,k}b^m$ for all $m \in \mathbb{N}$ and $k \in \{0, \dots, v_i - 1\}$. If, in addition, each φ_i is log-convex or log-concave, then $\varphi_i(n) = c_i r_i^n$ and $(\log r_1, \dots, \log r_d) \propto (v_1^{-1}, \dots, v_d^{-1})$.

Proof. Denote $S(x) := \sum_j \varphi_j(x_j)$. Then v-invariance implies:

$$\frac{\varphi_i(x_i)}{S(x)} = \frac{\varphi_i(x_i + v_i)}{S(x+v)}, \quad \frac{\varphi_j(x_j)}{S(x)} = \frac{\varphi_j(x_j + v_j)}{S(x+v)}.$$

Dividing these equalities and rearranging yields:

$$\frac{\varphi_i(x_i)}{\varphi_i(x_i + v_i)} = \frac{\varphi_j(x_j)}{\varphi_j(x_j + v_j)}.$$

As the left side is independent of x_j and the right side is independent of x_i, these ratios must equal some constant $b > 0$ independent of x_i, x_j. Hence for every $i \in [d]$ and $k \in \{0, 1, ..., v_i - 1\}$ we must have:

$$\varphi_i(mv_i + k) = \varphi_i(k) \cdot b^m, \quad \forall m \in \mathbb{N}.$$

Log-convexity implies for $k \in \{0, 1, ..., v_i - 2\}$:

$$\frac{\varphi_i(0)}{\varphi_i(1)} \leqslant \frac{\varphi_i(k)}{\varphi_i(k+1)} \leqslant \frac{\varphi_i(v_i)}{\varphi_i(v_i + 1)} = \frac{\varphi_i(0)b}{\varphi_i(1)b}.$$

If instead φ_i is log-concave, the direction of the inequalities all swap. Either way, we see these ratios are constant, and thus $\varphi_i(m) = c_i r_i^m$ where $r_i = b^{1/v_i}$. Thus we have

$$\frac{\log(r_i)}{\log(r_j)} = \frac{v_j}{v_i},$$

giving the desired result. □

Proof of Lemma 2.2.2

Lemma. If π is v-invariant, the projection-centered chain $\tilde{Y}_i^v(n) = v_d N_i(n) - v_i N_d(n)$ is irreducible, positive recurrent, and has a unique stationary distribution η on some subset of $\mathbb{Z}^{d-1}$.

Proof. Clearly the state-space of $\tilde{Y}^v$ is the subset of $\mathbb{Z}^{d-1}$ generated by nonnegative integer combinations of the vectors $\{v_d e_i\}_{i=1}^{d-1} \cup \{-v\}$. Of course, this does not always span Z^{d-1}, even when $\gcd(v_i) = 1$. For example, if $v_1 = 2, v_2 = 3, v_3 = 4$ then the first coordinate of $\tilde{Y}^v$ is always a multiple of 2.

Nonetheless, we show the chain is still irreducible on this state-space. To see why, suppose:

$$x = v_d \sum_{i=1}^{d-1} a_i e_i - a_d v, \quad a_i \geqslant 0.$$

Clearly we can go from 0 to x by sampling a_i times from index i for each $i \in [d-1]$ and sampling a_d times from index d. Conversely, we can go from x to 0 by sampling index i a total of $kv_i - a_i$ times, for some k large enough so $kv_d > a_d$ for all i. Then sample coordinate d a total of $kv_d - a_d$ times.

Lastly, we show the chain is positive recurrent by developing a suitable Lyapunov function and proving a Foster-Lyapunov drift condition. Let $\beta_i = v_i^{-1}$ and define:

$$Q_i := \beta_i N_i - \log(p_i(0)\beta_i), \quad \overline{Q} = \frac{1}{d}\sum_{i=1}^{d} Q_i.$$

The constants $\log(p_i(0)\beta_i)$ serve an analogous role to c_β in our diffusive limit: the shift the chain so it is centered at the zero-drift location. Note:

$$p_i = \frac{p_i(0)e^{-\beta_i N_i}}{\sum_j p_j(0)e^{-\beta_j N_j}} = \frac{v_i e^{-Q_i}}{\sum_j v_j e^{-Q_j}},$$

is a function of Q. Using this, define Lyapunov function:

$$V(Q) := \sum_{i=1}^{d} Z_i^2, \quad Z_i := Q_i - \overline{Q}.$$

Then V is a function of the projection-centered counts $v_d N_i - v_i N_d$ alone. Moreover, $V(Q) = V(Q + c1)$ for all $c \in \mathbb{R}$, so it suffices to consider the case $Q_d = 0$. We will show V has negative drift outside some finite set. First, note as $\sum Z_i = 0$::

$$V(Q + \Delta Q) = 2\sum_{i=1}^{d} Z_i \Delta Q_i + \sum_{i=1}^{d} (\Delta Q_i - \overline{\Delta Q})^2$$

Thus V has expected drift:

$$\mathbb{E}[\Delta V(Q) \mid Q = q] = 2\sum_{i=1}^{d} Z_i p_i(q)\beta_i + \frac{d-1}{d}\sum_{i=1}^{d} p_i(q)\beta_i^2$$

The second term is at most $C = \max \beta_i^2$. The first term is sufficiently negative on:

$$F := \left\{ Q \in \mathbb{N}^d : \max_i Q_i - \min_i Q_i > R, \quad Q_d = 0 \right\}.$$

To see why, suppose $M = \arg\max_i Q_i$ and $m = \arg\min_i Q_i$.

$$\begin{aligned}
\sum_{i=1}^{d} Z_i p_i \beta_i &= \frac{1}{d} \sum_{i<j} (q_i - q_j)(p_i\beta_i - p_j\beta_j). \\
&= \frac{1}{d \sum_j v_j e^{-q_j}} \sum_{i<j} (q_i - q_j)(e^{-q_i} - e^{-q_j}). \\
&\leqslant \frac{(q_m - q_M)(e^{-q_m} - e^{-q_M})}{d \sum_j v_j e^{-q_j}} \\
&\leqslant \frac{(q_m - q_M)(e^{-q_m} - e^{-q_M})}{d^2 e^{-q_m} \max_i v_i} \\
&\leqslant -\frac{R(1 - e^{-R})}{d^2 \max_i v_i}.
\end{aligned}$$

where in the first inequality we observed every term in the sum is nonnegative. The complement F^c is a finite subset of $\mathbb{Z}^{d-1}$, so we have the desired drift condition.

□

Proof of Lemma 2.2.3

Lemma. Suppose π is v-invariant and the associated projection-centered chain has stationary distribution η. If $y \sim \eta$ and $X \mid y \sim \pi(y)$, then $\mathbb{P}(X = i) = \frac{v_i}{\sum_j v_j}$ for each $i \in [d]$.

Proof. Suppose $\tilde{Y}^v \sim \eta$. Let $p_i := \mathbb{P}(X = i)$ and define $\Delta_n \tilde{Y}^v := \tilde{Y}^v(n+1) - \tilde{Y}^v(n)$. If $\tilde{Y}^v(n) \sim \eta$, then $\tilde{Y}^v(n+1) \sim \eta$ as well. Hence $\mathbb{E}\Delta_n \tilde{Y}^v = 0 \in \mathbb{Z}^{d-1}$. As discussed above, we can view $\tilde{Y}^v(n)$ as the image of $N(n)$ under the map

$$x \mapsto (v_d x_i - v_i x_d)_{i=1}^{d-1}.$$

Hence:

$$(\Delta_n \tilde{Y}^v)_i = \begin{cases} v_d & w.p. \quad p_i, \\ -v_i & w.p. \quad p_d, \\ 0 & w.p. \quad 1 - p_i - p_d. \end{cases}$$

Hence the i^{th} coordinate of this expectation is:

$$(\mathbb{E}\Delta_n \tilde{Y}^v)_i = v_d p_i - v_i p_d = 0.$$

Thus $p_i = \frac{v_i}{v_d} p_d$. Since $\sum_{i=1}^d p_i = 1$, we see:

$$\sum_{i=1}^{d} \frac{v_i}{v_d} = \frac{1}{p_d} \to p_d = \frac{v_d}{\sum_{j=1}^d v_j},$$

which yields the desired result. □

Proof of Proposition 2.2.4

Proposition. For every sequential sampling rule,

$$\mathbb{E}\left[\max_{1\le i\le d}\left|\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m)\right|\right] \lesssim \frac{d}{\sqrt{n}}.$$

If the projection-centered count process is irreducible and positive recurrent, then both $\mu_i(n)$ and $\frac{1}{n}\sum_{m=0}^{n-1} p_i(m)$ converge almost surely.

Proof. For a given $1 \leqslant i \leqslant d$, consider the sequence $M_i(0) = 0$, $M_i(n) := N_i(n) - \sum_{m=0}^{n-1} p_i(m)$. This is a martingale adapted to $\mathcal{F}_n := \sigma(N_i(1), \ldots, N_i(n))$, as

$$\mathbb{E}[M_i(n+1) - M_i(n) \mid \mathcal{F}_n] = \mathbb{E}[\mathbf{1}_{\{X_{n+1}=i\}} - p_i(n) \mid \mathcal{F}_n] = p_i(n) - p_i(n) = 0.$$

Moreover, these martingales each have bounded differences $|M_i(n+1) - M_i(n)| \leqslant 1$, so the Azuma-Hoeffding inequality gives

$$\mathbb{P}(|M_i(n)| \geqslant n\varepsilon) \leqslant 2e^{-n\varepsilon^2/2}.$$

This yields exponential concentration for $\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m) = M_i(n)/n$ around 0, so by the Borel-Cantelli theorem, the difference converges almost surely. And if the count process for a sequential sampling rule is irreducible and positive recurrent, then Birkhoff's ergodic theorem applied to the projection-centered count process guarantees almost sure convergence of $\frac{1}{n}\sum_{m=0}^{n-1} p_i(m)$.

For the expectation bound, first, a union bound gives

$$\mathbb{P}\left(\max_{1\leqslant i\leqslant d}\left|\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m)\right| \geqslant \varepsilon\right) \leqslant 2de^{-n\varepsilon^2/2}.$$

We conclude that

$$\begin{aligned}
\mathbb{E}\left[\max_{1\leqslant i\leqslant d}\left|\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m)\right|\right] &= \int_0^\infty \mathbb{P}\left(\max_{1\leqslant i\leqslant d}\left|\mu_i(n) - \frac{1}{n}\sum_{m=0}^{n-1} p_i(m)\right| \geqslant \varepsilon\right) d\varepsilon \\
&\leqslant \int_0^\infty 2de^{-n\varepsilon^2/2}\, d\varepsilon \\
&= d\sqrt{\frac{2\pi}{n}}.
\end{aligned}$$

□

Proof of Lemma 2.2.6

Lemma. If $Y_i(n) = N_i(n) - np_i^*$ and $Y(n) = (Y_1(n), \dots, Y_d(n))$, then for the self-balancing sampler one has $\sup_{n\geq 1} \mathbb{E}[\|Y(n)\|_1] < \infty$.

Proof. First, note that since

$$\exp(\beta_i n p_i^*) = \exp\left(\frac{n}{\sum_{j=1}^d \beta_j^{-1}}\right)$$

is independent of i, we see $p_i(N(n)) = p_i(Y(n))$. Hence the sample probabilities can be viewed as a function of the centered counts alone.

By Theorem 1 of [52], it suffices to prove that there exists a K such that $W_n := ||Y(n)||_1$ has negative drift whenever $W_n > K$. Note that:

$$\mathbb{E}[W_{n+1} - W_n \mid Y_n = y] = \sum_{i=1}^d [h_i(y_i) + p_i(y)g_i(y_i)],$$

where for each $i \in [d]$:

$$h_i(u) = \left|u - p_i^*\right| - |u| = \begin{cases} p_i^* & u \leqslant 0, \\ \text{linear} & u \in [0, p_i^*], \\ -p_i^* & u \geqslant p_i^*, \end{cases}$$

$$g_i(u) = \left|u + 1 - p_i^*\right| - \left|u - p_i^*\right| = \begin{cases} -1 & u \leqslant -(1 - p_i^*), \\ \text{linear} & u \in [-(1 - p_i^*), p_i^*], \\ 1 & u \geqslant p_i^*. \end{cases}$$

Let $p^*_{\max} := \max_i p_i^*$ and $p^*_{\min} := \min_i p_i^*$. First, if $\|y\|_1 \geqslant 2dp^*_{\max}$, then:

$$d \max_i y_i \geqslant \sum_{i:y_i>0} y_i = \frac{\|y\|_1}{2} \geqslant dp^*_{\max},$$

so for some j we have $y_j \geqslant p_j^*$ and hence $h_j(y_j) = -p_j^*$. Since $h_i(y_i) \leqslant p_i^*$:

$$\sum_{i=1}^d h_i(y_i) \leqslant \sum_{i\neq j} p_i^* - p_j^* = 1 - 2p_j^* \leqslant 1 - 2p^*_{\min}.$$

This controls the first term in the drift. Next we control $\sum_i p_i(y)g_i(y_i)$.

First, we show $m := \min_i \beta_i y_i \leqslant 0$ must be reasonably small. Since $\sum y_i = 0$:

$$dm \leqslant \sum_{i:y_i<0} \beta_i y_i \leqslant \beta_{min} \sum_{i:y_i<0} y_i \leqslant -\frac{\|y\|_1}{2} \cdot \beta_{min}.$$

Define the set S of coordinates that are not too small:

$$S := \{i \ : \ y_i > -1\}.$$

Then for $i \in S$:

$$p_i(y) = \frac{p_i(0)\exp(-\beta_i y_i)}{\sum_{j=1}^{d} p_j(0)\exp(-\beta_j y_j)} \leqslant \frac{p_{max}(0)}{p_{min}(0)} \cdot \frac{e^{-\beta_i y_i}}{e^{-m}} \leqslant \frac{p_{max}(0)}{p_{min}(0)} \cdot e^{m+\beta_{max}}.$$

By our above work, this is then upper bounded by:

$$\sum_{i \in S} p_i(y) \leqslant d\frac{p_{max}(0)}{p_{min}(0)} \cdot e^{-\frac{\|y\|_1}{2d}\beta_{min}+\beta_{max}} =: L(\|y\|_1).$$

Hence as $g_i(y_i) = -1$ for $i \in S^c$:

$$\sum_{i=1}^{d} p_i(y) g_i(y_i) \leqslant \sum_{i \in S} p_i(y) - \sum_{i \in S^c} p_i(y) = 2\sum_{i \in S} p_i(y) - 1 \leqslant 2L(\|y\|_1) - 1.$$

From here, we complete our drift bound:

$$\mathbb{E}[W_{n+1} - W_n \mid Y_n = y] \leqslant 1 - 2p^*_{\text{min}} + 2L(\|y\|_1) - 1 = 2L(\|y\|_1) - 2p^*_{\text{min}}.$$

For K large enough, if $\|y\|_1 > K$, then $L(\|y\|_1) \leqslant p^*_{\text{min}}/2$, giving us drift $\leqslant -p^*_{\text{min}}$. □

Proof of Theorem 2.2.7

Theorem. For the self-balancing sampler,

$$\|\mu(n) - p^*\|_{\text{TV}} \xrightarrow{a.s.} 0, \qquad \|\mu(n) - p^*\|_{\text{TV}} \lesssim \frac{d^2(\log n + O(1))}{n} \quad \text{a.s.}$$

Proof. The idea is to show that the log odds ratios of the sampling probabilities never get too large; this will force $\beta_i N_i \approx \beta_j N_j$, which will make $N_i/n \propto 1/\beta_i$. In particular, let

$$R_{i,j} := -\log\frac{p_i(n)}{p_j(n)} + \log\frac{p_i(0)}{p_j(0)} = \beta_i N_i - \beta_j N_j.$$

It is very unlikely for $R_{i,j}$ to exceed $2\log n$, as the following one-step analysis shows:

$$\begin{aligned}
\mathbb{P}(X_{n+1} = i, R_{i,j}(n) \geqslant 2\log n) &= \mathbb{E}[\mathbb{P}(X_{n+1} = i, R_{i,j}(n) \geqslant 2\log n \mid p(n))] \\
&= \mathbb{E}[p_i(n)\mathbf{1}_{\{R_{i,j}(n)\geqslant 2\log n\}}] \\
&\leqslant \mathbb{E}\left[p_j(n)\frac{p_j(0)}{p_i(0)}\frac{1}{n^2}\mathbf{1}_{\{R_{i,j}(n)\geqslant 2\log n\}}\right]
\end{aligned}$$

$$\lesssim \frac{1}{n^2}.$$

Therefore, by the Borel-Cantelli theorem, with probability 1, $R_{i,j}(n) \leqslant 2\log n + O(1)$ for all n. Applying this argument to $R_{j,i} = -R_{i,j}$, we get that $|R_{i,j}(n)| \leqslant 2\log n + O(1)$ for all n.

Now analyze the counts in terms of the $R_{i,1}$:

$$\frac{N_i}{n} = \frac{1}{\beta_i} \cdot \frac{\beta_1 N_1}{n} + \frac{R_{i,1}}{\beta_i n},$$

and summing over i gives

$$1 = \sum_{k=1}^{d} \frac{N_k}{n} = \frac{\beta_1 N_1}{n} \left(\sum_{k=1}^{d} \frac{1}{\beta_k} \right) + \frac{1}{n} \sum_{k=1}^{d} \frac{R_{k,1}}{\beta_k},$$

which we rearrange into

$$\frac{\beta_1 N_1}{n} = \frac{1}{\sum_{k=1}^{d} 1/\beta_k} \left(1 - \frac{1}{n} \sum_{k=1}^{d} \frac{R_{k,1}}{\beta_k} \right).$$

Plugging this back into the expression for N_i/n gives

$$\frac{N_i}{n} = \underbrace{\frac{1/\beta_i}{\sum_{k=1}^{d} 1/\beta_k}}_{p_i^*} \left(1 - \frac{1}{n} \sum_{k=1}^{d} \frac{R_{k,1}}{\beta_k} \right) + \frac{R_{i,1}}{\beta_i n}.$$

So the total variation distance is

$$\begin{aligned}
\|\mu(n) - p^*\|_{\mathrm{TV}} &= \frac{1}{2} \sum_{i=1}^{d} \left| \frac{N_i}{n} - p_i^* \right| \\
&= \frac{1}{2n} \sum_{i=1}^{d} \left| \frac{R_{i,1}}{\beta_i} - p_i^* \sum_{k=1}^{d} \frac{R_{k,1}}{\beta_k} \right| \\
&\lesssim \frac{d^2(\log n + O(1))}{n}.
\end{aligned}$$

□

Proof of Theorem 2.2.12

Theorem. If $C = \max_{1 \le i \le d} \min\{n \ge 1 : X_n = i\}$ and $p_i(0) = 1/d$, then

$$\mathbb{E}[C] = \sum_{i=1}^{d} \frac{\log(1 + \beta_i \log d)}{\beta_i} + O(d).$$

Proof. Writing $C = C_1 + \cdots + C_d$, where $C_i \coloneqq \{1 \leqslant n \leqslant C : X_n = i\}$ is the number of times we see outcome i before covering all the outcomes, we need only show that $\mathbb{E}[C_i] = \frac{\log(1+\beta_i \log d)}{\beta_i} + O(1)$.

The key insight is to embed the self-balancing sampler into a continuous time stochastic process with the sampling order determined by independent exponential clocks. For $1 \leqslant i \leqslant d$ and $j \geqslant 0$, let $W_{i,j} \sim \mathrm{Exp}(\alpha_i^j)$ be independent, and let $Y_{i,j} \coloneqq \sum_{k=0}^{j} W_{i,k}$. Then the order of the $Y_{i,j}$ and their associated i indices has the same distribution as the order of the X_n, e.g. $\mathbb{P}(Y_{1,0} < Y_{1,1} < Y_{6,0} < \cdots) = \mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 6, \ldots)$. Letting $\widetilde{C}_i(t) \coloneqq \max\{j \geqslant 0 : Y_{i,j} \leqslant t\}$, we claim that

$$\frac{\log(1+\beta_i t)}{\beta_i} \leqslant \mathbb{E}[\widetilde{C}_i(t)] \leqslant \frac{\log(1+\beta_i t)}{\beta_i} + 1.$$

The lower bound is equivalent to $1 + \beta_i t \leqslant e^{\beta_i \mathbb{E}[\widetilde{C}_i(t)]}$, which follows from

$$\begin{aligned}
e^{\beta_i \mathbb{E}[\widetilde{C}_i(t)]} &= 1 + \int_0^t \frac{d}{du} e^{\beta_i \mathbb{E}[\widetilde{C}_i(u)]}\Big|_{u=s} ds \\
&= 1 + \beta_i \int_0^t \frac{d}{du} \mathbb{E}[\widetilde{C}_i(u)]\Big|_{u=s} \cdot e^{\beta_i \mathbb{E}[\widetilde{C}_i(s)]} ds \\
&= 1 + \beta_i \int_0^t \mathbb{E}[e^{-\beta_i \widetilde{C}_i(s)}] e^{\beta_i \mathbb{E}[\widetilde{C}_i(s)]} ds \\
&\geqslant 1 + \beta_i \int_0^t e^{-\beta_i \mathbb{E}[\widetilde{C}_i(s)]} e^{\beta_i \mathbb{E}[\widetilde{C}_i(s)]} ds \\
&= 1 + \beta_i t,
\end{aligned}$$

where we used that the derivative of the expectation of a birth process is the expected birth rate. For the upper bound, we have

$$\begin{aligned}
e^{\beta_i \mathbb{E}[\widetilde{C}_i(t)]} &\leqslant \mathbb{E}[e^{\beta_i \widetilde{C}_i(t)}] \\
&= 1 + \int_0^t \frac{d}{du} \mathbb{E}[e^{\beta_i \widetilde{C}_i(u)}]\Big|_{u=s} ds.
\end{aligned}$$

By Kolmogorov's forward equation, $\frac{d}{du}\mathbb{E}[e^{\beta_i \widetilde{C}_i(u)}]$ is the expectation of the generator of the birth process $\widetilde{C}_i(t)$ applied to the function $f(z) = e^{\beta_i z}$.

$$\begin{aligned}
&= 1 + \int_0^t \mathbb{E}[e^{-\beta_i \widetilde{C}_i(s)}(e^{\beta_i(\widetilde{C}_i(s)+1)} - e^{\beta_i \widetilde{C}_i(s)})] ds \\
&= 1 + \int_0^t e^{\beta_i} - 1 \, ds \\
&= 1 + (e^{\beta_i} - 1)t.
\end{aligned}$$

This gives an upper bound of $\mathbb{E}[\widetilde{C}_i(t)] \leqslant \frac{\log(1+(e^{\beta_i}-1)t)}{\beta_i} \leqslant \frac{\log(1+\beta_i t)}{\beta_i} + 1$.

Now, we incorporate the random time in the continuous process which corresponds to the cover time in the self-balancing sampler: $M_{-i} := \max\{W_{j,0} : j \neq i\}$. Then $C_i \overset{d}{=} \max\{1, \widetilde{C}_i(M_{-i})\}$, so that

$$\mathbb{E}[C_i] = \mathbb{E}[\widetilde{C}_i(M_{-i})] + \underbrace{\mathbb{P}(W_{i,0} < W_{j,0}\,\forall j \neq i)}_{=1/d}.$$

Applying our previous inequalities gives

$$\left|\mathbb{E}[C_i] - \mathbb{E}\left[\frac{\log(1+\beta_i M_{-i})}{\beta_i}\right]\right| \leqslant 1 + \frac{1}{d},$$

and all that remains now is to show that $\mathbb{E}[\widetilde{C}_i(M_{-i})] \approx \mathbb{E}[\widetilde{C}_i(\log d)]$, where $\mathbb{E}[M_{-i}] = \sum_{k=1}^{d-1} \frac{1}{k} \approx \log d$. We have

$$\begin{aligned}\left|\mathbb{E}[C_i] - \frac{\log(1+\beta_i \log d)}{\beta_i}\right| \leqslant& \left|\mathbb{E}[C_i] - \mathbb{E}\left[\frac{\log(1+\beta_i M_{-i})}{\beta_i}\right]\right| \\ &+ \left|\mathbb{E}\left[\frac{\log(1+\beta_i M_{-i})}{\beta_i}\right] - \frac{\log(1+\beta_i \log d)}{\beta_i}\right|.\end{aligned}$$

The latter term can be bounded using the fact that $g(x) := \frac{\log(1+\beta_i x)}{\beta_i}$ is 1-Lipschitz, which follows from $|g'(x)| = |\frac{1}{1+\beta x}| \leqslant 1$ for all $x \geqslant 0$.

$$\begin{aligned}&\leqslant 1 + \frac{1}{d} + \mathbb{E}[|M_{-i} - \log d|] \\ &\leqslant 1 + \frac{1}{d} + \sqrt{\mathbb{E}[(M_{-i} - \log d)^2]} \\ &= 1 + \frac{1}{d} + \sqrt{(\mathbb{E}[M_{-i} - \log d])^2 + \mathrm{Var}(M_{-i})} \\ &= 1 + \frac{1}{d} + \sqrt{\left(\log d - \sum_{k=1}^{d-1} \frac{1}{k}\right)^2 + \sum_{k=1}^{d-1} \frac{1}{k^2}} \\ &\leqslant 1 + \frac{1}{d} + \sqrt{1 + \frac{\pi^2}{6}}.\end{aligned}$$

□

Proof of Theorem 2.2.9

Theorem. Let $L_\beta(n) = \frac{1}{2}\sum_{i=1}^d \beta_i(N_i(n) - np_i^*)^2$ and let $D_\beta(q) = \mathbb{E}[L_\beta(n+1) - L_\beta(n) \mid N(n), q]$. Then the optimization problem

$$\underset{q \in \Delta^d}{\arg\max}\big(H(q) - H(q, p(0)) - D_\beta(q)\big) = \underset{q \in \Delta^d}{\arg\min}\big(\mathrm{KL}(q \| p(0)) + D_\beta(q)\big)$$

has unique solution $q_i \propto p_i(0)e^{-\beta_i(N_i(n)+1/2)}$.

Proof. First, expand D to isolate the terms depending on q:

$$\begin{aligned}
D(q) &= \frac{1}{2}\sum_{i=1}^{d} \beta_i \mathbb{E}[N_i(n+1)^2 - 2(n+1)p_i^* N_i(n+1) + ((n+1)p_i^*)^2 \\
&\qquad\qquad - N_i(n)^2 + 2np_i^* N_i(n) - (np_i^*)^2 \mid N(n), q] \\
&= \frac{1}{2}\sum_{i=1}^{d} \beta_i [q_i(N_i(n)+1)^2 + (1-q_i)N_i(n)^2 - N_i(n)^2 \\
&\qquad\qquad - 2p_i^*((n+1)(N_i(n)+q_i) - nN_i(n)) + (2n+1)(p_i^*)^2] \\
&= \frac{1}{2}\sum_{i=1}^{d} \beta_i [2q_i N_i(n) + q_i - 2(n+1)p_i^* q_i - 2p_i^* N_i(n) + (2n+1)(p_i^*)^2] \\
&= \sum_{i=1}^{d} \left(\beta_i N_i(n) q_i + \frac{1}{2}\beta_i q_i + \frac{2n+1}{2}\beta_i (p_i^*)^2 - \frac{N_i(n)}{\sum_{k=1}^{d} 1/\beta_k} \right) - \frac{n+1}{\sum_{k=1}^{d} 1/\beta_k},
\end{aligned}$$

so it is equivalent to maximize $H(q) - \sum_{i=1}^{d} \beta_i N_i(n) q_i + \frac{1}{2}\beta_i q_i$. When all the β_i are equal, this just looks like maximizing $H(q) - \beta\langle N(n), q\rangle$.

Now solve the optimization problem using the Lagrange multiplier λ. The Lagrangian is

$$\mathcal{L}(q) := \sum_{i=1}^{d} \left(-q_i \log q_i + q_i \log p_i(0) - \beta_i N_i q_i - \frac{1}{2}\beta_i q_i + \lambda q_i \right) - \lambda.$$

Setting $\nabla_q \mathcal{L}(q) = 0$ gives

$$0 = -\log q_i - 1 + \log p_i(0) - \beta_i N_i - \frac{\beta_i}{2} + \lambda \qquad \Longrightarrow \qquad q_i = p_i(0) e^{-\beta_i N_i - \beta_i/2 + \lambda - 1}.$$

Therefore, $q_i \propto p_i(0) e^{-\beta_i (N_i(n) + 1/2)}$, as claimed. $\square$

Proof of Theorem 2.2.10

Theorem. If $L_\beta(x) = \frac{1}{2}\sum_{i=1}^{d} \beta_i (x_i - np_i^*)^2$, then

$$\operatorname*{arg\,min}_{q \in \Delta^d} \Big(\langle \nabla L_\beta(N(n)), q \rangle + \mathrm{KL}(q \| p(0)) \Big)$$

has unique solution $q_i \propto p_i(0) e^{-\beta_i N_i(n)}$.

Proof. Calculate the gradient:

$$\frac{\partial L_\beta}{\partial x_i}(x) = \beta_i (x_i - np_i^*).$$

so we aim to optimize

$$\sum_{i=1}^{d} \beta_i(N_i(n) - np_i^*)q_i + \sum_{i=1}^{d} q_i \log q_i - q_i \log p_i(0).$$

Define the Lagrangian

$$\mathcal{L}(q) := \left(\sum_{i=1}^{d} \beta_i(N_i(n) - np_i^*)q_i + q_i \log q_i - q_i \log p_i(0) + \lambda q_i \right) - \lambda$$

with Lagrange multiplier λ. Setting $\nabla_q \mathcal{L}(q) = 0$ gives

$$\begin{aligned} 0 &= \beta_i(N_i(n) - np_i^*) + \log q_i + 1 - \log p_i(0) + \lambda, \\ &= \beta_i N_i(n) - \frac{n}{\sum_{j=1}^{n} 1/\beta_i} + \log q_i + 1 - \log p_i(0) + \lambda, \end{aligned}$$

which we solve to get

$$q_i \propto p_i(0) e^{-\beta_i N_i(n)}.$$

□

Proof of Theorem 2.2.11

Theorem. The optimization problem

$$\arg\min_{q \in \Delta^d} \left(h \left\langle \beta_{X_n} e_{X_n} - \frac{1}{\sum_{j=1}^{d} \beta_j^{-1}} \mathbf{1},\ q \right\rangle + \mathrm{KL}(q \| p(n)) \right)$$

has unique solution $q = p(n+1)$, i.e. $q_i \propto p_i(0) e^{-h\beta_i N_i(n+1)}$.

Proof. Since $\langle \mathbf{1}, q \rangle = 1$, it is equivalent to minimize

$$h\langle \beta_{X_n} e_{X_n}, q \rangle + \mathrm{KL}(q \| p(n)) = h\beta_{X_n} q_{X_n} + \sum_{i=1}^{d} q_i \log q_i - q_i \log p_i(n).$$

Defining the Lagrangian

$$\mathcal{L}(q) := h\beta_{X_n} q_{X_n} - \lambda + \sum_{i=1}^{d} q_i \log q_i - q_i \log p_i(n) + \lambda q_i$$

with Lagrange multiplier λ, the condition $\nabla_q \mathcal{L}(q) = 0$ gives

$$0 = \begin{cases} h\beta_i + \log q_i + 1 - \log p_i(n) + \lambda & \text{if } X_n = i, \\ \log q_i + 1 - \log p_i(n) + \lambda & \text{if } X_n \neq i. \end{cases}$$

Solving for q_i gives

$$q_i \propto \begin{cases} p_i(n)e^{-h\beta_i} & \text{if } X_n = i, \\ p_i(n) & \text{if } X_n \neq i, \end{cases}$$

which is precisely the update rule for the self-balancing sampler. □

Proof of Proposition 2.3.1

Proposition. Fix $i \in [d]$. If $Z_0^h \to z_0$, then $Z^h \Rightarrow Z$ in $D([0,T],\mathbb{R})$ for each $T < \infty$, where Z solves

$$dZ_t = -\frac{\beta_i\beta_d}{\beta_i+\beta_d} Z_t\,dt + \sqrt{\beta_i\beta_d}\,dW_t, \qquad Z_0 = z_0.$$

Proof. For convenience, drop the subscript on c_β. Note:

$$\frac{p_i^{\beta h}}{p_i^{\beta h} + p_d^{\beta h}} = \frac{e^{-\beta_i h N_i}}{e^{-\beta_i h N_i} + e^{-\beta_d h N_d}} = \frac{e^{-Y_i^{\beta h}}}{1 + e^{-Y_i^{\beta h}}}$$

Since $Z_{nh}^h = z$ implies $Y_i^{h\beta}(n) = z\sqrt{h} + c_\beta$, we have increments:

$$\Delta Z_{nh}^h := Z_{(n+1)h}^h - Z_{nh}^h \mid \{Z_{nh}^h = z\} = \begin{cases} \beta_i\sqrt{h} & \text{w.p. } \frac{e^{-(z\sqrt{h}+c)}}{1+e^{-(z\sqrt{h}+c)}}, \\ -\beta_d\sqrt{h} & \text{w.p. } \frac{1}{1+e^{-(z\sqrt{h}+c)}}, \end{cases}$$

so since $e^{-c} = \frac{\beta_d}{\beta_i}$, our process has drift:

$$\begin{aligned} \mu_h(z) &:= \mathbb{E}[\Delta Z_{nh}^h \mid Z_{nh}^h = z] \\ &= \frac{\sqrt{h}}{1 + e^{-(z\sqrt{h}+c)}}\left[\beta_i e^{-(z\sqrt{h}+c)} - \beta_d\right] \\ &= \frac{\beta_d\sqrt{h}}{1 + \frac{\beta_d}{\beta_i}e^{-z\sqrt{h}}}\left[e^{-z\sqrt{h}} - 1\right]. \end{aligned}$$

Hence:

$$\mu_h(z) = -\frac{\beta_d}{1+\frac{\beta_d}{\beta_i}} zh + o(h) = -\frac{z}{\beta_d^{-1} + \beta_i^{-1}} h + o(h).$$

Similarly, for the variance:

$$\begin{aligned} \sigma_h^2(z) &:= \mathbb{E}[(\Delta Z_{nh}^h)^2 \mid Z_{nh}^h = z] - \mu_h(z)^2 \\ &= \frac{h}{1 + \frac{\beta_d}{\beta_i}e^{-z\sqrt{h}}}\left[\beta_i^2 \cdot \frac{\beta_d}{\beta_i}e^{-z\sqrt{h}} + \beta_d^2\right] + o(h) \\ &= \frac{\beta_i\beta_d + \beta_d^2}{1 + \frac{\beta_d}{\beta_i}} h + o(h) \end{aligned}$$

$$= \beta_i \beta_d h + o(h)$$

Lastly, since $|\Delta Z^h_{nh}| \leqslant \max(\beta_i, \beta_d)\sqrt{h} \to 0$, we have no large jumps.

As the coefficients of the diffusion

$$dZ_t = -\frac{\beta_i \beta_d}{\beta_i + \beta_d} Z_t dt + \sqrt{\beta_i \beta_d} \cdot dW_t, \quad Z_0 = z_0,$$

are globally Lipschitz, there exists a unique strong solution, namely just the OU process. Standard theory shows the desired convergence. See for example Section 8.7 of [16]. □

Proof of Proposition 2.3.2

Proposition. If $Z^h_0 \to z_0$, then $Z^h \Rightarrow Z$ as $h \downarrow 0$, where

$$dZ_t = -\frac{1}{S_\beta} Z_t \, dt + \Sigma^{1/2} dW_t, \qquad S_\beta = \sum_j \beta_j^{-1}, \qquad \Sigma = S_\beta^{-1} \operatorname{diag}(\beta) - S_\beta^{-2} \mathbf{1}\mathbf{1}^T,$$

and the limiting diffusion is supported on $\{z : \langle z, p^* \rangle = 0\}$.

Proof. For convenience, drop the subscript of c_β. Note since $\beta_i p^*_i$ is independent of i:

$$p_i^{h\beta} = \frac{e^{-h\beta_i N_i}}{\sum_j e^{-h\beta_j N_j}} = \frac{e^{-Y_i^{h\beta}}}{\sum_j e^{-Y_j^{h\beta}}}$$

so the transitions are a function of our mean-centered counts alone. The value c_β is chosen so that $e^{-c_i} = \kappa \beta_i^{-1}$ for $\kappa > 0$ independent of i, and so $\langle c_\beta, p^* \rangle = 0$ satisfies the same constraint as Y^β. Since $Z^h_{nh} = z$ implies $Y_n^{\beta h} = z\sqrt{h} + c$, this process has drift:

$$\begin{aligned}
(\mu_h(z))_i &:= \mathbb{E}[(\Delta Z^h_{nh})_i \mid Z^h_{nh} = z] \\
&= \beta_i \sqrt{h} \left[\frac{e^{-(z_i\sqrt{h} + c_i)}}{\sum_{j=1}^d e^{-(z_j \sqrt{h} + c_j)}} - p^*_i \right] \\
&= \sqrt{h} \left[\frac{e^{-z_i\sqrt{h}}}{\sum_{j=1}^d \frac{1}{\beta_j} e^{-z_j\sqrt{h}}} - \beta_i p^*_i \right] \\
&= \sqrt{h} \left[\frac{1 - z_i\sqrt{h}}{\sum_{j=1}^d \frac{1}{\beta_j}(1 - z_j\sqrt{h})} - \beta_i p^*_i + O(h) \right] \\
&= \sqrt{h} \left[\frac{\left(\sum_{j=1}^d p^*_j z_j - z_i \right) \sqrt{h}}{\sum_{j=1}^d \frac{1}{\beta_j}(1 - z_j\sqrt{h})} + O(h) \right]
\end{aligned}$$

$$= \frac{1}{\sum_{j=1}^{d} \frac{1}{\beta_j}} \left(\sum_{j=1}^{d} p_j^* z_j - z_i \right) h + o(h)$$

Similarly, we can calculate the second moments. First:

$$\begin{aligned}
\mathbb{E}[(\Delta Z_{nh}^h)_i^2 \mid Z_{nh}^h = z] &= h\beta_i^2 \left[(p_i^*)^2 + (1-2p_i^*) \cdot \frac{\frac{1}{\beta_i} e^{-z_i\sqrt{h}}}{\sum_{j=1}^{d} \frac{1}{\beta_j} e^{-z_j\sqrt{h}}} \right] \\
&= h\beta_i^2 \left[(p_i^*)^2 + (1-2p_i^*) \cdot \left(p_i^* + O(\sqrt{h}) \right) \right] \\
&= h\beta_i^2 \left(p_i^* - (p_i^*)^2 \right) + o(h)
\end{aligned}$$

Moreover, for $i \neq j$:

$$\begin{aligned}
\mathbb{E}[(\Delta Z_{nh}^h)_i (\Delta Z_{nh}^h)_j \mid Z_{nh}^h = z] &= h\beta_i\beta_j \left[p_i^* p_j^* - \left(\frac{\frac{p_i^*}{\beta_j} e^{-z_j\sqrt{h}} + \frac{p_j^*}{\beta_i} e^{-z_i\sqrt{h}}}{\sum_{\ell=1}^{d} \frac{1}{\beta_\ell} e^{-z_\ell\sqrt{h}}} \right) \right] \\
&= h\beta_i\beta_j \left[p_i^* p_j^* - \left(\frac{\frac{p_i^*}{\beta_j} + \frac{p_j^*}{\beta_i} - O(\sqrt{h})}{\sum_{\ell=1}^{d} \frac{1}{\beta_\ell} - O(\sqrt{h})} \right) \right] \\
&= -\beta_i\beta_j p_i^* p_j^* h + o(h).
\end{aligned}$$

so we have the desired covariance. As the coefficients of the limiting diffusion are globally Lipschitz, the associated SDE has a unique strong solution. Again standard theory, such as Section 8.7 of [16], gives the desired convergence result. This gives a limiting diffusion of the form

$$dZ_t = -\frac{1}{S_\beta}(Z_t - \bar{Z}_t)dt + \Sigma^{1/2} dW_t, \qquad \bar{Z}_t := \langle p^*, Z_t \rangle.$$

Lastly, we show that the limiting diffusion lives in the hyperplane

$$Z_t \in \{z : \langle z, p^* \rangle = 0\}.$$

Note that by construction

$$\langle Y^\beta(n), p^* \rangle = \langle c_\beta, p^* \rangle = 0.$$

and hence $\langle Z_t^h, p^* \rangle = 0$ for all h and t, from which the result follows. In particular, $\bar{Z}_t = 0$, so our above diffusion simplifies to the desired form. □

Proof of Theorem 2.5.1

Theorem. Suppose $\pi_i(x) \propto a_i e^{-H_i(x)}$, that $\pi(x) = \pi(x')$ whenever $(I - p^* 1^T)x = (I - p^* 1^T)x'$, and that there exist $c, C > 0$ such that for all y with $\sum_i y_i = 0$ and $y_j \leq y_i$,

$$H_i(y) - H_j(y) \geq c(y_i - y_j) - C.$$

Then the empirical distribution associated to π satisfies $\mathbb{E}\|\mu_n - p^*\|_{\mathrm{TV}} = O(n^{-1})$.

Proof. Analogous to Theorem 2.2.5, it suffices to show the mean-centered counts $Y(n) := N(n) - np^*$ are uniformly bounded in L^1. The proof is almost identical to that of Lemma 2.2.6. As before, we let $W_n := ||Y(n)||_1$, and we prove a drift condition. We decompose:

$$\mathbb{E}[W_{n+1} - W_n \mid Y(n) = y] = \sum_{i=1}^{d} h_i(y_i) + \pi_i(y)g_i(y_i),$$

for h_i, g_i as defined in Lemma 2.2.6. We showed above $\sum h_i(y_i) \leqslant 1 - 2p^*_{min}$, so we just must handle the $\sum \pi_i(y)g_i(y_i)$ term.

Let $m := \min y_i \leqslant 0$, $M := \max -H_i(y)$, and $S = \{i : y_i > -1\}$. Then $m \leqslant -||y||_1/2d$. For $i \in S$:

$$\pi_i(y) = \frac{a_i \exp(-H_i(y))}{\sum_j a_j \exp(-H_j(y))} \leqslant \frac{a_{max}}{a_{min}} \cdot e^{-c(y_i - m) + C}.$$

Hence by our above work we see:

$$\sum_{i \in S} \pi_i(y) \leqslant d \cdot \frac{a_{max}}{a_{min}} \cdot e^{-c||y||_1/2d + C + c} := L(||y||_1).$$

The result then follows identically to before. □

Proof of Lemma 2.3.3

Lemma. When $d = 2$ and $\beta_1 = \beta_2 = \beta > 0$, the projection-centered chain $\tilde{Y}(n)$ is reversible on $\mathbb{Z}$ with stationary distribution

$$\pi(z) \propto \frac{p_2(0) + p_1(0)e^{-\beta z}}{p_1(0) + p_2(0)} \left(\frac{p_1(0)}{p_2(0)}\right)^z e^{-\beta\binom{z}{2}}.$$

Proof. In the $d = 2$ case then $\tilde{Y}_n$ is a simple birth-death chain with birth probability:

$$\lambda_n = \frac{p_1(0)e^{-\beta n}}{p_2(0) + p_1(0)e^{-\beta n}}$$

and death otherwise. The detailed balance equations then imply:

$$\frac{\pi(n)}{\pi(n-1)} = \frac{\lambda_{n-1}}{1 - \lambda_n}.$$

Hence:

$$\pi(n) = \pi_0 \prod_{j=1}^{n} \frac{\lambda_{j-1}}{1 - \lambda_j} = \pi_0 \frac{p_2(0) + p_1(0)e^{-\beta n}}{p_1(0) + p_2(0)} \left(\frac{p_1(0)}{p_2(0)}\right)^n e^{-\beta\binom{n}{2}}$$

If $\beta > 0$ then this is normalizable as it has discrete Gaussian tails. □

Proof of Lemma 2.3.4

Lemma. For $d = 2$, and uniform β, as $\beta \to 0$:

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}\left(X_k \neq \arg\min_{i=1,2} X_i(k-1) \right) = \frac{1}{2} - \Theta(\sqrt{\beta}).$$

Proof. Note in the $d = 2$ case with uniform β, we have:

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}\left(X_k \neq \arg\max_{i=1,2} \pi_i(N(k-1)) \right) = \limsup_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_{y\sim\tilde{Y}(k)} \left[\frac{1}{1+e^{\beta|y|}} \right].$$

Since $Y(n)$ is ergodic, the ergodic theorem and Lemma 2.3.3 imply:

$$\mathbb{E}_{y\sim\pi} \left[\frac{1}{1+e^{\beta|y|}} \right] = Z^{-1} \sum_{y\in\mathbb{Z}} \frac{\cosh\left(\frac{\beta y}{2}\right)}{1+e^{\beta|y|}} \cdot e^{-\frac{\beta y^2}{2}} = \frac{1}{2Z} \sum_{y\in\mathbb{Z}} e^{-\frac{\beta(y^2+|y|)}{2}}.$$

for $Z = \sum_{z\in\mathbb{Z}} \cosh(\beta z/2) \cdot e^{-\beta z^2/2}$. While no simple closed-form exists, the simple bound:

$$\frac{1}{2} - \frac{x}{4} \leqslant (1+e^x)^{-1} \leqslant \frac{1}{2} - \frac{x}{4} + \frac{x^3}{48}, \quad \forall x \geqslant 0,$$

implies for small β our error-rate is of order:

$$\mathbb{E}_{y\sim\pi} \left[\frac{1}{1+e^{\beta|y|}} \right] = \frac{1}{2} - \frac{\beta}{4} \mathbb{E}_\pi |Y| + O\left(\beta^3 E_\pi |Y|^3\right).$$

To control this, note the stationary distribution of Y is a mixture of two (discrete) Gaussians

$$\cosh\left(\frac{\beta y}{2}\right) e^{-\beta y^2/2} \propto \frac{1}{2}\left[e^{-\frac{\beta}{2}(y-1/2)^2} + e^{-\frac{\beta}{2}(y+1/2)^2} \right].$$

centered at $\pm\frac{1}{2}$ with variance β^{-1}. For small β, the standard Riemann sum comparison implies the distribution of Y is well-approximated by the continuous counterpart. Hence $Y \approx \frac{\xi}{2} + \beta^{-1/2} G$ for a standard discrete gaussian G and Rademacher $\xi \sim \mathrm{Unif}\{\pm 1\}$.

We can then estimate $\mathbb{E}|Y|$ and $\mathbb{E}|Y|^3$ using the standard moments of the folded normal. In particular, as $\beta \downarrow 0$ the gaussian portion G dominates and we get the moments of the half-normal distribution:

$$\mathbb{E}|Y| \asymp \sqrt{\frac{2}{\pi\beta}}, \qquad \mathbb{E}|Y|^3 = O(\beta^{-3/2}).$$

Combining this with the above yields:

$$\mathbb{E}_{y\sim\pi} \left[\frac{1}{1+e^{\beta|y|}} \right] = \frac{1}{2} - c\sqrt{\beta} + O(\beta^{3/2}) = \frac{1}{2} - \Theta(\sqrt{\beta}).$$

as desired. $\square$

Part II

Developing Projects

Chapter 3

Inference under the PCH Model

Abstract

This chapter studies phylogenetic inference under the PCH model of [10], a stochastic model for the evolution of linguistic characters which explicitly incorporates polymorphism. The foundation of this model is a simple birth-death process which describes how states evolve down the underlying phylogenetic tree. In this chapter, we analyze various probabilistic properties of this process, translating them into measures of quartet support and candidate bipartitions relevant to phylogenetic inference. These results motivate ASTRAL-based inference procedures and provide preliminary sample-complexity bounds for recovering the underlying language phylogeny from independently evolving characters.

3.1 Introduction

In phylogenetics, we have a collection of objects whose ancestral history we hope to infer using only modern-day information. This shared history is summarized by a *phylogenetic tree*, a graphical tree whose leaves are labeled by these objects. Classically, these objects represented different biological species, the relevant data was their DNA, and the phylogenetic tree encoded the sequence of speciation events that occurred over time to produce the species we see today. Nonetheless, the tools of phylogenetics are far more broadly applicable: they have been used in medicine to better understand the progression of cancer [61], in astrophysics as a hierarchical clustering procedure for astral bodies [24], and in terrane analysis to study the geological history of specific regions [44].

In this chapter, we study a specific application to linguistics, where the objects of interest are different languages and the underlying data are *linguistic characters*. These characters encode features that vary across languages and are commonly divided into three broad categories: *phonological*, *morphological*, and *lexical*.

Phonological characters describe sound patterns, such as whether a language distinguishes tones or permits consonant clusters. Morphological characters describe grammatical structure, such as how past tense or plurality is expressed. Lexical characters describe the word forms used to express specific ideas, often by grouping languages according to whether their words for common concepts such as "fire," or "mother" descend from a common historical source. Lexical characters are particularly interesting as they can be multi-state—several different words can simultaneously share the same meaning—a property known as *polymorphism*. We are specifically interested in the model introduced in [10], which explicitly models this polymorphism. We refer to this model as the *Polymorphic CHaracter (PCH) model*, in a similar vein to the follow-up paper [20].

3.1.1 The PCH Model

The key feature of the PCH model is that it allows for polymorphism: a given character c may take on multiple different states simultaenously. To differentiate between the set-valued collection of states a character currently takes on and the individual states themselves, we call the former the *macro-state* of the character and the latter its individual *micro-states*. For example, if the macro-state of a character c at a given point of time is $\{\alpha, \beta\}$ then it currently exhibits two micro-states: α and β. Under the PCH model, it is assumed that micro-states may be born or die as the character evolves along the tree, so that the current *number* of micro-states present in the current macro-state evolves according to a standard birth-death process, which we refer to as the *cardinality process*.

More formally, suppose that $\mathcal{T}^*$ is the underlying rooted phylogenetic tree of languages. Given some collection of characters $\mathcal{C}$, associated to each character c is a set $\Sigma^{(c)}$ of possible states, birth/death rates $\{\lambda_i^{(c)}\}, \{\mu_i^{(c)}\}$ controlling the cardinality process, a probability measure $\nu^{(c)}$ on $\Sigma^{(c)}$ for selecting new states, and a probability measure $\rho^{(c)}$ on the set $\binom{\Sigma^{(c)}}{*}$ of nonempty finite subsets of $\Sigma^{(c)}$ for determining the macro-state of the root.

More specifically, let $[r, \ell]$ denote the unique path from the root r of $\mathcal{T}^*$ to some leaf $\ell \in \mathcal{T}^*$. For $x \in [r, \ell]$ we assume the macro-state $c(x)$ for the given character evolves down this path as follows

(i) The macro-state $c(r)$ of the root is sampled from $\rho^{(c)}$.

(ii) If $|c(x)| = i$, a new micro-state is born at rate $\lambda_i^{(c)}$ and an existing micro-state dies at rate $\mu_i^{(c)}$ respectively.

(iii) If a death occurs, a micro-state $\alpha \in c(x)$ is chosen uniformly at random to delete. If a birth occurs, a new micro-state $\alpha \notin c(x)$ is chosen proportional to $\nu^{(c)}$, namely via

$$\mu(A) := \frac{\nu^{(c)}(A \setminus c(x))}{\nu^{(c)}(\Sigma^{(c)} \setminus c(x))}$$

When a branch-point v of $\mathcal{T}^*$ is reached, conditional on $c(v)$ the macro-state evolves independently down each child branch starting from state $c(v)$.

A micro-state $\alpha \in \Sigma^{(c)}$ with $\nu^{(c)}(\{\alpha\}) > 0$ is called *homoplastic*. These states can evolve independently in different lineages, rather than from shared ancestry. We denote the set of homoplastic states and non-homoplastic states by $\Sigma_H^{(c)}, \Sigma_N^{(c)}$ respectively. Homoplastic states in a sense complicate the inference process, as two languages can share the homoplastic trait without necessarily having any recent shared ancestry.

Throughout this paper, we will assume characters follow the *immigration-death rates*

$$\lambda_i^{(c)} = \lambda^{(c)}, \quad \mu_i^{(c)} = i\mu^{(c)} \cdot 1\{i > 1\}. \tag{3.1.1}$$

for $i \geqslant 1$. Namely, the birth-rate is constant, and the death rate is proportional to the number of states, except when only one state is present. These assumptions ensure that each character has at least one micro-state associated to it at any point in $\mathcal{T}^*$ and that the number of states cannot blow-up in finite time.

Since the upcoming inference procedure is based solely on non-homoplastic states, we also make the following assumptions about $\nu^{(c)}$ and $\rho^{(c)}$.

Assumption (1)**.** Non-homoplastic states occur sufficiently often. Namely

(i) There is positive probability that the macro-state at the root contains at least one micro-state that is not homoplastic

$$\rho^{(c)}\left(\left\{A \in \binom{\Sigma^{(c)}}{*} \;:\; A \cap \Sigma_N^{(c)} \neq \emptyset\right\}\right) > 0.$$

(ii) There is positive probability of adding a micro-state that is not homo- plastic to the current macro-state when there is a birth

$$\nu^{(c)}(\Sigma_N^{(c)}) > 0.$$

The full PCH model additionally incorporates another mechanism of character transmission known as *borrowing* which allows characters to transfer laterally throughout the tree, analogous to horizontal gene transfer in biological contexts. In this paper, we only consider the case where no borrowing occurs, but interested readers can find further details in [10].

3.1.2 ASTRAL

To infer the underlying tree from many independently evolving characters, we adapt ideas from quartet-based species-tree estimation. In phylogenetic inference, one often wants to combine information from many sources, such as genetic information across a collection of genes or, in our case, linguistic characters. In *consensus methods*, each source is used to generate an estimate g_i of the true phylogenetic tree $\mathcal{T}^*$, and then these estimates $\{g_i\}_{i=1}^n$ are aggregated. In the phylogenetics literature, the trees g_i are called the *gene trees*, while $\mathcal{T}^*$ is referred to as the *species tree*. We will occassionally utilize this vocabulary.

ASTRAL [46] is one commonly used aggregation procedure. It scores candidate phylogenetic trees $\mathcal{T}$ based on the number of *quartets*, four-leaf induced subtrees, they share with the estimated trees $\{g_i\}$. Specifically, it solves the optimization problem

$$\hat{T} = \arg\max_{\mathcal{T}} \sum_{q \in \mathcal{Q}(\mathcal{T})} W(q), \tag{3.1.2}$$

where $W(q)$ is some meaure of the support for the quartet q. Commonly,

$$W(q) = \sum_{i=1}^{n} 1\{q \in g_i\},$$

the number of gene trees in which the quartet appears. Later on, we will introduce other measures of the quartet support specific to the PCH model.

Since the set of phylogenetic trees is super-exponentially large, it is intractable to solve (3.1.2) exactly for trees with more than 20 or so leaves. For larger trees, a restricted search space is required. One possible strategy is to limit the bipartitions of the leaves the candidate trees $\mathcal{T}$ can induce.

A *bipartition* of a set S is a partition of $S = A \cup B$ into two nonempty, disjoint subsets. Every edge $e \in E(\mathcal{T})$ of our tree induces a bipartition of the leaves by considering the two connected components of the graph $(V(\mathcal{T}), E(\mathcal{T}) \setminus \{e\})$. Moreover, Buneman's splits-equivalence theorem shows $\mathcal{T}$ is completely characterized by its set $\mathcal{X}(\mathcal{T})$ of bipartitions. Since each g_i is a noisy estimate of $\mathcal{T}^*$, we might expect that $\mathcal{X}(\mathcal{T}^*) \subseteq \mathcal{X}_g := \bigcup_i \mathcal{X}(g_i)$, suggesting the restricted search space

$$\Big\{\mathcal{T} \;:\; \mathcal{X}(\mathcal{T}) \subseteq \bigcup_{i=1}^{n} \mathcal{X}(g_i)\Big\}.$$

More generally, given any set $\mathcal{X}$ of bipartitions, we can solve

$$\hat{T} = \arg\max_{\mathcal{X}(\mathcal{T}) \subseteq \mathcal{X}} \sum_{q \in \mathcal{Q}(\mathcal{T})} W(q). \tag{3.1.3}$$

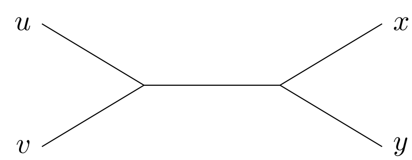
Using a clever dynamic-programming principle, ASTRAL efficiently solves this bipartition-constrained optimization problem. For polynomially-sized $\mathcal{X}$, the optimization problem (3.1.3) can be feasibly solved for trees with thousands of leaves. Moreover, when the trees g_i are generated under the multispecies coalescent (MSC) model [55], [29], its statistical properties are well-understood. In particular, it is asymptotically consistent whenever $\mathcal{X}_g \subseteq \mathcal{X}$.

The paper [59] showed for the exact version (3.1.2) that $n = \Theta(f^{-2}\log(k))$ independent gene trees g_i are required to recover $\mathcal{T}^*$ with high probability under the MSC model, where f is the minimum branch length in the tree. It has been empirically observed that the relaxed version (3.1.3) requires significantly more data to obtain similar recovery rates [59]. Thus there is a trade-off between computational and sample efficiency.

As discussed in [63], [41], the main shortcoming of the relaxed problem is that it is quite unlikely that $\mathcal{X}(\mathcal{T}) \subseteq \mathcal{X}_g$, since the bipartitions induced by edges near the root are very difficult to recover. Several heuristics have been proposed to extend the set $\mathcal{X}_g$ in an effort to improve its coverage properties and the empirical performance of the relaxed problem [45], [69], [14]. Alternatively [68] proposes a weighted version of (3.1.2) which incorporates quartet-support information from the gene-tree estimation stage, which down-weights possibly misleading quartets.

3.1.3 PCH-ASTRAL: Phylogenetic Inference in the PCH Model

For data generated under the PCH model, quartet support can be defined directly from the shared non-homoplastic states of the observed languages. Let $q = uv||xy$ denote the unrooted quartet topology that groups u with v and x with y.



Given a character c, define the support $M_q^{(c)}$ of the quartet $q = uv||xy$ via

$$M_q^{(c)} := \Big\{(\alpha,\beta) \in \Sigma_N \times \Sigma_N \;:\; \alpha \neq \beta, \quad \alpha \in U \cap V \cap \bar{X} \cap \bar{Y}, \quad \beta \in \bar{U} \cap \bar{V} \cap X \cap Y\Big\},$$

where $U := c(u)$, and so on. Namely, M_q consists of the pairs of non-homoplastic states that are seemingly "split" across the quartet q. For the true quartet topology, [20] shows this support is larger in expectation than the support assigned to the two alternatives:

Lemma 3.1.1. *If $\mathcal{T}_q^* = uv||xy$ is the quartet topology of the leaves (u, v, x, y) in $\mathcal{T}^*$ then*

$$\mathbb{E}M_{uv||xy} > \mathbb{E}M_{ux||vy} = \mathbb{E}M_{uy||vx},$$

so the correct topology has the highest expected support.

For each quartet q, they define its (empirical) support via

$$S_q := \frac{1}{n}\sum_{i=1}^{n} |M_q^{(c_i)}|. \tag{3.1.4}$$

For each $s = \{u, v, x, y\} \in \binom{[k]}{4}$ they then let

$$q_s := \arg\max_{q \in A_s} S_q, \tag{3.1.5}$$

where A_s is the set of three quartets with leaf-set s. Namely, q_s is quartet topology with maximal empirical support. By the law of large numbers and Lemma 3.1.1, the correct topology becomes overwhelmingly likely for large n. They then run ASTRAL on the quartet set $\{q_s\}$ to generate an estimate $\hat{\mathcal{T}}$ of $\mathcal{T}^*$. The authors of [20] call this algorithm *PCH-ASTRAL*, and additionally propose a weighted version, although we will only focus on the unweighted version described above.

Although not directly mentioned in [20], the proof of Lemma 3.1.1 directly implies the following. Recall, a phylogenetic tree (with branch lengths) is said to be *ultrametric* if every leaf is the same distance from the root. When this holds, the supports for the two incorrect topologies are identically distributed.

Corollary 3.1.2. *If $\mathcal{T}^*$ is ultrametric, then $M_{ux|vy} \overset{d}{=} M_{uy|vx}$.*

3.1.4 Summary of Contributions

In Section 3.2, we construct a coupling of the PCH model to a simple immigration-death process which allows for many direct computatins and will be the basis of most of our rigorous analysis. Using this, we develop some basic properties of both the original PCH model and of this coupled process.

Afterwords, in Section 3.3, we show how the search-space for PCH-ASTRAL can be significantly restricted while maintaining statistical consistency. This vastly improves the run-time of the underlying ASTRAL procedure, enabling phylogenetic analysis of hundreds of different languages simultaneously. While the restriction reduces the algorithms sample efficiency, we further introduce two heuristics meant to mitigate this degradation in performance.

Following this, in Section 3.4, we explore an alternative ASTRAL-based inference procedure, which we call *PCH-ASTRAL(sc)*. This procedure keeps character-level quartet information separate by converting each informative character-quartet pair into a single topology vote, rather than averaging support magnitudes across characters. We argue that PCH-ASTRAL(sc) may be more robust in heterogeneous or misspecified settings due to the heavy-tailed nature of the quartet-support distributions.

Lastly, in Section 3.5, we develop rudimentary sample complexity bounds for the two ASTRAL-based procedures when characters are sampled under the simplified immigration-death process $ID(\theta, \lambda, \mu)$. We show PCH-ASTRAL requires $O(L^{-2}\log(k) + L^{-1}\log^2(k))$ characters to recover $\mathcal{T}^*$ with high probability, where

$$L := \min_{s \in \binom{[k]}{4}} \min_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{E}\left[|M_{\mathcal{T}_s^*}| - |M_t|\right]$$

is the minimal margin the correct quartet topology has over incorrect ones. Similarly, PCH-ASTRAL(sc) requires $O(\Delta^{-2}\log(k))$ *informative* characters, where

$$\Delta := \min_{s \in \binom{[k]}{4}} \min_{q \in A_s \setminus \{\mathcal{T}_s^*\}} \left[\mathbb{P}_s(\mathcal{T}_s^*) - \mathbb{P}_s(q)\right], \quad \mathbb{P}_s(q) := \mathbb{P}\left(q = \arg\max_{q' \in A_s} |M_{q'}| \;\middle|\; \max_{q' \in A_s} |M_{q'}| > 0\right).$$

again gives some notion of the margin.

3.1.5 Notation

Let $\mathcal{T}^*$ denote the true species tree underlying our PCH model, which has k leaves (species/languages). Given any phylogenetic tree $\mathcal{T}$, let $\mathcal{Q}_{\mathcal{T}}$ denote its set of *quartets*, which are the subtrees induced by a collection of four leaves. Hence $|\mathcal{Q}_{\mathcal{T}}| = \binom{k}{4}$.

We will always use $q = uv||xy$ to denote a quartet, which inherently comes with some (unrooted) topology, and $s = \{u, v, , x, y\}$ to denote its underlying set of four leaves, which may not have a topology. Given a leaf set $s = \{u, v, x, y\}$, we will use

$$A_s := \{uv||xy, ux||vy, uy||vx\}$$

to denote the set of three different (unrooted) quartets on the same leaf set. We will use $\mathcal{T}_s^*$ to denote the quartet $q \in A_s$ which appears in the true phylogenetic tree $\mathcal{T}^*$.

Unless stated otherwise, assume the observed characters are independent and identically distributed, and assume any micro-states α, β referenced are non-homoplastic.

For ease of notation, when it is clear from context we will denote the macro-state $c(v)$ at a given point $v \in \mathcal{T}^*$ using upper-case letters. In particular, we will use $I := c(i)$, $J := c(j)$, and $K := c(k)$ down below.

Given two points $i, j \in \mathcal{T}^*$, let $[i, j]$ denote the unique path from i to j, and let ℓ_{ij} denote the length of this path.

3.2 Basic Results

3.2.1 Coupling to Immigration-Death Processes

The constraint $\mu_1 = 0$ makes the PCH model hard to analyze as it adds a certain dependence structure between the different micro-states of a specific character. If we drop this condition, we get a classical immigration-death (ID) process, which is quite tractable to analyze. This approximation is reasonable in the parameter regime where the number of characters in a given branch is rarely one, which occurs when λ is sufficiently large relative to μ.

Our key tool will be a coupling of the PCH model to a standard ID process run along the branches of our species tree $\mathcal{T}^*$. It relies on two simple observations about how the set of nonhomoplastic states evolves over time:

- Births always occur at rate at least $\lambda_N := \lambda\nu(\Sigma_N)$ and at most λ.
- Deaths always occur at rate at most μ per individual.

In particular, we can couple an ID process $(Y_s)_{s\in\mathcal{T}^*}$ to our PCH model $(X_s)_{s\in\mathcal{T}^*}$ so

$$Y_s \cap \Sigma_N \subseteq X_s \cap \Sigma_N, \quad \forall s \in \mathcal{T}^*.$$

In this way, we only reduce the number of nonhomoplastic states we observe in the leaves, which heuristically makes inference more challenging.

To produce this coupling, use a single $\tau \sim \text{Exp}(\lambda)$ clock for each branch of $\mathcal{T}^*$ to determine possible births. If τ rings at location $s \in \mathcal{T}^*$, sample a new state $\alpha \notin X_{s^-}$ for the PCH model as usual, namely according to the conditional distribution

$$\frac{\nu(\cdot \cap (\Sigma \setminus X_{s^-}))}{\nu(\Sigma \setminus X_{s^-})}.$$

Let $X_s = X_{s^-} \cup \{\alpha\}$, and update our ID process Y_s as follows: if $\alpha \notin \Sigma_N$, then let $Y_s = Y_{s^-}$. Namely, ignore the homoplastic states. Otherwise, let

$$Y_s = \begin{cases} Y_{s^-} \cup \{\alpha\} & \text{w.p.} \quad \nu(\Sigma \setminus X_{s^-}), \\ Y_{s^-} & \text{w.p.} \quad 1 - \nu(\Sigma \setminus X_{s^-}). \end{cases}$$

This ensures that the birth rate in (Y_s) along a given branch is always λ_N. Similarly, when a new state α is born in X_s, assign it an independent $\tau_\alpha \sim \text{Exp}(\mu)$ clock. If it rings at location $s \in \mathcal{T}^*$, let

$$X_s = \begin{cases} X_{s^-} \setminus \{\alpha\} & X_{s-} \neq \{\alpha\}, \\ \{\alpha\} & \text{otherwise.} \end{cases}$$

Then let $Y_s = Y_{s^-} \setminus \{\alpha\}$. Namely, (X_s) simply ignores the deaths causing the state set to become empty, while (Y_s) does not.

Heuristically, this coupling well-approximates the non-homoplastic states of the PCH model in the high-polymorphism regime where it is rare for there to be just a single state, and when there is little homoplasy so the renormalization of ν does not drammatically affect the birth-rate of non-homoplastic states over time.

In Appendix 3.A, we collect some properties of these immigration-death processes which will be useful for analyzing the PCH model. In particular, the following corollary of Lemma 3.A.3 shows bounding the quartet-supports reduces to bounding the means of some Poisson random variables.

Corollary 3.2.1. *Under the coupled $ID(\theta, \mu, \lambda)$ process, the quartet-supports equal*

$$|M_{uv||xy}| = N_{uv} \cdot N_{xy}, \quad |M_{ux||vy}| = N_{ux} \cdot N_{vy}, \quad |M_{uy||vx}| = N_{uy} \cdot N_{vx},$$

for $\{N_{uv}, ..., N_{vx}\}$ mutually independent Poisson random variables.

3.2.2 Properties of PCH Model

In this section, we derive some basic properties of the PCH model. The derivations often rely on the coupling we introduced in the previous section.

Given a path $[r, \ell]$ from the root r of $\mathcal{T}^*$ to some leaf ℓ, let $S_t = |c(t)|$ be the number of micro-states at location $t \in \mathcal{T}^*$. This is a simple birth-death process with birth rates $\lambda_i \equiv \lambda$ and death rates $\mu_i = i\mu \cdot 1\{i > 1\}$. Hence

Lemma 3.2.2. *The stationary distribution of S_t is $\pi = (Z|Z > 0)$ for $Z \sim \text{Poisson}(\lambda/\mu)$.*

$$\pi_i = \frac{(\lambda/\mu)^i}{i!(e^{\lambda/\mu} - 1)}, \quad \forall i \geqslant 1.$$

The ergodic theorem tells us as $T \to \infty$ the fraction of the time $\{S_t\}_{t \in [0,T]}$ spends in state 1 converges almost surely to $\pi_1 = \frac{\lambda/\mu}{e^{\lambda/\mu}-1}$. Heuristically, over long-branches our effective death-rate then becomes:

$$\mu_{\text{eff}} = \mu\,(1 - \pi_1) = \mu \cdot \frac{e^{\lambda/\mu} - \lambda/\mu - 1}{e^{\lambda/\mu} - 1}$$

If $\lambda \ll \mu$, then $\mu_{\text{eff}} \approx \frac{\lambda}{2}$ which is significantly less than μ in this low-polymorphism regime. If $\lambda \gg \mu$, then π_1 is exponentially small in m, so the process is rarely in state 1. Hence it becomes well-approximated by our ID process coupling.

To bound our quartet supports $|M_q|$, we will need two-sided bounds on the probability a given state α dies along a specific branch, which the next lemma provides.

Lemma 3.2.3 (Survival and Death Bounds)**.** *Let $p_{n,T}$ be the probability that, starting with n micro-states, a given state α dies along a branch of length T in the PCH model. Then*

$$\frac{1}{2}(1 - e^{-\lambda T/2})(1 - e^{-\mu T}) =: \xi_T \leqslant p_{n,T} \leqslant 1 - \eta_T := 1 - e^{-\mu T}$$

Thus ξ_T, η_T give lower bounds on the death and survival rates respectively. We will often be interested in applying these bounds to a specific path in the tree. Given two vertices i, j in the tree denote $\xi_{ij} := \xi_{\ell_{ij}}, \eta_{ij} := \eta_{\ell_{ij}}$ our corresponding lower bounds on death and survival along the path $[i, j]$.

Next, we aim to get some control over the tails of our supports $|M_q|$. For $q = uv||xy$, it is easy to see $M_q \subseteq (U \cap V) \times (X \cap Y)$. By Lemma 3.2.2, we roughly see then that $|M_q|$ is the product of two Poisson random variables, whose correlation depends on how closely related u, v are with x, y. This suggests tails $\mathbb{P}(|M_q| > t) \leqslant \exp(-O(\sqrt{t}\log t))$, namely slightly lighter tails than Weibull(1/2) essentially. The paper [37] develops many useful concentration inequalities for such random variables, which we rely on in Lemma 3.A.4 to prove an analogous statement for the ID process coupling.

Later on, we will need to develop some control over the expected margin between the species tree topology and the other topologies:

$$\Delta_q^{(c)} := \mathbb{E}D_t, \quad D_t := |M_{T_q^*}^{(c)}| - |M_t^{(c)}|, \quad t \in \mathcal{T}_q \setminus \{T_q^*\}$$

which quantifies the gap in Lemma 3.2.4. Let $\rho_N = \mathbb{E}|c(r) \cap \Sigma_N|$ be the expected number of non-homoplastic states present at the root. Note for a given (rooted) quartet, there are two possible topologies, as shown in Figure 3.1. As denoted in the figure, let i, j, k denote the most recent common ancestors of $\{u, v, x, y\}, \{u, v\}$, and $\{x, y\}$ respectively.

(a) Balanced quartet. (b) Caterpillar quartet.

Figure 3.1: The two rooted quartet topologies with split $uv||xy$.

Lemma 3.2.4 (Expected Margin Lower Bound)**.** *If q induces a balanced tree in $\mathcal{T}^*$, then*

$$\Delta_q^{(c)} \geqslant \left(\frac{\lambda_N}{\mu}\right)^2 (1-\eta_{ij})(1-\eta_{ik})\eta_{uv}\eta_{xy}.$$

If q induces a caterpillar tree in $\mathcal{T}^*$, then

$$\Delta_q^{(c)} \geqslant \frac{\lambda_N}{\mu}\left[\rho_N\eta_{ri} + \frac{\lambda_N}{\mu}(1-\eta_{ri})\right](1-\eta_{jk})\xi_{jk}\eta_{uv}\eta_{xy}.$$

More granular bounds can be found in the proof of Lemma 3.2.4 in the Appendix. The above highlights exactly the quartets that are hard to recover: those where u, v and x, y are far from their respective most recent common ancestors j, k, and where these common ancestors j, k are close to their least common ancestor i.

3.3 A Restricted Search-Space

The exact version of ASTRAL (3.1.2) becomes intractable to solve even for trees with a modest number of leaves. Instead, we must construct a set $\mathcal{X}$ of bipartitions which will limit the search space for the true tree $\mathcal{T}$, which is the content of this section.

To this effect, one useful class of sets when it comes to building $\mathcal{X}$ are the leaf sets

$$L_\alpha := \{\text{leaves } \ell : \alpha \in c(\ell)\}.$$

Say a subset $L \subseteq S$ of leaves is *contained* within a given bipartition $\varphi := (A|B)$ if

$$L \subseteq A \qquad \text{or} \qquad L \subseteq B.$$

If $\alpha \in \Sigma_N$ is born on an edge e inducing bipartition $\varphi_e = (A_e|B_e)$, then L_α is contained in φ_e. Moreover, if no deaths of α occur after its born then in fact $L_\alpha \in \{A_e, B_e\}$. Hence one choice for the candidate bipartitions is

$$\mathcal{X}_1 := \{L_\alpha^{(c)} : c \in C, \alpha \in \Sigma^{(c)}\}.$$

At stationarity, we typically have around $\lambda\mu^{-1}$ states in each leaf, so roughly we can see $|\mathcal{X}_1| \lesssim k|C|\lambda\mu^{-1}$ grows only linearly in k. Under mild assumptions this alone is sufficient for ensuring statistical consistency, since with positive probability a given non-homoplastic α is born along an edge e and does not die at all. More granullary

Theorem 3.3.1 (Bipartition Cover Bounds for PCH With No Borrowing)**.** *Let $\mathcal{B}$ be the set of bipartitions of a phylogenetic tree $\mathcal{T}^*$ with k leaves and minimum/maximum branch lengths f_{min}, f_{max} respectively. Suppose further we have a PCH model $\left\{\Sigma^{(c)}, \lambda^{(c)}, \mu^{(c)}, \rho^{(c)}\right\}_{c\in C}$ on $\mathcal{T}^*$ with no borrowing and whose rates follow an immigration-death process. Let*

$$C_N := \{c \in C \ : \ \nu^{(c)}(\Sigma_N^{(c)}) > 0\}$$

be the set of characters with positive chance of non-homoplasy. Furthermore, assume we have the following assumptions on the parameters of the PCH model:

(i) Nonvanishing Birth Rates: Suppose that $\lambda^{(c)}\nu^{(c)}(\Sigma_N^{(c)}) \geqslant b > 0$ for all $c \in C_N$.

(ii) Bounded death rates: Suppose $\mu^{(c)} \leqslant D < \infty$ for all $c \in C_N$.

Then we have exponential convergence of the coverage probability:

$$\mathbb{P}(\mathcal{B} \nsubseteq \mathcal{X}_1) \leqslant \sum_{\ell=2}^{k-2} q_\ell^{|C_N|} \leqslant (k-3)q_{k-2}^{|C_N|} \qquad q_\ell := \exp\left(-bf_{min}\exp\left[-2\ell D f_{max}\right]\right)$$

In particular, we have asymptotic consistency when $|C_N| \to \infty$:

$$\lim_{|C_N|\to\infty} \mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) = 1,$$

In particular, this implies if we run the ASTRAL as proposed in [20], restricted to the bipartition set $\mathcal{X}_1$, this is a statistically consistent estimate of the true underlying phylogenetic tree T, and that we require $\exp(O(k))$ characters to achieve a bipartition cover with high probability. This is essentially the same rate bipartition-covers achieve under the standard multispecies-coalescent model to which ASTRAL is typically applied [63], [41].

This result should be viewed as a baseline consistency guarantee rather than a practical sample-complexity result. It is clear that $\mathcal{X}_1$ is bad at generating large bipartitions. Namely, if $\varphi_e = (A|B)$ the probability a character born in edge e has no deaths is exponentially small in $|A|$. To rectify this, we propose a few heuristics for expanding the size of $\mathcal{X}_1$ so it has better coverage probabilities. A natural solution is to allow multiple states α to contribute:

$$\mathcal{X}_m := \left\{ \bigcup_{i=1}^{r} L_{\alpha_i}^{(c)} \ : \ c \in C, \alpha_i \in \Sigma^{(c)}, r \leqslant m \right\}$$

It is easy to see that $\mathcal{X}_n \subseteq \mathcal{X}_m$ for $n \leqslant m$, so in particular $\mathcal{X}_1 \subseteq \mathcal{X}_m$. Hence asymptotic consistency using $\mathcal{X}_m$ follows from that of $\mathcal{X}_1$. To avoid combining leaf sets from completely different regions in the tree, we can also introduce some sort of compatibility condition like $L_{\alpha_i}^{(c)} \cap L_{\alpha_j}^{(c)} \neq \emptyset$.

Alternatively, we can look for sets A so that L_α takes up "most" of A. Formally, defining some threshold τ we let

$$\mathcal{X}_\tau := \left\{ (A|S-A) \ : \ \exists \alpha \ \left(\frac{|A \setminus L_\alpha|}{|A|} < \tau \right) \right\}$$

Since $|A| < k$, this allows at most τk additions. Hence if we choose $\tau = r/k$ for $r < k$, we limit ourselves to at most r additions.

3.4 Alternative ASTRAL Procedure

Another inference method, closer in spirit to the original ASTRAL framework, is to let each character independently vote on the topology of each quartet. These character-level quartet votes then play the role of the quartet information induced by individual gene trees in the traditional MSC setting.

More formally, for each character c_i and each leaf set $s = \{u, v, x, y\}$, define

$$\tau_s^{(c_i)} := \arg\max_{q \in \mathcal{A}_s} |M_q^{(c_i)}|, \quad I_s^{(c_i)} := \left\{ \max_{q \in \mathcal{A}_s} |M_q^{(c_i)}| > 0 \right\},$$

with ties broken uniformly. We then solve (3.1.2) with quartet supports

$$W(q) = \sum_{i=1}^{n} 1\{\tau_s^{(c_i)} = q\} \cdot 1\{I_s^{(c_i)}\}.$$

The indicator $1\{I_s^{(c_i)}\}$ essentially just tells us to ignore characters which provide no information about the given quartet. We call this procedure *PCH-ASTRAL(sc)* for "separate characters". This inference scheme is only really reasonable if for each quartet, the correct topology is the most likely one. Namely

$$\mathbb{P}(\tau_s^{(c)} = \mathcal{T}_s^*) > \mathbb{P}(\tau_s^{(c)} = q) \qquad q \neq \mathcal{T}_s^*. \tag{3.4.1}$$

If property (3.4.1) holds for every quartet $\mathcal{T}_s^* \in \mathcal{Q}(\mathcal{T}^*)$, we say the PCH model is in the *anomaly-free zone*. When this occurs, the above ASTRAL-based procedure is asymptotically consistent and has sample-complexity essentially quadratic in the margin

Theorem 3.4.1 (Sample Complexity PCH-ASTRAL(sc))**.** *Suppose we have a collection of n iid characters $\{c_i\}$ in the anomaly-free zone. If*

$$n > \frac{9}{p_{min}\Delta^2} \log\left(\frac{4\binom{k}{4}}{\varepsilon}\right),$$

where

$$\Delta := \min_{s \in \binom{[k]}{4}} \min_{t \neq \mathcal{T}_s^*} \left[\mathbb{P}(\tau_s^{(c)} = \mathcal{T}_s^* \mid I_s) - \mathbb{P}(\tau_s^{(c)} = t \mid I_s)\right], \quad p_{min} := \min_{s \in \binom{[k]}{4}} \mathbb{P}(I_s),$$

then the above ASTRAL-based procedure recovers $\mathcal{T}^*$ with probability at least $1 - \epsilon$.

The proof is analogous to the proof of Theorem 2.1 in [59], which union bounds over quartets and utilizes simple concentration bounds for the multinomial distribution. In Appendix 3.A.1, we briefly discuss the anomaly-free zone.

3.5 Sample-Complexity Bounds

3.5.1 PCH-ASTRAL Sample-Complexity Bounds

In this section, we aim to understand the number of independent characters required by PCH-ASTRAL to recover the true species tree $\mathcal{T}^*$ with high probability. To attempt to do so, we prove a sample complexity result for the PCH-ASTRAL procedure when run on samples generated from an immigration-death process $ID(\theta, \mu, \lambda)$ run on the true species tree $\mathcal{T}^*$. It is a simple consequence of our concentration bound Lemma 3.A.4.

Lemma 3.5.1 (Sample Complexity PCH-ASTRAL under ID Process)**.** *Suppose we have a collection $\{c_i\}_{i=1}^n$ of iid characters whose multi-states are distributed according to the $ID(\theta, \mu, \lambda)$ process we discussed above ran on the tree $\mathcal{T}^*$. Let $\zeta = \theta \vee \frac{\lambda}{\mu}$ and*

$$\kappa := \max_{s \in \binom{[k]}{4}} \max_{uv \| xy \in A_s} \left(1 + \zeta\eta_{uv}\right)\left(1 + \zeta\eta_{xy}\right),$$

and define our worst-case margin

$$L := \min_{s \in \binom{[k]}{4}} \min_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{E}\left[|M_{\mathcal{T}_s^*}| - |M_t|\right].$$

Then if we have at least:

$$n \gtrsim \max\left\{\frac{\kappa^2}{L^2} t_\delta,\ \frac{\kappa}{L} t_\delta^2\right\}, \quad t_\delta := \log\left(\frac{4\binom{k}{4}}{\delta}\right),$$

characters, we recover the true species tree with probability at least $1 - \delta$.

As discussed in the proof, the parameter κ in a sense measures how heavy-tailed the support sizes $|M_{uv||xy}|$ are. It has the trivial upper bound $\kappa \leqslant (1+\zeta)^2$, which is a function merely of the expected number of states in a leaf of the $ID(\theta, \lambda, \mu)$ process. Hence the worst-case margin L is the main parameter of interest. Analogous to Lemma 3.2.4, under $ID(\theta, \mu, \lambda)$ we can lower bound L purely in terms of the underlying model parameters.

The worst-case margin L certainly depends on the topology of $\mathcal{T}^*$, although we can roughly analyze Lemma 3.2.4 to generate a heuristic topology-free bound. If f is the minimal branch length in $\mathcal{T}^*$ and

$$d := \max_{uv||xy \in \mathcal{Q}(\mathcal{T}^*)} \ell_{uv} \vee \ell_{xy},$$

is the maximal distance between any two leaves on the same side of a quartet, then the margin bounds in Lemma 3.2.4 roughly become

$$\Delta_q \geqslant \left(\frac{\lambda}{\mu}\right)^2 (1 - e^{-\mu f})^2 (e^{-\mu d})^2 \gtrsim \lambda^2 f^2 e^{-2\mu d},$$

when the induced subtree of q in $\mathcal{T}^*$ is balanced and similarly

$$\Delta_q \gtrsim \lambda^2 \mu f^3 e^{-2\mu d} \left(\rho_N \wedge \frac{\lambda}{\mu}\right),$$

when the induced subtree is a caterpillar. Thus the worst-case margins are dictated by its caterpillar trees, and for a tree of a fixed height the bound scales roughly like

$$n \gtrsim \max\left\{f^{-6}\log(k),\ f^{-3}\log^2(k)\right\},$$

significantly worse than the $\Theta(f^{-2}\log(k))$ of ASTRAL under the multi-species coalescent model [59]. Of course, our bound is naturally quite loose, and future improvements may significantly improve the rate.

3.6 Discussion and Concluding Remarks

In this chapter, we developed several potential candidates for bipartition covers which substantially reduce the search space of the original PCH-ASTRAL inference procedure. This modification plausibly expands the scope of PCH-ASTRAL from a few dozen languages to hundreds of languages. Although restricting the search space reduces sample efficiency, we proposed several heuristics designed to mitigate this loss in performance.

Beyond this, we derived preliminary sample-complexity bounds under the immigration-death process model $ID(\theta, \lambda, \mu)$, both for the original PCH-ASTRAL procedure and for the robustified procedure PCH-ASTRAL(sc). In both cases, the rates depend explicitly on a quartet-level margin measuring the advantage of the correct quartet topology over the incorrect ones. We further gave partial bounds relating these margins to the underlying model parameters.

There are several natural directions for future work. First, one would like to extend the sample-complexity results from the simplified $ID(\theta, \lambda, \mu)$ process to the full PCH model. The main obstacle is concentration. The immigration-death coupling provides a convenient lower bound on non-homoplastic state counts, which is enough for expected-margin estimates, but the concentration argument also requires an upper bound on macro-state sizes at the leaves. Establishing such an upper bound for the full PCH model would therefore be an important first step toward extending Lemma 3.5.1 beyond the ID approximation.

Second, since the true tree $\mathcal{T}^*$ is unknown at inference time, it would be useful to develop topology-free sample-complexity bounds depending only on simple geometric parameters of $\mathcal{T}^*$, such as the minimum branch length, analogous to the results of [59]. Under the ID approximation, the Poisson means $m_{uv} = \mathbb{E}N_{uv}$ are explicit functions of the model parameters, so more careful bookkeeping in Lemma 3.5.1 may lead to relatively explicit bounds.

Third, the anomaly-free condition for PCH-ASTRAL(sc) remains somewhat opaque. Under the ID approximation, the condition reduces to a comparison of independent product-Poisson random variables, and is therefore numerically checkable from the quartet-level means. However, we do not currently have simple geometric sufficient conditions ensuring that the anomaly-free property holds.

Finally, several refinements of the present analysis should be possible. The margin bounds used here depend on coarse survival and death probability estimates from Lemma 3.2.3; sharper estimates, for example using Laplace transforms of hitting times, may yield more granular constants.

Lastly, extensive empirical simulations would also be valuable, both to compare PCH-ASTRAL and PCH-ASTRAL(sc) under various model regimes and to assess how well the theoretical margins predict finite-sample behavior. Moreover, it would be interesting to study how well our proposed bipartition heuristics improve coverage properties.

3.A Immigration-Death Process Proofs

Let $\mathrm{ID}(\theta, \mu, \lambda)$ denote an immigration-death process Y_t with $|Y_0| \sim \mathrm{Poisson}(\theta)$, birth rate λ, and death rate μ per individual. We start by understanding the population size in such a process.

Lemma 3.A.1 (Population Sizes in ID Processes)**.** *If* $(Y_t)_{t\geqslant 0} \sim \mathrm{ID}(\theta, \mu, \lambda)$ *then:*

$$|Y_t| \sim \mathrm{Poisson}(m_\theta(t)), \quad m_\theta(t) = \theta e^{-\mu t} + \frac{\lambda}{\mu}(1 - e^{-\mu t}).$$

In particular, its stationary distribution is $\mathrm{Poisson}(\lambda/\mu)$*. Moreover for* $s < t$*:*

$$\mathbb{E}|Y_t \cap Y_s| = m_\theta(s) \cdot e^{-\mu(t-s)}$$

Proof. Kolmogorov's forward equations tell us:

$$\frac{d}{dt} p_k = \lambda p_{k-1} + \mu(k+1)p_{k+1} - (k\mu + \lambda)p_k, \quad p_k(0) = \frac{e^{-\theta}\theta^k}{k!}$$

It is easy to see the Poisson probabilities $p_k(t) = \frac{e^{-m(t)}m(t)^k}{k!}$ satisfy these equations, and hence $X(t) \sim \mathrm{Poisson}(m_\theta(t))$. The covariance formula follows from the fact:

$$|Y_t| = \mathrm{Binomial}(X_s, e^{-\mu(t-s)}) + |Z|, \quad |Z| \sim \mathrm{Poisson}(m_0(t-s))$$

where these two terms are independent. The formula follows from the law of total covariance. □

It will be important to know how much time the PCH model spends with just a single state. For this purpose, it will be important to know how long it takes to go down from n states to one. Since the dynamics of the ID process and PCH model essentially agree when there are more than one state, this reduces to studying the hitting times of our ID process.

Lemma 3.A.2 (Hitting Times in ID Process)**.** *Let:*

$$\tau_n := \inf\{t > 0 : |Y_t| = 1\}$$

conditional on $|Y_0| = n$*. Then:*

$$\mathbb{E}\tau_2 = \frac{\mu}{\lambda^2}(e^{\lambda/\mu} - 1 - \lambda/\mu)$$

Moreover, its Laplace transform satisfies:

$$\frac{2\mu}{\lambda + 2\mu + s} \leqslant \mathbb{E}e^{-s\tau_2} \leqslant \frac{\lambda + 2\mu}{\lambda + 2\mu + s}$$

Proof. Let $w_n := \mathbb{E}\tau_n$. Note by conditioning on the time of the first event we see:

$$w_n = \frac{1}{\lambda + n\mu}\left[1 + \lambda w_{n+1} + n\mu w_{n-1}\right], \quad w_1 = 0$$

Next control the Laplace transform $\varphi_n(s) := \mathbb{E}e^{-s\tau_n}$. We will rely on the classical relation between the Laplace transform of birth-death processes and continued fractions. Conditioning on the first event time (either a birth or a death) we see:

$$\varphi_n = \frac{\lambda}{\lambda + n\mu} \cdot \varphi_{n+1} + \frac{n\mu}{\lambda + n\mu} \cdot \varphi_{n-1}, \qquad \varphi_1 \equiv 1$$

Noting that $\varphi_3 \in [0,1]$ gives the desired bound on φ_2. More generally, this gives an upwards recurrence for the ratios $r_n := \frac{\varphi_n}{\varphi_{n-1}}$ where $r_2 := \varphi_2$:

$$r_n = \frac{n\mu}{\lambda + n\mu + s - \lambda r_{n+1}}$$

Iterating gives the continued fraction expansion:

$$r_n = \frac{n\mu}{\lambda + n\mu + s - \frac{\lambda(n+1)\mu}{\lambda+(n+1)\mu+s-\ldots}}$$

We can truncate this at even or odd convergents to get lower and upper bounds respectively on r_n. From these, we can bound $\varphi_n(s) = \prod_{k=2}^{n} r_k$. □

Note that the above implies in the low-polymorphism regime $\lambda \ll \mu$, we get $\mathbb{E}\tau_2 = \frac{1}{2\mu} + O(\lambda/\mu^2)$.

Proof of Lemma 3.2.2. This process is reversible, so by detailed balanced:

$$\pi_{i+1}(i+1)\mu = \lambda\pi_i, \quad \forall i \geqslant 1$$

Recursively applying this shows $\pi_i \propto \frac{(\lambda/\mu)^i}{i!}$ as desired. □

Proof of Lemma 3.2.3. Suppose along the given branch we spent a fraction $\tau \in [0,1]$ of the time with at least two states, whose distribution implicitly depends on n, T. Then if α is a state present at time zero:

$$\mathbb{P}(\alpha \text{ survives along branch} \mid \tau) = e^{-\mu T \cdot \tau}$$

Hence the survival probability depends only on the Laplace transform φ of τ:

$$\mathbb{P}(\alpha \text{ survives along branch}) = \varphi(\mu T)$$

Trivially $\tau \leqslant 1$, which gives the upper bound. For the lower bound, note that $p_{n,T}$ is increasing in n by a simple coupling argument. Hence WLOG we assume $n = 1$. One way for α to die in this setting is for another state β to be born at time $\gamma \sim \mathrm{Exp}(\lambda)$. The first of these two states dies by time T with probability $1 - e^{-2\mu(T-\gamma)}$ and the death is equally likely to be α or β. Hence:

$$p_{n,T} \geqslant \frac{1}{2} \cdot \int_0^T (1 - e^{-2\mu(T-\gamma)})\lambda e^{-\lambda\gamma} d\gamma = \frac{1}{2}\left[(1 - e^{-\lambda T}) - \frac{\lambda(e^{-\lambda T} - e^{-2\mu T})}{2\mu - \lambda}\right]$$

An even simpler bound is:

$$p_{n,T} \geqslant \frac{1}{2}(1 - e^{-\lambda T/2})(1 - e^{-\mu T})$$

which is the probability we have a birth of some state β by time $T/2$, a death of either α, β in the remaining time $T/2$, and that this death is α. □

The following shows that our quartet supports $|M_{uv||xy}|$ take a particularly simple form under the ID process coupling.

Lemma 3.A.3 (Quartet Supports under ID Process)**.** *Suppose $Y_t \sim ID(\theta, \lambda, \mu)$ on the underlying tree $\mathcal{T}^*$. For any subset $\mathcal{L}$ of the leaves of $\mathcal{T}^*$, for each $A \subseteq \mathcal{L}$ define*

$$N_A^{\mathcal{L}} := \left| \left(\bigcap_{\ell \in A} \{\alpha \in Y_\ell\} \right) \cap \left(\bigcap_{\ell \notin A} \{\alpha \notin Y_\ell\} \right) \right|$$

the number of states α which appear exactly in the set of leaves A, among those in $\mathcal{L}$. Then $\{N_A^{\mathcal{L}}\}_{A \subseteq \mathcal{L}}$ are mutually independent Poisson random variables. In particular, for $s = \{u, v, x, y\}$ this implies

$$|M_{uv||xy}| = N_{uv}^{(s)} \cdot N_{xy}^{(s)},$$

is the product of two independent Poisson random variables.

This Lemma makes analyzing the expected margin

$$\Delta_t := \mathbb{E}|M_{\mathcal{T}_s^*}| - \mathbb{E}|M_t|, \quad t \in A_s \setminus \{\mathcal{T}_s^*\},$$

particularly easy under the $ID(\theta, \lambda, \mu)$ process: we just must try to understand the means $m_{uv}^{(s)}$ of these Poisson random variables $N_{uv}^{(s)}$. Calculating these exactly can be challenging, but Lemma 3.A.1 gives a simple upper bound

$$m_{uv}^{(s)} \leqslant \left(\theta \vee \frac{\lambda}{\mu}\right) \eta_{uv}.$$

The first term accounts for the number of states reaching the most recent common ancestor of u, v, and the second term thins these to only those reaching both u, v.

Proof of Lemma 3.A.3. Fix a point $x \in \mathcal{T}^*$. For any states α born at x, we see that $\alpha \in N_A^{\mathcal{L}}$ for some $A \subseteq \mathcal{L}$. For fixed tree topology $\mathcal{T}^*$ and model parameters θ, λ, μ, the probability $p_x(\alpha \in N_A^{\mathcal{L}})$ it lands in a particular one of these sets depends only on the location x, and is independent of the birth-process itself and any other states.

The number of states born along the tree form a Poisson point process with intensity

$$\nu(dx) := \theta\delta_r(dz) + \lambda dz,$$

where r is the root and dz denotes length measure on the branches of $\mathcal{T}^*$. Kingman's marking theorem then implies that the marked process forms a Poisson point process on $\mathcal{T}^* \times \{A : A \subseteq \mathcal{L}\}$ with intensity $\nu(dz)p_x(dA)$. See, for example, the chapter on marked Poisson processes in [36]. The desired result then follows immediately. ☐

Lemma 3.A.4 (Tail Bounds for Quartet Supports). *Let $\zeta = \theta \vee \frac{\lambda}{\mu}$. Under $ID(\theta, \mu, \lambda)$*

$$\mathbb{P}(|M_{uv||xy}| > t) \leqslant 2\exp\left(-\zeta\min\{\eta_{uv}, \eta_{xy}\} \cdot h\left(\zeta^{-1}\sqrt{\frac{t}{\eta_{uv}\eta_{xy}}}\right)\right)$$

where $h(t) := (t\log(t) - t + 1)1\{t \geqslant 1\}$. Moreover, $M_{uv||xy}$ is sub-Weibull$(1/2)$ with Orcliz pseudo-norm associated to $\Phi_\alpha(x) = \exp(x^\alpha) - 1$ given by

$$||M_{uv||xy}||_{\Psi_{1/2}} \lesssim (1 + \zeta \cdot \eta_{uv})\,(1 + \zeta \cdot \eta_{xy})\,.$$

Proof of Lemma 3.A.4 (quartet-support tail bound). Note that:

$$M_{uv||xy} \subseteq \{(\alpha, \beta) : \alpha \in U \cap V, \beta \in X \cap Y\} = (U \cap V) \times (X \cap Y)$$

Suppose that w, z are the most recent common ancestors of u, v and x, y respectively. Lemma 3.A.1 implies that $|Z|, |W| \leqslant_{st} \text{Poisson}(\zeta)$. Then Poisson-thinning implies

$$|U \cap V| \leqslant_{st} \text{Poisson}\,(m_{uv})\,, \quad m_{uv} := \zeta\eta_{uv},$$

and analogously for $|X \cap Y|$. Chernoff's implies:

$$\mathbb{P}(\text{Poisson}(m) \geqslant mt) \leqslant \exp(-mh(t)),$$

By union bound, for any $\gamma > 0$:

$$\mathbb{P}(|M_{uv||xy}| > t) \leqslant \mathbb{P}(|U \cap V| \geqslant \gamma\sqrt{t}) + \mathbb{P}(|X \cap Y| \geqslant \gamma^{-1}\sqrt{t}).$$

Setting $\gamma = \sqrt{m_{uv}/m_{xy}}$ weights the two terms according to their means, giving:

$$\leqslant \exp\left(-m_{uv}h\left(\sqrt{\frac{t}{m_{uv}m_{xy}}}\right)\right) + \exp\left(-m_{xy}h\left(\sqrt{\frac{t}{m_{uv}m_{xy}}}\right)\right)$$

from which the desired bound directly follows.

To upper bound the Orcliz norm, we rely on the Hölder-type inequality appearing as equation (3.5) in [37]. This gives

$$||M_q||_{\Psi_{1/2}} \leqslant ||\text{Poisson}(m_{uv})||_{\Psi_1} \cdot ||\text{Poisson}(m_{xy})||_{\Psi_1}.$$

A simple calculation and the bound $\log(1+x) \geqslant \frac{x}{1+x}$ for $x \geqslant 0$ shows

$$||\text{Poisson}(m)||_{\Psi_1} = \frac{1}{\log(1+\log(2)/m)} \leqslant 1 + \frac{m}{\log(2)}$$

from which the result follows. □

3.A.1 The Anomaly-Free Zone

In this section, we briefly discuss the anomaly-free property (3.4.1). Lemma 3.1.1 shows $\mathbb{E}|M_t|$ is maximized over $q \in A_s$ when $q = \mathcal{T}_s^*$, but this does not imply $\mathcal{T}_s^*$ maximizes

$$\mathbb{P}\left(q = \arg\max_{t' \in A_s} |M_{t'}|\right),$$

as it is possible $|M_{\mathcal{T}_s^*}|$ is zero most of the time, but takes on large values rarely. To avoid this, we must show the supports $|M_{\mathcal{T}_q^*}|$ are reasonably concentrated. We focus on the simpler $ID(\theta, \lambda, \mu)$ setting.

Lemma 3.A.3 shows in this setting that the supports take the particular form

$$|M_{uv||xy}| = N_{uv}N_{xy}, \quad |M_{ux||vy}| = N_{ux}N_{vy}, \quad |M_{uy||vx}| = N_{uy}N_{vx},$$

where these six random variables $\{N_{uv}, ..., N_{vx}\}$ are mutually independent Poissons whose means $m_{ab} := \mathbb{E}N_{ab}$ are reasonably simple to compute. The main difficultly is that these means depend on the underlying parameters $(\mathcal{T}^*, \lambda, \mu, \theta)$ in complicated, inter-woven ways. For example, we know from Lemma 3.A.1 that $|I| \sim \text{Poisson}(m_\theta(\ell_{ri}))$. Hence if the true quartet topology is $q = uv||xy$ and it induces a balanced subtree in $\mathcal{T}^*$, then

$$m_{uv} = \eta_{uv}\left[m_0(\ell_{ij}) + m_\theta(\ell_{ri})\eta_{ij}(1 - \eta_{ik}(\eta_{ky} + \eta_{kx} - \eta_{kx}\eta_{ky}))\right],$$

$$m_{ux} = m_\theta(\ell_{ri})\eta_{ux}\left[(1 - \eta_{jv}) \cdot (1 - \eta_{ky})\right],$$

Even with these means known, these Poisson-products depend on these means in complicated ways. However, in the low-polymorphism regime $\lambda \ll \mu$, where these means are all close to zero, the latter issue simplifies drastically. In this regime, the supports are heavily concentrated on $\{0, 1\}$, and the maximum support is typically the unique nonzero one.

Lemma 3.A.5 (Low Polymorphism Implies Anomaly-Free Zone)**.** *For an $ID(\epsilon\theta, \epsilon\lambda, \mu)$ process on $\mathcal{T}^*$, let $s = \{u, v, x, y\}$ be any set of four leaves. If*

$$\omega_*^{(\epsilon)} = \mathbb{E}|M_{\mathcal{T}_s^*}^{(\epsilon)}|, \quad \{\omega_1^{(\epsilon)}, \omega_2^{(\epsilon)}\} = \{|M_{q'}^{(\epsilon)}| \ : \ q \in A_s \setminus \{\mathcal{T}_s^*\}\},$$

are the mean supports of the correct and two incorrect topologies respectively, then as $\epsilon \downarrow 0$

$$\mathbb{P}\left(q = \arg\max_{q' \in A_s} |M_{q'}^{(\epsilon)}| \,\middle|\, \max_{q' \in A_s} |M_{q'}^{(\epsilon)}| > 0\right) = \frac{\omega_q^{(1)}}{\omega_*^{(1)} + \omega_1^{(1)} + \omega_2^{(1)}} + O(\epsilon).$$

In particular, for small enough $\epsilon > 0$ the anomaly-free property (3.4.1) *holds.*

Proof. Let $N_{uv}^{(\epsilon)}, ..., N_{vx}^{(\epsilon)}$ be the corresponding Poisson random variables from Lemma 3.A.3, and $m_{uv}^{(\epsilon)}, ..., m_{vx}^{(\epsilon)}$ their means. Observe $m_{uv}^{(\epsilon)} = \epsilon m_{uv}^{(1)}$, and similarly for the other means, by a simple Poisson-thinning argument. Letting $\omega_q := \omega_q^{(1)}$ and $m_{uv} := m_{uv}^{(1)}$, we see

$$p_{uv||xy} := \mathbb{P}\left(|M_{uv||xy}^{(\epsilon)}| > 0\right) = (1 - e^{-\epsilon m_{uv}})(1 - e^{-\epsilon m_{xy}}) = \epsilon^2 \omega_{uv||xy} + O(\epsilon^3).$$

and likewise for the other supports. More specifically, using $a \geqslant 1 - e^{-a} \geqslant a - a^2/2$ for $a \geqslant 0$ and the fact that $m_{uv}^{(1)} \leqslant \zeta := \theta \vee \lambda\mu^{-1}$, we see

$$-\epsilon^3 \zeta \omega_q \leqslant p_q - \epsilon^2 \omega_q \leqslant 0.$$

Since the supports are independent by Lemma 3.A.3, we see

$$-\sum_{q' < q} p_q p_{q'} \leqslant \mathbb{P}(\max |M_q^{(\epsilon)}| > 0) - \sum_{q \in A_s} p_q \leqslant 0.$$

For $W := \omega_* + \omega_1 + \omega_2$, the quadratic term is bound by

$$\sum_{q' < q} p_q p_{q'} \leqslant \frac{\epsilon^4 W^2}{2}.$$

Letting

$$\tau^{(\epsilon)} = \arg\max_{q \in A_s} |M_q^{(\epsilon)}|, \quad I_s^{(\epsilon)} = \left\{\max_{q \in A_s} |M_q^{(\epsilon)}| > 0\right\}.$$

our above analysis shows

$$\begin{aligned}
\mathbb{P}\left(\tau^{(\epsilon)} = q \mid I_s^{(\epsilon)}\right) &= \frac{\epsilon^2 \omega_q + O(\epsilon^3)}{1 - \prod_{q' \in A_s}(1 - \epsilon^2 \omega_{q'} + O(\epsilon^3))} \\
&= \frac{\omega_q}{\omega_* + \omega_1 + \omega_2} + O(\epsilon),
\end{aligned}$$

as desired. □

Hence, the anomaly-free property holds in this low-polymorphism regime, at least under the simpler $ID(\theta, \lambda, \mu)$ model. Of course, this is also the regime where the immigration-death process approximation is the least accurate, so it is not clear how this result translates to the full PCH model. However, we see this result as evidence that the anomaly-free property is not a particularly restrictive condition, and is likely to hold in many practical settings.

3.B Additional Proofs

Proof of Theorem 3.3.1 (Bipartition Cover Bound). Fix an edge $e \in E(\mathcal{T})$ with k_e descendants and a given character c. For ease of notation, we drop all the (c) superscripts for now. If φ_e is the corresponding bipartition, we will lower bound the probability that $\varphi_e = (L_\alpha | S - L_\alpha)$ for some $\alpha \in \Sigma$.

First, note that as the birth-rate is constant the number N_e of non-homoplastic states born along the edge e satisfies:

$$N_e \sim \text{Poisson}(\lambda T_e \cdot \nu(\Sigma_N)) \succeq_{st} Poisson(\eta) \qquad \eta := \lambda T_{min} \nu(\Sigma_N)$$

where T_e is the length of the branch e and T_{min} the minimum branch length in the tree.

For a given non-homoplastic α born along the edge e, we can lower bound the probability it has no deaths below e. By Lemma 3.2.3 the probability α survives along a branch of length T is at least $\exp(-\mu T)$. There are $2k_e - 1 \leqslant 2k_e$ edges (including e) that α can die on. Each is at most length T_{max}, so we can lower bound the probability α has no deaths at all by:

$$\mathbb{P}(\alpha \text{ has no deaths } | \alpha \in \Sigma_N, \alpha \text{ born on e}) \geqslant \exp(-\tau_{k_e}) \qquad \tau_{k_e} := 2\mu k_e T_{max}$$

Using this result followed by $N_e \succeq_{st} \text{Poisson}(\eta)$ we see:

$$\mathbb{P}(\exists \alpha \; \varphi_e = (L_\alpha, S - L_\alpha)) \geqslant \mathbb{E}_{n \sim N_e} \left(1 - (1 - \exp(-\tau))^n\right) \geqslant 1 - \exp(-\eta e^{-\tau})$$

From here, we use the caterpillar bound from [41] applied to our $|C|$ characters implies:

$$\mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) \geqslant 1 - \sum_{\ell=2}^{k-2} \prod_{c \in C} \exp\left(-\eta^{(c)} e^{-\tau_\ell^{(c)}}\right) = 1 - \sum_{\ell=2}^{k-2} \exp\left(-\sum_{c \in C} \eta^{(c)} e^{-\tau_\ell^{(c)}}\right).$$

Note none of the above uses our boundedness assumptions on the birth or death rates $\lambda^{(c)}, \mu^{(c)}$. Using these assumptions on the birth and death rates, we see:

$$\mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) \geqslant 1 - \sum_{\ell=2}^{k-2} q_\ell^{|C_N|} \geqslant 1 - (k-3) q_{k-2}^{|C_N|}, \qquad q_\ell := \exp\left(-b T_{min} \exp\left[-2\ell D T_{max}\right]\right).$$

Hence we get the desired exponential convergence of the converge probability. $\square$

Proof of Theorem 3.4.1 (Sample Complexity PCH-ASTRAL(sc)). Note for a specific leaf set s, we have $n_s \sim \text{Binomial}(n, p_s)$ informative characters (characters where I_s occurs). Applying Chernoff's bound

$$\mathbb{P}\left(n_s < \frac{np_s}{2}\right) \leqslant \exp\left(-\frac{np_s}{8}\right) \leqslant \exp\left(-\frac{np_m}{8}\right).$$

Union bounding over all $\binom{k}{4}$ quartets gives

$$\mathbb{P}\left(\min_{s \in \binom{[k]}{4}} n_s < \frac{np_m}{2}\right) \leqslant \binom{k}{4} \exp\left(-\frac{np_m}{8}\right).$$

Thus, provided

$$np_m \geqslant 8 \log \left(\frac{\binom{k}{4}}{\varepsilon} \right),$$

we have, with probability at least $1-\varepsilon$, at least $n' = np_m/2$ informative characters for every quartet.

From here, we mimic the proof of Theorem 2.1 in [59]. First, if for every quartet $q \in \mathcal{Q}$ the modal topology among $\{\tau_q^{(c_i)}\}_{i=1}^{n'}$ is the true topology $\mathcal{T}_q^*$, then exact ASTRAL returns the true tree $\mathcal{T}^*$. Hence it suffices to union bound over quartets. Let E_q denote the event that the true topology is not modal for quartet q.

Fix a quartet q, and let t_1, t_2 denote the two incorrect topologies. Set

$$p_q := \mathbb{P}(\tau_q^{(c)} = \mathcal{T}_q^*), \qquad r_{t_i} := \mathbb{P}(\tau_q^{(c)} = t_i),$$

By the anomaly-free property, $p_q > r_{t_i}$. Let

$$\Delta_q := p_q - r_q, \qquad \Delta := \min_{q \in \mathcal{Q}} \Delta_q > 0.$$

For this fixed quartet, let

$$(N_*, N_1, N_2) \sim \mathrm{Mult}(n'; p_q, r_q, r_q)$$

count the number of characters assigning the true topology and the two incorrect topologies.

Fix any $w < \Delta_q/3$, and define

$$A_q := \left\{ \frac{N_*}{n'} \in (p_q - w, p_q + w) \right\},$$

and

$$B_{q,j} := \left\{ \frac{N_j}{n'} \in (r_q - w, r_q + w) \right\}, \qquad j = 1, 2.$$

If $A_q \cap (B_{q,1} \cup B_{q,2})$ occurs, then the true topology is modal. Hence, by a union bound and Hoeffding's inequality,

$$\mathbb{P}(E_q) \leqslant \mathbb{P}(A_q^c) + \mathbb{P}(B_{q,1}^c) \leqslant 4 \exp(-2n'w^2).$$

Taking w arbitrarily close to $\Delta_q/3$ gives

$$\mathbb{P}(E_q) \leqslant 4 \exp\left(-\frac{2n'\Delta_q^2}{9} \right) \leqslant 4 \exp\left(-\frac{2n'\Delta^2}{9} \right).$$

Finally, union bounding over all $\binom{k}{4}$ quartets,

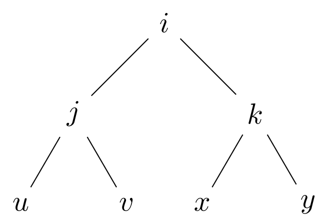
$$\mathbb{P}(\widehat{\mathcal{T}} \neq \mathcal{T}^*) \leqslant 4 \binom{k}{4} \exp\left(-\frac{2n'\Delta^2}{9} \right).$$

Thus, to recover $\mathcal{T}^*$ with probability at least $1 - \varepsilon$, it suffices that

$$n' > \frac{9}{2\Delta^2} \log\left(\frac{4\binom{k}{4}}{\varepsilon}\right).$$

$\square$

Proof of Lemma 3.2.4 (Expected support margin). Suppose first that the quartet $q = uv||xy$ induces a balanced tree in $\mathcal{T}^*$. Let i, j, k denote the internal vertices of this subtree like so:



Let $I := c(i)$ be the set of micro-states at vertex i, and likewise for J, K. Then:

$$\mathbb{E}[\Delta_q^{(c)} \mid J, K] = \sum_{(\alpha,\beta)\in\mathcal{A}(J,K)} \mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid J, K)$$

where:

$$\mathcal{A}(J, K) = \{(\alpha, \beta) : \alpha \neq \beta \in \Sigma_N, \alpha \in J, \beta \in K, \{\alpha, \beta\} \nsubseteq J \cap K\}$$

We see that:

$$\mathcal{B}(J, K) := \{(\alpha, \beta) : \alpha \in J \setminus K, \beta \in K \setminus J\} \subseteq \mathcal{A}(J, K)$$

For $(\alpha, \beta) \in \mathcal{B}(J, K)$, since we just need α, β to survive until the leaves, by Lemma 3.2.3:

$$\mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid J, K) \geqslant \eta_{ju}\eta_{jv}\eta_{kx}\eta_{ky}$$

Hence:

$$\mathbb{E}\Delta_q^{(c)} \geqslant \mathbb{E}|\mathcal{B}(J, K)| \cdot \eta_{ju}\eta_{jv}\eta_{kx}\eta_{ky}$$

Note $(\alpha, \beta) \in \mathcal{B}(J, K)$ can occur if:

(i) State α is born along the branch $(i, j]$ and survives until j. State β is born along the branch $(i, k]$ and survives until k.

(ii) State α is born before i, dies along $(i, k]$, and survives along $(i, j]$. State β is born along the branch $(i, k]$ and survives until k.

(iii) State β is born before i, dies along $(i, j]$, and survives along $(i, k]$. State α is born along the branch $(i, j]$ and survives until j.

(iv) State α is born before i, dies along $(i, k]$, and survives along $(i, j]$. State β is born before i, dies along $(i, j]$, and survives along $(i, k]$.

We focus solely on (i), as in this case the roles of J, K nicely decouple. By Lemma 3.A.1, the number of such α and β are at least $\mathrm{Poisson}(\mathrm{m}_0(\ell_{\mathrm{ij}}))$ and $\mathrm{Poisson}(\mathrm{m}_0(\ell_{\mathrm{ik}}))$ respectively, where $m_\theta(t) = \theta e^{-\mu t} + \frac{\lambda_N}{\mu} \cdot (1 - \eta_t)$, and these are clearly independent. This produces on average $m_0(\ell_{ij}) \cdot m_0(\ell_{ik})$ pairs (α, β).

The contributions from (ii), (iii), (iv) can similarly be calculated using the death-rate lower bound found in Lemma 3.2.3. If we let $\theta = \mathbb{E}|R \cap \Sigma_N|$ be the expected number of non-homoplastic states at the root, then by Poisson thinning we get contributions

(ii) $m_\theta(\ell_{ri})\xi_{ik}\eta_{ij} \cdot m_0(\ell_{ik})$.

(iii) $m_0(\ell_{ij}) \cdot m_\theta(\ell_{ri})\xi_{ij}\eta_{ik}$.

If we additionally assume $|R \cap \Sigma_N| \sim \mathrm{Poisson}(\theta)$, then

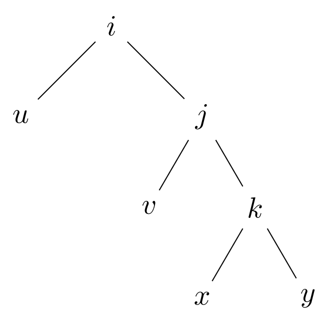
(iv) $m_\theta(\ell_{ri})^2\xi_{ik}\eta_{ij}\xi_{ij}\eta_{ik}$.

Formula (iv) comes from the fact that $|I| \sim \mathrm{Poisson}(m_\theta(\ell_{ri}))$, and so there are $\mathbb{E}|I|(|I|-1) = m_\theta(\ell_{ri})^2$ ordered pairs of distinct states on average at i. Hence when the subtree induced by the quartet is balanced:

$$\mathbb{E}|\mathcal{B}(J,K)| \geqslant [m_0(\ell_{ij}) + m_\theta(\ell_{ri})\eta_{ij}\xi_{ik}]\,[m_0(\ell_{ik}) + m_\theta(\ell_{ri})\xi_{ij}\eta_{ik}] \geqslant m_0(\ell_{ij})m_0(\ell_{ik}).$$

where the right hand side is always a valid lower bound even without the Poisson assumption on $|R|$. With our above work, this gives the desired bound on $\mathbb{E}\Delta_q^{(c)}$ when the induced subtree is balanced.

If the subtree induced by q is a caterpillar tree like so:



Then we saw:

$$\mathbb{E}[\Delta_q^{(c)} \mid I, J, K] = \sum_{(\alpha,\beta)\in\tilde{\mathcal{A}}(I,J,K)} \mathbb{E}(I_{uv||xy}(\alpha,\beta) \mid I, J, K)$$

where:

$$\tilde{\mathcal{A}}(I,J,K) = \{(\alpha,\beta) : \alpha \neq \beta \in \Sigma_N, \alpha \in I \cap J, \beta \in K, \{\alpha,\beta\} \not\subseteq J \cap K\}$$

Again, we see this set contains:

$$\tilde{\mathcal{B}}(I,J,K) := \{(\alpha,\beta) : \alpha \in (I \cap J) \setminus K, \beta \in K \setminus J\}$$

For $(\alpha,\beta) \in \tilde{\mathcal{B}}(I,J,K)$ we see $(\alpha,\beta) \in M_{uv||xy}$ if and only if:

- α survives on $(i, u]$ and $(j, v]$.
- β survives on $(k, x]$ and $(k, y]$

so again by Lemma 3.2.3:

$$\mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid I, J, K) \geqslant \eta_{iu}\eta_{jv}\eta_{kx}\eta_{ky}$$

Note $(\alpha, \beta) \in \tilde{B}(I, J, K)$ can occur only if α is born before i, survives along $(i, j]$ and dies on $(j, k]$, and β is born on $(j, k]$ and survives until k. Let r denote the root of the tree. Note for α, any non-homoplastic state born at the root survives to i with probability at least η_{ri}. Independently from this, we have at least $\text{Poisson}(\mathrm{m}_0(\ell_{\mathrm{ri}}))$ born along the path $[r, i]$. Of these, at least a $\eta_{ij}\xi_{jk}$ fraction survive until they reach j but die before reaching k. Independently of this, at least $\text{Poisson}(\mathrm{m}_0(\ell_{\mathrm{jk}}))$ non-homoplastic states β are born along $(j, k]$ and do not die on this branch. Hence if we let $\rho_N := \mathbb{E}|c(r) \cap \Sigma_N|$ be the expected number of non-homoplastic states present at the root

$$\mathbb{E}|\tilde{B}(I, J, K)| \geqslant \eta_{ij}\xi_{jk}\left[\rho_N \cdot \eta_{ri} + \frac{\lambda_N}{\mu}(1 - e^{-\mu\ell_{ri}})\right] \cdot \left[\frac{\lambda_N}{\mu}(1 - e^{-\mu\ell_{jk}})\right],$$

giving the desired bound. □

Proof of Lemma 3.5.1 (Sample Complexity PCH-ASTRAL). If for each leaf-set s the topology $q \in A_s$ maximizing the score S_q is the correct topology, then PCH-ASTRAL will trivially return the correct species tree $\mathcal{T}^*$, as the set of topologies $\{q_s\}_{s \in \binom{[k]}{4}}$ will exactly match those generated by the species tree $\mathcal{T}^*$. Hence if $\hat{\mathcal{T}}$ is the output of this inference procedure, by union bound:

$$\mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}^*) \geqslant 1 - \sum_s \mathbb{P}(q_s \neq \mathcal{T}_s^*).$$

Hence it suffices to upper bound $p_s := \mathbb{P}(q_s \neq \mathcal{T}_s^*)$. Note

$$p_s \leqslant \sum_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{P}(S_{\mathcal{T}_s^*} \leqslant S_t).$$

so it suffices to bound each term on the right hand side. For $t \in A_s \setminus \{\mathcal{T}_s^*\}$, define

$$D_i := |M_{\mathcal{T}_s^*}^{(c_i)}| - |M_t^{(c_i)}|, \qquad S_{\mathcal{T}_s^*} - S_t = \frac{1}{n}\sum_{i=1}^n D_i.$$

Since $L \leqslant \mathbb{E}D_i$ we have

$$\mathbb{P}(S_{\mathcal{T}_s^*} \leqslant S_t) \leqslant \mathbb{P}\left(\left|\sum_{i=1}^n (D_i - \mathbb{E}D_i)\right| \geqslant nL\right).$$

Lemma 3.A.4 implies for universal constants

$$||D_i - \mathbb{E}D_i||_{\Psi_{1/2}} \lesssim \Big||M_{\mathcal{T}_s^*}|\Big|_{\Psi_{1/2}} + \Big||M_t|\Big|_{\Psi_{1/2}} \lesssim \kappa,$$

Applying the sub-Weibull concentration inequality of Theorem 3.1 in [37]

$$\mathbb{P}\left(\left|\sum_{i=1}^{n}(D_i - \mathbb{E}D_i)\right| \gtrsim \kappa(\sqrt{nt} + t^2)\right) \leqslant 2e^{-t}.$$

Choose $t_\delta = \log(4\binom{k}{4}/\delta)$. Note our assumptions on n imply

$$nL \gtrsim \kappa(\sqrt{nt_\delta} + t_\delta^2)$$

and we have

$$\mathbb{P}(S_{\mathcal{T}_s^*} \leqslant S_t) \leqslant 2\exp(-t_\delta).$$

Unioning-bounding over quartets and the two incorrect topologies then gives

$$\mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}^*) \geqslant 1 - 2\binom{k}{4} \cdot 2\exp(-t_\delta) = 1 - \delta,$$

as desired. $\square$

Chapter 4

Symmetry-Maximizing Fair Values

Abstract

This chapter introduces a symmetry-based approach to estimating the fair value of an asset from a central limit order book. The bid and ask sides of the book are encoded as measures, and the fair value is defined as the point around which these measures are as symmetric as possible. We study this variational problem under Wasserstein metrics, derive explicit formulas in the W_2 case, incorporate quantity imbalances in a way that recovers the weighted mid price in a simple limiting case, and prove basic stability properties of the resulting estimator. We also discuss a rudimentary market-maker model which connects the symmetry objective to latent pairings between bids and asks.

4.1 Introduction

The *central limit order book (CLOB)* is one of the most common mechanisms for pairing buyers and sellers in electronic marketplaces, including major exchanges such as NYSE, NASDAQ, and LSE. At any point in time, the CLOB records the prices and quantities at which participants are willing to buy or sell an asset. Crossing orders are matched and removed, leaving bids and asks separated by a positive *spread*. The book changes as orders are matched, added, modified, or canceled, and its dynamics provide a rich source of information about market sentiment.

The bid side of the order book is typically encoded as list of prices $\{b_i\}$ at which orders are placed, where $b_0 > b_1 > ...$, and a list of total quantities $\{\beta_i\}$ at each price level. Likewise, the ask side has lists $\{a_i\}, \{\alpha_i\}$ for $a_0 < a_1 < ....$ The highest buy offer b_0 and lowest sell offer a_0 are called the *best bid* (BB) and *best ask/offer* (BO) respectively, and together they are called the BBO. The distance $s := a_0 - b_0$ is called the *spread*, and, since orders with crossing prices are matched and removed from the order book, it is strictly positive

Note in this view of the market, there is no one agreed upon price of the asset that people are willing to trade at. Instead, there is a range of prices and quantities people are willing to buy at, and a range they are willing to sell at. Nonetheless, these bids and offers still provide

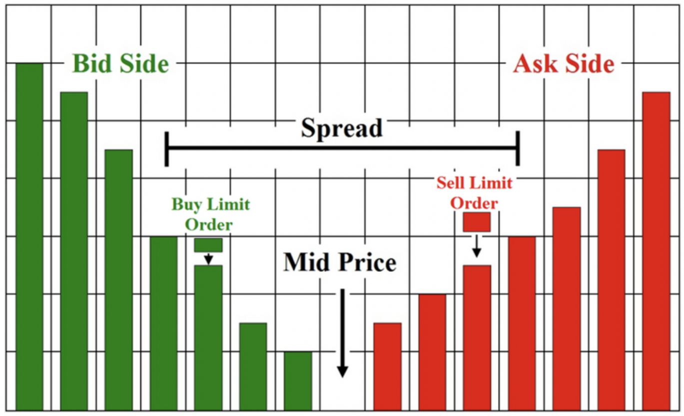


Figure 4.1: Limit order book.

some information about what market participants think the asset is reasonably worth, the asset's *fair value*: no reasonable person would place a bid higher than their fair value, nor an ask lower.

Many approaches have been proposed for modeling CLOB dynamics, ranging from classical point-process and stochastic differential equation models to more recent deep-learning and agent-based methods; see [30] for a recent survey. Although these models differ substantially, they often share the goal of extracting fair-value information from displayed liquidity. This chapter proposes one such estimator.

A typical naive model for the fair value is just the *mid price* $v_m = (b_0 + a_0)/2$, the average of the best bid and best offer. A more nuanced measure is the *weighted mid price*:

$$v_w = \frac{b_0 \cdot \alpha_0 + a_0 \cdot \beta_0}{\beta_0 + \alpha_0} = v_m + \frac{s}{2} \cdot \frac{\beta_0 - \alpha_0}{\beta_0 + \alpha_0},$$

which incorporates information about the imbalance in quantities available at the BBO. Heuristically, when there are many more bids than asks we might expect this buying pressure to force the value of the asset upwards, and adjust our fair value accordingly.

The main downside of these two fair-value estimates is that they only use the first price level. Other heuristics, such as volume-weighted averages across several levels or market-impact estimates based on walking the book, use more of the displayed depth but still lack a clear variational motivation.

We aim to develop a fair-value estimate that captures the *shape* of the order book, has a clear variational characterization, and recovers familiar heuristics under natural assumptions.

We first view the bid side and ask side of the order book as two separate distributions μ_b, μ_a. One way of doing this is to view each quantity (or order) as a unit mass on the order book (and then possibly normalize):

$$\mu_b \propto \sum_i \beta_i \cdot \delta_{b_i}, \qquad \mu_a \propto \sum_j \alpha_j \cdot \delta_{a_j}. \tag{4.1.1}$$

Instead of treating each quantity identically, we can also weight the orders according to some scheme. Since it seems likely that orders near the spread are more informed, it is natural to weight these orders more. Namely, by introducing a discount $\lambda \in (0,1]$, we could choose to add more mass to the larger bids and the smaller asks, the orders nearer to the spread:

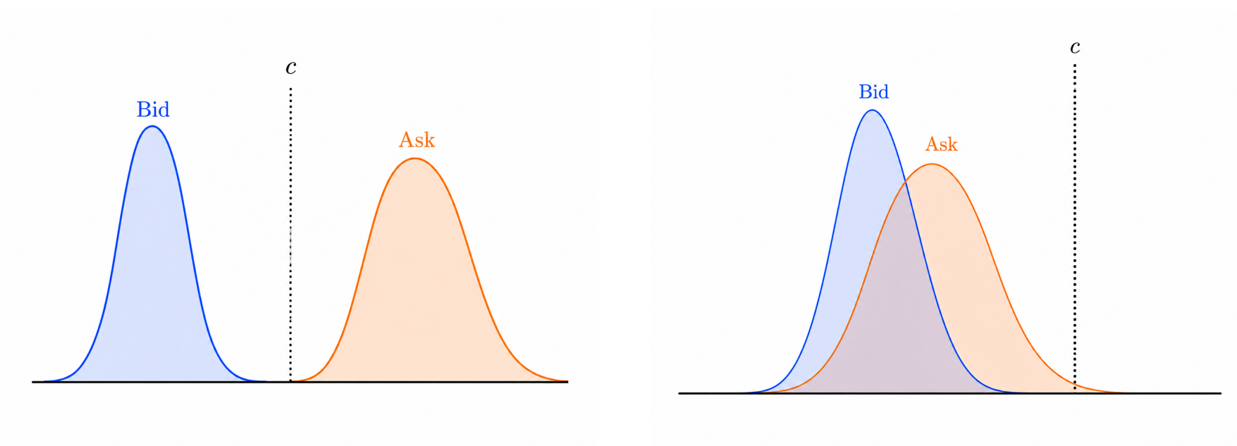
$$\mu_b \propto \sum_i \lambda^{-b_i} \cdot \beta_i \cdot \delta_{b_i}, \qquad \mu_a \propto \sum_j \lambda^{a_j} \cdot \alpha_j \cdot \delta_{a_j}. \tag{4.1.2}$$

In the rest of this note, we will not assume μ_a, μ_b take any particular form, but these two examples are good to have in mind.

To try to generalize our heuristics, consider the simple situation where the bid and asks books are perfect mirror images of each other, separated possibly by some spread. In this case, buying and selling pressures are exactly equal, so the mid price v_m is a natural candidate for the fair value. In particular, it is the point around which the two distributions are mirror images: reflecting the ask book across the mid price gives exactly the bid book. In other words, v_m solves:

$$v_s = \underset{c\in\mathbb{R}}{\arg\min}\ d(\mu_a(\cdot + c),\ \mu_b(c - \cdot)), \tag{4.1.3}$$

for any metric d on a suitable space of probability distributions. In general, we will not have such symmetry, but we can still seek to solve (4.1.3) for a fixed metric d. We will call the solution v_s the *symmetry-maximizing value*, and it implicitly depends on d. A visualization of this procedure is given in Figure 4.2.



(a) Original order book. (b) Reflected ask book across c.

Figure 4.2: Visualization of the symmetry-maximizing fair value. The ask book is reflected across the candidate fair value c, and the distance between the reflected ask book and the bid book is measured by some metric d. Then c is chosen to minimize this distance.

4.2 Incorporating Order Book Shape

There are many natural measures of "distance" between probability measures: total variation, Jensen-Shannon divergence, and χ^2-type distances. These are less suitable for the present application because they compare mass at fixed locations, making them sensitive to small shifts in order prices or quantities. Since financial markets are inherently noisy, we want a metric that is more robust under such perturbations, and which incorporates more of the *geometry* of the order book than these pointwise comparisons.

The Wasserstein distance is the natural candidate. In this section, we aim to study how the symmetry-maximizing fair value behaves when the underlying metric d is chosen to be this distance. In Theorem 4.2.1, we characterize the symmetry-maximizing fair value under the Wasserstein metric, and in Lemma 4.5.1 we quantify the stability of this fair-value with respect to small changes of the order book.

4.2.1 Wasserstein p-metric

This metric comes from the theory of optimal transport. It measures how difficult it is to convert one distribution into another by transporting it "piece by piece". The goal is to find a transportation plan that requires the least total distance the mass must be transported. Formally, its defined as:

$$d_{W_p}(\mu, \nu) := \left(\inf_{(X,Y) \in \Pi(\mu,\nu)} \mathbb{E}|X - Y|^p \right)^{1/p},$$

where $\Pi(\mu, \nu)$ is the space of all couplings of μ, ν, joint distributions of (X, Y) that have marginals μ, ν respectively. Namely, for each $x \in \mathbb{R}$ our coupling $\pi(x, \cdot)$ specifies where in ν the mass $\mu(x)$ must be sent, and we search for a way of doing this that minimizes the total transportation distance. The below theorem shows (4.1.3) takes the form of a simple convex optimization problem. First, we introduce some notation for equation (4.1.2).

$$b_i := \frac{\sum_{j=1}^{i} \lambda^{b_j} \cdot \beta_j}{\sum_{j=1}^{n} \lambda^{b_j} \cdot \beta_j}, \qquad a_i := \frac{\sum_{j=1}^{i} \lambda^{a_j} \cdot \alpha_j}{\sum_{j=1}^{m} \lambda^{a_j} \cdot \alpha_j}, \qquad a_0 = b_0 = 0,$$

$$z_k := \inf\{a_i : a_i > z_{k-1}\} \cup \{b_i : b_i > z_{k-1}\}, \qquad \Delta z_\ell := z_\ell - z_{\ell-1}, \qquad z_0 := 0.$$

Namely, the a_i, b_i keep track of our cumulative distribution functions of the bid and ask distributions as we move away from the best bid and best offer, and the z_k are the order statistics of these combined values. Moreover, define $f(k) = i$ if $[a_{i-1}, a_i)$ is the unique interval of this kind containing z_k. Likewise define $g(k) = i$ if $z_k \in [b_{i-1}, b_i)$. Now we are ready to relate these quantities to the symmetry-maximizing fair value.

Theorem 4.2.1. *For any $p \geqslant 1$, v_s solves the convex optimization problem:*

$$v_s = \arg\min_{c>0} \sum_{\ell=1}^{m+n} \Delta z_\ell \cdot |2c - b_{f(\ell)} - a_{g(\ell)}|^p. \tag{4.2.1}$$

Moreover, in the case $p = 2$ the explicit solution is:

$$v_s = \frac{1}{2}\Big(m_2(\mu_a) + m_2(\mu_b)\Big). \tag{4.2.2}$$

Remark 4.2.2. As discussed in Section 4.5 we see (4.2.2) is just the mean of the W_2 barycenter of μ_a, μ_b. Thus we can take the barycenter as an estimate of the full fair-value distribution rather than just taking its mean as this point estimate.

4.3 A Simple Model For Market Makers

The following model is not intended as a literal description of order-book formation. It is a stylized mechanism showing how a symmetry-based fair value can arise when market makers place bid and ask quotes around latent fair values. In the balanced case, a paired-order model takes bids and asks of the form

$$\text{bid} = v - h, \qquad \text{ask} = v + h,$$

where h is a latent half-spread reflecting uncertainty, inventory concerns, or compensation for supplying liquidity. With imbalanced buying and selling pressure, we instead allow asymmetric quotes:

$$\text{bid} = v - \theta^{-1}h, \qquad \text{ask} = v + (1-\theta)^{-1}h,$$

where $\theta \in (0,1)$ is an exogenous imbalance parameter. As θ approaches zero, the bid is moved farther from the fair value; as θ approaches one, the analogous effect occurs on the ask side. Formally, we study:

$$\begin{aligned} &v \sim \mu \in \mathbb{P}(\mathbb{R}) && \text{(fair value distribution)} \\ &h \sim \nu \in \mathbb{P}((0,\infty)) && \text{(half-spread distribution)} \\ &\theta \in (0,1) && \text{(imbalance).} \end{aligned}$$

$$\text{bid} = v - \theta^{-1}h, \qquad \text{ask} = v + (1-\theta)^{-1}h.$$

Note that the above implies:

$$v = \theta \cdot \text{bid} + (1-\theta)\cdot \text{ask}, \qquad h = \theta(1-\theta)\cdot(\text{ask} - \text{bid}).$$

Thus, if we knew which bid was paired with each ask, we could aggregate these to get reasonable estimates of the fair value and half-spread distributions:

$$\hat{\mu} := \frac{1}{n}\sum_{i=1}^{n} \delta_{\theta \cdot \mathrm{bid}_i + (1-\theta)\cdot \mathrm{ask}_i}, \qquad \hat{\nu} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\theta(1-\theta)(\mathrm{ask}_i - \mathrm{bid}_i)}.$$

We start by getting some simple bounds on the basic moments of these distributions:

Lemma 4.3.1. *Suppose that $bid_n \leqslant ... \leqslant bid_1 \leqslant ask_1 \leqslant ... \leqslant ask_n$. Then if we define μ_a, μ_b as in* (4.1.1)*, regardless of how bids and asks are paired up, we have:*

$$m_2(\hat{\mu}) = \theta \cdot m_2(\mu_b) + (1-\theta)\cdot m_2(\mu_a), \qquad m_2(\hat{\nu}) = \theta(1-\theta)(m_2(\mu_a) - m_2(\mu_b)).$$

where $m_2(\mu)$ is the standard mean. Moreover, if for a permutation τ of $[n]$ we let:

$$\mathrm{Cov}_\tau(bid, ask) := \frac{1}{n}\sum_{i=1}^{n} [bid_i - m_2(\mu_b)] \cdot [ask_{\tau(i)} - m_2(\mu_a)],$$

be the empirical covariance between the bid and ask prices under pairing τ, then:

$$\mathrm{Cov}_{\mathrm{id}}(bid, ask) \leqslant \frac{1}{2\theta(1-\theta)}\Big[\mathrm{Var}(\hat{\mu}_\tau) - (\mathrm{Var}(\theta\mu_b) + \mathrm{Var}((1-\theta)\mu_a))\Big] \leqslant \mathrm{Cov}_\rho(bid, ask).$$

$$\mathrm{Cov}_{\mathrm{id}}(bid, ask) \leqslant -\frac{1}{2}\Big[\frac{\mathrm{Var}(\hat{\nu}_\tau)}{\theta^2(1-\theta)^2} - (\mathrm{Var}(\mu_b) + \mathrm{Var}(\mu_a))\Big] \leqslant \mathrm{Cov}_\rho(bid, ask).$$

where $\rho(i) = n+1-i$ is the reverse permutation, and $\hat{\mu}_\tau, \hat{\nu}_\tau$ are the empirical distributions under the pairing $(bid_i, ask_{\tau(i)})_{i=1}^n$. Namely, the minimizer/maximizer are the "inside-out" and "left-to-right" pairings of bids, asks respectively.

Likewise, if we knew the pairings of bids and asks and were interested in the p^{th} mean $m_p(\mu) := \arg\min_c \mathbb{E}_\mu (X-c)^p$ we could approximate it by:

$$m_p(\hat{\mu}) = \arg\min_c \sum_{i=1}^{n} |c - \theta \cdot \mathrm{bid}_i - (1-\theta)\cdot \mathrm{ask}_i|^p.$$

As we will see later, this has the same form as our symmetric fair-value objective (4.4.3) once a pairing of bids and asks has been chosen. Since the identity of the issuer of each order is unknown, the main problem becomes choosing a reasonable pairing. Equivalently, the symmetry-maximizing value can be viewed as a relaxation of this latent pairing problem.

Of course, a real order book need not admit a literal one-to-one decomposition into bid/ask pairs. The paired model should instead be viewed as a finite approximation to a coupling between the bid-side and ask-side distributions.

4.3.1 Determining optimal bid/ask pairing

Suppose we have n bids $\text{bid}_1 \geqslant ... \geqslant \text{bid}_n$ and n asks $\text{ask}_1 \leqslant ... \leqslant \text{ask}_n$. Pairing these up amounts to choosing a permutation $\tau \in S_n$ and pairing bid_i with $\text{ask}_{\tau(i)}$. If we assume this is the true pairing, the above generative model specifies exactly its probability distribution. Finding the optimal way of pairing up asks and bids then amounts to solving:

$$\begin{aligned}
&\arg\max_{\tau \in S_n} \mathbb{P}_{v,h}\left((\text{bid}_i, \text{ask}_{\tau(i)})_{i=1}^n\right) \\
&= \arg\max_{\tau \in S_n} \sum_{i=1}^n \log\left(\mathbb{P}_{v,h}(\text{bid}_i, \text{ask}_{\tau(i)})\right) \\
&= \arg\max_{\tau \in S_n} \sum_{i=1}^n \log\left(\mathbb{P}_h(\theta(1-\theta)\cdot(\text{ask}_{\tau(i)} - \text{bid}_i)\right) + \log\left(\mathbb{P}_v(\theta\text{bid}_i + (1-\theta)\cdot\text{ask}_{\tau(i)})\right).
\end{aligned}$$

If we assume we have densities $\mathbb{P}_h = e^{f(x)}\,dx$ and $\mathbb{P}_v = e^{g(x)}\,dx$ then this reduces to:

$$= \arg\max_{\tau \in S_n} \sum_{i=1}^n f\Big(\theta(1-\theta)\cdot(\text{ask}_{\tau(i)} - \text{bid}_i)\Big) + g\Big(\theta\cdot\text{bid}_i + (1-\theta)\cdot\text{ask}_{\tau(i)}\Big).$$

The next lemma gives conditions under which this score is maximized when bids and asks are paired in the natural way: from inside the spread outwards. Towards this end, we introduce the discrete differences

$$\Delta_r f(z) := f(z+r) - f(z).$$

Concavity of f is equivalent to $\Delta_r f$ being nonincreasing for each $r > 0$, so the second discrete derivative $\Delta_t\Delta_r f$ is nonpositive. The following identifies the optimal pairing when one of the latent distributions is sufficiently more log-concave than the other.

Lemma 4.3.2. *Suppose $\mu = N(0, \sigma^2)$ and $\nu = e^{f(x)}$. If for all $z \in \mathbb{R}$ and $t, r > 0$:*

$$\Delta_t\Delta_r f(z) \geqslant -\frac{1}{\theta(1-\theta)\sigma^2}\cdot rt. \tag{4.3.1}$$

If $bid_n \leqslant ... \leqslant bid_1 \leqslant ask_1 \leqslant ... \leqslant ask_n$ then:

$$\text{id} = \arg\max_{\tau \in S_n} \sum_{i=1}^n f\Big(\theta(1-\theta)\cdot(ask_{\tau(i)} - bid_i)\Big) + g\Big(\theta\cdot bid_i + (1-\theta)\cdot ask_{\tau(i)}\Big).$$

Namely, the optimal pairing is moving from inside the spread outwards. Moreover, if the reverse inequality is true the optimal pairing is from left to right.

Note, ignoring the constant depending on θ, the right hand side of equation (4.3.1) is just the second discrete derivative of $g(x) = -\frac{x^2}{2\sigma^2}$, the log-density of the Gaussian. Thus, the condition is saying that our fair value distribution is more log-concave than our spread distribution in a strong way.

Remark 4.3.3. If f is twice differentiable, then a Taylor series expansion shows $\Delta_t \Delta_r f(z) \approx rt \cdot f''(z)$. Thus the assumption of the theorem roughly says the second derivative is bounded below by some sufficiently large constant.

Remark 4.3.4. In practice, since bids and asks can only be placed at discrete tick-sizes we could effectively relax the assumptions to only $z, t, r \in \mathbb{Z}_{>0}$.

Example 4.3.5. In particular, one natural choice is $\nu = \text{Gamma}(\lambda, k)$. Here:

$$f(x) = -\lambda x + (k-1)\log(x) + C_{\lambda,k} \quad \rightarrow \quad \Delta_t \Delta_r f(x) = (k-1)\log\left(1 - \frac{rt}{(x+r)(x+t)}\right).$$

In the exponential case ($k = 1$) or when $k < 1$ then assumption (4.3.1) is trivially satisfied because the second discrete derivative is nonnegative. Otherwise, if $k > 1$ then $\Delta_t \Delta_r f(x) \to -\infty$ as $x \downarrow 0$, so the assumptions are not met. In this regime, the Gamma distribution has sufficient mass away from zero, so large spreads are not as heavily penalized, and the optimal pairing is no longer necessarily from inside out.

Note that our above proof of Lemma 4.3.2 actually shows something slightly stronger.

Corollary 4.3.6. *If $bid_n \leqslant \ldots \leqslant bid_1 \leqslant ask_1 \leqslant \ldots \leqslant ask_n$ and for all $t, r > 0$:*

$$\Delta_{\theta t}\, \Delta_{(1-\theta)r}\, f(z) \geqslant \Delta_r \Delta_t g(w),$$

for $z \in [0, ask_n - bid_n]$ and $w \in [bid_n, ask_n]$, then we can see:

$$\text{id} = \arg\max_{\tau \in S_n} \sum_{i=1}^{n} f\Big(\theta(1-\theta) \cdot (ask_{\tau(i)} - bid_i)\Big) + g\Big(\theta \cdot bid_i + (1-\theta) \cdot ask_{\tau(i)}\Big).$$

Namely, the optimal pairing is moving from inside the spread outwards.

Under this simple model, the optimal pairing is one of two natural pairings: "inside-out," starting from the best bid and best offer, or "left-to-right," starting from the worst bid and best offer. The Wasserstein formulation in Section 4.2 can be viewed as a continuous relaxation of this pairing problem.

4.4 Introducing Quantity Imbalances

A natural criticism of our work in Section 4.2 is that we lose information about imbalance between the bid and ask sides of the order book, imbalances which fair-value estimates like the weighted mid v_w discussed earlier seek to exploit. Thus, it may only be a decent estimate of the fair value in situations when the total number of bids and total number asks are roughly equal. In this section, we discuss a way of expanding our previous approach to incorporate this kind of information.

In this section, we drop our assumption that μ_a, μ_b be probability measures, so they can take unnormalized forms like in equation (4.1.2). One simple measure of imbalance is then just the following ratio of the total masses m_a, m_b of these measures:

$$\theta = \frac{m_a}{m_a + m_b}. \tag{4.4.1}$$

In the case of the unnormalized measures 4.1.1, this is just the ratio of the total volume on the bid book and the total volume on the ask book. Using this, we can extend 4.1.3 in a way that incorporates the imbalances in the bid and ask books:

$$v_s^\theta = \arg\min_{c>0} d\left(\frac{\mu_b[(c-\cdot)/2\theta]}{m_b}, \frac{\mu_a[(\cdot+c)/2(1-\theta)]}{m_a}\right). \tag{4.4.2}$$

We then have the corresponding extension of Theorem 4.2.1, which in the case $p = 2$ just says we replace the standard average in Theorem 4.2.1 by an average weighted by the imbalance.

Theorem 4.4.1. *Let $\theta \in (0, 1)$. Then for any $p \geqslant 1$, v_s^θ solves the convex problem:*

$$v_s^\theta = \arg\min_{c>0} \sum_{\ell=1}^{m+n} \Delta z_\ell \cdot |2c - 2\theta \cdot b_{f(\ell)} - 2(1-\theta) \cdot a_{g(\ell)}|^p. \tag{4.4.3}$$

Moreover, in the case $p = 2$ the explicit solution is:

$$v_s^\theta = \theta \cdot m_2(\mu_b) + (1-\theta) \cdot m_2(\mu_a). \tag{4.4.4}$$

Remark 4.4.2. In the case there is only one price level and (4.1.1), or when μ_a, μ_b only put mass on the first price level proportional to the quantity, and we choose θ by (4.4.1) then $v_s^\theta = v_w$, the weighted mid price.

We can of course consider other imbalance measurements $\theta \in (0, 1)$ other than (4.4.1). Some other reasonable alternatives might compare recent trade volumes, higher moments of μ_a, μ_b, or possibly even the age of different orders on the book. The main point is that we can easily incorporate this kind of information into our framework by simply changing the way we warp the bid and ask distributions before comparing them.

4.5 Stability of the Fair Value

Recall the main motivation behind using Wasserstein metrics in the first place were that they provide a natural robustness to small perturbations in the underlying distribution. In this section, we extend classical stability results for the Wasserstein metric to our symmetrized fair value. This gives the following continuity property under d_{W_2}

Lemma 4.5.1. *The fair value v_s^θ under d_{W_2} in (4.4.4) satisfies for any $p \geqslant 1$:*

$$\left| v_s^\theta(\mu,\nu) - v_s^{\theta'}(\mu',\nu') \right| \leqslant \theta' d_{W_p}(\mu,\mu') + (1-\theta') d_{W_p}(\nu,\nu') + |\theta-\theta'| \cdot d_{W_p}(\mu,\nu). \tag{4.5.1}$$

In particular, if $\mu' = t^\# \mu, \nu' = s^\# \nu$ are pushforward measures then:

$$\left| v_s^\theta(\mu,\nu) - v_s^{\theta'}(\mu',\nu') \right| \leqslant \theta' ||t - id||_{L^p(\mu)} + (1-\theta') ||s - id||_{L^p(\nu)} + |\theta-\theta'| \cdot d_{W_p}(\mu,\nu). \tag{4.5.2}$$

The result is not exactly surprising: equation (4.4.4) shows the fair value is completely determined by the means of μ, ν, so if we only perturb μ, ν by a small amount, their means should not change by much either.

Under d_{W_p} for $p \neq 2$, things get a little more interesting. Such an explicit closed-form bound is no longer possible. However, intuitively, in equation (4.4.3) we are solving an optimization problem which is strictly convex in c for $p > 1$ and depends smoothly on the parameters z_ℓ, θ, a_i, b_j. Hence we still expect some level of stability, which the following makes precise.

Lemma 4.5.2. *For any $p > 1$, if $\mu_n \xrightarrow{W_p} \mu, \nu_n \xrightarrow{W_p} \nu$, and $\theta_n \to \theta$, then*

$$v_s^{\theta_n}(\mu_n, \nu_n) \to v_s^\theta(\mu,\nu).$$

The difficulty in obtaining a quantitative bound analogous to Lemma 4.5.1 is that, for $p \neq 2$, the p^{th}-mean is no longer a linear functional of the underlying distribution, so the movement of the minimizer depends on the local curvature of the objective near its minimum rather than just on the perturbation level.

4.6 Concluding Remarks and Future Directions

In this paper, we proposed a fair-value estimate which maximizes the symmetry of the central limit order book around it. This generalizes several common heuristics and gives a variational characterization of fair value. Under the Wasserstein distance, we derive explicit formulas and show stability under small perturbations of the order book.

Corollary 4.7.4 suggests a natural extension: in the W_2 case, the symmetrized fair value is the mean of the weighted barycenter of μ_a and μ_b, given by

$$\overline{\mu_\theta} := \arg\min_\nu \theta d_{W_2}(\nu, \mu_a) + (1-\theta) d_{W_2}(\nu, \mu_b).$$

This identifies the fair value with a Wasserstein barycenter and suggests estimating a full fair-value distribution rather than only a point estimate. The same idea could also incorporate several order books for the same asset across different venues. Instead of merging all bids and asks directly, one could first form bid-side and ask-side barycenters across venues, reflect the ask-side barycenter, and then barycenter the two resulting distributions. This is similar in spirit to the aggregation procedure in [54]; it would be interesting to identify a market model under which such an estimator is optimal.

Another natural direction is empirical. One could simulate order books under standard CLOB models, such as the point-process or agent-based models surveyed in [30], and compare the resulting symmetry-maximizing value to the mid price, weighted mid price, and other benchmarks. Useful diagnostics would include the smoothness of the resulting time series, its visual reasonableness relative to the displayed book shape, and its usefulness as a feature for predicting short-term price movements.

Finally, the estimator studied here uses a static snapshot of the order book. A dynamic version could instead maximize symmetry across time, which may be more robust to noise. It would be useful to understand when symmetry arises in standard CLOB dynamics and how the resulting estimator performs empirically.

4.7 Proof Appendix

We start by collecting some basic results about Wasserstein spaces which will be useful in our analysis. More details can be found in [51].

4.7.1 Properties of Wasserstein Spaces

Lemma 4.7.1. *If $\mu_n \xrightarrow{W_p} \mu$ then $\mathbb{E}_{\mu_n} X \to \mathbb{E}_\mu X$. More specifically:*

$$\left| \mathbb{E}_{\mu_n} X - \mathbb{E}_\mu X \right| \leqslant d_{W_p}(\mu_n, \mu).$$

Proof. Since for any coupling $\pi \in \Pi(\mu_n, \mu)$ we have for $p \geqslant 1$ by Jensen's inequality:

$$\left| \int_{\mathbb{R}} x \ \mu_n(dx) - \int_{\mathbb{R}} y \ \mu(dy) \right| \leqslant \left(\int_{\mathbb{R}^2} |x-y|^p \ d\pi \right)^{1/p}.$$

Taking the infimum over such couplings provides the desired result. □

In fact, more is true: as shown in [51], convergence in W_p is equivalent to weak convergence and convergence of p^{th} moments. Below, we show for distributions on $\mathbb{R}$ that the Wasserstein distance can be easily computed using the quantile function Q_μ. Again, this result can be found in [51].

Lemma 4.7.2. *For any $p \geqslant 1$ the map $\mu \mapsto Q_\mu$ from $W_p(\mathbb{R})$ to $L_p([0,1], \mathrm{Leb})$ is an isometry. Namely for $\mu, \nu \in W_p(\mathbb{R})$:*

$$d_{W_p}(\mu, \nu) = \left(\int_0^1 |Q_\mu(z) - Q_\nu(z)|^p \ dz \right)^{1/p}.$$

Lemma 4.7.3. *Given $\{\mu_i\}_{i=1}^n \in W_2(\mathbb{R})$ its (weighted) empirical barycenter satisfies:*

$$\overline{\mu} := \operatorname*{arg\,min}_{\nu \in W_2} \sum_{i=1}^n \omega_i \cdot d_{W_2}(\nu, \mu_i)^2 \quad \to \quad Q_{\overline{\mu}} = \sum_{i=1}^n \omega_i \cdot Q_{\mu_i}.$$

In particular, $\mathbb{E}_{\overline{\mu}} X = \sum \omega_i \cdot \mathbb{E}_{\mu_i} X$.

Proof. We adapt the proof of Proposition 3.1 in [54]. Apply Lemma 4.7.2 and the bias-variance decomposition to see $Q_{\overline{\mu}}$ must minimize:

$$\sum_{i=1}^n \omega_i \cdot ||Q_{\mu_i} - Q_{\overline{\mu}}||_{L^2}^2 = \sum_{i=1}^n \omega_i \cdot \left|\left| \sum_{i=1}^n \omega_i \cdot Q_{\mu_i} - Q_{\overline{\mu}} \right|\right|_{L^2}^2 + \left|\left| \sum_{i=1}^n \omega_i \cdot Q_{\mu_i} - Q_{\mu_i} \right|\right|_{L^2}^2.$$

For the latter result, just use $\mathbb{E}_\mu X = \int_0^1 Q_\mu(z) \ dz$ and linearity of expectation. □

4.7.2 Main Results

Proof of Theorems 4.2.1 ,4.4.1. By Lemma 4.7.2 it suffices to work with quantile functions. If $\mu = \sum_i p_i \delta_{x_i}$ is discrete, its quantile function is an increasing step function. Namely, assuming $x_1 < x_2 < ... < x_n$ it takes the form:

$$Q_\mu(z) = x_k \text{ for } z \in \left(\sum_{i=1}^{k-1} p_i, \quad \sum_{i=1}^{k} p_i \right].$$

Thus if $\mu = \sum_i p_i \delta_{x_i}$ and $\nu = \sum_i q_i \delta_{y_i}$ then:

$$|Q_\mu(z) - Q_\nu(z)| = |x_k - y_r| \text{ for } z \in \left(\sum_{i=1}^{k-1} p_i, \quad \sum_{i=1}^{k} p_i \right] \cap \left(\sum_{i=1}^{r-1} q_i, \quad \sum_{i=1}^{r} q_i \right],$$

is piecewise constant, with pieces delineated by endpoints $\sum_{i \leqslant k} p_i, \sum_{j \leqslant k} q_j$. Namely, if we let $z_1 \leqslant z_2 \leqslant ... \leqslant z_{m+n}$ be the order statistics of $\{\sum_{i=1}^{k} p_i \ : \ k \leqslant n\} \cup \{\sum_{i=1}^{r} q_i \ : \ r \leqslant m\}$ we can see:

$$d_{W_p}(\mu, \nu)^p = \sum_{\ell=1}^{m+n} \Delta z_\ell \cdot |x_{f(\ell)} - y_{g(\ell)}|^p, \qquad \Delta z_\ell := z_\ell - z_{\ell-1},$$

where $f : [m+n] \to [n]$ is the map that sends ℓ to k if $z_\ell \in \left[\sum_{i=1}^{k-1} p_i, \sum_{i=1}^{k} p_i \right)$. Likewise $g : [m+n] \to [m]$ sends ℓ to k if $z_\ell \in \left[\sum_{i=1}^{k-1} q_i, \sum_{i=1}^{k} q_i \right)$.

Thus in the setting of (4.4.2) for $\theta \in (0,1)$ we get:

$$d_{W_p}\left(\mu\left[\frac{\cdot + c}{2\theta} \right], \ \nu\left[\frac{c - \cdot}{2(1-\theta)} \right] \right)^p = \sum_{\ell=1}^{m+n} \Delta z_\ell \cdot \left| 2c - 2\theta \cdot x_{f(\ell)} - 2(1-\theta) \cdot y_{g(\ell)} \right|^p, \tag{4.7.1}$$

where $x_1 < ... < x_n$ and $y_1 > ... > y_n$. Note the reversal in the order of the y_i (and hence their corresponding q_i). Note that the z_ℓ depend only on p_i, q_i, not on c. In the case of W_2, by taking the gradient we can explicitly solve for the optimal c in (4.1.3):

$$\begin{aligned} v_s^\theta &= \sum_{\ell=1}^{m+n} \Delta z_\ell \cdot \left(\theta x_{f(\ell)} + (1-\theta) y_{g(\ell)} \right) \\ &= \theta \sum_{k=1}^{n} p_k x_k + (1-\theta) \sum_{k=1}^{m} q_k y_k \\ &= \theta \cdot m_2(\mu) + (1-\theta) \cdot m_2(\nu). \end{aligned}$$

where the second line follows by grouping terms where $f(\ell) = k$ and terms where $g(\ell) = k$ respectively. Replacing p_i, q_i, x_i, y_i by their analogs in (4.1.2) yields Theorem 4.4.1. Theorem 4.2.1 is just the special "balanced" case when $\theta = \frac{1}{2}$. □

We can re-state the above proof purely in terms of the quantile functions. Note if X has quantile function Q_X, then the transformed random variables $Y = c - 2\theta X$ and $Y' = 2(1-\theta)X - c$ have quantile functions (up to Lebesgue null sets) given by:

$$Q_Y(u) = c - 2\theta Q_X(1-u), \quad Q_{Y'}(u) = 2(1-\theta)Q_X(u) - c,$$

respectively. Using Lemma 4.7.2, then (4.4.2) becomes:

$$d_{W_p}\left(\mu\left[\frac{c-\cdot}{2\theta}\right],\ \nu\left[\frac{c+\cdot}{2(1-\theta)}\right]\right)^p = 2^p\int_0^1 [c - \theta Q_\mu(1-u) - (1-\theta)Q_\nu(u)]^p\ du.$$

Defining:

$$\rho(u) := \theta Q_\mu(1-u) + (1-\theta)Q_\nu(u),$$

we can up to constants the above is just $\int_0^1 |c-\rho(u)|^p\ du$. Hence we can view v_s^θ in (4.4.2) as the p^{th} mean of the random variable $Z = \rho(U)$ for $U \sim \text{Unif}[0,1]$. This naturally extends beyond discrete distributions as well.

In the special case of $p = 2$, the p^{th} mean is just the standard mean. By linearity

$$\mathbb{E}Z = \theta\mathbb{E}Q_\mu(1-U) + (1-\theta)\mathbb{E}Q_\nu(U) = \theta\mathbb{E}Q_\mu(U) + (1-\theta)\mathbb{E}Q_\nu(U) = \mathbb{E}Q_{\bar{\mu}_\theta}(U),$$

where $\bar{\mu}_\theta$ is the W_2 barycenter of μ, ν with weights $\theta, 1-\theta$, as discussed in Lemma 4.7.3. Thus, in the case of $p = 2$, our symmetrized fair value is just the mean of the (imbalance-weighted) W_2 barycenter of μ, ν. Namely

Corollary 4.7.4. *For any measures $\mu, \nu \in W_2(\mathbb{R})$:*

$$v_s^\theta = \arg\min_{c>0} d_{W_2}\left(\mu\left[\frac{\cdot + c}{2\theta}\right],\ \nu\left[\frac{c-\cdot}{2(1-\theta)}\right]\right) = \int_{\mathbb{R}} x\ \bar{\mu}_\theta(dx).$$

For $p \neq 2$, we lose the linearity of the p^{th} mean. However, we can still interpret ρ as a measure of the fair value at different "spreads/levels" u, and Z encodes the distribution these fair values can take on. The p^{th} mean then just aims to quantify the typical fair value. For example, when μ, ν are mirror images, then ρ is just some constant c, so every level u agrees on the fair value. In this case, the p^{th} is just c for every $p > 1$.

Now, we prove the above variance bounds for the estimated fair value and spread. The key point is that the only dependence on the pairing τ is in $\sum \text{bid}_i \text{ask}_{\tau(i)}$, which is maximized when bids and asks are paired in an increasing way, where the smallest bid is paired with the smallest ask and so on (our left to right pairing), and minimized when bids and asks are paired in a decreasing way, where the largest bid is paired with the smallest ask (our inside out pairing).

Proof of Lemma 4.3.1. The equations for the means is immediate from the definition and linearity of expectation. For the variances, some simple algebra shows that if $\hat{\mu}_n^\tau$ is the empirical distribution where bid_i is paired with $\text{ask}_{\tau(i)}$ then:

$$\begin{aligned}\text{Var}(\hat{\mu}_n^\tau) &= \mathbb{E}_{\hat{\mu}_n^\tau} X^2 - (\mathbb{E}_{\hat{\mu}_n^\tau} X)^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}[\theta \text{bid}_i + (1-\theta)\cdot \text{ask}_{\tau(i)}]^2 - (\theta m_2(\mu_b) + (1-\theta)m_2(\mu_a))^2 \\ &= \text{Var}(\theta\mu_b) + \text{Var}((1-\theta)\mu_a) + \frac{2\theta(1-\theta)}{n}\sum_{i=1}^{n}(\text{bid}_i\text{ask}_{\tau(i)} - m_2(\mu_b)m_2(\mu_a)).\end{aligned}$$

and if we define $\hat{\nu}_n^\tau$ similarly that:

$$\begin{aligned}\text{Var}(\hat{\nu}_n^\tau) &= \mathbb{E}_{\hat{\nu}_n^\tau} X^2 - (\mathbb{E}_{\hat{\nu}_n^\tau} X)^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}[\theta(1-\theta)(\text{ask}_{\tau(i)} - \text{bid}_i)]^2 - [\theta(1-\theta)(m_2(\mu_a) - m_2(\mu_b))]^2 \\ &= \theta^2(1-\theta)^2\Big[\text{Var}(\mu_b) + \text{Var}(\mu_a) - \frac{2}{n}\sum_{i=1}^{n}[\text{bid}_i\text{ask}_{\tau(i)} - m_2(\mu_b)m_2(\mu_a)]\Big].\end{aligned}$$

In both, the only dependence on τ is in $\sum \text{bid}_i\text{ask}_{\tau(i)}$. Trivially, this empirical correlation is maximized when bids and asks are paired in an increasing way, where the smallest bid is paired with the smallest ask and so on (our left to right pairing), and minimized when bids and asks are paired in a decreasing way, where the largest bid is paired with the smallest ask (our inside out pairing). □

Proof of Lemma 4.3.2. The proof is analogous to Lemma 4.1 of [5], which shows log-concavity is the crucial condition. Call the sum to be maximized $S(\tau)$ and the summand $V(\text{bid}_i, \text{ask}_{\tau(i)})$. Also shorten bid_i to b_i and ask_j to a_j. If $i < j$ but $\tau(i) =: u > v := \tau(j)$ we will show $S(\tau') \geqslant S(\tau)$ for $\tau' = (uv)\tau$. Namely, by correcting the order of i, j in the image we can improve the objective value. Note since μ is normally distributed:

$$g(x) = -\frac{x^2}{2\sigma^2} \qquad \to \qquad \Delta_r\Delta_t g(x) = -\frac{rt}{\sigma^2}.$$

Recall that $b_j \leqslant b_i \leqslant a_v \leqslant a_u$. Define $r := b_i - b_j \geqslant 0$, $z := a_v - b_i \geqslant 0$, and $t := a_u - a_v \geqslant 0$ to be the spacing between these four values and let $w := \theta b_j + (1-\theta)a_v$ to be the weighted average of the smallest ask and bid in this group. Then we see:

$$\begin{aligned}&S(\tau') - S(\tau) \\ &= [V(b_i, a_v) + V(b_j, a_u)] - [V(b_i, a_u) + V(b_j, a_v)] \\ &= -\Big[\Big(g(\theta b_i + (1-\theta)a_u) - g(\theta b_i + (1-\theta)a_v)\Big)\end{aligned}$$

$$
\begin{aligned}
&\qquad\qquad - \Big(g(\theta b_j + (1-\theta)a_u) - g(\theta b_j + (1-\theta)a_v)\Big)\Big] \\
&+ \Big[\Big(f(\theta(1-\theta)(a_u - b_j)) - f(\theta(1-\theta)(a_u - b_i))\Big) \\
&\qquad\qquad - \Big(f(\theta(1-\theta)(a_v - b_j)) - f(\theta(1-\theta)(a_v - b_i))\Big)\Big] \\
&= -\Delta_{\theta r}\Delta_{(1-\theta)t}g(w) + \Delta_{\theta(1-\theta)t}\,\Delta_{\theta(1-\theta)r}\,f(\theta(1-\theta)z) \\
&= \frac{\theta(1-\theta)}{\sigma^2}\cdot rt + \Delta_{\theta(1-\theta)t}\,\Delta_{\theta(1-\theta)r}\,f(\theta(1-\theta)z).
\end{aligned}
$$

Substituting $t' = \theta(1-\theta)t$ and likewise for the other variables we get:

$$
S(\tau') - S(\tau) = \frac{1}{\theta(1-\theta)\sigma^2}\cdot r't' + \Delta_{t'}\Delta_{r'}f(z').
$$

By assumption, this is nonnegative. Because we can go from any permutation to the identity through a series of transpositions, the result follows. □

Proof of Lemma 4.5.1. The first inequality follows almost immediately from (4.4.4) and Lemma 4.7.1. Just note

$$
\begin{aligned}
&\left|v_s^{\theta}(\mu,\nu) - v_s^{\theta'}(\mu',\nu')\right| \\
&= |\theta m_2(\mu) + (1-\theta)m_2(\nu) - \theta' m_2(\mu') - (1-\theta')m_2(\nu')| \\
&= \left|\theta'\big(m_2(\mu) - m_2(\mu')\big) + (1-\theta')\big(m_2(\nu) - m_2(\nu')\big) + (\theta-\theta')\big(m_2(\mu) - m_2(\nu)\big)\right| \\
&\leqslant \theta'\,|m_2(\mu) - m_2(\mu')| + (1-\theta')\,|m_2(\nu) - m_2(\nu')| + |\theta-\theta'|\,|m_2(\mu) - m_2(\nu)| \\
&\leqslant \theta' d_{W_p}(\mu,\mu') + (1-\theta')d_{W_p}(\nu,\nu') + |\theta-\theta'|d_{W_p}(\mu,\nu).
\end{aligned}
$$

Using the triangle inequality on the last term gives the desired result. For the second, note by Jemnsen's inequality

$$
m_2(\mu) - m_2(\mu') = \int_{\mathbb{R}} (x - t(x))\,\mu(dx) \leqslant ||\mathrm{id} - t||_{L^p(\mu)},
$$

so the above work gives the desired result again. □

Now, we prove the more general stability result under d_{W_p} for $p \neq 2$. The main idea is that the optimization problem in (4.4.3) is strictly convex in c for $p > 1$ and depends smoothly on the parameters.

Proof of Lemma 4.5.2. For $u \in [0,1]$, define

$$
\rho_n(u) := \theta_n Q_{\mu_n}(1-u) + (1-\theta_n)Q_{\nu_n}(u), \quad \rho(u) := \theta Q_\mu(1-u) + (1-\theta)Q_\nu(u).
$$

and the functions

$$f_n(c) := \int_0^1 |c - \rho_n(u)|^p \, du, \quad f(c) := \int_0^1 |c - \rho(u)|^p \, du.$$

By Lemma 4.7.2 and the definition of v_s^θ in (4.4.3), we have

$$v_s^\theta(\mu_n, \nu_n) = \arg\min_{c>0} f_n(c), \quad v_s^\theta(\mu, \nu) = \arg\min_{c>0} f(c).$$

We start by showing $\rho_n \to \rho$ in $L_p([0,1])$. By Lemma 4.7.2

$$\begin{aligned} ||\rho_n - \rho||_{L_p} &\leqslant ||\theta_n Q_{\mu_n}(1-\cdot) - \theta Q_\mu(1-\cdot)||_{L_p} + ||(1-\theta_n)Q_{\nu_n} - (1-\theta)Q_\nu||_{L_p} \\ &\leqslant |\theta_n - \theta| \cdot ||Q_\mu||_{L_p} + \theta_n ||Q_{\mu_n} - Q_\mu||_{L_p} + |\theta_n - \theta| \cdot ||Q_\nu||_{L_p} + (1-\theta_n)||Q_{\nu_n} - Q_\nu||_{L_p} \quad = |\theta_n - \theta \end{aligned}$$

As $\mu_n \xrightarrow{W_p} \mu, \nu_n \xrightarrow{W_p} \nu$ and $\theta_n \to \theta$, this vanishes, so $\rho_n \to \rho$ in L_p. Then, using the inequality

$$||a|^p - |b|^p| \leqslant p|a-b| \cdot (|a|^{p-1} + |b|^{p-1}),$$

we see:

$$\begin{aligned} \Big| |\rho_n(u) - c|^p - |\rho(u) - c|^p \Big| &\leqslant p \cdot |\rho_n(u) - \rho(u)| \cdot \left[(|\rho_n(u)| + c)^{p-1} + (|\rho(u)| + c)^{p-1} \right] \\ &\leqslant C_p \cdot |\rho_n(u) - \rho(u)| \cdot \left[c + |\rho_n(u)|^{p-1} + |\rho(u)|^{p-1} \right]. \end{aligned}$$

Hence for $R > 0$

$$\sup_{|c| \leqslant R} |f_n(c) - f(c)| \leqslant C_p \cdot ||\rho_n - \rho||_{L_1} \cdot \left[R + ||\rho_n||_{L^{p-1}}^{p-1} + ||\rho||_{L^{p-1}}^{p-1} \right].$$

Since $\rho_n \to \rho$ in L_p, for $p > 1$, we have $\sup_n ||\rho_n||_{L^{p-1}} < \infty$ and $||\rho_n - \rho||_{L_1} \to 0$, so the right hand side goes to zero. Hence $f_n \to f$ uniformly on compact sets.

Note that f_n, f are strictly convex for $p > 1$, and they achieve their minimum at a unique point. This and locally uniform convergence then imply these minimizers converge, so $v_s^\theta(\mu_n, \nu_n) \to v_s^\theta(\mu, \nu)$ as desired. □

Chapter 5

Diffusion Limits in Entropy-Regularized Stochastic Control

Goals

- Show convergence of the discrete-time entropy-regularized optimal control problem to the continuous-time exploratory formulation of [65].
- Can we develop a measure-valued notion of entropy-regularized optimal control?

5.1 Introduction

Suppose we have some discrete-time dynamical system on Euclidean space:

$$X_{t+1} = X_t + b(X_t, u_t) + \sigma(X_t, u_t) \cdot \xi_t,$$

where $u_t \in U$ is the input of some controller and $\xi_t \overset{iid}{\sim} \mathcal{N}(0, I)$ is the noise. In classical control theory, we have some loss function $\ell(x, u)$ and we wish to minimize the expected cummulative (discounted) loss over the entire trajectory $(X_t)_{t\in\mathbb{N}}$. Namely we aim to solve:

$$V(x) = \min_{(u_t)_{t\in\mathbb{N}}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \ell(X_t, u_t) \,\middle|\, X_0 = x\right]$$

for some discount $\gamma \in (0, 1)$. The value function V and optimal control (u_t) can be determined using the standard Bellman equation.

$$V(x) = \inf_{u\in U}\{\ell(x, u) + \gamma \mathbb{E} V(x + b(x, u) + \sigma(x, u) \cdot \xi)\}.$$

where the expectation is taken with respect to the noise ξ. Defining:

$$Q_\gamma(x, u) := \ell(x, u) + \gamma \mathbb{E} V(x + b(x, u) + \sigma(x, u) \cdot \xi),$$

then this implies the optimal control and value function satisfy the implicit relations:

$$u^*(x) \in \arg\min_{u\in U} Q_\gamma(x, u), \quad V(x) = Q_\gamma(x, u^*(x)). \tag{5.1.1}$$

We can make things more interesting by allowing for the control policy to select its control u_t at random, possibly depending on the current state. Namely, suppose there is some deterministic function $\pi(\cdot \mid x) : S \to \mathcal{P}(U)$ defining a distribution over the set of controls for each possible state. Then we can define a Markov chain by:

$$X_{t+1} = X_t + b(X_t, U_t) + \sigma(X_t, U_t) \cdot \xi_t, \quad U_t \sim \pi(\cdot \mid X_t).$$

Namely, we sample a control independently at each point of time according to our policy (π_t). This allows us to model a few different phenomena:

(i) *Imperfect Controls*: In many real-world settings, the controller does not have perfect control over the system, and the implemented actions may be subject to noise. By allowing for randomized control policies, we can model this inherent uncertainty in the control process.

(ii) *Adversarial or Strategic Environments*: In some settings, predictability can be exploited by an adversary or the environment. Entropy regularization encourages policies that remain sufficiently randomized, thereby mitigating worst-case or strategic responses to deterministic behavior.

(iii) *Exploration Versus Exploitation*: In reinforcement learning, we often want to encourage the agent to explore the state space in order to learn about the environment and the corresponding reward process. Randomized policies explicitly encode this exploration into the objective function.

(iv) *Robustness and Regularization*: Entropy regularization acts as a convex regularizer on the control problem, smoothing the optimization landscape and preventing degenerate (overly deterministic) solutions. This can significantly improve stability and sample efficiency in practice [27].

(v) *Information Constraints / Bounded Rationality*: Entropy regularization can also be interpreted as penalizing deviations from a reference policy via a Kullback–Leibler divergence. In this view, the controller faces a cost for using highly concentrated policies, reflecting limited information-processing capacity or a preference for staying close to a prior behavior ([50]).

The *entropy-regularized optimal control problem* seeks to encourage control distributions that simultaneously explore the state-space while maintaining a low cummulative loss. Its corresponding value function is defined by:

$$V(x) = \inf_{(\pi_t)} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \cdot (\ell(X_t, U_t) - \tau H(\pi(\cdot \mid X_t))) \,\middle|\, X_0 = x \right],$$

where $H(\mu) := -KL(\mu \parallel \nu)$ is the Kullback-Leibler divergence with respect to some reference measure ν on U and τ is a parameter controlling the regularization strength. In the case $U = \mathbb{R}^d$ and ν is the Lebesgue measure, $H(\mu)$ is the standard Shannon entropy. Again one-step analysis shows:

$$V(x) = \inf_{\mu \in \mathcal{P}(U)} \left\{ \int_U \left[Q_\gamma(x, u) + \tau \log\left(\frac{d\mu}{d\nu}\right) \right] \mu(du) \right\}. \tag{5.1.2}$$

Define the partition function:

$$Z(x) := \int_U \exp\left(-\frac{Q_\gamma(x, u)}{\tau}\right) d\nu(u),$$

Then if $Z(x) < \infty$ the optimal policy π^* and value function V satisfy:

$$\pi(u \mid x)\, d\nu \propto \exp\left(-\frac{Q_\gamma(x, u)}{\tau}\right), \quad V(x) = -\tau \log(Z(x)). \tag{5.1.3}$$

As we see, the optimal policy is naturally a soft-min / Gibbs distribution with Hamiltonian $-Q_\gamma(x, \cdot)/\tau$, and the value function is essentially the log-partition function of this distribution. Entropy-regularization thus serves to smooths the strict minimization problem in equation (5.1.1), and sending $\tau \downarrow 0$ in (5.1.3) morally recovers (5.1.1).

We would like to extend this notion of entropy-regularization to continuous time stochastic control. Given any fixed control sequence $(u_t)_{t\geqslant 0}$ we can define the diffusion:

$$dX_t = b(X_t, u_t)dt + \sigma(X_t, u_t)dW_t,$$

Under standard regularity conditions on b, σ, which we discuss more below, this SDE has a unique strong solution. Of course, sampling a control independently at each point in time is ill-defined in continuous time. The paper [65] overcomes this obstacle by defining "relaxed" or "averaged" dynamics induced by a policy π. Namely, let:

$$b_\pi(x) = \int_U b(x,u)\,\pi(du \mid x), \quad \sigma_\pi(x) := \sqrt{\int_U \sigma(x,u)\sigma^T(x,u)\,\pi(du \mid x)},$$

$$\tilde{\ell}_\pi(x) := \int_U \ell(x,u)\,\pi(du \mid x).$$

If we let:

$$dX_t^\pi = b_\pi(X_t^\pi)dt + \sigma_\pi(X_t^\pi)dW_t, \tag{5.1.4}$$

then we can define the value function of a given policy π by:

$$V_\pi(x) := \mathbb{E}\left[\int_0^\infty e^{-\rho t}\cdot(\tilde{\ell}_\pi(X_t^\pi) - \tau H(\pi(\cdot \mid X_t^\pi)))dt \,\middle|\, X_0^\pi = x\right]. \tag{5.1.5}$$

where the expectation is over the Brownian motion $\{W_t\}_{t\geqslant 0}$. Using this, we can solve the entropy-regularized stochastic optimal control problem:

$$V(x) := \inf_{\pi\in\mathcal{A}(x)} V_\pi(x), \tag{5.1.6}$$

where $\mathcal{A}(x)$ is the set of admissible controls, which possibly depends on the state x. [65] call this the *exploratory* formulation, and motivate it as the average local dynamics across many realizations of the Brownian motion paths. Again, suitable regularity on b, σ are required for the SDE 5.1.4 to have a unique strong solution.

In this short note, we show that the exploratory dynamics of [65] arise as the scaling limit of a natural sequence of entropy-regularized discrete-time processes that sample controls at increasingly fine intervals. This gives a possibly more natural interpretation of entropy-regularization in continuous time as todo. Our main result is the following:

Theorem 5.1.1 (Entropy-Regularized Diffusive Limits)**.** *Given drift $b : \mathbb{R}^d \times U \to \mathbb{R}^d$, covariance $\sigma : \mathbb{R}^d \times U \to \mathbb{R}^{d\times d}$, loss function $\ell : \mathbb{R}^d \times U \to \mathbb{R}$, discount $\gamma = e^{-\rho}$, and entropy-regularization parameter τ, define a family of processes $(Y_t^h)_{t\geqslant 0}$ by:*

$$X_{n+1}^h - X_n^h := h\cdot b(X_n^h, u_n^h) + \sqrt{h}\cdot\sigma(X_n^h, u_n^h)\cdot\xi_n, \quad \xi_n \overset{iid}{\sim} N(0, I_d),$$

$$Y_t^h := X_{\lfloor t/h\rfloor}^h.$$

where the control u_n^h is sampled from the optimal policy π^h in Equation (5.1.3) *for the entropy-regularized control problem with parameters $(hb, \sqrt{h}\sigma, h\ell, \gamma^h, h\tau)$. Assume that:*

(i) *aggregate proper conditions from lemmas.*

Then as $h \to 0$, we have $Y_t^h \to Y_t$ in distribution in the Skorokhod space $D([0,T], \mathbb{R}^d)$, where Y_t is the optimally controlled exploratory dyamics a diffusion process satisfying:

$$dY_t = b(Y_t, \pi)dt + \sigma(Y_t, \pi)dW_t, \tag{5.1.7}$$

5.2 Allowable Controls

In this section we discuss the conditions on the control policies π, drift b, and covariance σ to ensure that the stochastic optimal control problem (5.1.6) is well-posed. As [65] mention, any $\pi \in \mathcal{A}(x)$ in the set of admissible controls must at least satisfy:

(i) For each $t \geq 0$, $\pi_t \in \mathcal{P}(U)$ almost surely.

(ii) For each $A \in \text{Borel}(U)$, the process

$$\left\{ \int_A \pi_t(u)\, du \;:\; t \geq 0 \right\}$$

is $\{\mathcal{F}_t\}$-progressively measurable.

(iii) The exploratory SDE (5.1.4) admits a unique strong solution $X^\pi = \{X_t^\pi : t \geq 0\}$ when the control π is applied.

(iv) The value function $V^\pi(x)$ defined in (5.1.5) is finite for all $x \in \mathbb{R}^d$.

Zack comment: The below is more a note to myself.

Condition (ii) ensures π is measurable as a measure-valued stochastic process, and that the coefficients of (5.1.5) are progressively-measureable and hence the underlying Ito integrals are well-defined. Condition (iii) ensures the stochastic process $\tilde{X}_t^\pi$ and hence its corresponding value function V^π is well-defined.

Conditions (i)–(iv) ensure that each admissible relaxed control induces a well-defined controlled diffusion with finite objective value. However, these assumptions alone do not guarantee existence of an optimal control. For existence, one typically also requires compactness or tightness of the admissible control class together with continuity and lower semicontinuity properties of the coefficients and cost functional, so that minimizing sequences admit convergent subsequences whose limits remain admissible.

Zack comment: Possibly some sort of compactness is required for the space of admissible controls.

In the case of the standard linear-quadratic controller where $U = \mathbb{R}^d$ and:

$$b(x,u) = Ax + Bu, \quad \sigma(x,u) = Cx + Du$$

and quadratic loss. Given policy π, let:

$$\mu_t := \int_U u\pi_t(du), \qquad \sigma_t^2 := \int_U (u-\mu_t)^2 \pi_t(du).$$

As in [65], we will assume the following::

(a) Conditions (i) and (ii) above.

(b) For each $t \geqslant 0$, $\mathbb{E}_\pi \left[\int_0^t (\mu_s^2 + \sigma_s^2) ds\right] < \infty$.

(c) $\liminf_{T\to 0} e^{-\rho T}\mathbb{E}(X_T^\pi)^2 = 0$.

(d) $\mathbb{E}\left[\int_0^\infty e^{-\rho t}|L(X_t^\pi, \pi_t)|\, dt\right] < \infty$.

Zack comment: I guess in this paper, we are only interested in controls where π_t is a function of the current state alone, while in general the results are stated for any $\mathcal{F}_t$-progressively-measurable, measure-valued process (π_t)

5.3 Proof of Theorem 5.1.1

Recall given policy $\pi : \mathbb{R}^d \to \mathcal{P}(U)$, drift $b : \mathbb{R}^d \times U \to \mathbb{R}^d$ and covariance $\sigma : \mathbb{R}^d \times U \to \mathbb{R}^{d\times d}$, we define for each $h > 0$:

$$X_{n+1}^h - X_n^h := h \cdot b(X_n^h, u_n^h) + \sqrt{h} \cdot \sigma(X_n^h, u_n^h) \cdot \xi_n, \quad Y_t^h := X_{\lfloor t/h \rfloor}^h,$$

$$\xi_n \overset{iid}{\sim} N(0, I_d), \quad u_n^h \mid \{X_n^h = x\} \sim \pi(\cdot \mid x).$$

where the controls u_n^h are sampled independently of both ξ_n and X_n^h. Heuristically, this gives the natural discrete-time approximation to the exploratory dynamics (5.1.4) when we sample controls independently every h seconds.

We break the proof of Theorem 5.1.1 into some key lemmas.

5.3.1 Fixed Policy Convergence of Paths

First, we show for a fixed policy π, we get convergence to the exploratory dynamics (5.1.4).

Lemma 5.3.1 (Fixed-Policy Convergence to Exploratory Dynamics)**.** *Suppose π satisfies:*

$$\sup_{||y||\leqslant R} \left| \int_U b_i(y,u) b_j(y,u)\, \pi(du \mid y)\right| < \infty, \qquad \sup_{||y||\leqslant R} \left| \int_U \mathrm{Tr}(\sigma\sigma^T)^{1+\delta/2}\, \pi(du \mid y)\right| < \infty.$$

for some $\delta > 0$. Moreover, suppose $b_\pi(y), \sigma_\pi(y)$ are continuous and there is a unique strong solution X_t^π to the SDE (5.1.4). If $Y_0^h = x_h \to x$, then $Y_t^h \Rightarrow X_t^\pi$ with $X_0^\pi = x$ as $h \downarrow 0$ weakly in the Skorokhod space $D([0,T], \mathbb{R}^d)$ for any $T < \infty$.

Proof of Lemma 5.3.1. Let $\Delta Y_t^h := Y_{t+h}^h - Y_t^h$. Note by the integrability constraints:

$$\begin{aligned}
\mathbb{E}[\Delta Y_{nh}^h \mid Y_{nh}^h = y] &= \mathbb{E}_\xi \int \left(hb(y,u) + \sqrt{h}\sigma(y,u)\xi\right)\ \pi(du \mid y) \\
&= h\int b(y,u)\pi(du \mid y) + \sqrt{h}\int \sigma(y,u)\pi(du \mid y)\cdot \mathbb{E}_\xi[\xi] \\
&= hb_\pi(y)
\end{aligned}$$

Hence the drift always takes the correct form. Similarly we see:

$$\begin{aligned}
&\mathbb{E}[(\Delta Y_{nh}^h)(\Delta Y_{nh}^h)^T \mid Y_{nh}^h = y] \\
&= \mathbb{E}_\xi \int \left(hb(y,u) + \sqrt{h}\sigma(y,u)\xi\right)\left(hb(y,u) + \sqrt{h}\sigma(y,u)\xi\right)^T\ \pi(du \mid y) \\
&= h^2 \int b(y,u)b(y,u)^T\pi(du \mid y) + h\int \sigma(y,u)\sigma(y,u)^T\pi(du \mid y) \\
&\qquad\qquad\qquad\qquad + h^{3/2}\int b(y,u)(\sigma(y,u)\cdot \mathbb{E}_\xi[\xi])^T\pi(du \mid y) \\
&= h\sigma_\pi^2(y) + h^2\int_U b(y,u)b(y,u)^T\pi(du \mid y).
\end{aligned}$$

Hence the infinitesmal variance is correct up to an $o(h)$ correction. Our assumption on π imply this correction is bounded uniformly for y in compact sets. Moreover, the probability of a jump larger than ε is at most:

$$\mathbb{P}\left(||\Delta Y_{nh}^h||_2 > \varepsilon \mid Y_{nh}^h = y\right) \leqslant \mathbb{P}(||b(y,U_y)|| > \epsilon/h) + \mathbb{P}(||\sigma(y,U_y)\xi|| > \epsilon/\sqrt{h}) \tag{5.3.1}$$

for $U_y \sim \pi(\cdot \mid y)$. By Markov's inequality, the first term satisfies:

$$\mathbb{P}(||b(y,U_y)|| > \epsilon/h) \leqslant \frac{h^2}{\epsilon^2}\int_U ||b(y,u)||^2\pi(du \mid y).$$

which is $o(h)$ uniformly for $|y| \leqslant R$ by assumption. Defining $V_y := ||\sigma(y,U_y)\xi||$:

$$\begin{aligned}
\mathbb{E}V_y^{2+\delta} &= \int_U \mathbb{E}_\xi ||\sigma(y,u)\xi||^{2+\delta}\pi(du \mid y) \\
&\leqslant \mathbb{E}_\xi||\xi||^{2+\delta}\cdot \int_U ||\sigma(y,u)||_F^{2+\delta}\ \pi(du \mid y) \\
&= C_\delta \int_U \mathrm{Trace}(\sigma\sigma^T)^{1+\delta/2}\ \pi(du \mid y).
\end{aligned}$$

Hence again by Markov's inequality we see:

$$\mathbb{P}(||\sigma(y,U_y)\xi|| > \epsilon/\sqrt{h}) \leqslant \frac{h^{1+\delta/2}}{\epsilon^{2+\delta}}\cdot \mathbb{E}V_y^{2+\delta},$$

which is again $o(h)$ uniformly in $|y| \leqslant R$ by assumption. Hence by (5.3.1), the probability of large jumps is $o(h)$ uniformly for $|y| \leqslant R$.

By assumption, the limiting SDE (5.1.4) has a unique strong solution, and hence the associated martingale problem is well-posed. By the standard functional convergence results (e.g. Section 8.7 of [16])), we have $Y_t^h \Rightarrow X_t^\pi$ in $D([0,T], \mathbb{R}^d)$. □

By *the associated martingale problem is well-posed* we mean that for the generator:

$$Lf = \langle \nabla f, b_\pi \rangle + \frac{1}{2}\mathrm{Trace}(\sigma_\pi^T H f \sigma_\pi)$$

of our limiting Ito process, there exists a unique law on $C([0,T], \mathbb{R}^d)$ so that:

$$M_t := f(X_t) - f(x) - \int_0^t Lf(X_s)\, ds, \quad X_0 \overset{a.s.}{=} x,$$

is a (local) martingale for every $f \in C_b^2(\mathbb{R}^d)$. Durret provides a sufficient condition for this in Section 8.7 of [16], condition (A):

Assumption (A)**.** The coefficients σ_π^{ij} and b_π^i are continuous, and for each x there is a unique measure P_x on $(C, \mathcal{C})$ such that $P_x(X_0 = x) = 1$ and

$$X_t^i - \int_0^t b^i(X_s)\, ds, \qquad X_t^i X_t^j - \int_0^t \sigma_\pi^{ij}(X_s)\, ds$$

are local martingales.

The main benefit of (A) is it avoids reference to the domain $C_b^2(\mathbb{R}^d)$ of test functions, and allows us to work explicitly with the limiting infinitesmal drift and variance.

5.3.2 Fixed Policy Convergence of Value Function

Next, we show that again for a fixed policy π, the discrete-time value functions converge uniformly on compact sets to the analogous continuous-time value function defined by the exploratory dynamics under the proper scalings of the relevant parameters. Namely:

Lemma 5.3.2 (Convergence of Value Functions)**.** *Given policy $\pi : \mathbb{R}^d \to \mathcal{P}(U)$, loss $\ell : \mathbb{R}^d \times U \to \mathbb{R}$, discount $\gamma = e^{-\rho}$, and regularization parameter τ, define for each $h > 0$:*

$$V_\pi^h(x) = \mathbb{E}\left[\sum_{t=0}^{\infty} e^{-\rho h t} \cdot h g_\pi(X_t^h) \,\middle|\, X_0^h = x\right].$$

where $g_\pi(x) := \int_U \ell(x,u) + \tau \log(\pi(u \mid x))\ \pi(u \mid x) d\nu(u)$ is the instantaneous cost under policy π at state x. Suppose: Zack comment: I guess we are implicitly assuming the space U of controls has some metric structure.

(i) *Lipschitz Coefficients: The coefficients $b(x,u), \sigma(x,u)$ are Lipschitz in x,u:*

$$|b(x,u) - b(y,v)| + |\sigma(x,u) - \sigma(y,v)| \leqslant C|x-y| + D|u-v|.$$

where $|\sigma|^2 = \sum_{i,j} |\sigma_{ij}|^2$ is the Frobenius norm.

(ii) *Lipschitz Cost: The instantaneous cost $g_\pi(x)$ is Lipschitz.*

(iii) *Lipschitz Policy: The policy $\pi(\cdot \mid x)$ is Lipschitz in the Wasserstein space W_2:*

$$d_{W_2}(\pi(\cdot \mid x), \pi(\cdot \mid y)) \leqslant F|x-y|, \quad \textit{and} \quad \int_U |u|^2 \pi(u \mid x) < \infty \quad \forall x.$$

(iv) *Non-vanishing Variance: There exists $\lambda > 0$ so $\sigma_\pi^2(x) \succeq \lambda I$ for all x.*

(v) *Sufficient Discounting: The discount ρ is large enough. Namely, there exists a finite discount ρ so that the below conclusion holds.*

Then as $h \to 0$, we have $V_\pi^h \to V_\pi$ uniformly on compact sets, where V_π is given by the exploratory dynamics (5.1.5).

Zack comment: Possibly also have to assume the moment bounds from the convergence result? As we use below the fact that we have weak convergence of the discrete chain to the limiting diffusion.

First, observe that the above assumptions imply the relaxed coefficients b_π, σ_π and the policy π satisfy a natural global growth condition. For π, note by its W_2-Lipschitzness:

$$\int_U |u|^2 \; \pi(u \mid x) \leqslant C(1+|x|^2).$$

Using this, note:

$$|b_\pi(x)| \leqslant \mathbb{E}|b(x,U_x)| \lesssim 1 + |x| + \mathbb{E}|U_x| \lesssim 1 + |x|,$$
$$|\sigma_\pi(x)|^2 = \int_U \mathrm{Tr}(\sigma\sigma^T)\pi(u \mid x) = \mathbb{E}|\sigma(x,U_x)|^2 \lesssim 1 + |x|^2.$$

The latter of course implies $|\sigma_\pi(x)| \lesssim 1+|x|$. Moreover, $b_\pi(x), \sigma_\pi(x)$ are Lipschitz and locally Lipschitz respectively. For b_π, simply observe that:

$$\begin{aligned}
\Big|b_\pi(x) - b_\pi(y)\Big| &= \left| \int_U b(x,u)\pi(u \mid x) - \int_U b(y,v)\pi(v \mid y) \right| \\
&\leqslant \int_U |b(x,u) - b(y,u)|\pi(u \mid x) + \left| \int_U b(y,u)\pi(u \mid x) - \int_U b(y,v)\pi(v \mid y) \right| \\
&\leqslant C|x-y| + \mathbb{E}|b(y,U_x) - b(y,U_y)|.
\end{aligned}$$

for any coupling (U_x, U_y) of $\pi(\cdot \mid x), \pi(\cdot \mid y)$. Using the Lipschitzness of $b(y, \cdot)$ and π proves the claim. Similarly for $x, y \in K$ compact:

$$\left|\sigma_\pi(x) - \sigma_\pi(y)\right|^2 \leqslant \left|\sqrt{\int_U \sigma(x,u)\sigma(x,u)^T \pi(u \mid x)} - \sqrt{\int_U \sigma(y,u)\sigma(y,u)^T \pi(u \mid y)}\right|^2$$

which is only $\frac{1}{2}$-Hölder continuous without further assumptions as the square root map $A \mapsto A^{1/2}$ is not Lipschitz when A is close to singular. Assumption (v) essentially forces our relaxed covariance matrices σ_π^2 to be sufficiently non-singular, so they are restricted to a region where the square-root map is actually Lipschitz. Under this assumption, Cauchy-Schwarz and the Lipschitzness of σ, π imply:

$$\begin{aligned}
&\lesssim \frac{1}{\lambda}\left| \int_U \sigma(x,u)\sigma(x,u)^T \pi(u \mid x) - \int_U \sigma(y,v)\sigma(y,v)^T \pi(v \mid y)\right|^2 \\
&\leqslant \lambda^{-1} \sup_{x \in K} \mathbb{E}|\sigma(x, U^x)|^2 \cdot \mathbb{E}|\sigma(x, U^x) - \sigma(y, U^y)|^2 \\
&\lesssim |x-y|^2 \lambda^{-1} \sup_{x \in K} \mathbb{E}|\sigma(x, U^x)|^2
\end{aligned}$$

Again, Lipschitzness implies this supremum is finite, so σ_π is locally Lipschitz.

By standard SDE theory, the above ensures the SDE (5.1.4) has a unique strong solution, and hence V_π is well-defined. The above parameter choices — $e^{-\rho h}$ for the discount, $h\ell$ for the loss, and $h\tau$ for the entropy-regularization — are essentially chosen so that h steps of the discrete-time process X_n^h corresponds to one unit of time in the continuous-time process.

Proof of Lemma 5.3.2. We can express V_π^h in terms of Y_t^h and its transition semigroup P_t^h:

$$V_\pi^h(x) = \mathbb{E}\left[\sum_{t=0}^\infty e^{-\rho h t} \cdot h g_\pi(Y_{ht}^h) \,\middle|\, Y_0^h = x\right] = \int_0^\infty e^{-\rho \lfloor t/h \rfloor h} P_t^h g_\pi(x) dt.$$

Likewise if P_t is the transition semigroup associated to X_t^π:

$$V_\pi(x) = \int_0^\infty e^{-\rho t} P_t g_\pi(x) dt.$$

Pointwise convergence $V_\pi^h \to V_\pi$ follows almost immediately from the weak convergence of $Y_t^h \Rightarrow X_t^\pi$ that we develop in Lemma 5.3.1. Simply apply this result to the functional:

$$F : D([0,T], \mathbb{R}^d) \to \mathbb{R}, \quad F(\omega) = \int_0^T e^{-\rho t} g_\pi(\omega(t)) dt.$$

The above assumptions give us sufficient control on the tails $\int_T^\infty e^{-\rho t} |P_t g_\pi(x)| dt$ then to ensure the value functions converge. The main obstacle is Lemma 5.3.1 gives us no control on how this convergence behaves as we vary the initial condition, which we will require to prove uniform convergence. The key ingredient will be a stability result with respect to the initial condition, which shows that trajectories diverge at most exponentially fast:

$$\mathbb{E}|X_t^{\pi,x} - X_t^{\pi,z}| \lesssim |x-z|e^{Ct}.$$

This stability result follows from the Lipschitzness via classical arguments, which we will reiterate below. We will reduce the proof of the result to a few claims. Below, for $a, b \geqslant 0$ let $a \lesssim b$ denote that $a \leqslant Cy$ for some $C \geqslant 0$ depending only on the compact set K, time horizon T, and all the relevant Lipschitz constants. In some cases, such as the tail estimates of Claim 3 below, the dependence on T will be important, in which case we will specifically emphasize this.

Claim 1: For every compact $K \subset \mathbb{R}^d$ and $T < \infty$,

$$\sup_{t\in[0,T]} \sup_{x\in K} |P_t^h g_\pi(x) - P_t g_\pi(x)| \to 0 \quad \text{as} \quad h \downarrow 0.$$

Proof of Claim 1. Let $\{z_i\}$ be an ε-net of K, $x \in K$, and t fixed. Note:

$$|P_t^h g_\pi(x) - P_t g_\pi(x)| \leqslant |P_t^h g_\pi(x) - P_t^h g_\pi(z_i)| + |P_t^h g_\pi(z_i) - P_t g_\pi(z_i)| + |P_t g_\pi(x) - P_t g_\pi(z_i)|$$

The middle and last terms are easy to handle. For the middle, note since $(Y_t^h) \Rightarrow (X_t^\pi)$ weakly in $D([0,T],\mathbb{R}^d)$, Skorokhod's representation theorem implies we can construct versions $\tilde{Y}^h \to \tilde{X}^\pi$ converging almost surely in $D([0,T],\mathbb{R}^d)$. Since $\tilde{X}^\pi \in C([0,T],\mathbb{R}^d)$ almost surely, this convergence thus must be almost surely uniform. Thus as g_π is Lipschitz:

$$\sup_{t\in[0,T]} |P_t^h g_\pi(z_i) - P_t g_\pi(z_i)| \lesssim \sup_{t\in[0,T]} \mathbb{E}|\tilde{Y}_t^h - \tilde{X}_t^\pi| \leqslant \mathbb{E}\left[\sup_{t\in[0,T]} |\tilde{Y}_t^h - \tilde{X}_t^\pi|\right].$$

If we can show $Z^h := \sup_{t\in[0,T]} |\tilde{Y}_t^h - \tilde{X}_t^\pi|$ are uniformly integrable when $h \leqslant 1$, this tends to zero by Vitali's convergence theorem. To do so, we show they are uniformly bounded in L^2, which will follow from Lipschitzness and our growth conditions. Note:

$$\mathbb{E}Z_h^2 \leqslant 2\mathbb{E}\sup_{t\in[0,T]} |Y_t^h|^2 + 2\mathbb{E}\sup_{t\in[0,T]} |X_t^\pi|^2.$$

First, note Cauchy-Schwarz and the BDG inequality imply:

$$\begin{aligned}\mathbb{E}\sup_{t\in[0,T]} |X_t^\pi|^2 &\lesssim |x_0|^2 + \mathbb{E}\sup_{t\in[0,T]}\left(\int_0^t b\, ds\right)^2 + \mathbb{E}\sup_{t\in[0,T]}\left(\int_0^t \sigma dB_s\right)^2 \\ &\lesssim |x_0|^2 + T\mathbb{E}\int_0^T b_\pi^2\, ds + \mathbb{E}\int_0^T \sigma_\pi^2\, ds.\end{aligned}$$

Since b_π, σ_π have linear growth:

$$\mathbb{E}\sup_{t\in[0,T]} |X_t^\pi|^2 \lesssim |x_0|^2 + (T+1)\int_0^T (1 + \mathbb{E}|X_s^\pi|^2)\, ds$$

$$\lesssim 1 + |x_0|^2 + \int_0^T \mathbb{E} \sup_{t\in[0,s]} |X_t^\pi|^2 \, ds.$$

Grönwall's inequality then implies $E\sup_{t\in[0,T]} |X_t^\pi|^2 \lesssim 1 + |x_0|^2$. Similarly, for Y_t^h:

$$\begin{aligned}
\mathbb{E} \sup_{t\in[0,T]} |Y_t^h|^2 &= \mathbb{E} \max_{n\leqslant\lfloor T/h\rfloor} |X_n^h|^2 \\
&\lesssim |x_0|^2 + h^2 \cdot \mathbb{E} \sup_{t\in[0,T]} \left(\sum_{i=1}^{\lfloor T/h\rfloor} b \right)^2 + h\mathbb{E} \sup_{t\in[0,T]} \left(\sum_{i=1}^{\lfloor T/h\rfloor} \sigma\xi_i \right)^2 , \\
&\lesssim |x_0|^2 + hT \sum_{i=1}^{\lfloor T/h\rfloor} \mathbb{E}b^2 + h \sum_{i=1}^{\lfloor T/h\rfloor} \mathbb{E}\sigma^2 \\
&\lesssim 1 + |x_0|^2 + h \sum_{i=1}^{\lfloor T/h\rfloor} \mathbb{E} \max_{t\leqslant i} |X_t^h|^2,
\end{aligned}$$

by linear growth of b, σ and π. By discrete Grönwall's inequality:

$$\mathbb{E} \max_{n\leqslant\lfloor T/h\rfloor} |X_n^h|^2 \lesssim (1 + |x_0|^2) e^{Ch\lfloor T/h\rfloor} \lesssim 1 + |x_0|^2.$$

which finishes the proof of the desired uniform integrability.

For the last term, note our assumptions imply our limiting diffusion X_t^π is Feller continuous, which would suffice if g_π were bounded. Instead, we can show directly $P_t g_\pi(x)$ is Lipschitz in x. As g_π is globally Lipschitz:

$$|P_t g_\pi(x) - P_t g_\pi(z)| \lesssim \sqrt{\mathbb{E}|X_t^{\pi,x} - X_t^{\pi,z}|^2},$$

where $X_t^{h,x}$ denotes the unique strong solution started at initial condition x. Using the Lipschitzness of the relaxed coefficients, b, σ, a standard application of Ito's isometry and Grönwall's inequality implies:

$$\mathbb{E}|X_t^{\pi,x} - X_t^{\pi,z}|^2 \lesssim |x - z|^2.$$

See for example the proof of Theorem 5.2.1 in [49]. Zack comment: The stated proof requires σ to be globally Lipschitz rather than just locally; some truncation argument is required maybe. This gives:

$$|P_t g_\pi(x) - P_t g_\pi(z_i)| \lesssim \epsilon.$$

Finally, we handle the first term. As g_π is Lipschitz:

$$|P_t^h g_\pi(x) - P_t^h g_\pi(z_i)| \lesssim \mathbb{E}|X_t^{h,x} - X_t^{h,z_i}|.$$

To bound this expectation, let $D_n := X_n^{h,x} - X_n^{h,z}$ be the current difference between the coupled processes and $\Delta X_n^{h,x} := X_{n+1}^{h,x} - X_n^{h,z}$ the one-step increment in one of the processes. Observe for *any* coupling of $(X_t^{h,x}, X_t^{h,z})$:

$$\mathbb{E}\left[D_{n+1}^2 \mid \mathcal{F}_n\right] = D_n^2 + 2D_n \cdot \mathbb{E}\left[\Delta X_n^{h,x} - \Delta X_n^{h,z} \mid \mathcal{F}_n\right] + \mathbb{E}\left[(\Delta X_n^{h,x} - \Delta X_n^{h,z})^2 \mid \mathcal{F}_n\right]$$

$$= D_n^2 + 2D_n \cdot h\left(b(X_n^{h,x}) - b(X_n^{h,z})\right) + \mathbb{E}\left[(\Delta X_n^{h,x} - \Delta X_n^{h,z})^2 \mid \mathcal{F}_n\right]$$

Lipschitzness of b implies:

$$2D_n \cdot h\left(b(X_n^{h,x}) - b(X_n^{h,z})\right) \lesssim hD_n^2.$$

To handle the last term, first observe that if (U^x, U^y) are coupled according to the optimal W_2 coupling of $\pi(\cdot \mid x)$ and $\pi(\cdot \mid y)$ then the Lipschitzness of b, π imply:

$$\begin{aligned}\mathbb{E}|b(x,U^x) - b(y,U^y)|^2 &\leqslant 2\left[\mathbb{E}|b(x,U^x) - b(y,U^x)|^2 + \mathbb{E}|b(y,U^x) - b(y,U^y)|^2\right] \\ &\lesssim |x-y|^2,\end{aligned}$$

and similarly for σ. Hence the above suggests we couple $(X_t^{h,x}, X_t^{h,z})$ as follows. At every step, we have them share the same noise ξ_n and we couple the controls (U_n^x, U_n^y) according to the optimal W_2-coupling of $\pi(\cdot \mid X_n^{h,x}), \pi(\cdot \mid X_n^{h,z})$. Using $||Ax||_2 \leqslant ||A||_F ||x||_2$, this gives:

$$\begin{aligned}&\mathbb{E}\left[(\Delta X_n^{h,x} - \Delta X_n^{h,z})^2 \mid \mathcal{F}_n\right] \\ &\leqslant h^2 \mathbb{E}\left|b(X_n^{h,x}, U_n^x) - b(X_n^{h,z}, U_n^z)\right|^2 + h\mathbb{E}\left|\sigma(X_n^{h,x}, U_n^x) - \sigma(X_n^{h,y}, U_n^y)\right|^2 \\ &\lesssim h|X_n^{h,x} - X_n^{h,y}|^2. \\ &= hD_n^2.\end{aligned}$$

for $h \leqslant 1$. Combining with our above work, this implies:

$$\mathbb{E}\left[D_{n+1}^2 - D_n^2 \mid \mathcal{F}_n\right] \lesssim h.$$

It follows by discrete Grönwall's inequality that $\mathbb{E}D_n^2 \leqslant e^{Cnh} D_0^2 \leqslant e^{CT} D_0^2$ and hence:

$$\mathbb{E}|X_t^{h,x} - X_t^{h,z_i}| \lesssim \epsilon.$$

Combining our above results yields:

$$\sup_{t \in [0,T]} \sup_{x \in K} |P_t^h g_\pi(x) - P_t g_\pi(x)| \lesssim 2\epsilon + o(h).$$

where the $o(h)$ term does not depend on ϵ, giving the desired result. □

Claim 2: For every compact $K \subseteq \mathbb{R}^d$ and $T < \infty$:

$$\sup_{x \in K} \int_0^T e^{-\rho t} |P_t g_\pi(x)| \, dt < \infty.$$

Proof of Claim 2. We instead show the stronger statement:

$$\sup_{t \in [0,T]} \sup_{x \in K} |P_t g_\pi(x)| < \infty.$$

In Claim 1, we showed the map $x \mapsto P_t g_\pi(x)$ is Lipschitz in x, with Lipschitz constant uniform in t. Hence:

$$\sup_{x \in K} |P_t g_\pi(x)| \lesssim 1 + |P_t g_\pi(x_0)|,$$

for any fixed $x_0 \in K$. Thus it suffices to show $\sup_{t\in[0,T]} |P_t g_\pi(x_0)| < \infty$. As g_π is Lipschitz we must have $|g_\pi(x)| \lesssim (1 + |x|)$ and hence:

$$\sup_{t\in[0,T]} |P_t g_\pi(x_0)| \lesssim 1 + \sup_{t\in[0,T]} \mathbb{E}_{x_0}|X_t^\pi| \leqslant 1 + \sqrt{\mathbb{E}_{x_0} \sup_{t\in[0,T]} |X_t^\pi|^2},$$

which we show is finite in Claim 1 above. □

Claim 3: For every compact $K \subset \mathbb{R}^d$, if ρ is sufficiently large

$$\sup_{h\leq 1} \sup_{x\in K} \int_T^\infty e^{-\rho t} |P_t^h g_\pi(x)|\, dt \to 0 \quad \text{as } T \to \infty,$$

and similarly

$$\sup_{x\in K} \int_T^\infty e^{-\rho t} |P_t g_\pi(x)|\, dt \to 0 \quad \text{as } T \to \infty.$$

Proof of Claim 3. Kolmogorov's backwards equations and linear growth give:

$$\frac{d}{dt}\mathbb{E}_x |X_t^\pi|^2 = \mathbb{E}_x\left[2\langle b_\pi, X_t^\pi\rangle + |\sigma_\pi|^2\right] \lesssim (1 + \mathbb{E}|X_t^\pi|^2).$$

for a constant C not depending on t. Grönwall's implies:

$$\mathbb{E}_x |X_t^\pi| \lesssim (1 + |x|)\, e^{Ct},$$

Hence for $\rho > C$, as g_π has linear growth we see:

$$\sup_{x\in K} \int_T^\infty e^{-\rho t} |P_t g_\pi(x)| \lesssim \sup_{x\in K} (1 + |x|) \cdot \int_T^\infty e^{-(\rho - C)t}\, dt \to 0.$$

By considering the second moments of the discrete increments $X_n^h - X_{n-1}^h$, an identical argument handles the discrete case. □

Using the above claims, we will prove the main result. Fix a compact set $K \subset \mathbb{R}^d$. Then for $x \in K$ and $T < \infty$, we have:

$$V_\pi^h(x) - V_\pi(x) = I_1^h(x,T) + I_2^h(x,T) + I_3^h(x,T),$$

where:

$$I_1^h(x,T) := \int_0^T e^{-\rho h\lfloor t/h \rfloor} \left(P_t^h g_\pi(x) - P_t g_\pi(x)\right) dt,$$

$$I_2^h(x,T) := \int_0^T \left(e^{-\rho h\lfloor t/h\rfloor} - e^{-\rho t}\right) P_t g_\pi(x)\,dt,$$
$$I_3^h(x,T) := \int_T^\infty e^{-\rho h\lfloor t/h\rfloor} P_t^h g_\pi(x)\,dt - \int_T^\infty e^{-\rho t} P_t g_\pi(x)\,dt.$$

This separates $V_\pi^h - V_\pi$ into a semigroup error term I_1^h, a discounting error term I_2^h, and a tail error term I_3^h. We estimate each term uniformly on K.

For the first term, by Claim 1,

$$\sup_{x\in K} |I_1^h(x,T)| \le T \sup_{t\in[0,T]} \sup_{x\in K} |P_t^h g_\pi(x) - P_t g_\pi(x)| \to 0.$$

For the second term, since $0 \le t - h\lfloor t/h\rfloor < h$, we have:

$$|e^{-\rho h\lfloor t/h\rfloor} - e^{-\rho t}| \le (e^{\rho h} - 1)e^{-\rho t}$$

Hence Claim 2 gives:

$$\sup_{x\in K} |I_2^h(x,T)| \le (e^{\rho h} - 1)\cdot \sup_{x\in K} \int_0^T e^{-\rho t}|P_t g_\pi(x)|\,dt \to 0.$$

For the tail term, Claim 3 gives:

$$\sup_{h\le 1}\sup_{x\in K} |I_3^h(x,T)| \to 0 \qquad \text{as } T\to\infty.$$

Now let $\varepsilon > 0$. Choose T so large that:

$$\sup_{h\le 1}\sup_{x\in K} |I_3^h(x,T)| < \varepsilon.$$

Then choose h sufficiently small so that

$$\sup_{x\in K} |I_1^h(x,T)| + \sup_{x\in K} |I_2^h(x,T)| < \varepsilon.$$

It follows that

$$\sup_{x\in K} |V_\pi^h(x) - V_\pi(x)| < 2\varepsilon$$

for all sufficiently small h, which proves the result. □

A standard localization argument gives the following:

Corollary 5.3.3. *Suppose b, σ, g_π, and π are only locally Lipschtiz. If we additionally assume the global growth conditions:*

(i) $|b(x,u)| + |\sigma(x,u)| \leqslant C(1 + |u| + |x|)$.

(ii) $\int_U |u|^2 \, \pi(u \mid x) \leqslant C(1 + |x|^2)$.

(iii) $|g_\pi(x)| \leqslant C(1 + |x|^2)$.

then the conclusion of Lemma 5.3.2 still holds.

Proof. Fix a compact set K. For $R > \max_{x \in K} |x|$, let:

$$\tau_R := \inf \{t > 0 \; : \; |X_t^\pi| \geqslant R\}$$

be the first exit time of the ball of radius R. Analogously, define τ_R^h as the first exit time for the discrete chain. Our uniform L^2 bounds derived in Claim 1 above imply:

$$\mathbb{P}(\tau_R^h \leqslant T) = \mathbb{P}\left(\sup_{t \in [0,T]} |Y_t^h|^2 \geqslant R \right) \leqslant \frac{\mathbb{E}_x \sup_{t \in [0,T]} |Y_t^h|^2}{R^2} \lesssim \frac{e^{CT}}{R^2}.$$

uniformly in h. An analogous statement holds for τ_R. Moreover, the constants in these bounds are independent of the Lipschitz constants and T; they only depend on the growth conditions and K. This suggests we can reasonably truncate our process without too much error, and apply Lemma 5.3.2 to the truncated processes.

More formally, let $\varphi_R : \mathbb{R}^d \to \overline{B}(0, R)$ be the orthogonal projection onto the closed ball of radius R. Define new coefficients and policy:

$$b_R(x, u) := b(\varphi_R(x), u), \quad \sigma_R(x, u) := \sigma(\varphi_R(x), u), \quad \pi_R(\cdot \mid x) := \pi(\cdot \mid \varphi_R(x)).$$

For fixed R, these are globally Lipschitz, and satisfy all the conditions of Lemma 5.3.2. Let $V_\pi^{h,R}$ and V_π^R denote the value functions corresponding to these new coefficients/policy for the discrete and continuous processes respectively. It suffices then to show that the truncation error tends to zero uniformly in h, x as $R \to \infty$:

$$\sup_{h \leqslant 1} \sup_{x \in K} |V_\pi^h(x) - V_\pi^{h,R}(x)| \to 0, \quad \sup_{x \in K} |V_\pi(x) - V_\pi^R(x)| \to 0$$

Note by the growth bound on g_π and Cauchy-Schwarz:

$$\begin{aligned} |V_\pi(x) - V_\pi^R|^2 &\lesssim \left(\int_0^\infty e^{-\rho t} \mathbb{E}_x \left[(1 + |X_t^\pi|^2) \cdot 1\{\tau_R \leqslant t\} \right] \, dt \right)^2 \\ &\lesssim \int_0^\infty e^{-2\rho t} (1 + \mathbb{E}_x |X_t^\pi|^4) \, dt \cdot \int_0^\infty \mathbb{P}(\tau_R \leqslant t) \, dt \\ &\lesssim \frac{(1 + |x|^2)(1 + |x|^4)}{R^2} \int_0^\infty e^{-(\rho - C)t} dt \end{aligned}$$

todo; we can bound the first using Kolmogorovs backwards equations as before, and the second we bound using our above statement. Analogously, we can bound the discrete chain. ∎

5.3.3 Convergence of Optimal Policies

Now, we handle the convergence of the relevant optimal policies.

For the discrete chain X_n^h, if the optimal value function $V^h \in C^2$, it satisfies:

$$
\begin{aligned}
Q_\gamma^h(x,u) &= h\ell(x,u) + e^{-\rho h}\mathbb{E}_\xi V_\pi^h(x + hb(x,u) + \sqrt{h}\sigma(x,u)\xi), \\
&= V^h + h\left[\ell + V_x^h \cdot b + \frac{\sigma^2}{2} \cdot V_{xx}^h - \rho V^h\right] + o(h).
\end{aligned}
$$

Moreover, Bellman's equation (5.1.2) then implies:

$$
\rho V^h(x) = \inf_{\mu \in \mathcal{P}(U)} \left\{ \int_U \left[\ell + V_x^h \cdot b + \frac{\sigma^2}{2} \cdot V_{xx}^h + \tau \log\left(\frac{d\mu}{d\nu}\right) + o(1)\right] \mu(du) \right\}.
$$

Since we have uniform convergence of these to the limiting value function, some simple analysis of the discrete Bellman equation (5.1.2) suggests the form of the limiting Bellman equation. Namely:

$$
\rho V(x) = \inf_{\mu \in \mathcal{P}(U)} \left\{ \int_U \left[\ell + V_x \cdot b + \frac{\sigma^2}{2} \cdot V_{xx} + \tau \log\left(\frac{d\mu}{d\nu}\right)\right] \mu(du) \right\}.
$$

which recovers equation 14 in [65].

Lemma 5.3.4 (Convergence of Optimal Policy)**.** *Under some conditions we have pointwise convergence of the optimal policies:*

$$
\pi^h(u \mid x)\, d\nu = Z^h(x)^{-1} \exp\left(-\frac{Q_\gamma^h(x,u)}{h\tau}\right) \to Z(x)^{-1} \exp\left(-\frac{Q_\gamma(x,u)}{\tau}\right)\, d\nu = \pi^*(u \mid x)
$$

Proof. □

5.3.4 Linear-Quadratic Optimal Control

We should show the conditions/assumptions we make above naturally hold under the simple case of the LQR. Then we should just briefly review the results of [65] in terms of what the limit looks like, and what our corresponding discrete approximations look like.

Zack comment: Can we actually compute what the optimal policy is in the discrete chain case?

5.4 Appendix

We start by proving the optimal policy for the entropy-regularized control problem takes the form (5.1.3). As before, let:

$$Z(x) := \int_U \exp\left(-\frac{Q_\gamma(x,u)}{\tau}\right)\, d\nu(u)$$

denote the partition function.

Lemma 5.4.1 (Optimal Policy for Entropy-Regularized Control)**.** *If* $0 < Z(x) < \infty$ *then:*

$$\pi^*(u \mid x)\, d\nu = Z(x)^{-1} \exp\left(-\frac{Q_\gamma(x,u)}{\tau}\right), \quad V(x) = -\tau \log(Z(x)).$$

is the optimal policy for the entropy-regularized Bellman equation (5.1.2)*:*

$$L(\mu) := \int_U \left[Q_\gamma(x,u) + \tau \log\left(\frac{d\mu}{d\nu}\right)\right]\, d\mu.$$

Proof. Suppose $\mu \in \mathcal{P}(U)$ is any other policy. We can assume $\mu \ll \nu$ as otherwise $KL(\mu \mid \nu) = \infty$, so clearly μ cannot be optimal. Note:

$$Q_\gamma(x,u) = -\tau \log(Z(x)) - \tau \log\left(\frac{d\pi^*}{d\nu}\right), \quad L(\pi^*(\cdot \mid x)) = -\tau \log(Z(x)).$$

Hence:

$$\begin{aligned} L(\mu) - L(\pi^*(\cdot \mid x)) &= \int_U \left[Q_\gamma(x,u) + \tau \log\left(\frac{d\mu}{d\nu}\right)\right]\, \mu(du) + \tau \log(Z(x)) \\ &= \tau \int_U \left[\log\left(\frac{d\mu}{d\nu}\right) - \log\left(\frac{d\pi^*}{d\nu}\right)\right]\, \mu(du) \\ &= \tau KL(\mu \mid \pi^*(\cdot \mid x)) \\ &\geqslant 0 \end{aligned}$$

where the above is infinite if μ is not absolutely continuous with respect to $\pi^*(\cdot \mid x)$. Hence $\pi^*(\cdot \mid x)$ is optimal. $\square$

Chapter 6

Exchangeable Array Processes

Abstract

todo

6.1 Introduction

The goal of this paper is to give a framework for building large combinatorial objects in a "consistent" way and to attempt to extend the work of [2] which sought, among other aims, to develop a general theory of limits of combinatorial objects.

say more about why this may be useful/interesting, and the relationship between exchangeability and sampling.

Given a transient Markov chain $(X_n)_{n\in\mathbb{N}}$ with a countable state space S along with a base state $x_0 \in S$ from which the chain is irreducible, some of the goals of Martin boundary theory are:

(i) Complete S into a compact metric space by adding "points at ∞" to which the Markov chain is converging.

(ii) Characterize all the nonnegative/bounded harmonic functions $h : S \to [0, \infty)$.

(iii) Develop a notion of conditioning the chain "at infinity" and understanding the behavior of these conditioned chains.

Since the chain X_n is transient and discrete, it has no natural notion of convergence within S. Endowing it with a metric structure allows us to discuss convergence in a more topological sense, and compactness gives us the suitable regularity to ensure such convergence often exists. A review of the Doob-Martin boundary can be found in Appendix 6.6.

6.2 Exchangeable Array Processes

Given a set A let $A^{(d)}$ denote the set of all subsets of size d of A. A *d-dimensional (jointly) exchangeable array* is a collection of random variables $(X_e)_{e\in\mathbb{N}^{(d)}}$ indexed by $d-subsets$ of $\mathbb{N}$ whose distribution is invariant under simultaneous permutation of the coordinates. Namely, letting $Sym(\mathbb{N})$ be the set of finite permutations of $\mathbb{N}$ then

$$(X_{\sigma(e)})_{e\in\mathbb{N}^{(d)}} \overset{\text{law}}{=} (X_e)_{e\in\mathbb{N}^{(d)}}, \quad \forall\sigma \in \text{Sym}(\mathbb{N}),$$

where $\sigma(e) = (\sigma(i_1), ..., \sigma(i_d))$ for $e = (i_1, ..., i_d)$.

If $(X_e)_{e\in\mathbb{N}^{(d)}}$ is an exchangeable array there are two natural Markov processes associated to it: a "labeled" chain L_n and an "unlabeled" one W_n. Informally, the labeled chain L_n is produced by just revealing the entries with maximal coordinate n at step n. Namely, the finite labeled chain L_n is just the "top left corner" of the array: the restriction of the array to entries corresponding $[n]^{(d)}$. In this framework, the full array (X_e) plays the role of the projective limit of the labeled chain which often appears in Doob Martin theory.

The unlabeled chain M_n is produced from the labeled one by "forgetting the order" of the rows, columns, et cetera and just keeping track of the set of values appearing in each row, column, and so on. More formally:

Definition 6.2.1 (Exchangeable Array Processes)**.** Given an exchangeable array $(X_e)_{e\in\mathbb{N}^{(d)}}$ with entries in some Borel space S, define a graded Markov chain L_n whose state space is $E := \bigcup_{n\in\mathbb{N}} E_n$, where $E_n := S^{[n]^{(d)}}$ by:

$$(L_n)_e := X_e \quad \forall e \in [n]^{(d)}.$$

Define an equivalence relation $\sim$ on E_n by saying $A \sim B$ if there exists some $\sigma \in Sym([n])$ so that $A_e = B_{\sigma(e)}$ for all $e \in [n]^{(d)}$, where $\sigma(e) = (\sigma(i_1), \sigma(i_2), ..., \sigma(i_k))$ just permutes all elements of the subset e by the same permutation. With this, define the unlabeled chain:

$$M_n := [L_n]_\sim.$$

a Markov chain on the space of equivalence classes of finite $d - dimensional$ arrays.

In this paper, we will almost exclusively think of S as countable, and often finite. Note that L_n is trivially a Markov process in the sense that any sequence of random variables becomes a Markov process if the state space is just the entire history of the process. In general, projections of Markov chains are not Markovian, but in this case, exchangeability ensures that the Markov property is preserved.

Lemma 6.2.2 (Markov Property of the Unlabeled Chain)**.** *When the state space S is countable, the process M_n is a Markov chain.*

The above lemma is true for any initial distribution, but we are almost exclusively interested in the case M_0 is the empty array. The main benefit of this framework is that exchangeable arrays share a strong structural property: they have a representation through sampling from some hypercube.

Theorem 6.2.3 (Representation Theorem)**.** *An exchangeable array on d-subsets of* $\mathbb{N}$ *has a representation of the form:*

$$\{X_e\}_{e\in\mathbb{N}^{(d)}} =^d \left\{ f\left(U_\emptyset, (U_i)_{i\in e}, (U_a)_{a\in\binom{e}{2}}, ..., (U_a)_{a\in\binom{e}{d-1}}, U_e\right)\right\}_{e\in\mathbb{N}^{(d)}},$$

for some middle symmetric $f : [0,1]^{2^d} \to S$ *and iid uniforms. Moreover, any array of this form is exchangeable.*

A middle-symmetric function satisfy the following for any $\sigma \in Sym([d])$:

$$f\left(x_\emptyset, (x_i)_{i\in[d]}, (x_a)_{a\in[d]^{(2)}}, ..., x_{[d]})\right) = f\left(x_\emptyset, (x_{\sigma(i)})_{i\in[d]}, (x_{\sigma(a)})_{a\in[d]^{(2)}}, ..., x_{[d]})\right).$$

In words, you are allowed to apply the same permutation to the indices "at each level" separately. More details can be found in [3] or Chapter 7 of [32].

Throughout this paper, we will encounter many domains of the form $[0,1]^{2^k}$ for some k. For ease of notation, we will assume the coordinates are always indexed by subsets of $[k]$ as in the definition of a middle symmetric function above.

In the case of one dimensional arrays, this representation theorem reduces to the standard De Finetti's theorem which says an exchangeable sequence is a mixture of iid sequences and thus takes the form $(X_n) =^d (f(U, U_n))$. Moreover, two-dimensional arrays take the form $f(U, U_i, U_j, U_{ij})$ for some function f with $f(w, x, y, z) = f(w, y, x, z)$.

As [2] points out, this quantifies the fundamental relationship between exchangeability and sampling that is already so well known: if we can get a hold of the representing function f, it gives us a simple procedure for generating the corresponding combinatorial objects. Next, we now point out a simple way of generating new exchangeable arrays from old ones.

Lemma 6.2.4 (State-Space Projections of Exchangeable Arrays)**.** *If* (X_e) *is a exchangeable array with entries in* S *and* $g : S \to S'$ *is measurable then* $(g(X_e))$ *is an exchangeable array of the same dimension. Moreover, if* f *is the representing function of* (X_e)*, then* $g \circ f$ *is the representing function of* $(g(X_e))$*.*

Here we are essentially defining an equivalence relation on the entries of the array where we let $s \sim s'$ if $g(s) = g(s')$. We call this the *state-space equivalence.*

Combined with definition 6.2.1, this shows that there are essentially three layers of Markov processes associated to a given exchangeable array: (i) the initial "labeled" exchangeable array process corresponding to (X_e), then (ii) the "labeled" array process corresponding to $(g(X_e))$ where we apply an equivalence to the state space and lastly (iii) the "unlabeled" chain where we apply an equivalence relation to the coordinates of the array. See the example below on ballot sequences for an example where all three layers are utilized.

Example 6.2.5 (Polya's Urn)**.** In the case $d = 1$, an exchangeable array is just an exchangeable sequence. It is well-known the sequence of balls drawn from a Polya's urn is an exchangeable sequence. In this case, we can think of X_i of taking values in $[m]$ if we have m colors. The labeled chain L_n is the sequence of balls, and the unlabeled chain W_n just keeps track of the counts of balls of each color. This has representation function

$$f(x,y) = \sum_{i=1}^{m} i \cdot \mathbf{1}\left\{y \in [t_{i-1}, t_i)\right\} \qquad t_i := \sum_{k=1}^{i} p_k(x),$$

where $(p_1(U), ..., p_m(U)) \sim Dirchlet(\alpha_1, ..., \alpha_m)$ for α_i initial balls of color i.

Example 6.2.6 (Graphon Chain)**.** Given any graphon w there is a sequence of finite graphs G_n converging to w in the cut metric (equivalently the Doob-Martin topology by [26]) constructed via sampling $U_i \sim Unif[0,1]$ and then connecting vertex i to vertex j with probability $w(U_i, U_j)$. Thus a graphon can be represented as an "infinite adjacency matrix" which forms an exchangeable array with entries

$$X_e = \mathbf{1}\{U_e \leqslant w(U_i, U_j)\} \quad \text{ for } e = (i,j) \in \binom{\mathbb{N}}{2},$$

Here the representation function $f(x_\emptyset, x_i, x_j, x_{ij})$ does not depend on $x_\emptyset$.

Example 6.2.7 (d-Uniform Hypergraph Chain)**.** We can extend the graphon example to d-uniform hypergraphs, which have hyperedges between subsets of $[n]$ of size d, initially explored in [17]. Namely, define a *hypergraphon* as a middle-symmetric function $W : [0,1]^{2^d-2} \to [0,1]$ with coordinates indexed by proper, nonempty subsets of $[d]$. Then, analogous to graphons, this has representation function:

$$f\left(x_\emptyset, (x_i)_{i\in[d]}, ..., x_{[d]}\right) = \mathbf{1}\Big\{x_{[d]} \leqslant W\left((x_i)_{i\in[d]}, ..., (x_a)_{a\in[d]^{(d-1)}}\right)\Big\}.$$

As we will see in Corollary 6.5.5 below, these are in a sense the only binary array processes.

Example 6.2.8 (Ballot Sequences)**.** Given some total ordering of $I_0 := \bigcup_{i\in\mathbb{N}}\{a_i, b_i\}$ so $a_i < b_i$ for each i, there are are 6 possible "interweavings" of the letters:

$$a_i b_i a_j b_j \quad a_i a_j b_i b_j \quad a_i a_j b_j b_i \quad a_j a_i b_i b_j \quad a_j a_i b_j b_i \quad a_j b_j a_i b_i.$$

This ordering can be encoded by a 2-dimensional array X_{ij} with values in $\{1, ..., 6\}$ describing which of these relationships the pairs $(a_i, b_i), (a_j, b_j)$ share. If we equate the interweavings above which have the $a's$ and $b's$ in the same locations, the unlabeled chain generates *ballot sequences*: strings of $a's$ and $b's$ where every prefix has at least as many $a's$ as $b's$. In [13], the boundary is characterized through a pair of diffuse marginals (ζ, μ) encoding the relative density of $a's$ and $b's$ in the limit.

Exchangeable array processes also arise naturally in the the theory of exchangeable partitions through Kingman's paintbox theorem [3], random permutations through permutons [25], planar binary trees through Remy's chain [19], and growing random strings [12].

These and the preceeding examples give a common strategy for embedding combinatorial processes as random arrays. If a random structure consists of certain "parts" (e.g. vertices of a graph) and is completely determined by the relationships within each subset of d parts (e.g. which vertices are connected by an edge), then it can be encoded as a d-dimensional array whose entries encode the relationship. Exchangeability then enforces the constraint that the distribution of the structure is invariant under relabeling of the parts.

6.3 Doob-Martin Boundary of Exchangeable Array Processes

The backwards dynamics of the labeled chain are deterministic: given L_n you simply restrict to $e \in [n-1]^{(d)}$ to obtain L_{n-1}. Thus, its Doob-Martin boundary is almost trivial.

Lemma 6.3.1 (Boundary of the Labeled Chain)**.** *The boundary of L_n consists of the deterministic arrays $x \in \mathrm{supp}(X_e)$.*

Next, we discuss how state-space equivalence and unlabeling affect this boundary.

6.3.1 State-Space Equivalence

Given the exchangeable array (X_e) and measurable function $g : S \to S'$, we can generate a new exchangeable array $(g(X_e))$. By Lemma 6.3.1 the boundary of the corresponding labeled array processes consists of deterministic arrays on the relevant state spaces.

Zack comment: We know from the previous section the boundary of $g(X_e)$ is again just the deterministic arrays (y_e). But I want to relate this to the boundary of X_e in a nice way. The preimage $g^{-1}((y_e))$ of a deterministic array includes random arrays (X_e) so that $supp(X_e) \subseteq g^{-1}(y_e)$. Thus we get a whole collection of bridges of (X_e), some of which are extremal and some of which are mixtures of these extremal ones. I should go back to the ballot chain example to see how to possibly interpret this as a measure on the boundary of the labeled chain.

6.3.2 Boundary of the Unlabeled Chain

Now we seek to understand the boundary of the unlabeled chain M_n. Recall the correspondence (6.6.6) between infinite bridges of a Markov chain and its nonnegative harmonic functions via the Doob h-transform.

Lemma 6.3.2 (Exchangeable Bridges of the Labeled Chain). *Any infinite bridge of the labeled exchangeable array process is an array process, and it is exchangeable if and only if the corresponding harmonic function h is constant on the equivalence classes of $\sim$.*

An immediate consequence is that the harmonic functions of the unlabeled chain are in direct correspondence with the bridges of labeled chain which come from exchangeable arrays.

Corollary 6.3.3 (Bridges of the Unlabeled Chain). *If $M_n^{\tilde{h}}$ is an infinite bridge of the unlabeled chain M_n, then $M_n^{\tilde{h}}$ equals the unlabeled process associated to some exchangeable array (X_e^h). In particular, if $\tilde{h}$ is the harmonic corresponding to the bridge $M_n^{\tilde{h}}$, then $h(x) := \tilde{h}([x])$ is the harmonic corresponding to the array (X_e^h).*

The relation $\sim$ extends to infinite $d-$*dimensional* infinite arrays x in the obvious way: $x \sim y$ if there is $\sigma \in Sym(\mathbb{N})$ so $x = \sigma(y)$. We say a measurable collection I of arrays is ν*-invariant* if permutations do not change up up to ν-null sets:

$$I \overset{\nu-a.s.}{=} \tau I, \qquad \forall \tau \in Sym(\mathbb{N}).$$

Say an exchangeable array with distribution ν is *ergodic* if $\nu(I) \in \{0,1\}$ for all $\nu-$invariant sets I, and say it is *dissociated* if for any disjoint index sets $I_1, ..., I_m \subset \mathbb{N}$ the arrays $\{X_e^i\}_{e \in I_i^{(d)}}$ are independent. For example, the exchangeable array corresponding to a graphon is disassociated because the random subgraphs corresponding to disjoint collections of vertices are independent in the graphon model since edges are added independently between the selected vertices. The following shows these properties characterize the Doob-Martin boundary of the unlabeled chain.

Theorem 6.3.4 (Extremal Bridges and Ergodic Arrays). *For any infinite bridge of the unlabeled exchangeable array process M_n, the following are equivalent:*

1. *The infinite bridge is extremal.*
2. *Its corresponding exchangeable array is ergodic.*
3. *Its corresponding exchangeable array is dissociated.*
4. *Its corresponding representation function f does not depend on $U_\emptyset$ (almost surely).*

Remark 6.3.5. By Example 6.2.6, given a graphon w the representation function

$$f(q, x, y, u) := \mathbf{1}\{u \leqslant w(x, y)\},$$

of the associated binary exchangeable array process does not depend on q. Hence, Theorem 6.3.4 implies graphons correspond to the extremal infinite bridges, and hence the minimal Doob-Martin boundary, of the graphon chain, as noted in [26].

As a consequence, the representation function of any unlabeled bridge is a mixture of the extremal representation functions $f^{\tilde{K}(\cdot,\alpha)}$, which do not depend on the first coordinate by the previous theorem. More specifically, if we let $g : [0,1] \to \overline{E}$ be a function so that $g(U)$ has distribution $\mu_{\tilde{h}}$ this shows

$$f^{\tilde{h}}(U,(U_i),...,U_e) = f^{g(U)}((U_i),...,U_e)$$

6.4 Embedding Density of the Representation Functions

The idea of embedding a smaller structure into a larger one plays a pivotal role in the theory of combinatorial processes. Two of our principal examples, graphons and permutons, already come equipped with natural notions of embedding: subgraphs and subpermutations respectively. Somehow these embeddings encode the "local" structure of the larger combinatorial object. We can then compare two such objects by comparing these local structures at various different scales.

The aim of this section is to recast these notions in the language of exchangeable arrays, with particular emphasis on binary arrays, and to extend the framework developed in [7]. We begin by introducing some notation.

First, we introduce some notation. For a finite binary d-dimensional array F of size k let

$$\mathbf{1}(F) := \{S \in [k]^{(d)} : F_S = 1\}.$$

denote the collection of entries of our array which are one. Moreover, given a vector $x := (x_e)_{e\subseteq[n]} \in [0,1]^{2^n}$ and a subset $A \in [n]^{(d)}$ denote

$$x_{(A)} := (x_\emptyset, (x_i)_{i\in A}, (x_{ij})_{ij\in A^{(2)}}, \ \ldots \ , x_A) \in [0,1]^{2^d}.$$

Namely, if x is indexed by all subsets of $[n]$ then $x_{(A)}$ restricts to only subsets of A. Lastly, for a finite d-dimensional array of size k let $|x| := k$ denote its size.

6.4.1 Embeddings of Labeled States

Suppose we have arrays $(x_e)_{e\in[n]^{(d)}}$ and $(y_e)_{e\in[n+m]^{(d)}}$. Then we say *x is embedded in y* if $y_e = x_e$ for all $e \in [n]^{(d)}$ and denote this by $x \leqslant y$. Namely, we obtain x_e by looking at the "top left corner" of the array y_e. From the viewpoint of our labeled Markov chain L_n, this just means that the array (y_e) is reachable starting in state (x_e). This extends the standard notion of an induced subgraphs beyond binary arrays.

We say a binary array x is *weakly embedded* in binary array y, denoting this by $x \lesssim y$, if $y_e = 1$ whenever $x_e = 1$. This is the standard notion of a graph embedding.

6.4.2 Embeddings of Unlabeled States

Again suppose we have arrays $(x_e)_{e\in[n]^{(d)}}$ and $(y_e)_{e\in[n+m]^{(d)}}$. Say $[x]_\sim$ is embedded in $[y]_\sim$ if there exists $\sigma \in Sym([n+m])$ so $(x_e) \leqslant (y_{\sigma(e)})$. Equivalently, there exists an injective function $\varphi : [n] \to [n+m]$ so $(x_{\varphi(e)}) = (y_e)$.

In this case, there can be more than one embedding if there are several different choices of σ that produce an embedding. This suggests a notion of *embedding density* analogous to the theory of graphons and permutons.

$$\begin{aligned} emb(x, y) &:= \frac{|\{\sigma \in Sym([n+m]) \ : \ (x_e) \leqslant (y_{\sigma(e)})\}|}{(n+m)!} \\ &= \frac{|\{\varphi : [n] \xrightarrow{injective} [n+m] \ : \ (x_{\varphi(e)}) \leqslant (y_e)\}|}{(n+m)(n+m-1)\cdot \ldots \cdot (n+1)} \end{aligned} \tag{6.4.1}$$

for $x \in E_n, y \in E_{n+m}$. Note that the embedding density is constant on each equivalence class of x so this is well-defined as a function on these equivalence classes. Likewise we can define the *morphism density* by removing the injectivity constraint:

$$mor(x, y) := \frac{|\{\varphi : [n] \to [n+m] \ : \ (x_e) \leqslant (y_{\varphi(e)})\}|}{(n+m)^n}. \tag{6.4.2}$$

Analogously, for binary arrays we can define the *weak embedding/morphism density* emb^W and mor^W respectively for by replacing $\leqslant$ by $\lesssim$. These share the following relationships:

Lemma 6.4.1 (Embedding and Morphism Densities)**.** *For binary arrays x, y let x' be an array of the same size as x. Then:*

(i) $emb(x, y) \leqslant emb^W(x, y)$.

(ii) $emb^W(x, y) = \sum_{x' \geqslant x} emb(x, y)$.

(iii) $emb(x, y) = \sum_{x' \geqslant x} (-1)^{|1_x| - |1_{x'}|} \cdot emb^W(x', y)$.

(iv) $|mor^W(x, y) - emb^W(x, y)| \leqslant \frac{1}{|y|} \cdot \binom{|x|}{2}$.

(v) $|mor(x, y) - emb(x, y)| \leqslant \frac{1}{|y|} \cdot \binom{|x|}{2}$.

Given any notion $t(x, y)$ of the embedding density of x into y, we can topologize our collection of combinatorial objects: we simply say a sequence y_n converges if

$$\lim_{n\to\infty} t(x, y_n) \text{ exists}$$

for every fixed x. Thus Lemma 6.4.1 shows that all these notions of embedding are essentially topologically equivalent. In fact, such convergence is closely related to the Doob-Martin topology.

Lemma 6.4.2 (Doob-Martin Convergence and Embedding Densities). *The following are equivalent:*

(i) The sequence y_N converges in the Doob-Martin topology.

(ii) The embedding density $emb(x, y_N)$ converges for all x.

(iii) The morphism density $mor(x, y_N)$ converges for all x.

For binary arrays, convergence of the weak embedding/morphism density is also equivalent.

In the case of our graphon and permuton chains, the notion of embedding can be extended to the limiting object, graphons and permutons, via sampling. Namely, for a graphon w its embedding density $t(\cdot, w)$ is essentially defined so that for each n and finite graph F

$$t(F, w) = \mathbb{E}t(F, G_n(w)),$$

where $G_n(w)$ is the random graph generated by sampling n uniforms $U_1, ..., U_n$ and connecting vertices i, j with probability $w(U_i, U_j)$.

Analogously, we would like to extend this notion of an embedding density to our limiting objects: the representation functions f of our exchangeable arrays. This suggests

$$emb(x, f) := \mathbb{E}\; emb(x, X_n(f)), \qquad emb^W(x, f) := \mathbb{E}\; emb^W(x, X_n(f)), \tag{6.4.3}$$

where $X_n(f)$ is the array indexed by $[n]^{(d)}$ generated by:

$$(X_n(f))_e := f(U_\emptyset, (U_i)_{i\in e}, ..., U_e).$$

We will now show these expectations are independent of n and have an integral representation analogous to the embedding density of graphons.

Lemma 6.4.3 (Embedding Densities of Representation Functions). *Suppose $d \leqslant k \leqslant n$. If F is a size k array d-dimensional array and f the representing function of an exchangeable array X with then*

$$emb^W(F, f) = \int_{[0,1]^{2^k}} \prod_{S\in \mathbf{1}(F)} f(x_{(S)})\, d\lambda_{2^k} \tag{6.4.4}$$

$$emb(F, f) = \int_{[0,1]^{2^k}} \prod_{S\in [k]^{(d)}} \mathbf{1}\{f(x_{(S)}) = F_S\}\, d\lambda_{2^k}. \tag{6.4.5}$$

In particular, it is independent of n. Moreover we have concentration

$$\mathbb{P}\left(|emb^W(F, X_n(f)) - emb^W(F, f)| > \epsilon\right) \leqslant 2\exp\left(-\frac{\epsilon^2}{2k^2}n\right),$$

$$\mathbb{P}\left(|emb(F, X_n(f)) - emb(F, f)| > \epsilon\right) \leqslant 2\exp\left(-\frac{\epsilon^2}{2k^2}n\right).$$

Thus,

$$\lim_{n\to\infty} mor(F, X_n(f)) = \lim_{n\to\infty} emb(F, X_n(f)) \stackrel{a.s.}{=} emb(F, f),$$

and

$$\lim_{n\to\infty} mor^W(F, X_n(f)) = \lim_{n\to\infty} emb^W(F, X_n(f)) \stackrel{a.s.}{=} emb^W(F, f).$$

In particular, the results of Lemma 6.4.1 (i) - (iii) extend to $emb^W(F, f)$ and $emb(F, f)$.

This is perfectly analogous to the theory of graphons which shows

$$\mathbb{E}(emb(F, G_n(w)) = \int_{[0,1]^k} \prod_{(i,j)\in E_F} w(x_i, x_j)\, dx.$$

To see why, note when $d = 2$

$$\int_{[0,1]^{2^k}} \prod_{S\in\mathbf{1}(F)} f(x_{(S)})\, d\lambda_{2^k} = \int_{[0,1]^{2^k}} \prod_{(i,j)\in\mathbf{1}(F)} f(x_\emptyset, x_i, x_j, x_{ij})\, d\lambda_{2^k}.$$

For graphons the terms inside the product become $\mathbf{1}\{x_{ij} \leqslant w(x_i, x_j)\}$ so we can integrate out the coordinate $x_\emptyset$ and the coordinates x_{ij} (which each only appear in one term of the product) to get the familiar formula.

In general, we can see that the integral in (6.4.4) only relies on the $\sum_{i=0}^{d} \binom{k}{d}$ coordinates corresponding to subsets $e \subseteq k$ of size at most d, so many of the coordinates in the integral are superfluous. We leave them in for simplicity of notation.

6.5 Metrizing Convergence of Exchangeable Array Proceses

Zack comment: I should briefly summarize our results and the work of[70] discussing why our attempts at generalizing the cut metric seems to fail. Summarize our idea to extend to non-binary arrays via our projection method.

As Lemma 6.4.2 and [26] show, graphons and the Doob-Martin theory are topologically equivalent. While the latter is trivially metrizable as it is essentially just a Euclidean space, this metric is not really interpretable in terms of the underlying combinatorial process. The real success of the theory of graphons was the introduction of a metric, the cut metric, which is reasonably interpretable, explicitly encodes graph-theoretic properties, and allows for exact computation of many quantites of interest.

Firstly we will develop a metric for binary arrays and latter we will discuss how it can be extended to any exchangeable array process. As discussed above, d-dimensional binary arrays are essentially just d-uniform hypergraph chains. Thus we can view this first section as extending the cut metric to d-uniform hypergraphs and hypergraphons.

We saw in the above example that the graphon w of a graphon chain is closely related to the representation function f of its associated exchangeable array. Namely:

$$f(q, x, y, u) = \mathbf{1}\{u \leqslant w(x, y)\}.$$

Hence we hope to reinterpret the cut metric in terms of a metric on these representation functions f, and from there find a way to extend this metric to arbitrary exchangeable array processes. First, a simple observation.

$$\begin{aligned}\delta_{\square}(w, w') &:= \sup_{S,T \subseteq [0,1]} \left| \int_{S\times T} w(x, y) - w'(x, y)\, dx\, dy \right| \\ &= \sup_{R,S,T\subseteq[0,1]} \left| \int_{R\times S\times T} \int_0^1 f(q, x, y, u) - f'(q, x, y, u)\, du\, dy\, dx\, dq \right|.\end{aligned}$$

This suggests a natural generalization to our exchangeable array processes. Given representation function f, f' of d-dimensional exchangeable arrays define the following:

$$\delta(f, f') := \sup_{T_e\subseteq[0,1]} \left| \int_{\prod_e T_e} \int_0^1 f - f' \right|.$$

where the supremum is over proper subsets $e \subset [d]$.

Daniel comment: So far, it looks like maybe this should be the supremum over subsets of $[0, 1]^{2^d-1}$, rather than products of subsets of $[0, 1]$.

The asymmetry in the top coordinate may seem strange at first glance. We will show that, in fact, the topology is the same as the topology generated by the following symmetric metric:

$$\begin{aligned}\delta_s(f, f') &:= \sup_{T\subseteq[0,1]^{2^d}} \left| \int_T f - f' \right| \\ &= \delta_s(f, f') = \sup_{0\leqslant g\leqslant 1} \left| \int_{[0,1]^{2^d}} g(f - f') \right|.\end{aligned}$$

where the second equality is because indicator functions are the extreme points of $\{g \in L^2([0, 1]^{2^d}) : 0 \leqslant g \leqslant 1\}$.

Daniel comment: Can we slacken the requirement on the set T using the middle symmetry of f? I don't think we can in general say that it's a product of subsets of $[0, 1]$, but we can probably say *something*.

Zack comment: This makes sense for binary arrays certainly, or if the state space is like $[0, 1]$. But for our isomorphism class state spaces we would have to be a bit more careful. Sure we could label the classes as like $1, 2, ...$ and just integerate this, but we would want the distance to be invariant under different choices of the labeling scheme. Maybe a better idea is to use some sort of hamming metric where we instead integrate $\mathbf{1}\{f \neq f'\}$. The problem with this is that it is unsigned (e.g. with graphons the integrand can be both positive or negative) so the maximizing subsets will just be all of $[0, 1]$. This wouldn't quite reduce to the cut metric. Again you could try to associate some sort of signed hamming metric but it is not clear how to do so in a way that is invariant under relabelings of the states.

Zack comment: Still not quite sure this is the proper metric. We originally wanted to define the supremum over every integrand, but in the case of grahpons it does not reduce to the standard cut metric in this case. I gave the counterexample of the two grahphons $w \equiv 0.5$ and $w' = \mathbf{1}_{[0,\epsilon]^2}$ which is maximized on $[0, 1]^3 \times [0, 0.5]$

Zack comment: Just like how graphons are really equivalence classes modulo Lebesgue isomorphisms, we will have to think of our representation functions f in the same way, possibly where we can permute each "part" of the representation by a different Lebesgue isomorphism.

Another natural class of metrics is of course just the L^p norms. Namely:

$$\delta_p(f, f') := \left(\int_{\mathbb{R}^{2^k}} |\rho(f, f')|^p \right)^{1/p}.$$

where $\rho(f, f') = |f - f'|$ for $\mathbb{R}$-valued functions, or a hamming metric for isomorphism classes. Let's proceed with some examples.

Example 6.5.1 (Perfectly Dependent)**.** In the case every element of the array is just the same random variable, the representation function $f(U, U_i, ...)$ just takes the form $f(U)$ for some $f : [0, 1] \to S$. Thus, the metric reduces to:

$$\delta(f, f') = \sup_{T \subseteq [0,1]} \left| \int_T f(x) - f'(x)\, dx \right|.$$

Zack comment: δ seems analogous to sort of a TV distance with respect to the measures $f\, dx, f'\, dx$ but these are not probability measures in general.

Example 6.5.2 (IID Entries)**.** If the array has iid entries, its representation takes the form $f(U, U_i, ..., U_e) = f(U_e)$ for some $f : [0, 1] \to S$. In this case, the metric reduces to:

$$\delta(f, f') = \left| \int_0^1 f(x) - f'(x)\, dx \right|.$$

Zack comment: In the case of two Bernoulli random variables with success probabilities p, q this just reduces to $|p - q|$

Example 6.5.3 (Polya's Urn)**.** In the two-color case starting with n, m balls of each color we have $f(x, y) = \mathbf{1}\{y \leqslant f(x)\}$ where f is the quantile function of $Beta(n, m)$. Then the metric reduces to:

$$\delta(f, f') = \sup_{T \subseteq [0,1]} \left| \int_T f(x) - f'(x)\, dx \right|.$$

analogous to the perfectly dependent case.

For all three previous examples, our L^p norm reduces to:

$$\delta_p(f, f') = \left(\int^1 |f(x) - f'(x)|^p\, dx \right)^{1/p}.$$

In the case $S = \mathbb{R}$ then f is just the quantile function of the distribution. Thus, δ_p reduces to the p-Wasserstein distance.

Of course, the representation f of a given exchangeable array need not be unique. This is even apparent in the theory of graphons, which are defined as isomorphism classes of functions $w : [0,1]^2 \to [0,1]$ modulo Lebesgue isomorphisms: $w(\cdot,\cdot) \sim w(\lambda(\cdot), \lambda(\cdot))$. The idea is that applying a Lebesgue isomorphism to a uniform does not change its distribution, and thus will not influence the overall sampling distribution. More generally, the representation functions of exchangeable arrays have a similar uniqueness result "up to Lebesgue preserving maps".

The slight technicality is that we have to allow such Lebesgue preserving maps to depend on the preceding coordinates. Namely, say a map $T : [0,1]^{2^d} \to [0,1]$ *preserves λ in the last coordinate* if for fixed $x_,(x_i)_{i\in[d]}, ..., (x_e)_{e\in[d]^{(d-1)}}$ the map:

$$y \mapsto T(x_,(x_i)_{i\in[d]}, ..., (x_e)_{e\in[d]^{(d-1)}}, y),$$

preserves the Lebesgue measure.

For a proof of the following, see Theorem 7.28 in [32], which proves a more general statement that does not restrict the array to a finite dimension d. For something easier to parse, the case $d = 2$ can also be found in Theorem 2.1' of [31].

Theorem 6.5.4 (Uniqueness of Representation Functions)**.** *The following are equivalent:*

1. *$f, g : [0,1]^{2^d} \to S$ generate the same exchangeable array (in distribution).*
2. *There exists measurable functions $T_i, T_i' : [0,1]^{2^i} \to [0,1]$ for $i = 0, ..., d$ which preserve λ in the last coordinate so that a.s.:*

$$f\Big(T_0(x_\emptyset), (T_1(x_\emptyset, x_i))_{i\in[d]}, \ldots, T_d(x_\emptyset, (x_i)_{i\in[d]}, \ldots, x_{[d]})\Big)$$
$$= g\Big(T_0'(x_\emptyset), (T_1'(x_\emptyset, x_i))_{i\in[d]}, \ldots, T_d'(x_\emptyset, (x_i)_{i\in[d]}, \ldots, x_{[d]})\Big).$$

An immediate corollary of Theorem 6.5.4 is that, in a sense, the d-uniform hypergraph chain mentioned in the previous examples is the *only* binary exchangeable array process.

Corollary 6.5.5 (Binary Representation Functions)**.** *If $f : [0,1]^{2^d} \to \mathbb{Z}_2$ is the representation function of a binary exchangeable array, then up to the uniqueness of Theorem 6.5.4, the function f is of the form:*

$$f = \mathbf{1}\Big\{x_{[d]} \leqslant g\left(x_\emptyset, (x_i)_{i\in[d]}, ..., (x_e)_{e\in[d]^{(d-1)}}\right)\Big\}.$$

Again, we are primarily interested in the extremal representation functions, which by Theorem 6.3.4, are the ones depending trivially on $x_\emptyset$. Hence in the $d = 2$ case, for extremal representation functions we exactly recover the uniqueness statement for graphons discussed above.

Two arrays generated by equivalent representation functions have the same distribution, and any worthwhile metric on such representation functions ought to assign distance zero between such functions. This suggests if all we care about is the sampling distribution then, just as is done with graphons, we ought to equate all functions whose arrays have the same law and instead work with equivalence classes of representation functions. We formalize this with the following:

$$d(f, f') := \inf_{g\sim f, g'\sim f'} \delta(g, g') \quad = \inf_{g\sim f} \delta(g, f').$$

As with the theory of graphons, the notion of convergence induced by this metric is the same as that of the embedding densities as well as the Doob-Martin topology.

Theorem 6.5.6 (Convergence via the Cut Metric)**.** *For representation functions g_n, g, the following are equivalent.*

(i) $emb^W(F, g_n) \to emb^W(F, g)$ *as* $n \to \infty$ *for all finite arrays* F*.*

(ii) $\delta(g_n, g) \to 0$ *as* $n \to \infty$*.*

(iii) $\delta_s(g_n, g) \to 0$ *as* $n \to \infty$*.*

Daniel comment: I'll write down my attempt at proving the equivalence here:

First, we show that the embedding densities are Lipschitz continuous with respect to the cut metric, and thus that convergence in the cut metric implies convergence in the embedding densities. This shows the cut topology is finer than that of the embedding densities. Many of these proofs are direct extensions of those in [43] and [7].

Lemma 6.5.7 (Cut Metric Controls Embedding Densities)**.** *Let F be a finite, binary d-array of size $k \geqslant d$, and let $f, f' : [0,1]^{2^d} \to \{0,1\}$ be middle symmetric functions. Then*

$$|emb^W(F, f) - emb^W(F, f')| \leqslant |\mathbf{1}(F)|\delta(f, f') \leqslant |\mathbf{1}(F)|\delta_s(f, f').$$

The next step is to show that convergence in the embedding densities implies convergence in the cut metric . This follows the same proof strategy as the analogous theorem for graphons. Namely, there are a few layers of approximations:

1. Approximate the representation function f by a large finite (random) array $X_n(f)$ sampled from it.

2. Show when the embedding densities of f, g are close, the sampled arrays are close as well.

3. Approximate a representation function f by a "step function" with few parts.

The first step requires us to uniquely associate a specific representation function f_x to every finite array x. This allows us to embed our finite arrays in our space of representation functions. A natural candidate is just a continuous version of the hypergraph adjacency matrix, analogous to how we associate a graphon to a finite graph:

$$f_x(u_,(u_i)_{i\in[d]}, ..., u_{[d]}) := \mathbf{1}\left\{(\lceil |x| \cdot u_1 \rceil, ..., \lceil |x| \cdot u_d \rceil) \in edges(x)\right\}.$$

We call f_x the *sampling function* associated to a finite array x. For this embedding to respect the topology of our embedding densities, we require $emb^W(F, f_x) = emb^W(F, x)$. Namely, the value of the (weak) embedding density does not change whether we view x as an array or as its corresponding representation function. It is easy to verify that f_x satisfies this constraint:

$$\begin{aligned}
emb^W(F, f_x) &:= \int_{[0,1]^{2^k}} \prod_{S\in 1(F)} f_x(y_{(S)})\, d\lambda_{2^k}(y) \\
&:= \int_{[0,1]^d} \prod_{S\in 1(F)} \mathbf{1}\left\{(\lceil |x| \cdot y_{s_1} \rceil, ..., \lceil |x| \cdot y_{s_d} \rceil) \in edges(x)\right\}\ d\lambda_d(y) \\
&:= \int_{[0,1]^d} \mathbf{1}\left\{(i \mapsto \lceil |x| \cdot y_i \rceil) \in emb(F, x)\right\}\ d\lambda_d(y) \\
&= emb^W(F, x).
\end{aligned}$$

This gives us the desired embedding. We will come back to it later.

The following lemma and corollary handle step two above by relating the distribution of our sampled arrays to the embedding densities.

Lemma 6.5.8 (Sampling Probability Formula)**.** *Let A_n be a finite, binary array of size n sampled from f. Then*

$$\mathbb{P}(A_n = x) = \frac{n!}{|\operatorname{Aut}(x)|^2} \sum_{x' \geqslant x} (-1)^{|1_x| - |1_{x'}|} emb^W(x', f) = \frac{n!}{|\operatorname{Aut}(x)|^2} \cdot emb(x, f).$$

The next result shows that if the weak embedding densities of two representation functions are close, the finite sampling distributions are close as well.

Corollary 6.5.9 (Total Variation Bound for Samples)**.** *Let A_n, A'_n be finite, binary d-arrays of size n sampled from f, f', respectively. Then*

$$\| \operatorname{Law}(A_n) - \operatorname{Law}(A'_n)\|_{\mathrm{TV}} \leqslant \frac{n!}{2} \sum_{|x|=n} \sum_{x' \geqslant x} |emb^W(x', f) - emb^W(x', f')|,$$

where this sum is over pairs of arrays of size n.

Lemma 6.5.10 (Coupling of Samples)**.** *Let $(A_n)_n, (A'_n)_n$ be sequences of finite arrays sampled from f, f', and let $f_n = f_{A_n}, f'_n = f_{A'_n}$ be the associated sampling functions. Then there exists a coupling of A_n, A'_n such that*

$$\mathbb{E}[\delta_s(f_n, f'_n)] \leqslant \frac{n!}{2} 3^{\binom{n}{d}} \max_{|x|=n} |emb^W(x, f) - emb^W(x, f')|.$$

This completes the approximation specified in part two above. We now state a technical lemma which is at the heart of parts one and three, postponing its proof until later.

Lemma 6.5.11 (Approximation by Sampled Arrays)**.** *Let $(A_n)_n$ be a sequence of finite arrays of size n sampled from f, and let $f_n = f_{A_n}$ be the associated sampling functions. Then*

$$\mathbb{E}[\delta_s(f_n, f)] \leqslant \frac{5}{\sqrt{\log_2 n}}.$$

Daniel comment: This proof will require some more work, like a version of the graph regularity lemma.

Now we are ready to prove the main result. Note that no part of the proof above utilizes the specific definition of the cut norm or the sampling function associated to a finite array. All that is left now is to prove Lemma 6.5.11.

Daniel comment: Here, I'll collect the ingredients and arguments needed to prove Lemma 6.5.11. We can decide later how we want to organize all these results.

Given a partition $\mathcal{P}$ of $[0, 1]$, define the grid σ-algebra on $[0, 1]^d$ by $\mathcal{G}_{\mathcal{P}} := \sigma(\mathcal{P})^{\otimes d}$. Martingale convergence tells us that $\mathbb{E}[f \mid \mathcal{G}_{\mathcal{P}}] \to f$ as $\mathcal{P}$ becomes increasingly refined. We will, however, need a quantitative rate of convergence in terms of the number of pieces in the grid.

Lemma 6.5.12 (Step-Function Approximation)**.** *For any $\varepsilon > 0$, there exists a measurable partition $\mathcal{P}$ with $q \leqslant$ [placeholder] pieces such that*

$$\delta_s(\mathbb{E}[f \mid \mathcal{G}_{\mathcal{P}}], f) \leqslant \varepsilon \|f\|_{L^2([0,1]^{2^d})}.$$

Daniel comment: For the following proof, we need to have defined the metric as being the inf over all relabelings (i.e. Lebesgue isomorphisms), since the strategy is to find a relabeled version of g_n that matches very closely with g. In the proof, I'll just assume that we've defined δ_s, δ that way. We can sort out the notation later.

Zack comment: So it seems like there are two problems. The first is clearly proving Lemma 6.5.12 which says we can approximate our representation function f by a step function (which uses only the coordinates associated to the singletons). The second is proving that this step function actually well-approximates f in the cut norm. It seems the part showing the step functon

6.5.1 Cut metric for nonbinary arrays

Zack comment: This section is very rough for now.

First, we need to extend the notion of a weak embedding to the nonbinary case. For a subset of states $C \subseteq S$ define a *C-weak embedding* of x into y to be an injective map $\varphi : [|x|] \to [|y|]$ which preserves the states in C. Namely, if $s \in C$ then $x_e = s$ implies $y_{\varphi(e)} = s$. In the case $C = \{1\}$ and $S = \{0, 1\}$ this reduces to the standard notion of weak embeddings of a binary array. Denote by $emb_C(x, y)$ the set of all such C-weak embeddings and $emb^C(x, y)$ the corresponding embedding density. Then we have the following extension of Lemmas 6.4.1 and 6.4.2.

Zack comment: The above gives us a way of considering multiple states to be the same. Namely, if we clump all the states in C^c together it does not change the $C-weak$ embedding density.

Lemma 6.5.13 (Weak Embeddings for Nonbinary Arrays)**.** *For any $C \subseteq S$ then:*

(i) $emb^C(x, y) = \sum_{x' \geqslant_C x} emb(x', y)$

(ii) $emb(x, y) \leqslant \sum_{x' \geqslant_C x}$ *todo*

Moreover, if $C = S \setminus \{s\}$ then:

(iii) $emb(x, y) = \sum_{x' \geqslant_C x} (-1)^{|1_s(x)| - |1_s(x')|} \cdot emb^C(x', y)$

where $x' \geqslant_C x$ means x' can be obtained from x by changing some of the values x_e with $x_e \notin C$ and $1_s(x) = |\{e : x_e = s\}|$ is the number of entries of x which are s.

In particular, this shows that convergence of $emb(\cdot, y_N)$ implies convergence of $emb^C(\cdot, y_N)$ for any C, and the converse is true if $C = S \setminus \{s\}$.

Zack comment: Note no result (iii) can hold for smaller C because then $x' \geqslant_C x$ fails to be a partial order, so we cannot find the standard Mobius transform.

There is a simple way of encoding a nonbinary array with finite state space S as a collection of binary arrays of the same dimension: simply apply a one-hot encoding to the state space. More specifically, for each $A \subseteq S$ define the function $g_A : S \to \{0, 1\}$ by $g_A(s) = \mathbf{1}\{s \in A\}$. Thus each $d-$dimensional array with values in S is encoded as $2^{|S|}$ binary arrays of the same size. First, we show there is a relationship between embeddings in the original state space and embeddings of the corresponding binary arrays.

Zack comment: This might be overkill. Maybe just sets of the form $A = \{s\}$ suffice.

Lemma 6.5.14 (Binary Projections of Nonbinary Arrays)**.** *For d-dimensional arrays x, y with values in some finite set S then:*

$$ind(x, y) = \bigcap_{s \in S} ind(g_s(x), g_s(y)).$$

Moreover, $emb(\cdot, y_N) \to emb(\cdot, f)$ as $N \to \infty$ if and only if $emb(\cdot, g_A(y_N)) \to emb(\cdot, g_A \circ f)$ for all $A \subseteq S$.

Given an exchangeable array process (X_e) represented by f, lemma 6.2.4 implies the array process $(g_A(X_e))$ is a binary exchangeable array process represented by $g_A \circ f$. Thus it is natural to define:

$$\delta(f, f') := \frac{1}{2^{|S|}} \sum_{A \subseteq S} \delta(g_A \circ f,\ g_A \circ f'). \tag{6.5.1}$$

whenever f, f' share the same state space S.

Zack comment: Can we define it when f, f' have different state spaces? Maybe just extend the smaller state space to the larger one by adding states it never takes on.

Daniel comment: You an do that, but I think the distance will always be 1. But if you can make sense out of there being some states which are considered the same between the two combinatorial structures, that might be interesting.

Daniel comment: One thing that might be a good idea, as well, is to cover the case of structures with interactions of different orders, thinking of a product of arrays of dimension 1, 2,...,d (with a product space metric). Maybe that's not necessary, since there might be a way to encode that in the state space by enlarging S, but if we did a product of arrays of dimension d for all $d \in \mathbb{N}$, we could represent hypergraphs that way. The metric could probably be taken to be something like $\sum_{d=1}^{\infty} 2^{-d}\mathrm{dist}_d$.

6.6 Background on Doob-Martin Boundary Theory

Here we aim to provide some of the essentials of Doob-Martin boundary theory. We restrict our attention to the case of discrete Markov chains relevant to this paper. The primary reference on this topic is [15] but [57] and [19] also provide good overviews for the discrete case in more modern terminology.

Suppose $(X_n)_{n\in\mathbb{N}}$ is a transient Markov chain with a countable state space E and transition kernel p. Define the *Green's kernel* or *potential kernel* $g : E \times E \to \mathbb{R}_{\geqslant 0}$ by

$$g(x,y) := \sum_{n\geqslant 0} p^n(x,y) = \mathbb{E}_x(\text{number of times } X_n \text{ hits y}). \tag{6.6.1}$$

Transience implies $g(x,y) < \infty$. Assume further there exists a base state x_0 from which the chain is irreducible, namely $g(x_0,y) > 0$ for all $y \in S$. For this paper, x_0 will typically be the empty array. Moreover, we will typically be interested in *graded* chains. These are Markov chains in which the state space $E = \bigsqcup E_n$ can be partitioned so transitions occur only from E_n to E_{n+1}. Here

$$g(x,y) = \mathbb{P}_x(X_n \text{ hits } y).$$

Define the *Doob-Martin kernel* $K : E \times E \to \mathbb{R}_{\geqslant 0}$ with respect to base state x_0 via

$$K(x,y) := \frac{g(x,y)}{g(x_0,y)} = \frac{\mathbb{P}_x(X_n \text{ hits y})}{\mathbb{P}_{x_0}(X_n \text{ hits y})}. \tag{6.6.2}$$

Observe for each fixed y the function $K(\cdot,y)$ is "almost harmonic" in the sense

$$\sum_{z\in S} p(x,z)K(z,y) = K(x,y) \quad x \neq y.$$

This suggests if we let y "go off to infinity" we ought to get a truly harmonic function in the limit. The Doob-Martin boundary will make this intuition precise.

Note for fixed x, the kernel $K(x,y)$ is a bounded function of y. To see why, note irreducibility implies $p^m(x_0,x) > 0$ for some m so

$$0 \leqslant K(x,y) := \frac{g(x,y)}{g(x_0,y)} \leqslant \frac{g(x,y)}{p^m(x_0,x)g(x,y)} = \frac{1}{p^m(x_0,x)} =: C_x.$$

Hence we can view the collection of functions $\{K(\cdot,y)\}_{y\in E}$ as a Euclidean subset

$$\{K(\cdot,y)\}_{y\in E} \subseteq \bigtimes_{x\in E} [0,C_x] \subseteq \mathbb{R}^E_{\geqslant 0},$$

so $\{K(\cdot,y)\}_{y\in E}$ is precompact under the product topology. Moreover, as $K(\cdot,y) \neq K(\cdot,y')$ for $y \neq y'$ we can identify E with $\{K(\cdot,y)\}_{y\in E}$. The closure $\overline{E} \subseteq \mathbb{R}^E_{\geqslant 0}$ under this identification is compact. We call $\overline{E}$ the *Doob-Martin compactification* of E and its boundary $\partial E := \overline{E} - E$ the *Doob-Martin boundary*.

In summary, we have turned E into a metric space by identifying it with a subset of $\mathbb{R}^E_{\geqslant 0}$ which carries a natural metric structure, and used this identification to compactify E.

Remark 6.6.1. By construction, a sequence $\{y_n\} \in E$ converges to a point $\alpha \in \overline{E}$ if and only if $\{K(\cdot, y_n)\}$ converges in $\mathbb{R}^E_{\geqslant 0}$, namely if and only if $K(x, y_n)$ converges in $\mathbb{R}_{\geqslant 0}$ for all $x \in E$. Thus for fixed x the function $K(x, \cdot) : E \to \mathbb{R}_{\geqslant 0}$ extends continuously to $\overline{E}$.

There is a close relationship between the Doob-Martin kernel K and the backwards transition probabilities of the chain. Namely, for $x \in E_n, y_N \in E_N$ Bayes rule implies

$$K(x, y) = \frac{\mathbb{P}(X_N = y_N \mid X_n = x)}{\mathbb{P}(X_N = y_N)} = \frac{\mathbb{P}(X_n = x \mid X_N = y_N)}{\mathbb{P}(X_n = x)}.$$

Thus $K(x, y_N)$ converges as $N \to \infty$ for all x if and only if the conditional probabilities $\mathbb{P}(X_n = x \mid X_N = y_N)$ converge for all x.

6.6.1 Minimal Boundary and Representation of Harmonics

Given a Markov chain X_n on state-space S, we say a function $h : S \to \mathbb{R}$ is *harmonic* if

$$h(x) = \mathbb{E}[h(X_{n+1}) \mid X_n = x].$$

Note $K(\cdot, \alpha)$ is harmonic for every $\alpha \in \partial E$. The *minimal boundary* is the subset

$$\partial_m E := \{\alpha \in \partial E : K(\cdot, \alpha) \text{ is minimal harmonic}\}. \tag{6.6.3}$$

A *minimal harmonic* function is a harmonic function $u \geqslant 0$ so for any other harmonic w satisfying $0 \leqslant w \leqslant u$ we must have $w = cu$ for some constant $c \geqslant 0$.

The set $\mathbb{H}_{1,+}$ of nonnegative harmonics with $h(x_0) = 1$ is a compact, convex set whose extreme points are precisely the minimal harmonics. Clearly $K(\cdot, \alpha) \in \mathbb{H}_{1,+}$, so the Doob-Martin boundary can be thought of as a subset of $\mathbb{H}_{1,+}$, and the minimal boundary is precisely just the elements of that subset which are extreme points of $\mathbb{H}_{1,+}$. Any nonnegative harmonic function u can be represented as an integral over these extreme points:

$$u(x) = \int_{\partial_m E} K(x, \alpha) \mu_u(d\alpha). \tag{6.6.4}$$

Moreover, as $K(\cdot, \alpha)$ is a harmonic for $\alpha \in \partial_m E$ and $|K(x, \alpha)| \leqslant C_x$, equation (6.6.4) defines a nonnegative harmonic function u for any finite measure μ_u, and u satisfies $u(x_0) = 1$ if and only if μ_u is a probability measure.

In words, nonnegative harmonic functions are characterized by their distribution on the minimal boundary $\partial_m E$. For the special case of $u = K(\cdot, \alpha)$ for $\alpha \in \partial_m E$ clearly we have $\mu_u = \delta_\alpha$. In some cases such as the Rémy chain studied in [19], $\partial E = \partial_m E$, but in general this is not always the case. A full proof of these facts can be found in [57].

6.6.2 Using h-processes to Condition at Infinity

Given a harmonic function $h \in \mathbb{H}_{1,+}$, irreducibility and harmonicity imply in fact that $h > 0$ and so we can define a new transition mechanism p^h which "tilts" the dynamics using h:

$$p^h(x, y) := \frac{h(y)}{h(x)} p(x, y), \quad x, y \in E. \tag{6.6.5}$$

Thus the new chain reweights transitions of the original chain by a factor of $h(y)/h(x)$, favoring transitions towards states where h is large. When $h \equiv 1$ we recover the original chain.

We call the new Markov chain $X_n^{(h)}$ with transition probabilities p^h the *Doob h-process* or *Doob h-transform* of X_n corresponding to $h(x)$. The h-processes $X_n^{(K(\cdot,\alpha))}$ are of natural interest because they characterize the distributions of all other h-processes, including the original chain:

$$\mathbb{P}^h_{x_0}(B) = \int_{\partial_m E} \mathbb{P}^{K(\cdot,\alpha)}_{x_0}(B)\, \mu_h(d\alpha). \tag{6.6.6}$$

for the unique measure μ_h supported on the minimal boundary as given in (6.6.4). In particular, these h-processes converge a.s. in the Doob-Martin topology to the minimal boundary and have limiting distributions:

$$\mathbb{P}_{x_0}(\lim X_n^{(h)} \in A) = \int_{A \cap \partial_m E} K(x, \alpha) \mu_h(d\alpha), \tag{6.6.7}$$

Namely, an h-process is merely a chain conditioned to have distribution μ_h on the minimal boundary. Again, if $\alpha \in \partial_m E$ then $\mu_{K(\cdot,\alpha)} = \delta_\alpha$ so $\lim X_n^{K(\cdot,\alpha)} = \alpha$ almost surely. Thus we can think of the chain $X_n^{(K(\cdot,\alpha))}$ to be our original Markov chain conditioned to equal $\alpha \in \partial_m E$ "at infinity".

6.6.3 Infinite Bridges

A finite-step Markov process $(Y_n)_{n=1}^N$ with $Y_0 = x_0$ which has the same backwards transition probabilities as $(X_n)_{n \in \mathbb{N}}$ is called a (finite) *bridge* for X_n:

$$\mathbb{P}(X_n = i \mid X_{n+1} = j) = \mathbb{P}(Y_n = i \mid Y_{n+1} = j), \quad \forall i, j \in S, n < N.$$

Similarly, if the index set is $\mathbb{N}$ we call $(Y_n)_{n \in \mathbb{N}}$ an *infinite bridge*. Let $(X_0^{(y)}, X_1^{(y)}, ..., X_{N(y)}^{(y)})$ be the finite-step chain obtained from X_n by conditioning on $X_{N(y)} = y$. Then

$$\begin{aligned}
\mathbb{P}(X_n^{(y)} = i | X_{n+1}^{(y)} = j) &= \frac{\mathbb{P}_{x_0}(X_n = i, X_{n+1} = j, X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_{n+1} = j, X_{N(y)} = y)} \\
&= \frac{\mathbb{P}_{x_0}(X_n = i, X_{n+1} = j) \cdot \mathbb{P}_j(X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_{n+1} = j) \cdot \mathbb{P}_j(X_{N(y)} = y)}
\end{aligned}$$

$$= \mathbb{P}_{x_0}(X_n = i | X_{n+1} = j).$$

Namely, $X_n^{(y)}$ is a finite bridge of X_n. Moreover,

$$\begin{aligned}
\mathbb{P}(X_{n+1}^{(y)} = i | X_n^{(y)} = j) &= \frac{\mathbb{P}_{x_0}(X_n = j, X_{n+1} = i, X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_n = j, X_{N(y)} = y)} \\
&= \frac{\mathbb{P}_{x_0}(X \text{ hits j}) \cdot p(j,i) \cdot \mathbb{P}_i(X \text{ hits y})}{\mathbb{P}_{x_0}(X \text{ hits j}) \cdot \mathbb{P}_j(X \text{ hits y})} \\
&= K(j,y)^{-1} \cdot p(j,i) \cdot K(i,y).
\end{aligned}$$

This is essentially the definition of an h-process with $h = K(\cdot, y)$, except for the fact that $K(\cdot, y)$ is not harmonic for $y \in E$. However, as before, intuition suggests we ought to take the limit as y "goes to infinity" to get a truly harmonic function $K(\cdot, \alpha)$. Namely, by Remark 6.6.1 we know a sequence (y_k) in E converges to a point $\alpha \in \partial E$ if and only if $K(x, y_k)$ converges in $\mathbb{R}_{\geqslant 0}$ to $K(x, \alpha)$ for all $x \in E$ which of course implies

$$\mathbb{P}(X_{n+1}^{(y_k)} = i | X_n^{(y_k)} = j) \xrightarrow{k \to \infty} K(j, \alpha)^{-1} \cdot p(j,i) \cdot K(i, \alpha) =: p^{K(\cdot,\alpha)}(j,i).$$

As observed in [22] such a sequence converges in the Doob Martin topology if and only if finite initial segments of the corresponding bridges converge in distribution, namely

$$(X_0^{(y_k)}, X_1^{(y_k)}, ..., X_M^{(y_k)}) \Rightarrow (X_0^{K(\cdot,\alpha)}, X_1^{K(\cdot,\alpha)}, ..., X_M^{K(\cdot,\alpha)}) \tag{6.6.8}$$

as $k \to \infty$, for every finite M. More generally, given any nonnegative harmonic function with $h(x_0) = 1$, its Doob h-transform is an infinite bridge:

$$\begin{aligned}
\mathbb{P}(X_n^h = x \mid X_{n+1}^h = y) &= \frac{\mathbb{P}(X_{n+1}^h = y \mid X_n^h = x) \cdot \mathbb{P}(X_n^h = x)}{\mathbb{P}(X_{n+1}^h = y)} \\
&= \frac{\frac{h(y)}{h(x)}\mathbb{P}(X_{n+1} = y \mid X_n = x) \cdot \frac{h(x)}{h(x_0)}\mathbb{P}(X_n = x)}{\frac{h(y)}{h(x_0)}\mathbb{P}(X_{n+1}^h = y)} \\
&= \mathbb{P}(X_n = x \mid X_{n+1} = y).
\end{aligned}$$

Conveserly, given any infinite bridge X_n^∞ then (6.6.6) tells us there exists a measure μ_h so that X_n^∞ is a mixture of the extremal bridges $\mathbb{P}^{K(\cdot,\alpha)}$ and the corresponding harmonic h is given by (6.6.4).

Call an infinite bridge *extremal* if it has a trivial tail σ-field, or equivalently if it cannot be written as a nondegenerate mixture of other infinite bridges. By (6.6.6) and the correspondence between bridges and harmonic functions, all infinite bridges are just mixtures of the extremal bridges. Hence characterizing the extremal bridges allows us to characterize all infinite bridges.

Our observations from Section 6.6.2 tell us that for $\alpha \in \partial_m E$ that the conditioned chain $X_n^{K(\cdot,\alpha)}$ is extremal as it converges almost surely to the constant α. Moreover, the converse is true: $X^{K(\cdot,\alpha)}$ is extremal only if $\alpha \in \partial_m S$. Hence characterizing the extremal bridges will allow us to characterize the minimal boundary, and thus by (6.6.4) characterize all nonnegative harmonic functions for this chain.

In summary, we have several equivalent characterizations of the Doob-Martin boundary:

(i) Limits of sequences in E which converge in the Doob-Martin topology.

(ii) Sequences in E whose conditional distributions converge in distribution, by (6.6.8).

(iii) Extremal infinite bridges.

(iv) Extremal nonnegative harmonic functions.

6.7 Proof Appendix

Proof of Lemma 6.2.4 (State-Space Projections of Exchangeable Arrays). The result follows immediately from applying g coordinatewise. If (X_e) is exchangeable, then for each finite permutation σ,

$$(g(X_{\sigma(e)}))_{e\in\mathbb{N}^{(d)}} \stackrel{\text{law}}{=} (g(X_e))_{e\in\mathbb{N}^{(d)}}.$$

Moreover, if $X_e = f(U_\emptyset, (U_i)_{i\in e}, \dots, U_e)$ in distribution, then $g(X_e)$ is represented by $g\circ f$. □

Proof of Lemma 6.2.2 (Markov Property of the Unlabeled Chain). Firstly note that the definition of an exchangeable array implies:

$$\mathbb{P}(L_n = x) = \mathbb{P}(L_n = x') \qquad \forall x \sim x'.$$

Moreover if $x \sim x'$ are related by $\sigma \in Sym(\mathbb{N})$ then again exchangeability implies:

$$\begin{aligned}
\mathbb{P}(M_{n+1} = [y]_\sim,\ L_n = x) &= \sum_{y'\sim y} \mathbb{P}(L_{n+1} = y',\ L_n = x) \\
&= \sum_{y'\sim y} \mathbb{P}(L_{n+1} = \sigma(y'),\ L_n = \sigma(x)) \\
&= \sum_{z\sim y} \mathbb{P}(L_{n+1} = z,\ L_n = x') \\
&= \mathbb{P}(M_{n+1} = [y]_\sim,\ L_n = x').
\end{aligned}$$

Thus we see:

$$\mathbb{P}(M_{n+1} = [y]_\sim \mid L_n = x) = \mathbb{P}(M_{n+1} = [y]_\sim \mid L_n = x') \qquad \forall x \sim x', \quad [y]_\sim \in [E_{n+1}]_\sim.$$

Thus the transition probability from x into the class $[y]_\sim$ depends only on the equivalence class $[x]_\sim$. The projected transition kernel is therefore well-defined on equivalence classes, and hence M_n is a Markov chain. □

Proof of Lemma 6.3.1 (Boundary of the Labeled Chain). Clearly any deterministic array cannot be written as a mixture of other arrays and hence such arrays are extremal. Moreover, any nondeterministic array is a mixture (over its own distribution) of determinsitic arrays and hence is not extremal. □

Proof of Lemma 6.3.2 (Exchangeable Bridges of the Labeled Chain). Suppose the harmonic $h \geqslant 0$ with $h(x_0) = 1$ is the one corresponding to this particular bridge. If L_n is the original labeled exchangeable array process and L_n^h is the infinite bridge:

$$\mathbb{P}(L_0^h = x_0, ..., L_n^h = x_n) = h(x_n) \cdot \mathbb{P}(L_0 = x_0, ..., L_n = x_n).$$

The right hand side is zero if the states $x_0, ..., x_n$ are not nested in the sense of arrays, $x_m = (x_n)|_{[m]^{(d)}}$ for $m < n$, and thus the same is true of the left hand side. Thus, by taking a projective limit, the chain L_n^h a.s. defines an array (X_e^h) of the same dimension as the original exchangeable array (X_e) of which it is the corresponding array process.

Now if $\sigma \in Sym(\mathbb{N})$ let $\tilde{x} := \sigma(x)$ be the array generated by permuting the coordinates of x. Then the definition of the Doob h-transform and the exchangeability of the original array imply:

$$\begin{aligned}\mathbb{P}(L_0^h = \tilde{x}_0, ..., L_n^h = \tilde{x}_n) &= h(\tilde{x}_n) \cdot \mathbb{P}(L_0 = \tilde{x}_0, ..., L_n = \tilde{x}_n) \\ &= h(\tilde{x}_n) \cdot \mathbb{P}(L_0 = x_0, ..., L_n = x_n) \\ &= \frac{h(\tilde{x}_n)}{h(x_n)} \cdot \mathbb{P}(L_0^h = x_0, ..., L_n^h = x_n).\end{aligned}$$

These two are equal if and only if $h(\tilde{x}_n) = h(x_n)$. Since x_n and σ were arbitrary, the result follows from the fact that the finite dimensional distributions of our array (X_e^h) are determined from the distribution of (L_n^h). $\square$

Proof of Corollary 6.3.3 (Bridges of the Unlabeled Chain). The bridges of (M_n) are in correspondence with harmonic functions $\tilde{h}$ of this chain. Define $h(x) := \tilde{h}([x])$ on the labeled state space. Since the transition kernel of M_n is obtained by summing the transition probabilities of L_n over equivalence classes, harmonicity of $\tilde{h}$ gives

$$h(x) = \tilde{h}([x]) = \sum_{[y]_\sim} p_M([x]_\sim, [y]_\sim)\tilde{h}([y]_\sim) = \sum_y p_L(x, y)h(y).$$

Thus h is a harmonic function for the labeled chain and is constant on the equivalence classes of $\sim$. Applying Lemma 6.3.2 gives the corollary. $\square$

Proof of Theorem 6.3.4 (Extremal Bridges and Ergodic Arrays). The last three statements are equivalent by Lemma 7.35 of [32]. For the implication (2) $\to$ (1), suppose for contraposition that we have a nonextremal bridge $M_n^{\tilde{h}}$ associated to the harmonic $\tilde{h}$. Then equation (6.6.6) applied to the Doob-Martin kernel $\tilde{K}$ of our unlabeled chain implies:

$$\mathbb{P}((M_n^{\tilde{h}}) = [x]_\sim) = \int_{\partial_m \tilde{E}} \mathbb{P}^{\tilde{K}(\cdot,\alpha)}([x]_\sim)\, \mu_{\tilde{h}}(d\alpha),$$

for some nondegenerate measure $\mu_{\tilde{h}}$ on the minimal boundary $\partial_m \tilde{E}$ of the unlabeled chain. By construction of the Doob-Martin boundary, the functions $\tilde{K}(\cdot, \alpha)$ are harmonic for the unlabeled chain. Thus by Corollary 6.3.3 the distributions $\mathbb{P}^{\tilde{K}(\cdot,\alpha)}$ correspond to the unlabeled chains associated to exchangeable arrays (X_e^α). Likewise, $M_n^{\tilde{h}}$ corresponds to some exchangeable array $(X_e^{\tilde{h}})$ as well. Thus we can rephrase the previous equation in terms of the associated exchangeable arrays:

$$\mathbb{P}((X_e^{\tilde{h}}) \in [x]_\sim) = \int_{\partial_m \tilde{E}} \mathbb{P}(X_e^\alpha \in [x]_\sim)\, \mu_{\tilde{h}}(d\alpha).$$

These probabilities are constant on the equivalence classes by exchangeability. Hence:

$$\mathbb{P}((X_e^{\tilde{h}}) = x) = \int_{\partial_m \tilde{E}} \mathbb{P}(X_e^\alpha = x)\ \mu_{\tilde{h}}(d\alpha).$$

Thus as $\mu_{\tilde{h}}$ is nondegenerate, $(X_e^{\tilde{h}})$ is a nondegenerate mixture of other exchangeable arrays, and hence not ergodic.

For the implication (1) → (2) suppose that we have an extremal bridge M_n^α associated to the minimal harmonic $\tilde{K}(\cdot, \alpha)$, where α is in the minimal boundary $\tilde{E}$. As mentioned in Section 3, this is equivalent to the fact that this bridge has a trivial tail σ-field almost surely. By Corollary 6.3.3 there is an exchangeable array (X_e^α) associated to this bridge. Suppose this array has distribution ν and that I is a measurable ν-invariant event. Let r_n denote the function from infinite d-dimensional arrays to size n arrays that just restricts the entries of (X_e^α) to $e \in [n]^{(d)}$. Then for every finite choice of $N \in \mathbb{N}$ because the labeled chain L_n^α is just the size n "top-left corner" of the array X^α and because I is $\nu - invariant$:

$$\begin{aligned}\{(X_e^\alpha) \in I\} &= \bigcap_{n \geqslant N} \{L_n^\alpha \in r_n(I)\} \\ &\overset{\nu-a.s.}{=} \bigcap_{n \geqslant N} \bigcup_{\tau \in Sym([n])} \{L_n^\alpha \in \tau(r_n(I))\}.\end{aligned}$$

Lastly, because $\{M_n = [x]_\sim\} = \{L_n \in [x]_\sim\}$ by definition of our unlabeled/labeled chains:

$$\begin{aligned}&= \bigcap_{n \geqslant N} \{M_n^\alpha \in [r_n(I)]_\sim\} \\ &\in \sigma(M_N^\alpha, M_{N+1}^\alpha, \ldots).\end{aligned}$$

As $N \in \mathbb{N}$ was arbitrary, this implies $\{(X_e^\alpha) \in I\}$ lies in the tail σ-field of our bridge process (M_n^α), up to ν-null sets. By assumption of bridge extremality, this tail σ-field is trivial and thus $\mathbb{P}((X_e^\alpha) \in I) \in \{0, 1\}$ as desired. Hence the array (X_e^α) is ergodic. □

Proof of Lemma 6.4.1 (Embedding and Morphism Densities). These proofs are simple extensions of the proofs in [43] of analogous results in the theory of graphons. Firstly (i) is immediate as any embedding is automatically a weak embedding. For (ii) note that if $\sigma \in Sym(\mathbb{N})$ is such that $x \lesssim \sigma(y)$ then there is precisely one $x' \geqslant x$ so that $x' \leqslant y$, namely x' is the array $\sigma(y)$ restricted to the same size as x. For (iii) if for each e where $x_e = 0$ we let A_e be the set of weak embeddings of $x' = x + \delta_e$ into y then:

$$\{\text{embeddings x into } y\} = \{\text{weak embeddings } x \text{ into } y\} \setminus \bigcup_{e\,:\,x_e = 0} A_e.$$

Then the formula follows from standard inclusion-exclusion. The proofs of (iv) and (v) can be found in the Appendix 6.7. □

Proof of Lemma 6.4.2 (Doob-Martin Convergence and Embedding Densities). The equivalence of (ii) and (iii) follows immediately from Lemma 6.4.1. The equivalence of (i) is almost an immediate result of exchangeability which implies that:

$$\mathbb{P}(L_n = x \mid M_n = [x]_\sim) = \frac{\mathbb{P}(L_n = x)}{\mathbb{P}(M_n = [x]_\sim)} = \frac{1}{\mid [x]_\sim \mid}.$$

As L_n has deterministic backwards dynamics as we already noted above, this implies the unlabeled chain M_n is uniformly backwards exchangeable where the labeling just associates $[x]_\sim$ to one element of the equivalence class. Hence, by Theorem 3.1 of the other paper we can see the appropriate notion of embedding density is:

$$\frac{\mid [y]_\sim \cap \pi_n^{-1}([x]_\sim) \mid}{\mid [y]_\sim \mid} \quad x \in E_n, y \in E_{n+m},$$

where $\pi_n^{-1}([x]_\sim)$ is the collection of all arrays of the same size as y that can be made by extending some array $\tilde{x} \in [x]_\sim$. To see that this equals 6.4.1 note if $y = \sigma(y)$ for any $\sigma \in Sym([n+m])$ then the numerator and denominator of the above equation both scale by same amount; in case this is true for no σ we have immediate equality. □

Proof of Lemma 6.4.3 (Embedding Densities of Representation Functions). First we prove the expectation formulas, starting with emb^W. Let $\varphi : [k] \to [n]$ be an injective function. Then

$$\mathbb{P}(\varphi \text{ is an array embedding}) = \int_{[0,1]^{2^n}} \prod_{S \in \mathbf{1}(F)} f(x_{(\varphi(S))})\, d\lambda_{2^n}$$

By the injectivity of φ we can apply a change of coordinates,

$$= \int_{[0,1]^{2^n}} \prod_{S \in \mathbf{1}(F)} f(x_{(S)})\, d\lambda_{2^n}$$

After integrating out all the superfluous coordinates,

$$= emb^W(F, f).$$

So, writing $emb^W(F, X_n(f))$ in terms of the indicator functions $\mathbf{1}_{\{\varphi \text{ is an array embedding}\}}$,

$$\mathbb{E}[emb^W(F, X_n(f))] = emb^W(F, f).$$

The proof for emb is identical.

For the concentration results, we follow exactly the proof of Theorem 2.5 in [39]. Namely, for $m \leqslant n$ let X_m denote the (random) array induced by restricting $X_n(f)$ to subsets $e \subseteq [m]^{(d)}$. Since $\sigma(X_m) \subseteq \sigma(X_{m+1})$ since the former is a subarray of the latter, this can be used to define a Doob martingale:

$$B_m := \mathbb{E}(emb^W(F, X_n(f)) \mid X_m).$$

Bounding the increments using a bound on the number of injective functions $\varphi : [k] \to [n]$ and applying Azuma's inequality as in [39] gives the desired inequalities. Then apply an identical procedure for *emb*.

The convergence then follows immediately from Borel-Cantelli. Lemma 6.4.1 (iv) and (v) allow the convergence to be extended to the morphism densities as well. □

Proof of Lemma 6.5.7 (Cut Metric Controls Embedding Densities). The second inequality follows just by optimizing over the last coordinate, so we only need to prove the first inequality.

$$\text{LHS} = \left| \int_{[0,1]^{2^k}} \prod_{S \in \mathbf{1}(F)} f(x_{(S)}) - \prod_{S \in \mathbf{1}(F)} f'(x_{(S)})\, d\lambda_{2^k} \right|$$

Enumerate $\mathbf{1}(F) = \{S_1, S_2, \ldots, S_\ell\}$. Using the triangle inequality, interpolating between the products by changing one factor at a time, we get

$$\leqslant \sum_{m=1}^{\ell} \left| \int_{[0,1]^{2^k}} \left(\prod_{i=1}^{m-1} f(x_{(S_i)}) \right) \left(f(x_{(S_m)}) - f'(x_{(S_m)}) \right) \left(\prod_{i=m+1}^{\ell} f'(x_{(S_i)}) \right) d\lambda_{2^k} \right|$$

To bound each of the ℓ terms, first integrate out all $2^k - 2^d$ variables not included in $x_{(S_m)}$. What remains after integrating the two product terms over those variables is a function g_m of coordinates $\{x_A : A \subsetneq S_m\}$, so

$$\begin{aligned} &\leqslant |\mathbf{1}(F)| \sup_{0 \leqslant g \leqslant 1} \left| \int_{[0,1]^{2^d - 1}} \int_0^1 g(x_{-S})(f(x_{-S}, x_S) - f'(x_{-S}, x_S))\, dx_S\, d\lambda_{2^d - 1}(x_{-S}) \right| \\ &= |\mathbf{1}(F)| \delta(f, f'). \end{aligned}$$

□

Proof of Lemma 6.5.8 (Sampling Probability Formula). From Lemma 6.4.1, we have that

$$\begin{aligned} emb^W(x, f) &= \mathbb{E}[emb^W(x, A_n)] \\ &= \sum_{|y|=n} \mathbb{P}(A_n = y) emb^W(x, y) \\ &= \sum_{|y|=n} \sum_{x' \geqslant x} \mathbb{P}(A_n = y) emb(x', y) \end{aligned}$$

Since x and y are both of size n, an embedding is the same as an automorphism of the finite array. In particular, $emb(x', y) = 0$ if x', y are not isomorphic, and $emb(x', y) = |\mathrm{Aut}(x')|/n!$ if they are isomorphic.

$$\begin{aligned} &= \sum_{x' \geqslant x} \sum_{y \text{ iso. to } x'} \mathbb{P}(A_n = x') \frac{|\mathrm{Aut}(x')|}{n!} \\ &= \sum_{x' \geqslant x} \mathbb{P}(A_n = x') \frac{|\mathrm{Aut}(x')|^2}{n!}. \end{aligned}$$

By Möbius inversion, we get

$$\frac{|\mathrm{Aut}(x)|^2}{n!}\mathbb{P}(A_n = x) = \sum_{x' \geqslant x} (-1)^{|1_x| - |1_{x'}|} emb^W(x', f),$$

and the result follows. □

Proof of Corollary 6.5.9 (Total Variation Bound for Samples). By Lemma 6.5.8, for any fixed array x of size n,

$$\begin{aligned}|\mathbb{P}(A_n = x) - \mathbb{P}(A'_n = x)| &\leqslant \frac{n!}{|\mathrm{Aut}(x)|^2} \sum_{x' \geqslant x} |emb^W(x', f) - emb^W(x', f')| \\ &\leqslant n! \sum_{x' \geqslant x} |emb^W(x', f) - emb^W(x', f')|.\end{aligned}$$

Summing over all x, we get

$$\sum_{|x|=n} |\mathbb{P}(A_n = x) - \mathbb{P}(A'_n = x)| \leqslant n! \sum_{|x|=n} \sum_{x' \geqslant x} |emb^W(x, f) - emb^W(x', f')|.$$

The left hand size is twice the total variation distance between the laws of A_n, A'_n. □

Proof of Lemma 6.5.10 (Coupling of Samples). If we can show that

$$\|\mathrm{Law}(A_n) - \mathrm{Law}(A'_n)\|_{\mathrm{TV}} \leqslant \frac{n!}{2} 3^{\binom{n}{d}} \max_{|x|=n} |emb^W(x, f) - emb^W(x, f')|,$$

then there exists a coupling of A_n, A'_n such that $\mathbb{P}(A_n \neq A'_n) \leqslant []$. Since $\delta_s(f_n, f'_n) = 0$ when $A_n = A'_n$ and $\delta_s(f_n, f'_n) \leqslant 1$, this will give

$$\begin{aligned}\mathbb{E}[\delta_s(f_n, f'_n)] &= \mathbb{E}[\delta_s(f_n, f'_n)\mathbf{1}_{\{A_n \neq A'_n\}}] \\ &\leqslant \mathbb{E}[\mathbf{1}_{\{A_n \neq A'_n\}}] \\ &\leqslant \frac{n!}{2} 3^{\binom{n}{d}} \max_{|x|=n} |emb^W(x, f) - emb^W(x, f')|.\end{aligned}$$

To prove the total variation bound, we use Corollary 6.5.9:

$$\begin{aligned}\|\mathrm{Law}(A_n) - \mathrm{Law}(A'_n)\|_{\mathrm{TV}} &\leqslant \frac{n!}{2} \sum_{|x|=n} \sum_{x' \geqslant x} |emb^W(x', f) - emb^W(x', f')| \\ &\leqslant \frac{n!}{2} 3^{\binom{n}{d}} \max_{|x|=n} |emb^W(x, f) - emb^W(x, f')|.\end{aligned}$$

because for each of the $\binom{n}{d}$ entries of our arrays x', x we have three choices that ensure $x' \geqslant x$: $(x_e, x'_e) \in \{(0,0), (0,1), (1,1)\}$. □

Proof of Theorem 6.5.6 (Convergence via the Cut Metric). (iii) $\implies$ (ii): This follows by definition, as $\delta(g_n, g) \leqslant \delta_s(g_n, g)$.

(ii) $\implies$ (i): This follows from Lemma 6.5.7.

(i) $\implies$ (iii): For any given N, let A_N^n, A_N' be random arrays of size N, sampled from g_n, g, respectively, and let f_N^n, f_N' be the associated sampling functions. The triangle inequality gives

$$\delta_s(g_n, g) \leqslant \mathbb{E}[\delta_s(g_n, f_N^n)] + \mathbb{E}[\delta_s(f_N^n, f_N')] + \mathbb{E}[\delta_s(f_N', g)],$$

where we have coupled f_N^n, f_N' as in Lemma 6.5.10. Using Lemmas 6.5.11 and 6.5.10, we get

$$\leqslant \frac{10}{\sqrt{\log_2 N}} + \frac{N!}{2} 3^{\binom{N}{d}} \max_{|x|=N} |emb^W(x, g_n) - emb^W(x, g)|.$$

Letting $n \to \infty$, since there are only finitely many possible arrays of size N, we get

$$\limsup_{n\to\infty} \delta_s(g_n, g) \leqslant \frac{10}{\sqrt{\log_2 N}}.$$

Letting $N \to \infty$ completes the proof. □

Proof of Lemma 6.5.12 (Step-Function Approximation). Let $f_{\mathcal{P}} := \mathbb{E}[f \mid \mathcal{G}_{\mathcal{P}}]$ □

Proof of Lemma 6.5.11 (Approximation by Sampled Arrays). Let $\varepsilon > 0$, the value of which will be determined later. First, we will approximate f by $g := \mathbb{E}[f \mid \mathcal{G}_{\mathcal{P}}]$, where $\mathcal{P}$ is as given in Lemma 6.5.12. We also approximate f_n by g_n, where g_n is the (random) sampling function corresponding to a random array of size n, sampled from g.Then

$$\begin{aligned}\mathbb{E}[\delta_s(f_n, f)] &\leqslant \mathbb{E}[\delta_s(f_n, g_n)] + \mathbb{E}[\delta_s(g_n, g)] + \delta_s(g, f)\\ &\leqslant \mathbb{E}[\delta_s(f_n, g_n)] + \mathbb{E}[\delta_s(g_n, g)] + \varepsilon.\end{aligned}$$

To control $\delta_s(g_n, g)$, we will find a version of g_n for which the partition lines up well with $\mathcal{G}_{\mathcal{P}}$. Writing $\mathbb{P} = \{I_1, \ldots, I_q\}$, let $U_1, \ldots, U_n \overset{\text{iid}}{\sim} \text{Unif}[0,1]$, and let $N_i = |\{j : U_j \in I_i\}|$, so that $N_i \sim \text{Binomial}(n, \lambda(I_i))$. Then define a version of g_n as follows: Partition $[0,1]$ into $J_1, \ldots, J_q$ in such a way that $\lambda(J_i) = N_i/n$ (note that $N_1 + \cdots + N_q = n$) and $J_i \cap I_i = \min(\lambda(I_i), \lambda(J_i))$; that is, we line up the empirical partition as well as we can with $\mathcal{P}$. Then, letting $g(I_{i_1} \times \cdots \times I_{i_d})$ denote the value that g takes on $I_{i_1} \times \cdots \times I_{i_d}$, we set $\widetilde{g}_n(x_1, \ldots, x_d)$ to equal the value g takes on $I_{i_1} \times \cdots \times I_{i_d}$ when $x_1 \in J_{i_1}, \ldots, x_d \in J_{i_d}$. Using this version of g_n,

$$\begin{aligned}\delta_s(g_n, g) &\overset{d}{=} \delta_s(\widetilde{g}_n, g)\\ &\leqslant \int_{[0,1]^d} |\widetilde{g}_n - g| \, d\lambda_d\end{aligned}$$

These functions both take values in $[0, 1]$, so we can upper bound the integral by the measure of the set where the functions disagree.

$$\begin{aligned}
&\leqslant \lambda(\{x : \widetilde{g}_n(x) \neq g(x)\}) \\
&\leqslant 1 - \left(\sum_{i=1}^{q} \min(\lambda(I_i), \lambda(J_i))\right)^d
\end{aligned}$$

Using $1 - x^d = (1-x)(1 + x + \cdots + x^{d-1}) \leqslant d(1-x)$ for $0 \leqslant x \leqslant 1$,

$$\leqslant d\left(1 - \sum_{i=1}^{q} \min(\lambda(I_i), \lambda(J_i))\right)$$

Using the standard total variation identity for finite partitions,

$$\begin{aligned}
&= \frac{d}{2}\sum_{i=1}^{q} |\lambda(I_i) - \lambda(J_i)| \\
&\leqslant \frac{d\sqrt{q}}{2}\sqrt{\sum_{i=1}^{q}(\lambda(I_i) - \lambda(J_i))^2}.
\end{aligned}$$

Taking the expectation of both sides of the inequality, we get

$$\begin{aligned}
\mathbb{E}[\delta_s(g_n, g)] &\leqslant \frac{d\sqrt{q}}{2}\mathbb{E}\left[\sqrt{\sum_{i=1}^{q}\left(\lambda(I_i) - \frac{N_i}{n}\right)^2}\right] \\
&\leqslant \frac{d\sqrt{q}}{2}\sqrt{\mathbb{E}\left[\sum_{i=1}^{q}\left(\lambda(I_i) - \frac{N_i}{n}\right)^2\right]} \\
&= \frac{d\sqrt{q}}{2}\sqrt{\sum_{i=1}^{q}\mathrm{Var}(N_i/n)} \\
&\leqslant \frac{d\sqrt{q}}{2n}\sqrt{\sum_{i=1}^{q} n\lambda(I_i)} \\
&= \frac{d}{2}\sqrt{\frac{q}{n}}.
\end{aligned}$$

To bound $\delta_s(f_n, g_n)$, we combine Lemma 6.5.10 with Lemma 6.5.7:

$$\mathbb{E}[\delta_s(f_n, g_n)] \leqslant \frac{n!}{2}3^{\binom{n}{d}} \max_{|x|=n} |emb^W(x, f) - emb^W(x, g)|$$

$$\leqslant \frac{n!}{2} 3^{\binom{n}{d}} \binom{n}{d} \delta_s(f,g)$$
$$\leqslant \frac{n!}{2} 3^{\binom{n}{d}} \binom{n}{d} \varepsilon.$$

□

Proof of Lemma 6.5.13 (Weak Embeddings for Nonbinary Arrays). The proof is analogous to that of Lemmas 6.4.1 and 6.4.2. Namely, for (i) note that if φ is a C-weak embedding there is exactly one choice of $x' \geqslant_C x$ so that φ is an embedding of x' into y. Namely, $x'_e := y_{\varphi(e)}$ whenever $x_e \notin C$.

For (ii), note any C-weak embedding that is not an embedding must have $y_{\varphi(e)} \neq x_e$ for some $x_e \notin C$. Thus:

$$emb(x,y) = emb^C(x,y) \setminus \bigcup_{e:\, x_e \notin C} \bigcup_{t \neq x_e} \{\varphi \in emb^C(x,y) \ : \ y_{\varphi(e)} = t\}.$$

Let $A_{e,t}$ denote the set inside the union. Note

$$emb^{C\cup\{t\}}(x',y) \subseteq A_{e,t} \subseteq emb^C(x',y) \subseteq emb^C(x,y),$$

where x' is obtained from x by replacing x_e by t. This is because the first term requires $y_{\varphi(e)} = t$ for all $x'_e = t$, the second requires it for only our special index e, and the last term does not require it at all. If $t \in C$ the first three terms are equal but the last two are not, and if $t \notin C$ then the last two terms are equal but the first three are not. More generally for distinct e_i:

$$emb^{C\cup\{t_1,\ldots,t_m\}}(x',y) \subseteq \bigcap_{i=1}^{m} A_{e_i,t_i} \subseteq emb^C(x',y) \subseteq emb^C(x,y) \tag{6.7.1}$$

where x' is obtained from x by replacing x_{e_i} by t_i. By inclusion-exclusion:

$$|\mathrm{ind}(x,y)| = \sum_{m\geqslant 0} \sum_{\substack{e_1,\ldots,e_m \\ distinct \\ t_1,\ldots,t_m}} (-1)^m \left| \bigcap_{i=1}^{m} A_{e_i,t_i} \right|$$
$$= \sum_{x' \geqslant_C x} (-1)^{d_{ham}(x,x')} \, |A_{x,x'}|,$$

where $d_{ham}(x,x')$ is the hamming distance, the number of entries x, x' differ, and:

$$A_{x,x'} = \bigcap_{e\,:\, x_e \neq x'_e} A_{e,x'_e},$$

The desired result then follows from applying inequality (6.7.1).

For (iii), note that $t \neq x_e = s$ by assumption so we always fall into the case where $t \in C$ and thus $A_{e,t} = emb^C(x',y)$. Now we can just apply inclusion-exclusion as in the previous part. □

Proof of Lemma 6.5.14 (Binary Projections of Nonbinary Arrays). Suppose $\varphi : |x| \to |y|$ is injective. Then by construction of our one-hot encoding $x_e = y_{\varphi(e)}$ if and only if $g_s(x_e) = g_s(y_{\varphi(e)})$ for all $s \in S$, and hence the first statement follows.

Now suppose that $emb(\cdot, y_N)$ converges and that $S = \{0, ..., n\}$. If $A = S$ then $g_A(y_N)$ is the array of all ones for every N, so the result is trivial. Thus without loss of generality assume $0 \notin A$. By Lemma 6.4.2, it suffices to show $emb^W(b, g_A(y_N))$ converges for an arbitrary binary array b. Define a set of finite arrays of the same size of b by:

$$\mathcal{X}_A^b := \{x \in S^{[|b|]^{(d)}} \;:\; x_e \in A \text{ if } b_e = 1 \text{ and } x_e = 0 \text{ otherwise}\} \subset g_A^{-1}(b).$$

Namely, $\mathcal{X}_A^b$ is $g_A^{-1}(b)$ restricted to arrays with entries only in $A \cup \{0\}$. We aim to show:

$$emb^W(b, g_A(y_N)) = \bigsqcup_{x \in \mathcal{X}_A^b} emb^{S\setminus\{0\}}(x, y_N) \tag{6.7.2}$$

To see why the latter sets are disjoint, suppose that $\varphi \in emb^{S\setminus\{0\}}(x, y_N) \cap emb^{S\setminus\{0\}}(x', y_N)$. Note by the definition of $\mathcal{X}_A^b$ that $x_e = 0$ iff $b_e = 0$ iff $x'_e = 0$. Furthermore, for each e where $x_e, x'_e \neq 0$ we have $x_e = (y_N)_{\varphi(e)} = x'_e$. Thus $x = x'$.

For the forward inclusion, suppose $\varphi \in emb(b, g_A(y_N))$. Define a finite array x by:

$$x_e := \begin{cases} (y_N)_{\varphi(e)} & b_e = 1 \\ 0 & b_e = 0 \end{cases}.$$

If $b_e = 1$ then $g_A(y_N)_{\varphi(e)} = 1$ and thus $(y_N)_{\varphi(e)} \in A$. Thus $x \in \mathcal{X}_A^b$, and by construction $\varphi \in emb^{S\setminus\{0\}}(x, y_N)$.

For the reverse inclusion, suppose $\varphi \in emb^{S\setminus\{0\}}(x, y_N)$ for some $x \in \mathcal{X}_A^b$. Then for each e with $x_e \neq 0$ we have $(y_N)_{\varphi(e)} = x_e \in A$. Since $x_e \neq 0$ if and only if $b_e = 1$ this implies $b_e = g_A(y_N)_{\varphi(e)}$ as desired. Thus $\varphi \in emb^W(b, g_A(y_N))$. This completes the proof of equation (6.7.2).

Dividing both sides of (6.7.2) by the number of injective functions $\varphi : |x| \to |y_N|$ yields:

$$emb^W(b, g_A(y_N)) = \sum_{x \in \mathcal{X}_A^b} emb^{S\setminus\{0\}}(x, y_N). \tag{6.7.3}$$

The result then follows by applying Lemmas 6.5.13 and 6.4.2.

TODO Conversely, suppose $emb(\cdot,\ g_A(y_N))$ converges for all $A \subseteq S$ and let x be a S-valued array.

Zack comment: Choosing $A = \{s\}$

Zack comment: Hopefully we can just reverse equation (6.7.2) to show that $emb^{S\setminus\{0\}}$ converges and then apply lemma 6.5.13 □

6.8 TODO

1. DONE! Understand what infinite bridges of exchangeable array processes look like. I suspect they themselves can be framed as some kind of exchangeable array process.

2. DONE! Understand what the extremal infinite bridges of an exchangeable array process looks like, and hence the Doob Martin boundary of such chains. Following ideas in [19] and the ballot chain paper, I imagine these are just the infinite bridges that are ergodic in the sense of exchangeable arrays. In terms of their representation functions f I feel these ought to be the ones where f depends trivially on the first coordinate. For example, with our graph sequences these are the ones corresponding to graphons $f(u, x, y, z) := \mathbf{1}\{z \leqslant w(x, y)\}$.

3. DONE! Determine how the Doob-Martin topology relates to the embedding density, which is a natural notion in both exchangeable array processes and our backwards exchangeable chains. Namely, we would like to say you converge in the Doob-Martin topology iff the embedding density converges for every fixed input (e.g. for graphons this is the morphism density).

   - DONE: Develop a notion of embedding density for our candidate limits: the representation functions of our exchangeable arrays. Analogous to graphons, $t(F, f)$ ought to have the property that it equals the average embedding density of F into a finite array sampled from f.

4. Can we metrize this topology like graphons manage to do using the cut metric? Does the cut metric make sense for *any* array (beyond just the adjacency matrices of our graph chains; possibly we have to restrict to binary arrays?)

   - Namely, it seems like we want to create some sort of metric on the sampling representations f of the exchangeable array. Since the f from the sampling representation is not unique per say we will have to define some sort of equivalence classes, analogous to how graphons are really only defined up to Lebesgue isomorphisms. Steve says a similar "unique up to isomorphism" statement is true for exchangeable arrays representations f and that the details are in Kallenberg (Theorem 7.28 in his book on Symmetry).
   - **GOAL:** relate the cut metric to the sampling representation for graphons we deduced above. Maybe this can give us an idea of how to define the metric in a more general setting. It may depend on what the state space E is of the array of course, so maybe we assume this is some kind of metric space in its own right.
   - **GOAL:** Show that convergence in this extended cut metric is equivalent to convergence of the embedding densities (and hence also in the Doob-Martin topology).

  - Find a suitable notion of embedding a finite array x into our space of representation functions, analogous to how finite graphs are embedded in the space of graphons just by discretizing the adjacency matrix. This should be done in such a way so that the embedding density $t(F, x)$ is the same whether we view x as a finite array or as a representation function.

- This would be really cool if we could do this, as this could feasibly give us a way of comparing the exchangeable arrays corresponding to completely different combinatorial processes.

  Daniel comment: This is an interesting idea. Maybe this would depend on the realization as an exchangeable array.

5. Once we generalize the cut metric, can we extend any pre-existing theory of graphons to our new exchangeable array processes (and hence other combinatorial stochastic processes?). e.g. Can we extend theory of large deviations of random graphs to large deviations of other combinatorial structures? Can we interpret certain graph properties like the existence of independent sets, cliques, etc in these new settings? This aim may seem a bit vague, but we have a lot of pre-existing theory of graphons to work with. Review how the cut metric and graphon convergence has been used to prove results in extremal graph theory or spectral graph theory, and try to extend some of these results to other combinatorial stochastic processes.

6. Try to relate the sampling representation of exchangeable arrays to some of the sampling representations we have so far in Doob-martin theory (e.g. Remy's chain, ballot chain, Polya's urn, etc.). Can we say something more in general on what kind of measures are induced?

7. It is easy to see every exchangeable array process is backwards exchangeable. Try to determine the proper constraints for the converse to hold: which backwards exchangeable chains can be encoded as an exchangeable array process, and try to find a general scheme for doing so. So far all the examples we have seen above fall into both frameworks, so the class of such chains must be fairly large.

8. Not every combinatorial stochastic process can be encoded by the "k-wise" relationships between its parts. Daniel gave the example of the hypergraph chain. This seems to require a theory of "infinite dimensional" exchangeable arrays. One avenue to explore would be how to make this rigorous, and if there is a similar theory of representation functions for such arrays.

   - Zack comment: As far as I can tell, Kallenberg's theory is not restricted to arrays of a fixed dimension d, unlike what is stated in Tim Allen's exchangeable arrays notes. Theorem 7.22 In [32] is stated for any random array indexed by ALL subsets of $\mathbb{N}$. Tim Austin has some notes in a paper called Exchangeable Random Measures which makes it a bit more clear what the representation function looks like (Theorem 1.2). It just seems like you require one middle-symmetric function for each dimension, and you reuse the randomness from the uniform random variables of the lower dimensions.

9. What we have done above extends the theory of graphons of dense graphs to some notion of "dense" combinatorial stochastic processes. The *graphex model* and the notion of *sampling convergence* discussed in [8] extends these notions to sparse graphs. We could try to do some of the same if we want.

10. One sort of vague idea I had was it seems is that we often design these Markov chains in order to generate the combinatorial objects uniformly. The relation to exchangeable arrays comes from the fact that the dynamics of our chain has some sort of invariant that is perserved as the chain progresses (e.g. ancestry in Remy chain), use that to encode my process as an exchangeable array, and develop my limit theory from there. Could we in some sense work backwards? Like if I had some sort of invariant, could I define an array using it. Such arrays already have a natural Markov chain associated to it which we talk about. Could we say this generates the combinatorial objects uniformly, or maybe at least that some bridge of this chain can do this? Under what conditions does such a bridge exist?

11. The paper [23] gives a representation of exchangeable hierarchies in terms of exchangeable arrays. It would be interesting to explore what the associated markov chain is, and what its notion of embedding would be. They also in a sense characterize the extreme points of the limiting object as those exchangebale hierarchies on $\mathbb{N}$ which are independently generated. Can we deduce this immediately from our characterization of the extremal exchangeable arrays (the ergodic ones; specifically the equivalent independence condition stated in theorem 6.3.4).

12. In the papers [23] and [19], they encode their exchangeable processes as exchangeable arrays, and use these to build real trees from which the processes are encoded as sampling from these real trees. The former paper relies on a simple stick-breaking construction, and I wonder if this would be possible to do in general. This would allow us to view all exchangeable array processses as some sort of sampling process from a real tree. The array they use is:

$$X_{ij} = 1 - \lim_{n\to\infty} \frac{\#\text{leaves below mrca of i,j}}{n}.$$

similar to the distance function steve uses in his Remy chain paper. If we define a 3-dimensional binary array $X_{ijk} = \mathbf{1}\{\text{k is below mrca(i,j)}\}$ then this basically corresponds to taking an average of the last coordinate. Maybe we could do the same in general for a 3D binary array, or maybe any binary array?

13. Is there a way of encoding the randomness in exchangeable array processes to ensure the underlying combinatorial objects are generated uniformly? For example, in the Remy chain, [19] the underlying algorithm generates trees uniformly in an iterative fashion, and [40] extends this to d-ary trees.

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